

Microstructural Financial Modelling: Point Processes and Reinforcement Learning

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Dedicated to the journey before the destination.

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List of Publications

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This thesis has produced the following research papers (in chronological order):

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- **Nick Firoozye:** Academic supervision; guidance on research direction; critical feedback on methodology, analysis, and presentation.
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These publications were presented at several venues during the course of this PhD for soliciting feedback and critical reviews. Some selected talks are mentioned below:

- (a) An Impulse Control Approach to Market Making in a Hawkes LOB Market
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 - Bridging Stochastic Control And Reinforcement Learning Workshop (poster), Cambridge, Nov’25
 - SIAM Financial Mathematics Conference, Miami, Jul’25
- (b) No Tick-Size Too Small: A General Method for Modelling Small Tick Limit Order Books
 - ECO2STAT Workshop, Cagliari, Jul’25
 - Quantitative Finance Workshop, Palermo, Apr’25
 - Capital Fund Management Seminar, Paris, Jan’25
 - Bachelier Colloquium, Metabief, Jan’25
 - London-Oxford-Warwick Workshop, Oxford, Jan’25
 - Algo-Defi Workshop, Milan, Nov’24
 - SFMES Workshop, ICAIF Conference, New York, Nov’24
- (c) Limit Order Book Dynamics & Order Size Modelling Using Compound Hawkes Process
 - Oxford SML Finance Seminar, Oxford, May’24
 - Quantitative Finance Workshop, Bologna, Apr’24

- CFM Workshop on Market Microstructure (poster), London, Dec'23
- UCL Unwinding Complexity Conference (poster), London, Jan'24

Abstract

This thesis explores the intersection of high-frequency market microstructure modelling, stochastic optimal control and reinforcement learning (RL) for optimal decision making in electronic trading. Specifically, it focuses on developing realistic, data-driven models of Limit Order Book (LOB) dynamics using point processes, mathematically formulating impulse control settings for algorithmic trading tasks in a pure jump-driven market, and designing RL algorithms capable of learning robust strategies within such stochastic and reactive environments. The research addresses the challenge of building adaptive trading agents in low signal-to-noise ratio (SNR) settings—where traditional RL methods often exhibit poor sample efficiency, convergence to local minima, and limited generalizability.

Conducted in collaboration with JP Morgan Chase & Co., this work investigates how realistic jump-driven microstructure modelling and impulse control formulations can bridge the gap between theoretical stochastic control and practical algorithmic trading. Among the various LOB modelling philosophies in the literature explored in this thesis, we especially focus on and make use of the Hawkes process methodology. This is due to the memoryfulness, high adaptivity and mathematical tractability of this class of LOB models. One of the core tenets of this thesis is to rely on minimal assumptions on the nature of financial markets, perform scientifically rigorous studies in a data-driven way, and to be cognizant of the practical applicability of the trading strategies developed during the course of this research.

The thesis is structured around three core investigations:

1. **Limit Order Book Simulation:** The first study constructs a realistic LOB simulator to serve as the training environment. We ensure empirical consistency by validating against multiple stylized facts such as market impact and spread dynamics. The simulator employs a multidimensional compound Hawkes process to model order arrivals, cancellations, and executions.
2. **Generalizing the Simulator: Small-Tick LOBs:** The second study extends the Hawkes-based model to capture how microstructural properties evolve with varying relative tick-sizes. We uncover power-law relations in spreads, price change size, and market depth against *relative tick-size*, and propose a unified Hawkes process model that reproduces sparsity, multi-tick price moves, and

humped volume profiles observed in small-tick assets.

3. **Impulse Control and Reinforcement Learning solutions to Market Making:**

The final study leverages the realistic simulator to train first deep learning approximations to the impulse control problem, and second, RL agents that approximate the stochastic optimal control problem of market making. The proposed RL framework demonstrates adaptability to endogenous market feedback and is able to perform with very high sharpe-ratios with a shallow neural network architecture.

This thesis contributes to science by:

1. **Jump-process Stochastic Modelling:** Creating a realistic LOB dynamics simulator matching empirical stylized facts across relative tick-sizes. Specifically, we develop a novel parameter estimation strategy for the long-memory Hawkes process.
2. **Adapting the Modelling for High-Dimensional Stochastic Processes:** Proposing a unified Hawkes-based model capturing several stylized facts across assets of various relative tick-sizes. We build a modelling framework which is fully tractable mathematically, with non-diffusive quantities of interest (such as mid-prices), and exhibits phase transitions for the three classes of assets we examine.
3. **Impulse Control and its Approximation in purely Jump-Driven Environments:** Formulating the market making objective in the LOB as an impulse control problem, and solving the associated value function using a deep learning method. Next we approximate the impulse control problem by developing an RL framework that integrates stochastic control theory to learn adaptive market making strategies.

Impact Statement

Electronic trading constitutes more than 92% of all trading activity in today's markets - most of which is done using the Limit Order Book mechanism. With the rise of fully automatic trade execution engines, the need for intelligent decision making algorithms like RL is high. Some of the pitfalls of traditional limit order placement engines are heavy dominance of human expert driven heuristics and failure to adaptive to changing market conditions. My proposed research would have the following impact on the current electronic trading research:

1. **Cross-Asset Stylized Facts Study:** We quantify how LOB properties evolve with changing relative tick-size, uncovering power-law relations in various stylized facts, and providing a systematic classification of large-, medium-, and small-tick assets.
2. **Limit Order Book Simulator:** We develop a realistic LOB simulator with a focus on matching several important stylized facts of the empirical data. This will be beneficial to the practitioners in backtesting their trading strategies in a environment which is reactive to their trading activities.
3. **Modelling High Dimensional Point Processes:** We introduce a unified Hawkes-based LOB model that captures sparsity, multi-tick moves, and realistic volume shapes, enabling simulations that smoothly transition between large- and small-tick market regimes.
4. **Reinforcement Learning in Stochastic Environments:** We showcase the applicability of a novel reinforcement learning algorithm to approximate the stochastic optimal control problem involved with market making in the lit limit order book.

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Chapter 1.

Introduction

This thesis investigates first, the modelling of high-frequency financial data using pure jump dynamics, specifically using point processes, and second, the creation of a novel Reinforcement Learning (RL) technique specifically aimed at tackling highly stochastic environments by taking inspiration from the existing RL techniques as well as stochastic optimal control theory. Reinforcement Learning algorithms were built with a Limit Order Book simulator as its experimental environment. Efforts were made to ensure that the simulator can match the empirical stylized facts with a degree of empirical fidelity. A particular focus was on creating a simulator that has the price impact of orders as a feature. This enables us to reasonably determine that our RL agent will perform appropriately in the real-world setting as well, and not just in the simulator.

The focus of the first two experiments in this thesis is to first, identify the most prominent patterns and statistics of high-frequency, specifically tick-by-tick, equities data - more commonly known as ‘stylized facts’ of the data (Chapter 3). This tick-by-tick data consists of Limit Order Book (LOB) snapshots and event messages from the LOBSTER data provider (LOBSTER academic data - powered by NASDAQ OMX). Building on these observed stylized facts, the first experiment (Chapter 4) deals with using the compound Hawkes process - a memoryfull point process - to model the large-tick limit order book. An exact classification rule of large- vs small-tick stocks has been provided in Chapter 3. To generalize this model to work in the high-dimensional setting of the small-tick LOB, we perform an approximation by forming the Meta-Queue Hawkes process model in the second experiment (Chapter 5). We tackle the objective of market making - i.e. providing liquidity in the LOB with several round trip trades which are profitable on average - in Chapter 6. We pose the market making objective as a stochastic optimal control, specifically an impulse control problem and try to solve the resulting optimal policy and value function using two methods. We form the Hamilton-Jacobi-Bellman PDE and the associated Quasi-Variational Inequality (HJB-QVI) and try solving it using a mesh-free deep learning based numerical method. Secondly, we approximate this

HJB-QVI’s solution using reinforcement learning and use policy gradients to learn directly the optimal policy. In our experiments we see that the reinforcement learning market maker learns effectively and quickly converges to a highly profitable strategy while being conservative in its inventory holdings throughout the market making episode.

The majority of this investigation was conducted in collaboration with J.P. Morgan Chase & Co. under the supervision of Prof. Philip Treleaven, Dr. Nick Firoozye and Dr. Jonathan Kochems, facilitating expert guidance and validation, and confirming the market demand for this investigation.

1.1 Research Motivation

Machine Learning and Reinforcement Learning techniques have proven to be one of the most promising prospects in designing autonomous agents. Typically the success of these techniques have been prominent in areas of image classification, natural language processing and autonomous agents in game theoretical environments like role-playing games, chess, and Atari Games. All of these tasks come into the category of high signal-to-noise ratio (SNR) tasks. Reinforcement Learning (RL), particularly, has seen many challenges when applied on low SNR tasks such as applications in the financial domain.

The major culprit of almost all the problems with conventional RL, when taken into the context of a stochastic environment, is the extremely noisy map between environment states and optimal actions (see for instance Gueant [2023]). The RL technique, at its core, tries to estimate this map either directly, using value functions, or indirectly, using policy gradients. When applied to the low SNR environment of trading securities these techniques are in large part unable to estimate this map well. In an extensive review of RL’s usages in finance, Hambly et al. [2023] demonstrate that the general pattern is retro-fitting state-of-the-art RL models that have seen success in high SNR environments to finance. Reinforcement Learning, though a very powerful technique to build a trading agent, suffers with problems of poor sample efficiency (problems with scalability), local minimas (complexity) and over dependence on a fine tuned set of hyperparameters (lack of robustness).

One promising candidate for such a low SNR environment is the Lit Limit Order Book (LOB) market. The problem of trading in the LOB is quite suitable for an RL game-like setup - there are several competing agents, the rewards, though stochastic and sparse, are observable in monetary terms, the structure of the state-space can be constructed in a way such that it is Markovian (although it is partially observable) and automation is extremely useful for limit order placements.

Today, most Equities markets implement an electronic, continuous-time double-auction mechanism based on the essential idea of a limit order book. This is where essentially all trading happens, particularly in equities single-stocks, - buyers and sellers form a demand-supply ecosystem. They look at what the market perceives of the tradeable entity by analysing the current state of the LOB, they interact with and thereby alter the LOB by placing orders in it and, if a buyer-seller match happens this trade forms the price of the tradeable at the very fundamental level. Limit orders pegged at a wide range of different prices can arrive or be cancelled at a blindingly high frequency. Incoming market orders match with limit orders and remove them from the order book. There also exists a feedback loop between liquidity and order flow - the state of the LOB affects the arrival and cancellation rates of the limit and market orders (Bouchaud et al. [2018], Kissell [2021]).

There are a number of challenges in modelling the LOB ranging from issues related to model complexity, difficulty in replicating the statistical properties of empirical data, and several mechanical issues stemming from the internal working of the exchanges such as halts in trading, open, intraday and close auctions, hidden orders, queue priority and dark pools. For an in-depth analysis of the challenges faced in LOB modelling, we refer the reader to Gould et al. [2013]. Despite, or due to, these challenges, modelling and simulating LOBs is of quite high importance for researchers and practitioners alike. An especially noteworthy use-case of a LOB simulator is for backtesting (or training) algorithmic trading strategies. The reason being having a simulator enables the availability of a richer data set for the strategy to run on and be refined on. Since each security's price has had just one realization of the various possible time evolutions of its LOB dynamics, if the trading strategy were to be fitted on just this one trajectory, issues of overfitting and thereafter lack of true out-of-sample performance will be apparent (White [2000], Sullivan et al. [1999]). Avoiding overfitting ideally can be done by adding more data to the training set though generating more data is not trivial without knowing the generating process of the time series. One possible solution to this problem could be synthetic data using simulators. We note that training the strategies purely on a simulator might induce biases in the trading strategy since the simulator can never be perfectly representative of real data. This challenge can potentially be solved in two ways. The first and foremost is making sure the simulator is representative enough of the statistical properties of real world observed phenomena (i.e. 'stylized facts'). This in itself encompasses the entire field of Order Book simulations - can a simulator be built which can replicate the distributions of the stylized facts observed in nature and at the same time is parsimonious? The second is the usage of real world data to do

true out-of-sample testing of the trading strategy, or alternatively, combining the simulated data with real world data in the training data of the trading strategy. This will enable the practitioner to effectively ground their strategies in reality and avoid the pitfalls of backtesting on purely historical data as well as avoid inducing biases because of the simulator’s lack of realism.

In summary, one of the major challenges in the real-world applicability of trading strategies is twofold. First, much of the financial mathematics literature relies on strong modeling assumptions, such as continuous controls and stationary, continuous market dynamics, which are often at odds with the discrete and highly non-stationary nature of real markets. Second, in the computational finance literature, many proposed solutions take the form of black-box models, with issues such as lack of interpretability and susceptibility to overfitting and look-ahead bias. This thesis aims to partially address these challenges by developing models that are closer to observed market microstructure while remaining mathematically tractable. By grounding the proposed methods in explicit stochastic control formulations and transparent model dynamics, the approach seeks to balance realism with interpretability. While not claiming to fully resolve these issues, the thesis contributes towards more explainable and robust frameworks for optimal decision-making in limit order book environments. Solving the problem of optimal participation in the highly stochastic environment of a limit order book has benefits beyond monetary gains. Despite its complexity and the large number of interacting market participants, such a setting provides a natural testbed for decision-making under uncertainty. Progress on this problem would improve our understanding of reinforcement learning, particularly in highly stochastic environments with sparse rewards, as well as contribute to the theory and practice of optimal control.

Prominent RL paradigms employed in finance are summarised in the following. For a detailed description of these techniques we refer the reader to the book by Capponi and Lehalle [2023]. Actor–Critic (AC) algorithms became popular due to their estimations of both the policy gradient and value function together. They have been used for continuous control as well. Twin Delayed Deep Deterministic Policy Gradient (TD3) have been used for noise-resilient continuous actions. Offline RL has been gaining popularity in finance for training on fixed datasets with a behavioral policy. Model Predictive Control (MPC) is another interesting field and is used primarily for dynamically adapting to changing environments. Online RL for continuous updates, contextual bandits for partial feedback settings, and entropy-regularized SOC formulations that embed exploration directly into the control objective are some other directions of research in reinforcement learning that seems

to show some promise. Despite these advancements, several fundamental issues persist when applying traditional RL methods to LOB trading:

1. **Partially Observable Markovian or Non-Markovian State Space:** The LOB exhibits long memory effects driven by self- and cross-excitation among order flows.
2. **Non-Stationarity:** (a) Endogenously induced through feedback effects such as market impact; (b) Exogenously driven by volatility regimes, macroeconomic announcements, and news events.
3. **Simulation–Reality Gap:** RL agents trained in synthetic environments often fail to generalize to live market conditions.
4. **Ill-Defined Reward Structures:** Profit-maximization objectives often omit risk, leading to unstable and economically unrealistic strategies.
5. **Time Asynchronicity:** Financial markets evolve in event time rather than fixed time intervals, leading to asynchronous state transitions.

To address these challenges, the present thesis proposes the following methodological steps:

1. **RL on a Hawkes Process LOB:** The proposed framework leverages the endogenous self- and cross-exciting nature of Hawkes processes to mitigate non-stationarity¹ and approximate Markovianity within the LOB dynamics.
2. **Risk-Aware Reward Design:** A quadratic inventory penalty is incorporated into the reward function to capture risk aversion and control inventory.
3. **Asynchronous Learning Dynamics:** Value and Q-function updates are performed *after* each action to reflect the asynchronous nature of market evolution and reaction delays.

1.2 Research Objectives

The major objectives of this research are:

¹The Hawkes Process is a stationary point process however it still is able to capture the endogenous feedback cause of the general non-stationarity of the LOB markets. For exogenous events, one must model these shocks as well. This is beyond the scope of this thesis however the interested reader may refer to Rambaldi et al. [2015] for instance.

1. **LOB High-Fidelity Simulator:** Develop a novel limit order book dynamics simulator which is both able to replicate several important stylized facts of the empirical order book and is reactive to an agent's trading activities (i.e. market impact is present).
2. **Realistic Stochastic Control:** Solving the market making problem in the context of a Hawkes Process model of the market order flow and setting up the stochastic optimal control problem.
3. **Tradeable Reinforcement Learning Solutions:** Development of a novel methodology of approximating stochastic optimal control solutions using reinforcement learning techniques.

1.3 Research Experiments

This thesis consists of the following three investigations:

1. **Experimental Setup: Limit Order Book dynamics simulator:** The first study forms the experiment environment for the RL agent to train on. To ensure real world applicability of the RL agent, we made efforts to create a realistic simulator by testing against several stylized facts including presence of a concave market impact. We use a novel formulation of the multidimensional compound Hawkes process to model the behaviour of the orders.
2. **Generalizing the simulator:** In the second study we tackle the challenging problem of working with small-tick equities data in the Hawkes process setting. The key challenge is the high dimensionality and therefore low calibration quality of the Hawkes process setup for these equities. We develop a meta-queue model to approximate the LOB dynamics at a meta level and showcase several capabilities of the simulator with this setup.
3. **Reinforcement Learning in a limit order book market:** The final study makes use of the simulator as its training environment. Market-making strategies are sought from the environment by using a novel technique of combining stochastic optimal control with modern RL techniques. We test the agent's success both in the simulator and against several baselines.

1.4 Scientific Contributions

This thesis presents the following contributions to science:

1. **Jump-process Stochastic Modelling:** Creating a realistic LOB dynamics simulator matching empirical stylized facts across relative tick-sizes. Specifically,

we develop a novel parameter estimation strategy for the long-memory Hawkes process.

2. **Adapting the Modelling for High-Dimensional Stochastic Processes:** Proposing a unified Hawkes-based model capturing several stylized facts across assets of various relative tick-sizes. We build a modelling framework which is fully tractable mathematically, with non-diffusive quantities of interest (such as mid-prices), and exhibits phase transitions for the three classes of assets we examine.
3. **Impulse Control and its Approximation in purely Jump-Driven Environments:** Formulating the market making objective in the LOB as an impulse control problem, and solving the associated value function using a deep learning method. Next we approximate the impulse control problem by developing an RL framework that integrates stochastic control theory to learn adaptive market making strategies.

1.5 Thesis Structure

This thesis is organized as follows. Chapter 1 introduces the motivation, research objectives, experimental design, scientific contributions, and provides an overview of the thesis structure. Chapter 2 presents a comprehensive background and literature review, covering foundational concepts in LOB simulation, stylized facts, the role of market impact and reinforcement learning and stochastic control applications in market making. Chapter 3 investigates the stylized facts of limit order books, with a particular focus on the role of relative tick size and its influence on key market characteristics such as bid-ask spread, mid-price changes, and order book shape.

Chapter 4 details the methodology and empirical results of simulating LOBs using the Compound Hawkes Process, including model calibration and comparative studies. Chapter 5 extends the simulation framework to small-tick LOBs, introducing a meta-queue Hawkes model and analyzing its properties through simulation studies. Chapter 6 explores reinforcement learning and stochastic optimal control in LOB environments, formulating the market making problem as an impulse control task and presenting both deep learning and RL-based solution approaches.

Finally, Chapter 7 summarizes the main findings and outlines directions for future research. The appendices provide additional illustrations and technical details for the models and assumptions discussed throughout the thesis.

Chapter 2.

Background and Literature Review

This chapter summarizes the current state of the art in firstly, limit order book modelling, and secondly, RL and Stochastic Optimal Control techniques used in market-making. The discussion centres on themes like algorithmic trading and market making using automated trading engines.

The thesis is organized into three interconnected studies, each progressively deepening the analysis of high-frequency market dynamics and reinforcement learning–based decision making. The first study develops a realistic simulator of Limit Order Book (LOB) dynamics; the second extends the model to analyze small-tick regimes; and the final study integrates reinforcement learning (RL) techniques to solve the optimal market making problem within this simulated environment. Each study builds upon the findings and methodological foundations of the preceding one, bringing together a cumulative narrative that bridges microstructural modelling and algorithmic learning.

Given the shared theoretical and empirical foundations underlying all three studies—particularly in stochastic control, Hawkes process modelling, and market microstructure—the ‘Background and Literature Review’ chapter is structured by research theme rather than by experiment. This organization avoids repetition and provides a unified conceptual base that supports the development of all subsequent chapters. In addition, each study includes a focused literature section that expands upon specific strands of research introduced in the broader review, discussing prior work most relevant to the respective methodological and experimental contribution

2.1 Limit Order Book Simulation

This section provides an overview of the existing approaches to modelling Limit Order Book (LOB) dynamics, with an emphasis on their methodological foundations and empirical validation. We begin by classifying the literature according to the underlying modelling paradigm—ranging from point process–based formulations and agent-based models to deep learning architectures and stochastic differential equation (SDE)–driven frameworks. In particular, recent advances in generative modelling,

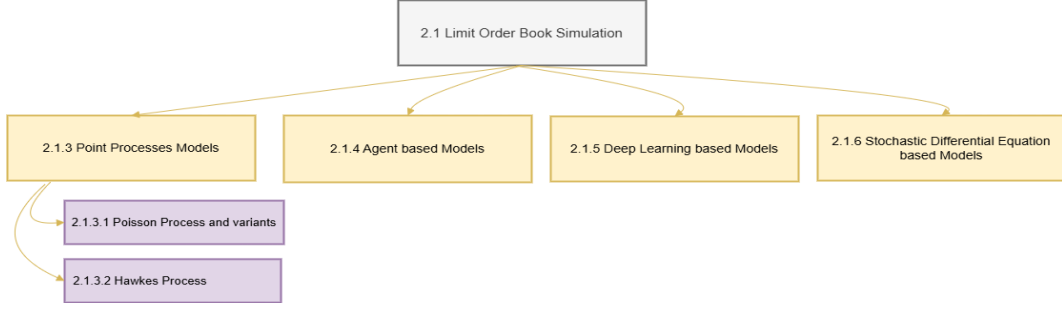


Figure 2.1: LOB Modelling Taxonomy

such as Generative Adversarial Networks (GANs) Goodfellow et al. [2014], Mirza and Osindero [2014], Arjovsky et al. [2017] and Large Language Models (LLMs) Nagy et al. [2023a], have led to the development of novel data-driven simulators capable of reproducing complex market microstructures. The flowchart in Figure 2.1 shows the taxonomy presented in this section.

The review then examines a wide range of stylized facts—empirically observed statistical regularities of LOB data—that serve as priors or calibration targets in the design of such models. Based on this survey, we identify a subset of stylized facts that are most relevant for applications in algorithmic and high-frequency trading. Furthermore, we compare the quality-of-fit tests employed in different simulation frameworks, evaluating how well each model reproduces real-world microstructural patterns.

Finally, an important dimension along which these simulators are assessed concerns their responsiveness to exogenous trading activity. This property, often referred to as market impact awareness, is critical for practical deployment. A simulator that neglects the feedback effects of external trades—effectively assuming zero market impact—can exhibit strong in-sample calibration yet fail to generalize out of sample, leading to poor performance in live trading conditions Biais et al. [1999], Foucault et al. [2005], Cont et al. [2014].

2.1.1 Limit Order Books

In this section we provide a brief mathematical description on Limit Order Books and specifically how the time evolution of the order book dynamics can be described in a mathematical fashion. For a detailed overview, we refer the reader to Abergel et al. [2016]. The order book consists of discrete price levels at one ‘tick’ difference from each other. This minimum difference in price levels is known as ‘tick-size’ and is often specified by the regulators for each exchange. Each price level can have a non-negative integer number of ‘orders’ resting there with different ‘sizes’ and

different ‘sides’. Orders here refer to the unexecuted (or ‘unmatched’) limit orders at that price level. Each order shows the intention of a market participant to trade at the specified price level, a quantity equalling the order size, and the direction of trade (buy or sell) specified by their order side. Since the order book consists of unmatched orders, the orders with intentions to buy are always at lower price levels than the order with intentions to sell. The buy section is known as ‘bid’ and the sell section is known as ‘ask’. In Figure 2.2, for example, we show the aggregate order sizes at each price level for Apple at a random time of the day.

The price level at best bid is known as ‘bid-price’: P^B and similarly at best ask, we have ‘ask-price’: P^A . The distance in price units between bid and ask is known as ‘spread’: $S := P^A - P^B$. The ‘mid-price’ is a theoretical price that signifies the average between P^A and P^B : $P := \frac{P^A + P^B}{2}$. There are three categories of orders that can be placed in an order book: limit orders which are a buy (resp. sell) side order which has a price level equal or lower (resp. higher) than the lowest ask (resp. highest bid), market orders which are orders which match the requested bid/ask in the order book and remove the pre-existing limit order and finally cancel orders which are cancellations of limit orders without any execution. Note that in many markets, not all of the liquidity (i.e. unexecuted orders) is displayed in the LOB and we have the possibility of hidden orders and hidden executions or hidden trades.

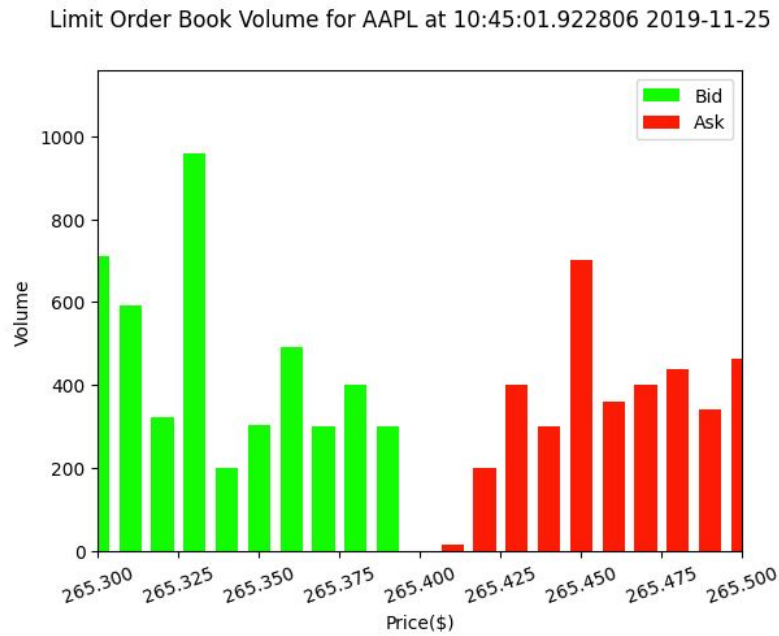


Figure 2.2: Snapshot of a Limit Order Book for Apple on NASDAQ

The order book is not stationary in time - it evolves with the three order types mentioned previously arriving randomly across the day. These orders evolve the

order book which in turn evolves the price processes of the security. Hence, the LOB dynamics is the most granular level of price process formation. In Figure 2.3, we portray the top 10 levels of the LOB on each side and their time evolution for 5 minutes. The darkest hue corresponds to the best bid/ask and the lighter hues reflect the deeper levels. We also provide marks for trades (both visible and hidden).

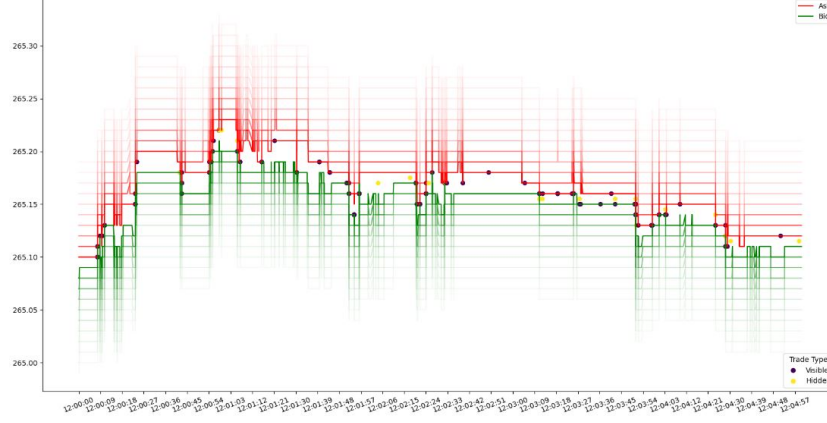


Figure 2.3: Time evolution of the top 10 levels of a Limit Order Book for Apple on NASDAQ from 12:00 to 12:05 PM on 2019-11-25

Dynamics: We refer to, and adapt from, Horst and Paulsen [2017] in their treatment of the order book's time evolution by defining the order book state as:

$$S_t(x) := (B_t, A_t, v_{b,t}(x), v_{a,t}(x))$$

where B_t and A_t are the best bid/ask prices respectively and $v_{b/a,t}(x)$ is the order book volume at x (in units of the resp. currency) distance away from the mid-price P_t . t here is the discrete time index. They define 8 types of events (bid/ask) which can change the state of an order book corresponding to Market Orders (bid/ask: **A/E**), Limit Order (in-spread) (bid/ask: **B/F**), Cancel Order ¹ (bid/ask: **C/G**) and Limit Orders (not in-spread) (bid/ask: **D/H**). The evolution of the order book at the event level can be described by the following:

$$S_{t+1} = S_t + \mathcal{D}_t(S_t)$$

where $\mathcal{D}_t(\cdot)$ is a random operator which depends on the dynamics that each 8 kinds of events induce on S_t . The dynamics induced are for example, if a queue-clearing buy/sell market order arrives, the ask/bid price moves up/down by one tick (or

¹Horst and Paulsen assume Market Orders which do not deplete a price level are the same as Cancel Orders. Here we do not make that assumption.

multiple ticks if the price levels near the best are empty), if a smaller market order arrives, it will change $v_{b/a,t}$ but not the prices, if an in-spread bid/ask limit order arrives, the bid/ask price moves up/down by one tick (or multiple ticks if the in-spread order is placed farther from best bid/ask) and so on for all 8 types of events.

The LOB state is partially observable and is often modelled as a Markovian system. Several statistical properties of interest of the order books are studied in the literature and we provide a brief description on some of them relevant to order book modelling in the following section.

2.1.2 Stylized Facts

Stylized facts are the various observed statistical properties of the order book or one of the order book's features such as mid-price, spread, etc. Since the order book state itself is partially observable, studying the statistical properties of the order book dynamics is quite useful while designing simulators - both as priors for the model as well as empirical ground truth to test the goodness of fitness of the simulator. We enlist some of the most commonly used stylized facts below.

In their seminal paper Bouchaud et al. [2002] outline two of the most important stylized facts in literature - order flow statistics and average order book shape. Cont [2011]'s expansive survey lays down a number of other observations of the market:

1. Price changes, on a small enough timescale, are autocorrelated negatively at the first lag and then uncorrelated in further lags.
2. Trading volumes are heterogenous and strongly autocorrelated.
3. Trading volumes exhibit intraday seasonality strongly.
4. Order flow is clustered in time. This implies durations between orders are autocorrelated and there is positive cross-correlation among arrivals of different order types.

These observations can be detected by measuring certain statistical properties of the LOB dynamics which are the so-called stylized facts. More recently, Vyetenko et al. [2020] provide a list of recommended stylized facts for order book simulators. We briefly describe some stylized facts below:

1. **Empirical Distributions:** densities of price, returns (particularly of note is the long-tailed distribution of returns), order volume, arrival rates, joint density of bid and ask queue sizes, time-to-fills.

2. **Autocorrelation of returns:** $\text{corr}(r_{t+\tau, \Delta t}, r_{t, \Delta t})$ is the general formulation where Δt is the time step of return calculation (which exhibits behaviours like vanishing autocorrelation of returns at larger time scales), alternatively the autocorrelation of absolute returns is another interesting aspect where we see a slower decay. Autocorrelation of the squared returns: $\text{corr}(r_{t+\tau, \Delta t}^2, r_{t, \Delta t}^2)$ is generally taken as a measure to detect ‘Volatility Clustering’.
3. **Correlations:** Volatility and Volume have positive correlations while Volatility and Returns have a negative correlation in empirical data.
4. **Intraday seasonality:** Volumes have a signature U-shaped intraday seasonality with increased trading at Open and Close compared to mid-day.
5. **Signature Plots:** This is defined as the relation between Volatility and Sampling Frequency - generally the empirical observation is that the signature plot decays quite slowly.
6. **Average Shape of the Book:** The mean of volumes at each price level with respect to mid-price is the quantity of interest here. Generally we observed the so-called ‘M’ shaped average shape for high spread stocks while a inverted ‘V’ shape is observed for low spread stocks.
7. **Price Paths:** The mid-price/ask-price/bid-price from the simulated order book is plotted against time in this stylized fact for a number of independent trials and comparisons are made to the empirical price paths observed.

2.1.3 Point Processes Models

The order book in its essence can be thought of, mathematically, as an aggregate of several individual orders arriving at different point of times. It is hence quite natural to think of the order book as a queueing system and so there has been a plethora of models using Point processes to model the individual orders’ arrivals. A point process has an associated counting process ($\equiv N_{t, t+\Delta t}$) which is the number of events occurring in $(t, t + \Delta t]$. With the usual conditions defined and satisfied on a complete probability space $(\Omega, \mathcal{F}, P)$, we define the intensity function λ_t by:

$$\lambda_t := \lim_{\Delta t \rightarrow 0} \frac{P(N_{t, t+\Delta t} > 0 | \mathcal{F}_t)}{\Delta t} \quad (2.1)$$

2.1.3.1 Poisson Process and variants

Poisson Process modelling assumes that the order arrivals are independent of each other. There are several applications of the Poisson Process in the queueing systems literature so it is a very natural choice to model the LOB with a collection of Poisson Processes as well. Usually, the Poisson Process is classified as a ‘zero-intelligence model’ i.e. a model which uses no prior information about the financial market or expert heuristics. However, as we will see later in this section, several ways of adding in priors have been formulated in the literature to make the Poisson Process models richer and more representative of the empirical observations.

Zero-Intelligence Models: Bouchaud et al. [2002] show that using a zero-intelligence model, they are able to match the two stylized facts, which are the shape of the order book and order flow arrival statistics, they investigate quite well. Smith et al. [2003] treat the order flow of Limit Orders and Market Orders as sampling from a uniform probability distribution with Cancels occurring with a constant probability per unit time. They further develop a stochastic model of accumulated volumes from this order flow.

Luckock [2003] showed results of first order approximations’ of the asymptotic behaviour of the Poisson arrival model. They show derivations of expressions for depth of the order book, time to fill and optimal order type. In Cont et al. [2010], the authors develop a model for the LOB by assuming Poisson arrival processes for Limit Orders, Market Orders and Cancellations. The arrival rates for Limit Orders & Cancels depend inversely on the distance from opposite side’s best quote, Cancels further depend on the number of outstanding shares at a level and Market Orders’ arrival rate is considered to be constant. The authors show, using Laplace Transform, the probabilities for quantities of interest like direction of price move, making the spread and filling time conditional on the current state of the order book can be calculated analytically. Building on Cont et al. [2010], Gao and Deng [2018] use fluid approximations of the above model and form a law of large numbers for the order book shapes. Similarly Kelly and Yudovina [2018] also use fluid approximations to study a model with orders having Poisson arrivals with random price drawn from a stationary distribution for orders. Further in Cont and de Larrard [2013], the authors show that with a Poisson arrival queueing system for quote dynamics, the price dynamics of the security can be thought of as a sum of independently and identically distributed (IID) random variables which, under the central limit theorem, forms a diffusion process. Finally, Abergel and Jedidi [2011] show a detailed analysis of price dynamics converging to a Brownian motion with a Poisson queueing system

model for the LOB.

Variable order intensity Poisson models: Moving away from the zero-intelligence approach, one way of adding priors to the Poisson model is to have non-constant order intensities. Hult and Kiessling [2010], for example, use a Poisson arrival model with Limit and Cancel orders' intensities being dependent on distance from mid-price while Market Orders' intensity is kept constant. Further to sample the orders' sizes, they use a stationary exponential distribution. They use this model to building a Markov chain model of the LOB and further they create optimal trading strategies using it.

Huang et al. [2015] study a variety of models for ask and bid queues around a fixed reference price - the first model they study is a Poisson Process model. They assume independence between ask and bid queues - here the arrival rates depend on the current queue size instead of being constant. This is what they call a 'queue-reactive' model which is quite popular in practice. Lu and Abergel [2018b] build on the above model and propose a non-Markovian order flow dynamic, albeit still using Poisson arrivals, by considering the order flow intensities to be dependent on not only the current state but also the previous history of order flow which led to this current state. They also address the limitations of having unit order size by consider limit orders sizes following a geometric distribution, cancels a truncated geometric and market orders a mixture of geometric distributions with Dirac delta functions for multiples of 50 to account for traders' preference over round numbers. They further proposed that in case of a queue depletion, the new limit order not only depends on the side of the cleared queue but also on the past removal events.

Discussion: Despite their simplicity and vast variability, Poisson arrivals do not fit well with some of the observed stylized facts. For example, the duration between orders are autocorrelated which leads to a clustering effect which Poisson processes are unable to explain. The core issue seems to be the assumption that all orders are independent which is generally contradictory to the practitioners judgement. Queue-reactive models do relax those assumptions to some extent but are still lacking in some other dimensions which we will highlight in the next section such as the endogeneity of the order arrivals. For a more detailed analysis, we refer the reader to Cont [2011] and Abergel and Jedidi [2011]. However due to their explainability and their mathematically convenient behaviour under scaling limits, Poisson models remain quite popular.

2.1.3.2 Hawkes Process

Hawkes Process as a way of modelling the LOB queueing system proves to be solving some of these challenges that Poisson Processes have. In their comprehensive review and tutorial, Bacry et al. [2015b] outlay the major ideas of the Hawkes Process, its mathematical theory, some of its crucial properties and finally applications including a detailed review over the Order Book models. Furthermore, they provide insights into calibration methodologies for empirical fitting and testing. The major two areas of significant improvement we see in Hawkes Process methods compared to Poisson methods is first, volatility clustering effect is observed in Hawkes Models and second, the Epps Effect which is the zeroing down of covariance between two assets as well limit the timescale to zero. We note that Hawkes Processes inherently have endogenously excited order flow as well as an implicit form of market impact. A more detailed study on the market impact of Hawkes process models is presented in Section 9. More recently Hawkes [2018] reviews the financial applications of Hawkes Processes.

Mathematical overview: Hawkes processes relax the assumption of independent increments in Poisson and instead use the fact that order flow is endogenously excited in its modelling. A multi-dimensional Hawkes process can also have cross excitation terms between the different dimensions (for e.g., a 2D Hawkes Process of (ask volume, bid volume) can have 4 excitation terms - ask→ask, bid→bid, ask→bid and bid→ask). Let us briefly describe mathematically the Hawkes Process formulation here. The intensity function of the Hawkes process contains the self and mutual excitation terms mentioned previously. For a d -dimensional Hawkes process the intensity of the process $\lambda_t^{(i)}$ and the associated counting process $N_t^{(i)}$ for $i = 1, \dots, d$ is defined as:

$$\lambda_t^{(i)} = \mu_t^{(i)} + \sum_{j=1}^d \sum_{t_j \in T_j} \phi^{(j \rightarrow i)}(t - t_j) \quad (2.2)$$

where $T_j := \{t_j : t_j \leq t\}$ denotes the set of past event times in the j dimension of the Hawkes Process. Here, $\mu_t^{(i)}$ is the exogenous intensity of the i -th dimension and $\phi^{(j \rightarrow i)}(t - t_j)$ is the excitation term from j -th dimension to i -th dimension. The excitation terms are a function of the time since the event (generally a decaying function in time like exponential decay or power law decay). An alternate but similar

formulation is the following:

$$\lambda_t^{(i)} = \mu_t^{(i)} + \sum_{j=1}^d \int_0^t \phi^{(j \rightarrow i)}(t-s) dN_s^{(j)} \quad (2.3)$$

There have been several discussions in the literature on the choice of the kernel functions. Nyström and Zhang [2022] show that a power law kernel fits much better to the empirical data than the exponential kernels. Da Fonseca and Zaatour [2014] also show using a Q-Q plot comparison between the distributions of empirical inter-arrival times to the exponential distribution that exponentially decaying kernels are probably insufficient in representing the empirical data.

n-dimensional Hawkes Process: In recent years there has been a significant increase in LOB models using Hawkes Process albeit with vastly different formulations. Toke [2010] creates a two-agent based model where liquidity takers (Market Orders), and liquidity providers' Limit Orders are each modelled as 1D Hawkes processes. Cancels are modelled to be Poisson arrivals and the price for Limit Orders and Cancels are sampled from a probability distribution. They compare performance improvement of using Hawkes processes against Poisson processes and show that Hawkes Processes with the following excitations: Limit & Market Orders' self-excitation and Market Orders exciting future Limit Orders give better fits to empirical data. Bacry et al. [2016] divide the events in an order book into two categories - those which change the mid-price and those which do not. They use an 8-dimensional Hawkes process with orders changing the mid price being modelled by one dimension, and for events which don't change the mid price are modelled by 3 dimensions (Market Orders, Limit Orders and Cancellations). This is done for both bid and ask sides giving us a total of 8 dimensions. Previously, Large [2007] followed a similar technique by using a 10D Hawkes Process: $\{\{\text{Limit Order, Market Order}\} \times \{\text{change mid, don't change mid}\}, \{\text{Cancel Orders}\}\} \times \{\text{bid, ask}\}$. They formalize the 'resiliency' of the order book which is the ability of the order book to replenish after being depleted by a large trade. Kirchner [2017] proposed an alternative, non-parameteric way of estimating the Hawkes process and showed the applicability of their method in LOB data. Particularly noteworthy is the techniques they show in doing model-selection and their usage of the AIC statistic to optimize the hyperparameters.

Da Fonseca and Zaatour [2014] propose an alternate strategy to fit the Hawkes Process by using a generalized method of moments to fit the first four moments of various quantities of interest in the Hawkes Process. This method is claimed to be

much faster than the traditional maximum likelihood estimation (MLE) methods used in the literature. They show weak convergence results of the fitted parameters to the true unknown parameters of the Hawkes Process. They use several key stylized facts to test the realism of their simulations. They also compare the fitted parameters to the MLE baseline.

Constrained Hawkes Process: Zheng et al. [2014] create a 4 dimensional Hawkes Process for Level 1 Order Book simulation with two for each of ask and bid queues and they construct a spread process to control events where bid becomes greater than ask. They thus create and provide analysis for a Hawkes Process with constraints. Lee and Seo [2022] create a 4D Hawkes Process for level 1 LOB simulation similar to Zheng et al. [2014] but instead of constructing a separate spread process, they propose that the exogenous intensity for spread-narrowing events is a function of spread relative to the price. This implies that at 1 tick wide spread, the exogenous intensities of spread-narrowing events is zero. Further they also let the decay kernels' (for self and cross excitation) to be again dependent on spread but also stochastic. In this manner they ensure that the intensity of spread-narrowing events is exactly zero when the spread is 1 tick wide. They derive some properties of the price and spread process. Further, they provide some techniques of checking the estimator's bias and also provide a comprehensive empirical study and derive several economic explanations for the observed phenomena.

Other variants: Kaj and Caglar [2017] model the order book events in the following manner: Market Orders as queue-reactive Poisson arrivals, Limit Orders as Hawkes with excitation from Market Orders and Cancels as constant intensity Poisson arrivals. They call this model a 'Buffer-Hawkes' Process. Morariu-Patrichi and Pakkanen [2022] develop a state-dependent Hawkes process where two types of states are considered: first, on the basis of spread being one tick or more than one tick, and second, on the basis order flow imbalance. They conclude that the excitation effects are observed to be highly dependent on the current state. They perform a number of analyses on their fitted Hawkes process to infer economic rationale behind observed effects in the kernels of the Hawkes process. Kirchner and Vetter [2022] also formulate a marked state-dependent Hawkes Process but they use non-parameteric methods to estimate the excitation kernels shape (although they use power laws to fit the shape later). They use the current imbalance as the state indicator and also work towards creating a parsimonious model by zero-ing out the smaller excitations they observe in the data. Another state-dependent model using Hawkes Process was proposed by Mucciante and Sancetta [2023] where they consider the intensity as a product of a Hawkes Process driven intensity with a linear

function on some observables in the market environment. We note that a key feature of this paper is the incorporation of time of day into its modelling - it is well known that order arrival intensities are non-stationary intraday and therefore most models clip the data to exclude open and close effects. Wu et al. [2019] take inspiration from Huang et al. [2015] queue reactive model to build a Hawkes Process with exogenous intensity being queue-reactive in one model, and a queue reactive multiplier on top of the the Hawkes process intensity in the second model. They show that adding queue-reactiveness improves the goodness of fit of the models against Huang et al.'s model as well as the non-queue reactive Hawkes model. Rambaldi et al. [2017] show that the order size (and not just order count) of each individual order is important in the excitation of future orders. They show that using a marked Hawkes Process, with the marks corresponding to various bins of the order sizes, they are able to follow Bacry et al. [2016]'s non-parametric estimation technique to create a more realistic simulator.

Scaling limits: In their influential paper, Abergel and Jedidi [2015] create a model of the full LOB where each level's Market Order and Limit Order intensities are modelled as Hawkes Processes with Cancels being modelled as having a queue-reactive Poisson intensity. They show through a mathematical analysis that this model can be used to create Stochastic Differential Equations (SDE) for aggregate features. Horst and Xu [2019] further showed that under some scaling limits, the Hawkes model for an LOB converges to an SDE for ask and bid prices while the intraday volume follows a system of Ordinary Differential Equations (ODE). They also show that the stationary intensities of the different types of events form Volterra-Fredholm Integral Equations.

Non-linear Hawkes Process: Lu and Abergel [2018a] create a 12D Hawkes process: $\{\text{Limit Order, Market Order, Cancels}\} \times \{\text{change mid, don't change mid}\} \times \{\text{bid, ask}\}$. They compare the performance of a Linear and Non-Linear Hawkes Process with the non-linear one having novel *inhibiting* kernels for negative excitation. They floor the intensities of the non-linear Hawkes model to zero. Another notable novelty in their research is the use of sum of exponential functions with varying half-lives as their kernels. Mounjid et al. [2019] also create a non-linear Hawkes Process to simulate the order book with the non-linear transformation being dependent on the event type, the current state of the LOB, the current time and a sum over past events' excitations. These excitation kernels further are allowed to depend on the event type and the current state of the order book. They perform a mathematical analysis comparing this framework with different kinds of intensity models: Poisson, queue-reactive Poisson, Hawkes and Quadratic Hawkes processes.

Neural Hawkes Process: More recently, Kumar [2021] developed a Deep Neural Hawkes Process for Market Making in a simulated order book. The order book is simulated through a combination of agent based traders and sampling a Hawkes Process fitted to historical data. They use LSTMs to improve upon the Neural Hawkes Process proposed by Mei and Eisner [2017]. Their hypothesis being that LSTMs are able to capture the more complex dynamics of feedback loops between various orders in the market since they inherently have this feature in their structure. A more detailed analysis of the various agents considered by the author is presented in Section 6. Shi and Cartlidge [2022] develop a neural Hawkes process with each order type's intensity being modelled by continuous time LSTM units. The process's intensity rates evolve in a way such that the current market state influences it. They draw the price and the size of the order from stationary distributions in their simulations.

Discussion: Hawkes process, with their high adaptability, provide a more comprehensive point process methodology to model the order book arrivals without having to necessarily model individual traders' behaviours in the market. Most importantly, their ability to reproduce important microstructural details like volatility clustering and Epps effect make them great candidates for the LOB models. It is of note that since these models (point processes) are mathematically descriptive, they are fully explainable in their nature and hence are suitable for applications where black-box solutions are not preferred. Recently, Bacry et al. [2017] have published a Python library for calibrating Hawkes process. The key challenge that the practitioner might face in using Hawkes Process is the difficulty in the calibration of these models. This stems from the fact that the likelihood function is quite complex. In addition to that the choice of Kernels in the Hawkes Process can make quite a large difference in the model's predictive power. Further, the question of model parsimony becomes quite relevant here since the number of kernels scale in $O(n^2)$ for an n -dimensional Hawkes Process.

2.1.4 Agent based Models (ABMs)

Considering that the order book is constituted by a large number of heterogeneous agents, examples of heterogeneity being differences in their trade frequency, trading objectives, access to financial data and access to low latency trading hardware, the key idea in this category of LOB models is that each agent needs to be modelled in a separate category. For example, Cont et al. [2023a] make use of clustering techniques to showcase from anonymized trade execution data that there exists at least four distinct clusters of types of agents in the lit LOB. The usual set of categories in

addition to the usual informed v/s uninformed traders include high frequency traders, trend followers, mean reverters, noise traders and algorithmic traders. We refer the reader to the expansive review by Chakraborti et al. [2011] and Chapter 5 in the book by Abergel et al. [2016] on Agent Based LOB models and review the newer research in the following.

Recent Work: Paddrik et al. [2012] use order speed and order placement as the differentiating characteristics to identify and model various types of agents. They use the May 6, 2010 *Flash Crash* of E-Mini S&P Futures to support their claim that many agents behave in a correlated manner. They create six categories of traders ranging from fundamental traders trading at very low frequency to market makers to HFTs. They model each category to be zero-intelligence Poisson processes. Huang et al. [2015] in their second model assume that institutional agents post their limit orders at the top of the book while HFTs, market makers and arbitrageurs post it in deeper levels. Hence they propose that the order arrival intensity of a level depends on whether the level is the best bid/offer or not. Further they enhance this model by adding order arrival rate's dependency on opposite side queue size by discretising the opposite side queue size into 4 categorical quantiles. In their queue-reactive model they further relax assumptions by letting the mid-price or reference price change by one tick with some constant probability, and also having a reinitialization constant probability event. Byrd et al. [2019] propose a software framework for simulating tens of thousands of agents with various types of objectives and trading patterns. They also introduce latency and an exchange agent for transactions to make the simulator more realistic. Belcak et al. [2020] too create a software package like ABIDES and provide the a Python API with C++ backend of the simulations. They further study a number of simulation statistics and also provide a methodology to measure market impact with permanent and temporary components.

Combining ABMs with other methods: Lehalle et al. [2011] describe the drawbacks of both the ABM (computational constraints and lack of analytical results) and Point Process Modelling (stationarity assumption and imperfect representation of stylized facts) approaches and propose using a mixed model. They create a zero-intelligence model (conditioned on a distance metric between the investor's view of the order book and the real order book) 'pegged' to an ABM with scaling limits taken as a Mean Field Game. As detailed in the previous section, Kumar [2021] uses a hybrid approach to modelling the LOB with Hawkes Process as their background process for several different types of agents to interact with. They propose to segregate the market participants into the following classes: the fundamental trader who follows a mean-reversion strategy, the chartist trader who follows a momentum strategy, a

noise trader and three kinds of market-makers - one which uses their proposed Deep Hawkes Process to quote bid-ask orders, second which uses the Neural Hawkes Process proposed by Mei and Eisner [2017], and the third being a probabilistic market maker whose order placement is based on their view of the fundamental price of the security. Shi and Cartlidge [2023] show that combining a stochastic model for a background simulator of the LOB with multi-agent simulation build on top of this background simulator has benefits over pure ABMs or pure stochastic models. They create a Neural Hawkes Process for the background simulator and use the ABIDES platform (Byrd et al. [2019]) for the multi-agent simulation. They perform studies on price impact and observe herding behaviours in their simulations.

Discussion: The interplay between the plethora of market participants has naturally led the LOB to be modelled in a statistical physics way. Agent Based Models of the LOB rise from the popularity of Econophysics modelling. The common difficulties with this kind of modelling is the heavy usage of heuristics in defining the behaviour of an individual or a class of agents. In addition to that, the computational cost of simulating individual agents is generally higher than other alternatives. Although, Mean Field Games analysis of the ABM system does help in the analytical tractability of this category of models. The usage of ABMs in combination with a background model is a promising area of future research since that combines the best out of both these contrasting modelling techniques.

2.1.5 Deep Learning based Models

Owing to the several sources of possible complexities and non-linearities in the time evolution of the order book as well as the distribution of the prices/returns and volumes, large parameterised models like deep learning networks have found recent surge in popularity in LOB simulation. There has been a significant amount of research done to use the predictive power of neural networks for predicting the mid-price, the volatility and the direction of price moves. Some of the more popular architectures being used are Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), Recurrent Neural Networks (RNNs), and Generative Adversarial Networks (GANs). For a detailed description of these architectures, we refer the reader to the text by Goodfellow et al. [2016]. For a focused review of machine learning applications, encompassing both the so-called traditional machine learning models and deep learning models, in finance, we refer the reader to the text by Capponi and Lehalle [2023].

Mid-price prediction from LOB: Sirignano and Cont [2018] use deep learning techniques like LSTMs to model the price formation mechanism with historical

price and order flow as inputs. They show that their price dynamics is highly path-dependent since increasing performance was observed with increased history. Although they do not model the limit order book state explicitly but rather model the next mid-price which is just a property of the order book as a whole, their universality results show the promise of deep learning in ingesting tick data. Zhang et al. [2019] use Deep Learning structures like CNNs coupled with LSTM and Inception modules to predict future price movements from the current state of the order book. More recently, Zhang et al. [2021] use Deep Learning on Market by Order data (Level 3 data) to predict the future price movements' category among up, down or flat. A detailed comparative analysis on price prediction from LOB states is presented by Briola et al. [2020].

Recurrent Neural Networks: Shi et al. [2021] make use of Recurrent Neural Network (RNN) structures like the Gated Recurrent Unit (GRU) and an ODE-RNN (Ordinary Differential Equation-RNN) to predict the volume at different levels of the order book. They use top of the book data (Level 1) to simulate five levels of data (Level 2). The authors claim that the ODE-RNN usage here is of particular importance since the traditional RNNs are unable to handle non-uniform time intervals in their history. Further they use transfer learning to show that the parameters learnt by training the network with one security's data can be fine tuned to a different security's data to get reasonably good performance. Further in Shi and Cartlidge [2021], they propose the usage of exponential decay kernels instead of the ODE kernels to make the model more parsimonious and to reduce the computational cost. They enrich their testing universe by using a wider set of stocks and they remove look-ahead biases from their previous model. They find that the order volume prediction accuracy decreases with increase in volatility.

As mentioned in Sections 5 and 6, Kumar [2021] used LSTMs in their Hawkes Process model to capture more complex feedback loops dynamics which exist in the various event types in the market.

Generative Networks: Takahashi et al. [2019], Wiese et al. [2020] and Ni et al. [2022] use GANs and its variants to generate financial time series. Particularly noteworthy is Wiese et al. [2020]'s use of the DY Metric (Drăgulescu and Yakovenko [2002]) to test the performance of the generative network. Li et al. [2020] use conditional Wasserstein GANs to create their Stock-GAN model which simulates orders in the market by conditioning on some finite window of historical orders. They use an LSTM to encode the history and claim that the time dependence of order flow intensity is captured by this recurrent network. They further add a continuous double-auction approximation neural network to evolve the order book from the order

streams that are simulated. Lim and Gorse [2021] use Sequence GANs (SeqGANs) to model the order flow. They argue that the SeqGAN architecture is a better choice in handling discrete sequences of events like order flow than conventional GANs. They model the order book data using SeqGAN and further apply this simulation to do an analysis of the macro-level mid-price movements in this model. Prenzel et al. [2022] created a methodology to calibrate GANs for order flow data dynamically for different market conditions instead of a single calibration over the entire dataset. They assume the order flow is following Poisson arrivals but rather than setting the intensity to a constant value, they use GANs to estimate the probability distribution function of the intensities for different conditions such as time of day and market volatility. Cont et al. [2023b] model the transitions between two consecutive LOB snapshots using Conditional Wasserstein GANs (conditional on current state of the LOB, therefore making the simulation Markovian). They particularly focus on creating a model which has implicit market impact in its order book transitions (more details in Section 9). They provide a thorough analysis of their model by comparing the simulation's and real world's empirical facts (more details in Section 10).

Coletta et al. [2022] make use of Conditional GANs (CGANs) to create an LOB model ('world model') which is compared against a baseline ABM. They train the CGAN to generate the next trading action given the current features of the state of the order book. They further perform 'Adversarial Attacks' on the CGAN model in Coletta et al. [2023] to highlight the dependence of the model on its input features.

Large Language Models: Nagy et al. [2023a] use the recently popular autoregressive generative models on order book message data to simulate order flow. They tokenize the LOB messages and treat sequences of these tokens to simulate order flow as a Large Language Model (LLM) would treat words in a language to create a comprehensible sentence. They perform several out-of-sample tests on their simulator to test its efficacy.

Discussion: Deep Learning models for simulating the order book are natural candidates to solve for the vast complexity of the order book dynamics. The ability of deep neural networks to model convoluted time evolution of Markov processes coupled with the astronomical increase in recent years in ease of training such models has popularized this category of models. We note that a number of these models are able to reproduce the stylized facts quite well in their simulations. Some of them also exhibit the concave Market Impact characteristic that practitioners observe in the real world. However these models, like any other deep learning model, have many challenges such as the lack of explainability owing to their black-box nature, high sensitiveness to carefully calibrated hyperparameters and a very high model

complexity with millions (or even billions) of parameters. A more parsimonious, explainable way of modelling the order book could be to model the transition of the order book state itself by taking the continuous time limit and creating a differential equation evolution of the stochastic process.

2.1.6 Stochastic Differential Equations Based Models

Since the LOB transitions are probabilistic in nature, the time evolution of an LOB state could be modelled as a set of differential equations. Here, usually, a continuous approximation is made of the state transition time which although is quite different from the reality of discrete time steps, but in larger time scales is an appropriate approximation since the frequency of LOB events is generally quite high. A major focus of this set of models is studying the long time dynamics of the order book - the so-called steady state (or the absence of it) is a major point of interest. Some of the more popular types of components used in the differential equations include diffusion and convection. These set of models are particularly important if one is concerned about the explainability of the LOB simulation. They also provide good segues into utilizing the modelled dynamics for optimal control problems like portfolio management, wealth management, optimal liquidation and market making.

Continuous limits of Point process models: Korolev et al. [2015] model the order flow arrivals as a Cox process (doubly stochastic Poisson arrivals) and form stochastic differential equations for order flow imbalance. Lakner et al. [2016] study one-sided order books assuming Poisson arrival of limit orders with intensity conditional on current best price. They provide weak limits of the price process and LOB process and show that the limit order book process is a solution to a stochastic differential equation. Huang and Rosenbaum [2017] extend their previous Poisson arrival based framework Huang et al. [2015] to create a more general stochastic dynamic model. They consider two separate jump processes for the order book state and a reference price which is not the mid-price but rather the so-called ‘fair-value’ of the security as perceived by the traders. They create a state transition matrix based on a queue-reactive flow assumption and further also incorporate a reinitialization probability of the reference price attributing it to exogenous jumps. They perform tests of ergodicity and conclude with proving that under certain scaling limits, the price dynamics converges to that of a Brownian motion. More recently, Cont et al. [2023c] form a more general LOB model with the order flow modelled as a point process and the trade execution is modelled as a deterministic mass transport operator. In certain scaling limits, they show that their framework generalizes a number of other LOB models and show that they are in fact special cases of their framework.

Volume of orders as a stochastic process: Cont and de Larrard [2011] use heavy-traffic limits to show that depending on the scaling behaviour of the order flow, LOBs can act as deterministic under the fluid limit and stochastic under the diffusion limit. They find that the diffusion limit case occurs much more often in the empirical data. They come to the conclusion that the diffusion limit's differential equations' solution can be approximated by a two-dimensional Brownian motion. They further derive analytical solutions to questions like price dynamics, duration between price moves and probability of a price move. Building on the model in Cont and de Larrard [2013], Chávez-Casillas and Figueroa-López [2017] allow for variable spread in the simulated dynamics as well as allow for in-spread orders. They further formulate diffusion limits of the price process. In their influential paper, Cont and Muller [2021] develop a continuous limit of volume at a time t and price p by using a volume density $v(t, p)$. They centre the volume density to $u_t(x) := v(t, S_t + x)$ where S_t is the mid-price. They follow a data-driven approach to model order book dynamics - they categorize the cancellations into deletions and modifications. Further, modifications are bifurcated into symmetric (i.e. cancel and place at a nearby level) and antisymmetric (cancel and replace near mid-price). They model the symmetric modifications as a diffusion process, and the antisymmetric as a convection process.

Probabilistic properties under scaling limits: Horst and Paulsen [2017], by assuming generalized time-dependent order arrival intensities, develop limit theorems for price and volume densities at bid and ask. They conclude that given some regularity conditions, the two processes converge to a coupled ODE-PDE system of equations until scaling limits. Horst and Kreher [2017] further generalize the previous work to develop a weak law of large numbers by considering the order flow as Markovian dynamics which are state dependent. Specifically they conjecture that the type of order, and its size & price, all are a function of the price and standing volume of the order book. In Horst and Kreher [2018], the authors further show that the previous model in Horst and Kreher [2017] can be used as first order approximation of liquidity in the order book to construct optimal liquidation trajectories. They now develop second order approximations to obtain confidence intervals around these trajectories. To that extent, they formulate two scaling limits accounting for the empirical fact that price change fluctuations are much slower than order arrival and cancellation fluctuations. More recently, Horst et al. [2023] create a 2nd order approximation with a single scaling instead of the previous two. They do so by assuming an unvarying Market order to Limit order ratio. They show that the price-volume process in this limit converges to a solution of a infinite-dimension

SPDE. Ma et al. [2014], Ma and Noh [2022] in their papers show that a one-sided order book dynamics can be modelled as an SPDE and equilibrium characteristics can be calculated for them. They further take the limit of traders in the market to infinity and show that a Mean-Field Game for the trader can be constructed using this SPDE, and under some conditions be solved using viscosity solutions of the Hamilton-Jacobi-Bellman (HJB) equations. On the other hand, Rojas et al. [2020] develop a law of large numbers, a central limit theorem and large deviations for a stressed order book - in their case they look at liquidity fluctuations.

Connecting various timescales: Hambly et al. [2020] build a set of models to connect the dynamics at various timescales - microscopic, then mesoscopic and finally, macroscopic. In the microscopic model they assume a Poisson arrival model for all order types with two intensities based on the frequency of trading: a common intensity for all order types is used for high frequency while an intensity dependent on queue price, mid-price, and the number of price changes before the current order is used at lower frequencies. Further they add a diffusion dynamic of orders diffusing to nearby price levels. They show that as the order arrival rate goes to infinity and the volume size of each order goes to zero, which is a continuous limit they formulate by looking at very small timescales, they can form a Markov diffusion process which is described by a system of reflected Stochastic Partial Differential Equations (SPDEs). They look at the time of price changes of these SPDEs to create the mesoscopic model of SPDEs. Finally, they take the limit of tick sizes going to zero to create a macroscopic continuous price process SPDE from the above model.

Discussion: SPDEs provide a mathematically tractable formulation of the time evolution of the order book. This feature of this category of models makes them attractive but at the same time also have brings out certain issues. More often than not, the SPDEs do not have an explicit solution. There are several approximations made in the literature to circumvent this problem like using viscosity solutions, considering the similarities to the heat equation in physics, and even some inspiration from energy models from statistical physics. Despite the difficulty in finding exact solutions to the system of equations, these models can be used to build a simulator since it is readily possible to evolve a stochastic variable with a set of SPDE for its dynamics. We recommend to the practitioner that they be careful of the core assumptions of the SPDE model while they're using it. Some challenges in using these models include high model complexity and requirement of a high amount of computing resources to perform the simulations. We note that the class of point process models can be scaled for large timescales to produce a set of SPDEs of the model. Another desirable property of this category of models is that since the

dynamics are analytically tractable, much like Hawkes Process models, we can use optimal control theory to participate in the simulator's trading.

2.1.7 Responsiveness to trades: Market Impact

Introduction: Market Impact or Price Impact is defined as the price movement due to one's own trading. Suppose a large market order is submitted by an agent on the buy side and it depletes a few levels of prices in the order book, the new best ask price will be a few ticks higher than the previous best ask. This suggests that the agent's trade 'walked the order book' and moved the price in the opposite direction. If the same agent wishes to buy again, they will have to pay a higher price. Similarly one can think of posting of limit orders shows the market one's intentions to trade at that price which gives *information* to the market which can react against the agent. Reducing Market Impact has been one of the pillar stones of all agency and electronic trading activities with years of research spent on building algorithms to reduce the impact of large orders. Common techniques to reduce market impact include batching of orders, following the market (i.e. targeting the Volume Weighted Average Price) and using alpha signals to place orders at a *smarter* price. Market Impact is one of the phenomena in the markets which cannot be measured immediately (at least not in any meaningful sense) - the agent will probably wait for some time for the market to *settle down* or in other words, wait for the price to come back to the previous levels. This makes Market Impact one of the harder aspects of the LOB to model. Not only does it depend on the supply and demand of the order book at that point of time but also the volatility of the security. The importance of market impact has been highlighted in the literature since a long time (Biais et al. [1999], Foucault et al. [2005], Cont et al. [2014]). Indeed Market Impact is, at the minutest of scales, quite closely related to the process of Price Formation (Lillo [2023]). Cont et al. [2014] show that the price impact can be explained by order imbalance and with a scaling they show that the 'square-root law' heuristic that traders in practice have can be derived quite easily.

Zero-Intelligence Models: Smith et al. [2003] posit that the instantaneous price impact function $\phi(\omega, t)$ is nothing but the inverse of the cumulative depth profile $N(p, t)$ of the order book where ω is the order size. They show that by using Taylor's expansion on $\omega(\delta p)$, i.e. order size needed to move the mid price by δp , they get an analytical price impact function. Their results show good matching with observed price impact when the Taylor series is expanded to two degrees. Further they assume order arrival intensities are dependent on the distance from mid of a price level. In addition to that, they add an additive noise from the rapid submissions

and deletions of High-Frequency traders in the form of a Brownian motion dependent on the centred volume density. They show that if the mid-price is an Arithmetic Brownian Motion, the volume density converges to a SPDE with a moving boundary problem's solution. They generalize this methodology to formulate an SPDE of the centred volume density and create a 2-factor model by creating SPDEs for both bid and ask side queues. They show that the long-term order book shape can be explicitly be solved for and they show a first order approximation of the shape in their results.

Poisson process: Huang et al. [2015] study the market impact of VWAP liquidation and exponential scheduling liquidation in their simulated model. They see concavity in their market impact observations against time and volume both. This shows that their model has intrinsic market impact and it matches some real-world behaviour of the markets. The same characteristics are observed in the model by Lu and Abergel [2018a]. In general, except for certain queue-reactive and state-dependent variants, Poisson models do not have market impact as a feature since the core assumption in a Poisson model is that each order event count is independent in increments.

Hawkes process: As pointed out by Lillo [2023], given a reference price P_t , the time evolution of the price is a deterministic function of the order flow point process. They show that Market Impact can be modelled by the Transient Impact Model (i.e. trading velocity impacts the price in a decaying function of time) if the order flow is considered to be exogenous of the price process. However they argue that empirical data shows some correlation of price movement with future order flow and hence the assumption that order flow is exogenous to price impact is probably not correct. Hawkes process (with price and order flow as its dimensions) relaxes this assumption by considering cross excitation of order flow from price movement and vice-versa.

Hawkes models are by definition reactive to the past events and hence the order intensities are influence by any past orders happening. This can be thought of like an implicit form of market impact in these models. Bacry et al. [2015a] study the so-called Hawkes Impact Model by considering a simple 2D Hawkes process of price. They conclude that while liquidating a meta-order (i.e. an order made of multiple child-orders which are individual market/limit orders), they observe a concave market impact followed by a convex relaxation of the price after the agent has stopped trading. This behaviour is very much in line with the expectation of traders. Also noteworthy is the recent work by Lee and Seo [2017] where they study the market impact using a Hawkes process in a more realistic sense by including the tick-size discretization of price levels. They further develop formulae for realized volatility with this and compare it to empirical realized volatility.

Agent Based Models: A recent study on how different categories of agents have different kinds of price impact has been done by Giamouridis [2023]. Paddrik et al. [2012] provide an interesting case - they do not explicitly study the market impact of an exogenous order but they perform a similar study to replicate the circumstances around the Flash Crash of 2010 by placing a large exogenous trade that *moves* the market. However we note that this impact is due to the behaviour of the High Frequency Traders and Market Makers modelled in the ABM and their reaction to the changing liquidity of the order book. Byrd et al. [2019] also show that agent based modelling can be utilised to study and estimate market impact models. They perform a case study with one exogenous trader placing orders, with varying proportions of Market and Limit orders, and they observe the price evolution during and after the trading is done. They perform comparisons of price paths with and without this exogenous trader and show that the price has meaningfully changed with the trading activity. The market impact they observe in their model is concave and is a decreasing function of the proportion of market orders in the trading strategy. Shi and Cartlidge [2023] follow the same framework to test the market impact in their model. Coletta et al. [2022] too follow ABIDES framework of studying Market Impact. Further in Coletta et al. [2023] they breakdown the impact into the impact of Market Orders and Limit Orders and they compare it to historical replay method of order book simulations. Belcak et al. [2020] study their model's market impact by analysing the average spread, variance of the spread and variance of the best price as a function of time since a large market order happened in the past. They also report a concave shaped market impact curve.

Stochastic PDEs based Models: Horst and Kreher [2018] show that using their 2nd order approximation, two different forms of Market Impact are found under different scaling limits - they term them to be temporary and permanent forms of impact.

Deep Learning based Models: In Cont et al. [2023b], MI study is done by analysing the price paths observed while executing a varying quantity of orders using three strategies: Market Order TWAP (time weighted average price is the target price), Limit Order TWAP and POV (percentage of volume i.e. the trader targets maintaining their traded volume to be a constant ratio of the market volume). There are clear trend lines observed in all three and comparisons are made to Poisson and Hawkes in the former two where it is shown that there is no clear trend in Poisson or Hawkes. We see a general rise in order book models being sensitive to the subject of price impact however there still is room for improvement. We once again stress the importance of being aware of market impact in the order book simulators especially

if the aim of building the simulator is to perform backtests of algotrading strategies. The ABIDES framework from Byrd et al. [2019] proves to be quite useful in the study of market impact since it is available in an open source code repository.

2.1.8 Comparative study

As we have seen above, there exists a rich literature spanning a number of modelling methods to model and simulate the order book. More often than not, we see researchers investigating some of the stylized facts observed in empirical data and using their reasonings for the observed distributions as priors in their modelling technique. It makes sense that building from those priors, they use these stylized facts as goodness-of-fit metrics as well. Table A.1 in the Appendix A enumerates the stylized facts being used in the models we review here. We also provide certain comments on how the authors have utilized the stylized facts in testing their model's efficacy against real data.

The vast majority of researchers make use of empirical probability distribution functions of various properties of the order book, the most popular ones being inter-order arrival times, spread and volumes, as their primary stylized fact. The technique for testing against this stylized fact is usually a qualitative test where the two distributions (empirical data and simulations) are plotted against each other. While this method is useful to test whether the general shape of the distribution is matching between the two datasets, we also note the usage of the far more sensitive to tail events method of Q-Q plots in several papers. This technique not only matches the dense regions of the distribution but also the lower and higher quantiles. Indeed, most of the Hawkes Process methodology works we have mentioned in Table A.1 make use of the Q-Q plot on the inter-order arrival times stylized fact to refute the Poisson method. Some notable observations are mentioned below:

Farmer et al. [2005] test the model proposed by Smith et al. [2003] by using empirical data from the London Stock Exchange. Notably their zero-intelligence model is able to reproduce a concave Market Impact function. Cont et al. [2010] test their model's quality of fit by comparing the average LOB profile and realized volatility against real-world data from the Tokyo Stock Exchange. They further show that the conditional probability estimates from their model matches empirical frequencies observed of direction of price moves and one-step transition. Abergel and Jedidi [2011] demonstrate a series of tests to compare simulation results to real-world data. These included comparisons of average depth profile, probability distributions of spread in ticks and price changes, autocorrelation of price changes and Q-Q plot of mid-price changes.

Paddrik et al. [2012] show that their agent based approach does show volatility clustering phenomenon which is quite remarkable. It seems like a mixed timescales approach to modelling the LOB naturally leads to volatility clustering. Interestingly they're also able to simulate crashes in the market when a large sell order is traded and HFTs and Market-makers withdraw from the market.

Huang et al. [2015]'s three models are tested against empirical data for two high spread-in-tick stocks in the French exchange. The zero-intelligence Model I fits better to the asymptotic order distribution of these stocks compared to the constant arrival rate models. Interestingly, in their third model with moving mid-price, without a reinitialisation event, they find that the realized volatility in simulations was much lower than the empirical realized volatility. They conclude that their model probably suffers from a mean-reverting behaviour to mid-price which is not necessarily true in practice.

Li et al. [2020] perform tests on distributions of price, quantity, inter-arrival time and spectral bid/ask prices. This is of note since instead of using the qualitative tests methodology we commented on previously, they make use of statistical tests to provide a more robust testing methodology. Another notable mention is Kirchner [2017]'s method of model selection. They provide some hyperparameters in their model to tune for each usecase and show examples of using the AIC metric in doing model selection in choosing these hyperparameters. An impressive study on the calibrated results is performed by Lee and Seo [2022] where they perform stationarity checks on their parameters and also compare the calibrated parameters against a baseline model. They also perform model-selection by comparing the five different proposed models. Mucciante and Sancetta [2023] showcase a testing methodology in which they measure the speed of convergence in their fitting method as well as show how sensitive their parameters are to various conditions to check the robustness of their model. They also perform tests on out-of-sample data. Comparisons to baseline models are also quite beneficial when the practitioner wishes to justify the incremental performance gain against simpler, more explainable methods. A good example are the papers Shi and Cartledge [2021], Shi et al. [2021] where the authors compare their method with a number of low complexity baselines. They also perform several studies to validate their use of ODE-RNNs instead of the more traditional LSTMs/GRUs and further provide an ablation study on the parameters.

As can be seen from Table A.1 Poisson models, generally, are successful in representing a number of 'first order' stylized facts like distribution of spread, volumes, average depth, average order book profile. However Abergel and Jedidi [2011] show that several other key stylized facts such as Autocorrelation functions of

price changes, signature plots and long term volatility are insufficiently replicated in a Poisson model. They show that a self-exciting process model like Hawkes Process could be one of the candidate solutions to these.

Hawkes Process models are seen to be much better than Poisson in representing the above mentioned stylized facts. They also fit the tails of the distribution of inter-order arrival time quite well. It has been shown Morariu-Patrichi and Pakkanen [2022] that the residuals after fitting the Hawkes process should follow the Exponential distribution however we see in Morariu-Patrichi and Pakkanen [2022] that this is generally not the case with the generic Hawkes Process. Therefore more complex models such as state-dependent Hawkes, Neural Hawkes Process etc are proposed. A frequent property we see being tested across all Hawkes Process models is the nature of the excitation kernel. The most usual choice of Exponential Kernels have been shown to be insufficient Nyström and Zhang [2022] and hence Power Law Kernels have become more popular Bacry et al. [2016] although they are harder to calibrate. A separate class of Hawkes models is the non-parametric estimation Hawkes models where the excitation kernels are estimated without any prior shape assigned to it. A number of authors report the shape of these kernels in time (or log-time).

In the Deep Learning category of models, we see the training and validation losses being generally reported in the papers which is the general industry standard. Also notable, particularly in the models using GANs, is the usage of simulated price paths and their qualitative comparisons to the empirical data. The Stochastic Partial Differential Equations category of models generally use the stylized facts to create priors in their differential equation dynamics and we see a general trend towards testing the first order features and long term asymptotic features.

Among the many mentioned stylized facts below, we suggest the reader to choose or formulate the ones which are the most important for their use-case. It is generally quite useful to do exploratory data analysis on your dataset before performing any data study, and calculating the various stylized facts is a good way to do the same. We have listed some of the most popular ones in Section 3. While these stylized facts are universal in their importance, the specific distribution of these stylized facts varies across securities, asset classes and regions. We also stress the importance of comparing the model's efficacy against both real world data and simpler baselines as well as the current state of the art in the relevant category. This would enable the user to further understand the reason their model performs better or worse than other models. Finally, robustness of the model's calibrated parameters should be checked against initial conditions, market volatility and other quantities which can bring in elements of non-stationarity in the probability space one is estimating.

2.1.9 Recent Developments (2023–2025)

Since the original version of this review was written in 2023 (Jain et al. [2024a]), several new research directions have emerged in the modelling and simulation of limit order books. A major development has been the application of deep generative models, particularly diffusion-based architectures, to high-frequency data. Backhouse et al. [2025] introduce an image-based diffusion framework for generating future LOB states, overcoming the error accumulation issues of autoregressive models and achieving state-of-the-art results on the newly proposed LOB-Bench. The benchmark itself, introduced in Nagy et al. [2025], provides the first standardized evaluation suite for generative LOB models, combining statistical, structural, and event-driven metrics. These advances signal a shift toward reproducible, metric-driven assessment of simulators and have accelerated progress in generative microstructure modelling.

At the same time, adjacent research has expanded the scope of LOB modelling toward multi-modal and causally informed approaches. Narrative and text–market interactions have been studied through causal embedding dynamics in Drinkall et al. [2025], highlighting how semantic shifts relate to market shocks. Large language models have also begun to influence financial modelling, with Nie et al. [2024] surveying their applications in forecasting, simulation, and agent-based reasoning. Complementary contributions include deep survival methods for estimating fill probabilities Arroyo et al. [2024] and machine-learning-driven execution algorithms Nagy et al. [2023b]. Together, these developments broaden the modelling landscape beyond pure order-flow simulation and point toward integrated, data-driven, and high-fidelity approaches to modern market microstructure research.

2.1.10 Conclusion

The field of order book simulations is growing in step with the modelling techniques themselves. For instance, the recent rise in Deep Learning’s popularity has meant a number of deep architectures being specifically build for mimicking the order book and its properties. In this review we present a classification of the order book simulator models by their core modelling methodology. The view that the order book is a stochastic process with several components and that intensities of these components can be modelled either on their own (zero-intelligence models) or with interacting terms (Hawkes process) is the basis of the Point Process class of models. This class of models look at an aggregate view of the order book dynamics and are not concerned with the motivation or purpose behind each order. The dynamics are purely dependent on the current state of the LOB and the history. The polar opposite view is taken by the Agent Based Modelling set of models. Here the research

focuses on mimicking the behaviour of the various kinds of agents in the real world interacting with each other and with the central limit order book. The challenges here are of course the vast variability of the kinds of agents real markets have as well as dependence of the simulator's realism on the assumptions made about the agent's expected behaviour. A third kind of modelling technique is to abandon forming any priors whatsoever on the data and let a universal approximator such as a deep learning network perform the estimation. The two major kinds of deep learning architectures we see in LOB modelling are firstly, the recurrent neural networks which predict the next order book state conditional on the history of the order book by performing non-linear transformations on the so-called memory of the order book states to produce the next one, and secondly, the generative networks where the underlying probability distribution function or the state transition probability is directly estimated. This technique, although quite powerful, has been shown to have difficulties in training and have high dependence on the hyperparameters. Deep learning's success does come with a major drawback - the lack of model explainability. Finally, the most explainable class of models are the ones which, using priors from observed stylized facts or expert judgement, use partial differential equations to model the dynamics of the order book states. These are known as the stochastic partial differential equation models and though they possess mathematical tractability which gives the user the ability to perform asymptotic analyses and solve optimal control problems in the order book framework itself, these models generally are seen to be overdependent on their prior assumptions.

We provide a concentrated study on some of the more important statistical properties of the order book and further provide a comparative analyses of how the models we review are making use of these properties in both their model building as well as testing. Finally we provide an in-depth analysis of the phenomena of market impact and how the simulators we study are sensitive to exogenous trades. Despite having such a wide variety of simulators, there is a lack of a parsimonious, explainable, analytically tractable model which has a good representation of most of the stylized facts the model is aiming to track.

2.2 Market Making, Stochastic Optimal Control and Reinforcement Learning

This section shifts focus from modelling of the LOB to optimal participation in the LOB market. Specifically we focus on a high-frequency algorithmic trading task, market making, in this modelled environment of the LOB.

Market Making in Limit Order Books (LOBs) is a high frequency trading task, where liquidity is provided with the goal of capturing the bid-ask spread. While many classical formulations rely on continuous-time dynamics and control, the true microstructure of the LOB is inherently discrete and driven by a pure jump process. In this section we will first define market making, then we discuss the common methodological gaps in the literature for market making, and finally we provide a literature review on two approaches to optimal market making - stochastic optimal control (SOC) and reinforcement learning (RL). We will provide a brief description of these optimization techniques in their respective sections as well.

2.2.1 Market Making

The distinction between liquidity provision and liquidity consumption represents a fundamental dichotomy in modern market microstructure theory. As noted in Guéant [2016], while optimal execution strategies primarily focus on liquidity-taking behavior—the efficient liquidation of large positions or implementation of hedging strategies—market making fundamentally concerns itself with the provision of liquidity to the marketplace. This shift in perspective necessitates a fundamentally different analytical framework, one that accounts for the unique risks and opportunities inherent in standing ready to transact at posted prices.

Market makers, defined as traders who continuously provide liquidity by quoting bid and ask prices, serve as the backbone of price discovery mechanisms across diverse market structures (Guéant [2016]). The heterogeneity of market making arrangements reflects the varied institutional frameworks within which these participants operate. On order-driven markets, official market makers such as Designated Market Makers (DMMs) on the NYSE operate under explicit contractual obligations to maintain fair and orderly markets, including mandatory participation in opening and closing auctions and adherence to National Best Bid and Offer (NBBO) quoting requirements. Conversely, proprietary market makers, including high-frequency trading firms, provide liquidity without formal obligations, seeking to profit from the bid-ask spread while maintaining flexibility in their market participation.

The fundamental economic proposition underlying market making involves the temporal arbitrage of buying at the bid price and selling at the ask price, with the bid-ask spread representing the market maker's expected compensation for providing immediacy services. However, this seemingly straightforward profit mechanism is complicated by the inevitable inventory risk that market makers must bear. The asynchronous nature of buy and sell transactions means that market makers typically hold non-zero inventory positions, exposing them to adverse price movements during

the holding period (Guéant [2016]).

2.2.2 Two Lenses: SOC and RL

Optimal Market Making (MM) in Limit Order Book (LOB) settings has been approached primarily through two methodological lenses: stochastic optimal control and reinforcement learning. We briefly provide an overview of the two fields of research in the following.

2.2.2.1 Stochastic Optimal Control:

Stochastic Optimal Control (SOC) provides a mathematical framework for sequential decision-making in continuous time under uncertainty. A concise introduction of the field is provided in Horst [2018] and Øksendal [2003]. For more details, we refer to the books Fleming and Rishel [1975], Fleming and Soner [2006], Yong and Zhou [1999], Pham [2009], Bertsekas and Shreve [1978]. We review briefly the mathematical formulations of various kinds of SOC techniques in the following.

Consider a controlled diffusion

$$dX_t = b(X_t, u_t) dt + \sigma(X_t, u_t) dW_t, \quad X_0 = x,$$

where $X_t \in \mathbb{R}^d$ is the state, $u_t \in \mathcal{U}$ is the control process, b and σ denote drift and diffusion coefficients, and $(W_t)_{t \geq 0}$ is a d -dimensional Brownian motion. For a given policy $u = (u_t)_{t \geq 0}$, the performance criterion is

$$J(x; u) = \mathbb{E} \left[\int_0^T \ell(X_t, u_t) dt + g(X_T) \right],$$

where ℓ is a running reward (or cost) and g is a terminal reward. The value function is

$$V(x) = \sup_u J(x; u),$$

and Dynamic Programming yields that V solves the Hamilton–Jacobi–Bellman (HJB) equation

$$\partial_t V(t, x) + \sup_{u \in \mathcal{U}} \left\{ \ell(x, u) + b(x, u) \cdot \nabla_x V(t, x) + \frac{1}{2} \text{Tr} [\sigma \sigma^\top(x, u) \nabla_x^2 V(t, x)] \right\} = 0,$$

with terminal condition $V(T, x) = g(x)$. The HJB provides a necessary and sufficient optimality condition when V is sufficiently regular, and the maximizer of the Hamiltonian gives the optimal control.

Stochastic Maximum Principle: An alternative to Dynamic Programming is the

Stochastic Maximum Principle (SMP), which provides pointwise optimality conditions in terms of forward–backward stochastic differential equations. Introducing the Hamiltonian

$$H(x, u, p, q) = \ell(x, u) + p^\top b(x, u) + \text{Tr}[q^\top \sigma(x, u)],$$

the adjoint processes (p_t, q_t) satisfy a backward SDE whose terminal condition is $\nabla g(X_T)$. Optimal controls maximize H almost surely:

$$u_t^* \in \arg \max_{u \in \mathcal{U}} H(X_t, u, p_t, q_t).$$

The SMP is particularly useful when differentiability of V is unavailable or when the control appears in the diffusion coefficient.

Optimal Stopping: In optimal stopping the controller chooses a stopping time τ to maximize

$$V(x) = \sup_{\tau \leq T} \mathbb{E}[g(X_\tau)].$$

No running reward or continuous-time control appears; the decision is *when* to stop. The value function solves a variational inequality,

$$\max \left\{ \mathcal{L}V(x), g(x) - V(x) \right\} = 0,$$

where $\mathcal{L}$ is the generator of X . The *continuation region* $\{V > g\}$ evolves according to $\mathcal{L}V = 0$, while the *stopping region* enforces $V = g$.

Impulse Control: Impulse control generalizes optimal stopping by allowing *repeated* discrete interventions. Let $\{\tau_k\}$ be intervention times and $\{\zeta_k\}$ corresponding impulses. The state follows

$$X_{\tau_k} = X_{\tau_k^-} + \Gamma(X_{\tau_k^-}, \zeta_k),$$

where Γ describes the jump induced by an impulse. The objective function is

$$J(x) = \sup_{\{\tau_k, \zeta_k\}} \mathbb{E} \left[\int_0^T \ell(X_t) dt + \sum_k K(X_{\tau_k^-}, \zeta_k) + g(X_T) \right],$$

with K the impulse reward or cost. The value function solves a Hamilton–Jacobi–

Bellman Quasi-Variational Inequality (HJB–QVI)

$$\max \left\{ \mathcal{L}V(x) + \ell(x), \sup_{\zeta \in Z} \{V(x + \Gamma(x, \zeta)) + K(x, \zeta)\} - V(x) \right\} = 0,$$

encoding that at each state the controller chooses between *continuing* (first term) or *intervening* (second term). Impulse control thus unifies continuous control and optimal stopping, providing a natural formalism for systems where decisions occur at discrete, strategically timed interventions.

2.2.2.2 Reinforcement Learning:

Reinforcement Learning (RL) provides a framework for sequential decision-making in systems whose dynamics are unknown or only partially understood. We consider a discrete-time controlled process with state $x_t \in \mathcal{X}$, control (or action) $u_t \in \mathcal{U}$, and stochastic disturbance e_t , evolving according to

$$x_{t+1} = f(x_t, u_t, e_t),$$

where f is an unknown transition map. At each time step the agent receives a reward

$$r_t = R(x_t, u_t),$$

with $R : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}$ known or specified by design. A *policy* is a (deterministic or randomized) mapping π taking states to actions, $u_t = \pi(x_t)$, or more generally a mapping from histories to actions. The agent’s objective is to select a policy π that maximizes the expected discounted return

$$J(\pi) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right],$$

where $\gamma \in (0, 1)$ is the discount factor and the expectation is taken over trajectories induced by π and the disturbance process.

Model-Based RL: Model-based methods estimate a surrogate transition function $\hat{f}$ from data. Given observed samples $\{(x_t, u_t, x_{t+1})\}$, one fits $\hat{f}$ (e.g., via least-squares regression of x_{t+1} on (x_t, u_t)). A controller is then synthesized under the nominal model, leveraging “certainty-equivalent” optimal control. This approach is statistically efficient because each transition provides vector-valued information about the dynamics. Its performance depends on the fidelity of $\hat{f}$ and on how model uncertainty is treated.

Approximate Dynamic Programming (ADP): Model-free approaches do not estimate f directly. Instead they approximate value functions. The *state–action value function* (or Q-function) is defined as

$$Q^\pi(x, u) = \mathbb{E} \left[\sum_{s=t}^{\infty} \gamma^{s-t} r_s \middle| x_t = x, u_t = u \right].$$

The optimal Q-function satisfies the Bellman equation

$$Q^\star(x, u) = R(x, u) + \gamma \mathbb{E} \left[\max_{u' \in \mathcal{U}} Q^\star(x', u') \middle| x, u \right],$$

where x' denotes the next state. Temporal Difference (TD) learning and Q-learning update parameterized approximations of Q using sampled TD-errors. These methods avoid explicit system identification but typically require many samples, since each transition yields only a scalar temporal-difference signal.

Direct Policy Search: Policy search treats RL as optimizing $J(\pi)$ over a parameterized policy class $\{\pi_\theta : \theta \in \Theta\}$. Introducing a distribution over trajectories $\tau = (x_0, u_0, x_1, u_1, \dots)$ and using the log-likelihood gradient identity yields

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta} [R(\tau) \nabla_\theta \log p_\theta(\tau)],$$

where $R(\tau) = \sum_{t \geq 0} \gamma^t r_t$ is the return of a trajectory. Algorithms such as REINFORCE and actor–critic methods stem from this construction. Policy search accommodates general function classes but may exhibit high variance in gradient estimates and reduced sample efficiency.

In summary, RL unifies learning and control through the optimization of long-run rewards under unknown dynamics, with model-based, value-based, and policy-gradient methods providing complementary strategies for continuous-control problems. A concise introduction of the field is provided in Recht [2019]. For more technical details and tutorials, we refer to the books Bertsekas [2019], Sutton and Barto [2018].

2.2.2.3 Discussion:

Early stochastic control models often assume Markovian mid-price dynamics with exogenously specified order flow. The seminal work of Avellaneda and Stoikov [2008] established the theoretical foundation for modern market making models by formulating the problem as a continuous-time stochastic control problem. Their framework, along with the closed-form approximations developed by Fernandez-

Tapia et al. [2016], provides elegant analytical solutions for optimal bid and ask quote placement.

Reinforcement Learning (RL) has emerged as a compelling alternative, capable of handling partial observability, path-dependence, and model uncertainty. Despite RL's flexibility, fundamental challenges remain—Guéant [2016] highlights issues such as non-stationarity, the absence of feedback loops in historical data, and the inherent difficulty in designing risk-sensitive reward structures. These limitations continue to motivate hybrid models that blend the tractability of control theory with the adaptability of learning-based methods.

2.2.3 Control in Limit Order Books

Continuous-time models, especially in the context of SOC approaches, face significant limitations when applied to actual order-driven markets. This is particularly noticed in their treatment of price discreteness arising from tick size constraints and their inability to capture the granular dynamics of limit order book evolution. While most of the works in this area of research rely on analytically tractable models, they tend to assume known dynamics, stationarity, and full observability, which limit their applicability to real-world markets.

The discrete nature of price movements in modern electronic markets, coupled with the complex interaction dynamics within limit order books, necessitates a more sophisticated modeling approach. While continuous-time models remain well-suited to quote-driven markets such as corporate bond trading, where dealers face pricing problems analogous to those originally modeled by Ho and Stoll [1981], equity markets require explicit modeling of the discrete limit order book structure. A more structural representation of the LOB dynamics is provided by Cont and Muller [2021], who model order flow evolution using stochastic partial differential equations (SPDEs). Their framework reproduces several empirical stylized facts of market microstructure, such as order imbalance, average trade durations, and quote time statistics. These features are precisely those that can inform and enrich the state representation of an RL agent. The intricate interaction between liquidity providers and takers thus motivates a modelling philosophy rooted in statistical mechanics and game-theoretic reasoning.

The work of Guilbaud and Pham [2013] represents a significant advancement in this direction, providing a discrete-time framework that more accurately captures the microstructural realities of modern electronic trading venues. In the context of RL, the importance of a realistic simulator cannot be overstated; for instance, Cartea et al. [2021] employ Double Deep Q-Networks (DDQNs) for FX trading in currency

triplets and model LOB evolution through recurrent latent variable Markov models, providing a data-driven yet structured perspective.

2.2.4 Stochastic Optimal Control Approaches to Market Making

A number of studies extend the SOC-based formulation of Avellaneda and Stoikov [2008] by explicitly modelling order arrivals as point processes. Guéant [2017] extend the market making problem originally formulated in the optimal control framework by Avellaneda and Stoikov [2008] by introducing stochastic optimal control (SOC) techniques. They derive optimal strategies under various market scenarios and thereby establish the foundational link between SOC and market making in limit order book (LOB) settings. Their subsequent work Guéant and Manziuk [2019] provides numerical solutions to the associated control equations and even illustrates the application of deep reinforcement learning (RL) methods such as actor–critic algorithms in credit markets. Cont and Kukanov [2017] model the limit order placement problem as a convex optimization problem, expressing the cost as a convex function that can be minimized efficiently. However, their model lacks adaptability to rapidly changing market conditions, a limitation that also affects the seminal Avellaneda and Stoikov [2008] framework, which relies on simplified reward formulations to maintain convexity at the expense of dynamic responsiveness. Guo et al. [2017] propose a closed-form solution for optimal order placement under a correlated random walk with mean reversion in the mid-price. This assumption, while analytically tractable, proves unrealistic in empirical settings and underperforms in simulation-based evaluations. For instance, Guilbaud and Pham [2013] model the mid-price as a continuous-time Markov chain and the bid-ask spread as a discrete Markov process, allowing the MM to place limit orders, market orders, and even aggressive internalizing strategies. These controls are optimized through a hybrid control framework involving a Hamilton-Jacobi-Bellman (HJB) equation.

More recent efforts have focused on incorporating more realistic LOB dynamics. Ricci [2014] consider a Hawkes-like LOB, where market orders impact a latent alpha signal and fill probabilities, making them informative. The MM, restricted to placing only limit orders, solves a stochastic control problem in this non-Markovian environment. Similarly, Jusselin [2021] study a quote-driven market in which market takers' actions are modeled via Hawkes processes. They derive HJB solutions in the exponential kernel case and extend to more realistic power-law kernels via approximations. A more discrete approach is taken by Abergel and Jedidi [2011], who model LOB dynamics using a state-dependent Poisson process and then extend the model to an exponential Hawkes process, solving the corresponding Markov

Decision Process (MDP). Building on this idea, Abergel et al. [2020] introduce a state-dependent Poisson LOB, formulating the problem as a Markov Decision Process (MDP) and extending it to the case of exponential Hawkes intensities. The resulting MDP is suitably adjusted and solved under this new setting.

The transition from continuous to discrete modeling frameworks raises fundamental questions about the nature of control in high-frequency market making environments, see for example Law and Viens [2019]. One recent related formulation of this problem is the jump decision process (or Piecewise Deterministic Decision Process for example see Bandini and Keller [2024], Bäuerle and Rieder [2011]) framework which assumes that controls can be adjusted instantaneously in response to jumps (i.e. market events). This assumption becomes increasingly unrealistic as trading speeds approach the physical limits of electronic order processing. The latency constraints inherent in real-world trading systems, combined with the discrete nature of order book updates, suggest that impulse control frameworks may provide a more realistic representation of market maker behavior. Impulse control has been a well studied area of stochastic control with several applications, for instance see Chen et al. [2025], Aïd et al. [2020], Lv and Xiong [2022], Davis et al. [2010], Chen and Guo [2013], Bayraktar et al. [2013].

2.2.5 Reinforcement Learning in Market Making

In recent years, reinforcement learning has emerged as a natural extension of stochastic optimal control, offering the capacity to learn policies directly from data in complex and partially observed environments. Gašperov et al. [2021] provide a comprehensive review of RL applications in market making, critically examining the assumptions of Avellaneda and Stoikov [2008]. They propose two taxonomies: the first distinguishes between (i) inventory-based methods, (ii) information-based methods addressing adverse selection, and (iii) model-uncertainty-based methods that mitigate model risk; the second differentiates analytical, non-deep (tabular RL, Q-learning), and deep learning approaches. To address RL's sample inefficiency, simulators are frequently employed, and roughly half of the studies reviewed rely on historical replay. Commonly used state variables include inventory, spread, profit-and-loss, imbalance, volatility, and relative strength index (RSI).

Gueant's chapter Gueant [2023] in Capponi and Lehalle [2023] delineates the challenges specific to financial RL applications. In contrast to standard RL domains, financial systems exhibit partial observability, non-stationary state transitions, data scarcity, time asynchronicity, and a lack of explicit risk considerations in the reward function. Moreover, RL methods trained solely on historical data lack the endogenous

feedback mechanisms that characterize live market interactions.

Empirical work by Gašperov and Kostanjčar [2022] utilizes an eight-dimensional Hawkes process, inspired by Bacry et al. [2016], as a baseline simulator. The action space includes both limit and market orders, and the RL problem is cast in a uni-episodic setting where each action is followed by a reward observation and a reset of prior placements. A Soft Actor–Critic (SAC) agent is benchmarked against an SOC-derived policy, achieving superior performance. The state representation includes inventory, spread, and recent market order trends. Alternative algorithms such as DQN and TD3 are also evaluated but exhibit weaker stability and convergence properties.

Spooner et al. [2018] propose a simplified RL setup based on historical replay with a discrete action space, omitting market impact modelling. In contrast, Gašperov and Kostanjčar [2021] introduce adversarial training to enhance robustness and employ signal-generating components to improve the agent’s situational awareness. Their framework also relies on historical replay.

2.2.6 Conclusion

This section has reviewed the problem of market making in limit order books through the complementary lenses of stochastic optimal control and reinforcement learning. Beginning from the microstructural foundations of liquidity provision, market making was framed as a high-frequency decision problem characterized by discrete price dynamics, stochastic order flow, and persistent inventory risk. This perspective highlights the fundamental mismatch between the inherently jump-driven nature of limit order books and classical continuous-time control formulations.

Stochastic optimal control offers a mathematically rigorous framework for market making, providing structural insights through Hamilton–Jacobi–Bellman equations, stochastic maximum principles, and extensions such as optimal stopping and impulse control. These approaches yield analytically tractable strategies under stylized assumptions and have played a central role in the development of the theoretical literature. However, their reliance on known dynamics, stationarity, and simplified representations of order flow limits their applicability in realistic electronic markets, where discreteness, latency, and endogenous interactions dominate.

Reinforcement learning, by contrast, relaxes the need for explicit model specification and enables policy learning directly from data or simulated environments. The surveyed literature demonstrates its flexibility in handling partial observability, path dependence, and complex state representations derived from limit order book features. At the same time, RL faces significant challenges, including sample inefficiency,

sensitivity to simulator design, non-stationarity, and the difficulty of encoding risk preferences and market feedback effects within reward functions. These limitations are particularly pronounced when training relies solely on historical replay without endogenous interaction.

Across both paradigms, the importance of realistic limit order book modeling emerges as a unifying theme. Discrete-time and event-driven frameworks, Hawkes-based order flow models, and stochastic PDE representations provide richer environments in which both control-theoretic and learning-based methods can be meaningfully deployed. The discussion highlights a gradual shift away from purely continuous formulations toward discrete and impulse-based control models that better reflect the operational constraints of high-frequency trading.

Overall, the literature suggests that neither stochastic optimal control nor reinforcement learning alone provides a complete solution to the market making problem in modern limit order books. Instead, the interaction between tractable control frameworks and data-driven learning methods motivates hybrid approaches that combine structural modeling with adaptive policy optimization. This synthesis remains an active area of research, driven by the need to reconcile mathematical rigor with empirical realism in high-frequency market making.

Chapter 3.

Limit Order Books: A Stylized Facts Study

This chapter provides an empirical analysis of how key stylized facts of the limit order book (LOB) vary with relative tick size. Using one year of high-frequency, tick-by-tick data for 15 large-capitalization U.S. equities with a common absolute tick size but widely differing price levels, we quantify systematic changes in LOB dynamics as the relative tick size decreases. By documenting how standard microstructural statistics evolve across large-, medium-, and small-tick regimes, this study highlights the structural differences induced by relative tick size and clarifies the challenges that arise when modeling LOB dynamics in the small-tick limit.

3.1 Introduction

This study presents visualizations of key stylized facts in market microstructure and seeks to quantify the specific relationship between decreasing relative tick size and the observed trends. To the best of our knowledge, this is the first work which defines quantifiable statistics of the LOB dynamics and depict their variation with relative tick-size of several assets. We showcase the fact that with the constant tick-size of the US securities and with a wide range of associated prices of the security, the LOB behaves significantly differently with varying relative tick-size. The aim of this discussion is to then highlight the challenges of modelling LOBs in a situation when the relative tick-size is very small.

To assess the variations in microstructural properties of the Limit Order Book (LOB) across different tick-sizes, we analyze several key aspects of LOB states over a one-year period, and report their empirical characteristics. These characteristics are commonly referred to in the literature as stylized facts, which represent consistent empirical regularities observed in financial markets (Cont [2001], Bouchaud et al. [2018]). The selection of stylized facts is intended to ensure that the most salient features of the LOB microstructure are captured within this statistical framework, thereby facilitating a comprehensive understanding of its dynamic behavior. In order

for our study to cover a large variety of dynamics, we chose a collection of assets with a wide range of average mid-price varying from $\langle P_{mid} \rangle = 6\$$ (SIRI) to $\langle P_{mid} \rangle = 1800\$$ (BKNG). The mid-price average was computed over the whole period averaging on all the mid-price change events. In that sense it corresponds to an event-average quantity as per convention in the literature¹

In this chapter, we present an analysis of several stylized facts for 15 assets over a whole year's LOB data (2nd January 2019 to 30th December 2019) i.e. 250 trading days. Some basic statistics related to each asset of this collection are shown in Table 3.1). For each asset, high-resolution, tick-by-tick LOB data obtained from the LOBSTER provider (LOBSTER academic data - powered by NASDAQ OMX) is employed. Let us point out that the assets in our collection (as all the US equities with large capitalization) have all a constant tick size $\delta = 1$ cent ².

Asset	Avg Spread ($\bar{s}$) (in ticks)	Avg Mid-Price ($\langle P_{mid} \rangle$) (in \$)	Relative Tick-Size (ϵ) (in bps)	Category
SIRI	1.03	6.07	16.47	Large-tick
BAC	1.06	29.40	3.40	Large-tick
INTC	1.09	45.50	2.19	Large-tick
CSCO	1.09	51.31	1.94	Large-tick
ORCL	1.10	54.05	1.85	Large-tick
ABBV	2.30	76.86	1.30	Medium-tick
PM	2.61	81.89	1.22	Medium-tick
MSFT	1.25	129.44	0.77	Medium-tick
IBM	3.37	137.94	0.72	Medium-tick
AAPL	1.97	208.62	0.48	Medium-tick
TSLA	12.66	270.94	0.36	Small-tick
CHTR	23.76	394.76	0.25	Small-tick
GOOG	57.98	1186.57	0.08	Small-tick
AMZN	46.81	1777.35	0.05	Small-tick
BKNG	162.39	1859.92	0.05	Small-tick

Table 3.1: Data Description: We make use of 15 assets which are the constituents of S&P500 as of 2nd January 2019 with a wide range of average mid-prices, average spread and relative tick size. We use tick-by-tick data for one calendar year (2nd January 2019 to 30th December 2019) from LOBSTER academic data - powered by NASDAQ OMX. The assets are categorized into Large- ($\epsilon \in [1.5, \infty)$), Medium- ($\epsilon \in [0.4, 1.5)$) and Small-tick ($\epsilon \in [0, 0.4)$) according to the relative tick size ϵ .

The chapter is organized as follows. In order to study tick size related stylized

¹Let us note that, throughout the chapter, given a time dependent quantity $a(t)$, $\langle a \rangle$ will denote the corresponding event-average (as for the average mid-price $\langle P_{mid} \rangle$) and $\bar{a}$ will denote the corresponding time-weighted average (as for the average spread $\bar{s}$), see Eq. (3.2).

²Some US equities have a tick size of one-hundredth of a cent however their market capitalization is very small compared to the equities considered in this study.

facts, we have to choose a universal proxy that we will be used for comparing cross-asset tick-sizes. We first introduce the relative tick size as a universal, dimensionless proxy that enables meaningful cross-asset comparisons of tick-size effects. We then examine the behavior of the bid–ask spread, analyzing both its mean value and full distribution as functions of the relative tick size. Next, we study trade-driven mid-price changes and document how their probability distributions evolve across different tick-size regimes. We subsequently analyze the shape of the limit order book by investigating the average volume profile and the location of its maxima, and relate these features to changes in relative tick size. The role of sparsity is then addressed by quantifying the prevalence of empty price levels and assessing their impact on the instantaneous shape of the LOB. We further explore LOB event dynamics by examining order arrivals, cancellations, and trades, highlighting structural similarities and differences in their endogenous excitation across tick-size regimes. The chapter concludes with a summary of the main empirical findings and a discussion of their implications for modeling limit order books, particularly in the small-tick limit.

3.2 The relative tick size : a universal proxy to compare cross assets tick-sizes

Traditionally, in order to compare cross asset tick size, one generally uses the average bid-ask spread value expressed in ticks (Bouchaud et al. [2018], Briola et al. [2024]). The bid-ask spread $s(t)$ of an asset at a given time t is defined by the difference between the best ask price and the best bid price at time t . Thus it is defined (when expressed in ticks) by :

$$s(t) := \frac{p_{\text{ask}}(t) - p_{\text{bid}}(t)}{\delta} \quad (3.1)$$

The average spread can then be defined as a time-weighted average denoted $\bar{s}$ as follows :

$$\bar{s} := \frac{\sum_{i=1}^N s(t_i) \Delta t_i}{T}, \quad (3.2)$$

where the t_i 's correspond to the timestamp of the jumps of the $s(t)$ process and $\Delta t_i = t_{i+1} - t_i$ and T the total observed duration. The average time-spread $\bar{s}$ of each asset of our assets collection are displayed in Table. 3.1. One sees that it covers a very wide range of value from 1.3 ticks (SIRI) to 162.39 ticks (BKNG). However, although this quantity is commonly used as a proxy for comparing cross-asset tick

size, it has two main drawbacks. First of all, it does not take directly into account the actual size (in the currency) of the tick, which, on some markets, could vary a lot even for very liquid and highly capitalized assets (e.g., London Stock Exchange, Xetra Frankfurt, etc.). It is natural to expect that this actual size has some impacts on actual statistical properties of the LOB dynamics. Secondly, the average spread size is mechanically floored by one. So, it prevents from comparing the tick size of two different large-tick assets.

Thus, in our study, as a proxy for comparing the tick size of the various assets, we prefer to use the relative tick-size (measured in basis points, bps) defined by

$$\epsilon := \frac{\delta}{\langle P_{mid} \rangle} \times 10^4. \quad (3.3)$$

As shown in Table 3.1, the relative tick size of the assets of our collections covers a very wide range of values from 16.47 bps (for SIRI) to 0.05 bps (for BKNG).

Following this proxy, for the sake of simplicity, we decide to regroup the assets of our collection into 3 groups :

- Large-tick assets ($\epsilon \geq 1.5$) : SIRI, BAC, INTC, CSCO, ORCL
- Medium-tick assets ($\epsilon \in [0.4, 1.5[$) : ABBV, PM, MSFT, IBM, AAPL
- Small-tick assets ($\epsilon < 0.4$) : TSLA, CHTR, GOOG, AMZN, BKNG

We are now ready to present the various tick size related stylized facts.

3.3 Bid-Ask Spread : Mean and Distribution vs Relative Tick-Size

In Figure 3.1, we plot, for each asset of our collection, the average spread $\bar{s}$ against the relative tick size ϵ on a log-log scale to explore the relationship between relative tick-size and spread. Unsurprisingly, we observe a gradual increase in the mean bid-ask spread as the relative tick-size decreases across assets in the dataset. For instance, the mean spread for INTC, a large-tick asset, is 1.02 ticks, whereas for AMZN, a small-tick asset, it reaches 47.18 ticks. The figure clearly reveals two regimes. A first regime for large-tick assets, for which, unsurprisingly (as already mentioned above, this is due to the constraint that the spread cannot be smaller than 1), the average spread is basically constant and equal to 1. A second regime for medium and small-tick assets, for which $\bar{s}$ seems to follow a power law as a function of ϵ (in our case we find a power-law exponent of the order of -1.23). Thus, we state

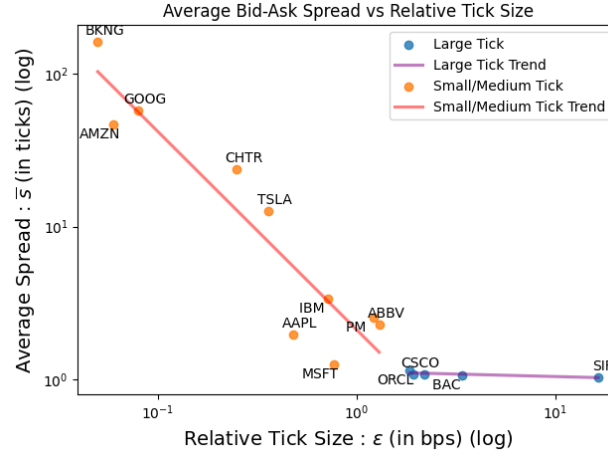


Figure 3.1: Average Spread (in ticks) vs Relative Tick Size(in bps) in a log-log scale : Remarkably, the average spread appears to be highly correlated with the relative tick size. The figure reveals clearly two regimes : a regime for large-tick assets (purple plain-line) for which the average spread is basically constant and equal to 1 tick (the minimum constrained value for the spread), and a regime for small and medium-tick assets (orange plain-line) for which (as already pointed out in Bouchaud et al. [2018]) the spread seems to be a power-law (Eq. (3.4)) of the relative tick-size with an exponent close to 1 (in our case the slope of the orange plain line is -1.23).

the following approximate relations for some constant k_s :

$$\bar{s} \simeq \begin{cases} 1, & \text{for large-tick assets} \\ k_s \epsilon^{-1}, & \text{for small- and medium-tick assets} \end{cases} \quad (3.4)$$

Let us point out a similar relation (to the latter relation) was found in Bouchaud et al. [2018]. Appendix C (Figure C.1) focuses on the distribution of the instantaneous spread $s(t)$ for 3 different assets (resp. large, medium and small-tick). The shape appears to be very different depending on the tick-size. In order to characterize these shapes, we make use of the Index of Dispersion metric (noted D hereafter) defined as the ratio of the variance of the distribution (denoted by σ^2) with the mean of the distribution (denoted by μ) i.e. $D := \frac{\sigma^2}{\mu}$. This choice is motivated by the empirical observation that with decreasing relative tick-size, one observes the mean shifting from one tick to multiple ticks however the tail of the distribution dies quicker as well. Therefore the mean increases while the variance does not seem to increase on an equivalent scale. We empirically measures the Index of Dispersion (D) one one calendar year. Figure 3.2 displays D as a function of the relative tick-size in a log-log scale. One can clearly observe a definitive linear trend as we decrease the

3.4. Trade-driven mid-price changes: Probability Distribution vs Relative Tick-Size⁷²

relative tick-size with a slope of -1.36 , so we have the relation

$$D \propto \epsilon^{-1.36} \quad (3.5)$$

Thus the shape of the spread distribution becomes less and less dispersed when

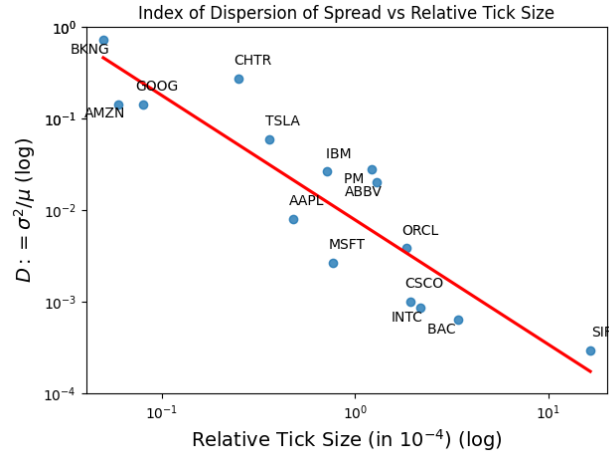


Figure 3.2: Spread Distribution's Index of Dispersion vs Relative Tick-size on a log-log scale : We observe that the index of dispersion D decreases while the relative tick-size increases. There seems to be a single regime for both small and large-tick assets implying a smooth transition of spread distribution's shape. The slope of the linear regression fit is -1.36 corresponding to the power-law regime of Eq. (3.5).

increasing the relative tick-size of an asset. Let us point out that, unlike the previous figure, there does not seem to be a regime change when we move from large to small-tick assets.

3.4 Trade-driven mid-price changes: Probability Distribution vs Relative Tick-Size

Mid-price changes can occur either due to a bid/ask order at a new best price or due to a market or cancellation order depleting the best bid or best ask queue. Since the price of an asset is restricted to be a multiple of the tick-size, the mid-price change has a minimum constrained value minimum of half a tick. In this section, we are focusing on mid-price changes triggered by trades. We ignore the mid-price changes due to cancellations and limit orders placed inside the spread due to two reasons. First, we want to decouple the effect of the current bid-ask spread to the mid-price changes. Therefore limit orders placed inside the spread are excluded. Secondly, in

3.4. Trade-driven mid-price changes: Probability Distribution vs Relative Tick-Size⁷³

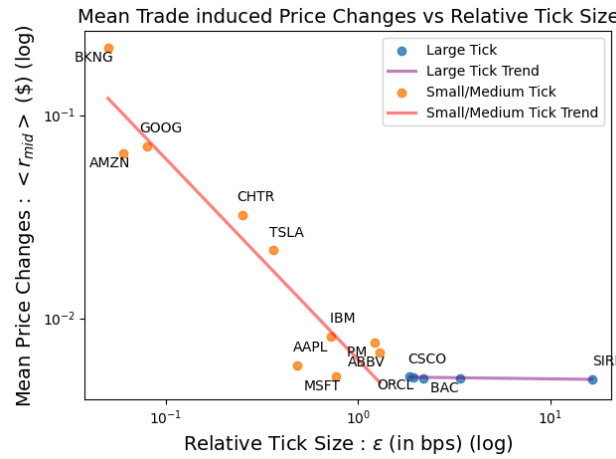


Figure 3.3: Mean Mid-Price Changes (in \$) vs Relative tick-size (in bps) in a log-log scale: The figure reveals clearly two regimes. For large-tick assets mid-price basically always change by half a tick, this is not the case for small and medium-tick assets. For the latter, a linear behavior (slope of -0.955) is found. Thus the trade induced mean price changes are inversely proportional to the relative tick-size (Eq. (3.7)). As explained in the text, it supports the use of Geometric Brownian Motion models of the price process.

order to filter spoofing³ driven mid-price changes, we exclude the cancellations as well. More precisely, we study the mean absolute trade induced mid-price moves defined by

$$\langle r_{mid} \rangle := \mathbb{E}_{(trades)} |\Delta p_{mid}(t_{trade})| \quad (3.6)$$

where $p_{mid}(t)$ is the instantaneous mid-price at time t and $\Delta p_{mid}(t_{trade})$ the variation of the mid-price directly induced by the trade occurring at time t_{trade} .

Figure 3.3 displays $\langle r_{mid} \rangle$ versus the relative tick-size on a log-log scale. Again, the figure clearly reveals two regimes. For large-tick assets, $\langle r_{mid} \rangle$ is basically constant and equal to its minimum constrained value ($1/2$ tick = $1/2$ cent), whereas for medium and small-tick assets, the two quantities seem to be linked, in a good approximation, by a power law (the purple plain line has a slope of ≈ -0.955). Thus we can therefore state the following approximate relation for some constant k_r :

$$\langle r_{mid} \rangle \simeq \begin{cases} \frac{1}{2}, & \text{for large-tick assets} \\ k_r \epsilon^{-1}, & \text{for small- and medium-tick assets} \end{cases} \quad (3.7)$$

³Spoofing is defined as a market manipulative practice where a trader intentionally places large orders to buy or sell an asset on one side of the order book, with the sole purpose of creating a false impression of market demand or supply, and then quickly cancels those orders before they are executed.

3.4. Trade-driven mid-price changes: Probability Distribution vs Relative Tick-Size⁷⁴

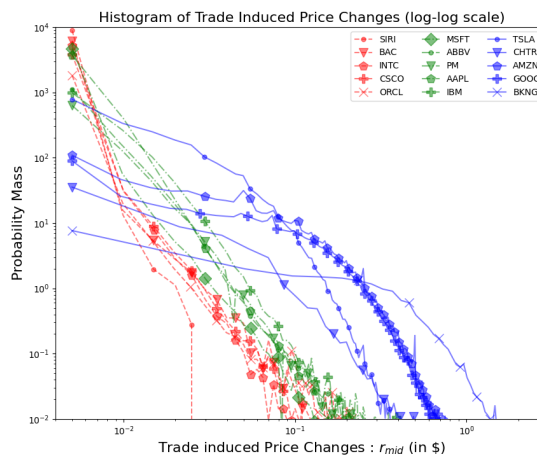


Figure 3.4: Histogram (log-log) of Trade induced Mid-Price Moves (in \$): Notably the tail of all large (red) and medium (green) tick assets quickly dies quickly after half a tick however the distribution has a higher variance and a higher (second) mode for small-tick assets (blue). This implies that it is not just the mean of the mid-price changes that vary with relative tick-size but the shape of the distribution changes significantly as well.

Since the relative tick-size is inversely proportional to the price of the security (see Eq. (3.3)), we therefore conclude that the mean of trade drive mid-price changes is almost linearly related to the current price of the security. This experiment supports the use of geometric scaling for price process models (the most popular one being the Geometric Brownian Motion). As mentioned previously, this is not true for large-tick assets due to the flooring of mid-price changes at half-ticks.

Actually, not only the mean of mid-price change induced by market orders depends strongly on the relative tick-size but the distribution too. In Figure 3.4, we overlay the density of mid-price moves resulting from a market order in the LOB. We note that the tail of the distributions become fatter as we decrease ϵ . Further we can see clearly that while the mode of all distributions is at 1/2 cents, the second mode increases rapidly with decreasing ϵ . These observations imply that the distribution of mid-price changes is heavily shaped by the asset's relative tick-size. For large-tick assets like SRI, the probability of a price move exceeding one tick (e.g., one cent on the x-axis) is almost zero. Conversely, for small-tick assets, price moves of hundreds of ticks (i.e. one dollar) are relatively common.

These results are not very surprising. Indeed, on the one hand for large-tick assets, market participants that send market orders are generally reluctant to deplete more than one level of the LOB at once. The induced “loss” of 1 (or more) tick for part of the order is significant, since the relative tick-size is large. Thus most market

orders induce a change of 1/2 tick in the mid-price. On the other hand, when the relative tick-size is small (or medium), a “loss” of 1 tick for a market participant is not that much significant, thus market participants are much less reluctant to deplete several queues with a single market order. As shown in Appendix C (see Fig. C.2), this behavior can be confirmed by studying the average market order sizes compared to the best queue size as a function of the relative tick-size of the asset.

Let us finally point out that the overall phenomenon is reinforced by the fact that the LOB of small-tick assets is much sparser than the one of a large-tick asset (and sparsity of the LOB, induces larger move in the mid-price). See Section 3.6 for sparsity related stylized facts.

3.5 Shape of the LOB: Average Volume Distribution and its Maxima vs Relative Tick-Size

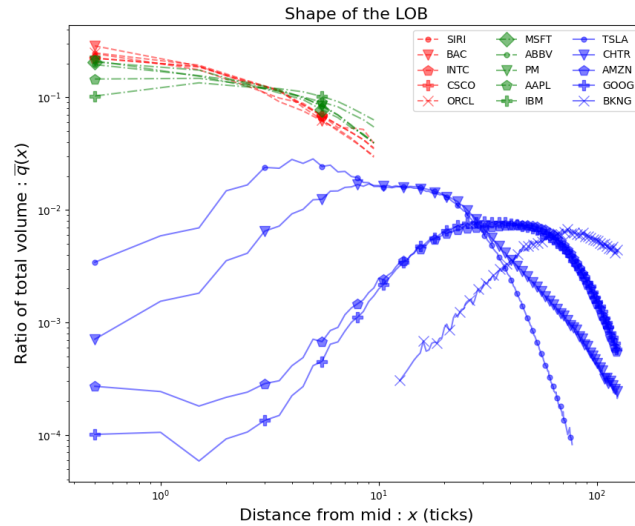


Figure 3.5: Shape of the LOB - empirical density (log-log): We plot the distribution of liquidity across price levels in this figure (Eq. (3.9)). The tails of large (red) and medium (green) tick assets is insignificant beyond 10 ticks and hence have been omitted. For all large- and most of the medium-tick assets, most of the liquidity is at the top of the LOB. As we decrease the relative tick-size, the distribution shifts progressively from near-top concentrated distribution to flatter distributions and finally to much deeper concentrated distributions. Actually, for small-tick assets (blue), we can clearly see a shift of the maxima to deeper levels as well as consistent shapes of the LOB.

Let the volume resting at a price level p at time t be denoted as $Q(p, t)$, which refers to the outstanding quotes at that price. We define the ‘instantaneous’ shape of

Asset	1st Quartile	2nd Quartile	3rd Quartile
SIRI	1.1 ± 0.54	2.7 ± 0.88	5.2 ± 1.10
BAC	0.7 ± 0.42	2.2 ± 0.52	4.6 ± 0.62
INTC	1.3 ± 0.40	2.6 ± 0.35	5.1 ± 0.53
CSCO	1.3 ± 0.39	2.5 ± 0.29	5.0 ± 0.53
ORCL	1.1 ± 0.45	2.5 ± 0.33	4.6 ± 0.43
MSFT	1.5 ± 0.15	2.9 ± 0.43	5.5 ± 0.33
AAPL	1.8 ± 0.31	3.8 ± 0.43	6.5 ± 0.51
ABBV	1.6 ± 0.38	3.5 ± 0.55	6.1 ± 0.70
PM	1.6 ± 0.40	3.6 ± 0.61	6.4 ± 0.84
IBM	2.5 ± 0.66	4.9 ± 1.04	7.9 ± 1.36
TSLA	7.7 ± 2.63	15.0 ± 4.37	23.4 ± 5.80
CHTR	11.8 ± 2.51	20.4 ± 4.44	33.6 ± 9.27
AMZN	32.1 ± 6.32	49.2 ± 8.52	67.0 ± 10.45
GOOG	31.2 ± 6.20	47.6 ± 8.41	66.9 ± 10.23
BKNG	59.3 ± 8.01	78.6 ± 7.52	98.8 ± 5.95

Table 3.2: Shape Quartiles (in ticks): For the 15 stocks we study, we report the three quartiles of the average shape of the LOB ($\bar{q}(x)$) in this table. As we can see the 1st and 2nd quartile for large-tick assets remains constant at one tick and three tick distance respectively from the mid-price. This underlines the fact that the available volume for these assets is concentrated at the top of the LOB. This fact is not true for medium and small-tick assets as can be seen by the almost monotonous shifting of the quartiles deeper in the LOB. Therefore the liquidity in medium- and small-tick assets is not concentrated at the top but rather is distributed across various levels deeper in the LOB.

the LOB at time t as the normalized distribution of the volume $Q(p, t)$ as a function of the distance of the price level p from the mid-price $p_{\text{mid}}(t)$ and denote it by $q(x, t)$ where $x := \frac{p - p_{\text{mid}}(t)}{\delta}$ is the distance from mid-price in ticks. Further we define the average shape of the LOB as the time-weighted average of $q(x, t)$ denote it by $\bar{q}(x)$. Similar to Equation (3.2), we have for index $i = 1, \dots, N$ and time $0 \leq t_i \leq T$, LOB volume distribution $q(x, t_i)$ lasting for Δt_i seconds, so the calculation of $\bar{q}(x)$ is:

$$q(x, t) := \frac{Q(p_{\text{mid}}(t) + \delta x, t)}{\sum_y Q(p_{\text{mid}}(t) + \delta y, t)} \quad (3.8)$$

$$\bar{q}(x) := \frac{\sum_{i=1}^N q(x, t_i) \Delta t_i}{T} \quad (3.9)$$

The average shape of the LOB for all 15 assets are displayed in Figure 3.5. It clearly shows that the distribution of liquidity shifts from near-top concentrated distribution for large-tick assets to flatter behavior for medium-tick asset and finally to deeper

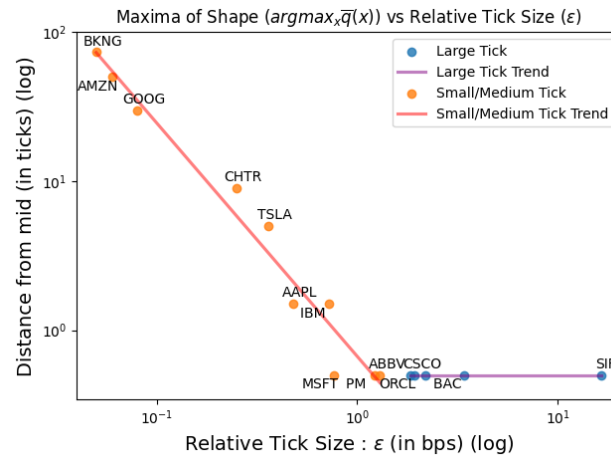


Figure 3.6: Depth with Maximum Liquidity of the LOB (log-log): Again we observe two regimes - the large-tick assets have the maximum liquidity at the top of the LOB (which is at a distance of $1/2$ tick of the mid-price) for the small-tick assets, the maximum liquidity is correlated with the current relative tick-size (Eq. (3.10)). This correlation is power law with an exponent over relative tick-size of approximately $-\frac{3}{2}$.

concentrated distribution for small-tick assets. Table 3.2 displays the corresponding quartiles of these distributions. One of the primary observations we have from this table is that around 50% of liquidity of the entire LOB is in the first ~ 2.5 price levels (measured from the mid-price) for large-tick assets. This along with the fact that the spread is almost always one tick in these assets implies that around half the liquidity of the LOB lies within the top two price levels of a large-tick LOB. This is clearly not the case for medium- and small-tick assets where the liquidity is more spread out across several deeper price levels. Further, the table clearly shows that basically the quartiles are increasing with decreasing relative tick-size, indicating that a “shift” of the mass of the distribution on the right side as the relative tick-size decreases. This result is confirmed by Fig. C.3 in Appendix C.3 which illustrates this “shift” by displaying some per asset daily and averaged over the year shapes of the LOB for a large, medium and small-tick asset.

The same “shift” can be illustrated directly plotting the average shape of the LOB profile (i.e. $\arg \max_x \bar{q}(x)$) as a function of the relative tick-size as done in Fig. 3.6. As usual, the figure reveals two regimes. On the one hand, large-tick assets have the maxima of the shape uniformly at half a tick away from the mid-price (fit shown in purple line). On the other hand we see an almost power-law trend wherein the maximum liquidity shifts deeper in the LOB as we decrease the relative tick-size (fit shown in red line). The slope of the red line is -1.513 , therefore the relation

is approximately power law with an exponent of $-\frac{3}{2}$ over the relative tick-size. For some constant k_q , these results can be summarized as :

$$\arg \max_x \bar{q}(x) \simeq \begin{cases} \frac{1}{2}, & \text{for large-tick assets} \\ k_q \epsilon^{-\frac{3}{2}}, & \text{for small- and medium-tick assets} \end{cases} \quad (3.10)$$

Let us point out that Bouchaud et al. [2018] mention the same fact that most liquidity in small-tick assets actually lies much deeper in the LOB however to the best of our knowledge, our work is the first to form a statistical relation between this proxy of ‘market depth’ and relative tick-size or mid-price. Since x is measured from the mid-price, and since the spread is scaling with the exponent of -1 , the maximum volume’s distance from the respective top should scale in a square-root manner with the inverse of relative tick-size from our observations. In Appendix C.3, we also plot the empirical density of the instantaneous maxima to analyze the depth at which maximum liquidity lies in the LOB in Figure C.4. We observe that not only the maxima of the average shape of the LOB is shifting deeper in the LOB but also the distributions of maxima of the instantaneous shape of the LOB are remarkably different. As we decrease ϵ , we clearly see the instantaneous maxima’s modes shifting deeper as well. We can therefore conclude that the top of the LOB is not necessarily the best place to post a passive limit order for these categories of assets since the maximum liquidity most likely lies much deeper in the LOB.

3.6 Sparsity of the LOB: Empty Price Levels and their effect on the instantaneous LOB’s Shape

A well known feature in small-tick assets’ LOB is that not every price level in the LOB will have liquidity. In this section we focus on the sparsity of the LOB as a function of the relative tick-size.

The most straightforward way to measure the sparsity of an LOB is to count the number of empty levels between successive quotes. Following this idea, let $n_e(i, t)$ denote the number of empty levels between the two successive quotes ranked i and $i + 1$, at time t (for each rank i , we add up the numbers on bid and ask sides). Figure 3.7 displays the empirical distribution (log-scaled) of $n_e(1, t)$ (number of empty levels between the best quote and the second best quote) sampled at each time t . Notice that as the relative tick-size decreases, the distribution becomes flatter and heavier-tailed. This clearly implies that while for large-tick and medium-tick assets, the LOB has liquidity in almost all levels near the top of the LOB, for small-tick

assets, we can have dozens of price levels empty between the best and the second best quote. In Appendix C.4, we confirm that this pattern persists for deeper levels in the LOB (Figure C.5).

However these measurements can be biased by insignificant very small orders standing alone in their queue. To avoid these biases, as a sparsity proxy, we choose to measure the time average (and the corresponding variance) of the Wasserstein distance (or Earth-Mover distance) $l(\bar{q}, q(., t_i))$ between the average shape of the LOB $\bar{q}(x)$ and the shape of the LOB $q(x, t_i)$ at a timestamp t_i :

$$l(\bar{q}, q(t_i)) := \sum_x |\bar{q}(x) - q(x, t_i)| \Delta x \quad (3.11)$$

Not surprisingly, Figure 3.8 shows that roughly, the larger Wasserstein distance the smaller the relative tick-size. However, let us point out the relation between the two is not linear and there are a few outliers (e.g., BKNG, MSFT).

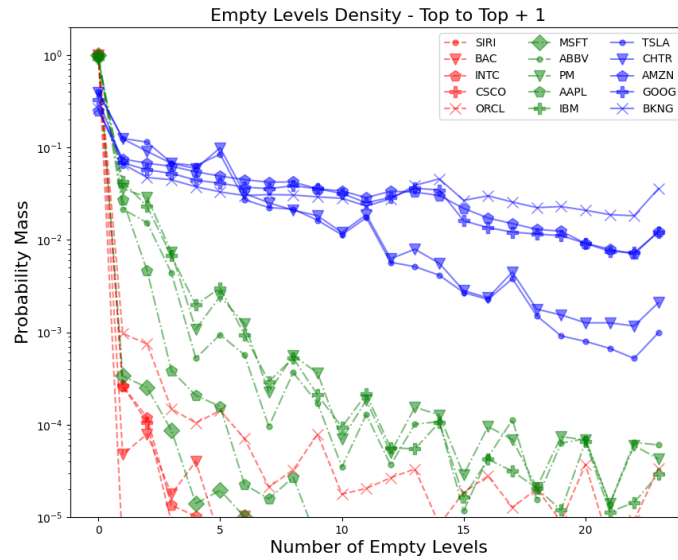


Figure 3.7: Empty Levels between Top and Top + 1 (log density) : This depicts that the distribution of the number of empty levels becomes flatter as we decrease the relative tick-size. Indeed we can clearly observe that for large-tick assets (red), the distribution is very narrow and most of the mass is at 0. However for small-tick (blue), there is a significant probability of more than 10 empty levels in the LOB.

It is worthwhile discussing a few phenomenological explanations of this observed sparse LOB. A sparse LOB likely results from market participants' preferences and strategies. Traders often queue behind existing liquidity rather than posting more

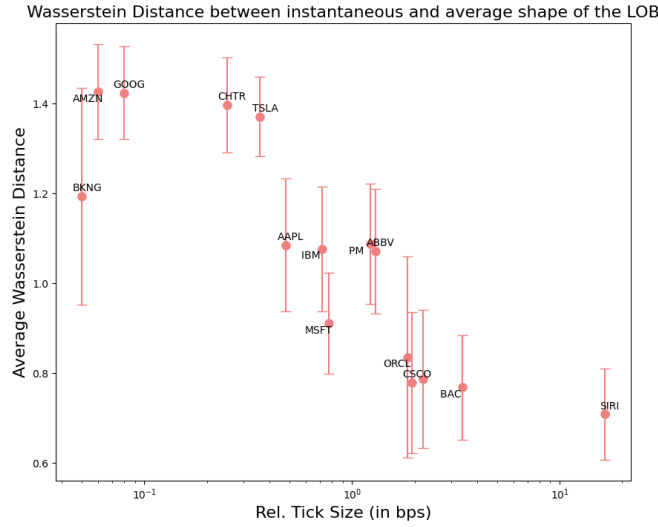


Figure 3.8: Typical vs Average Shape of the LOB: This measure of Wasserstein Distance (Eq. (3.11)) indicates how different, on a time-weighted average scale, is the instantaneous shape of the LOB from the average shape. This difference indicates whether the shape of the LOB remains persistent in the LOB or does the LOB’s liquidity vary a lot owing to the sparse levels in the LOB. As we can see clearly here, there is an increasing trend as we decrease the relative tick-size however this relation seems to be non-linear. The error bars denote the variance of the Wasserstein Distance observed in our dataset.

aggressive prices to avoid adverse selection, despite lower fill probability. They may also place orders deeper in the LOB or at round number offsets from the best quote. This behavior, which exposes them to execution risk i.e. the sharp decrease in execution probability for deeper orders, contributes to empty price levels and overall sparseness in the LOB, particularly in small-tick assets.

3.7 LOB Events and their Endogenous Excitation:

Structural Similarities across Relative Tick-Sizes

As previously discussed, a Limit Order Book can be modeled as a queuing system, where various types of events contribute to the formation and evolution of these queues. Empirical observations suggest that the realizations of these events persistently exhibit both cross-correlation with other events and autocorrelation with themselves. Notable works in this direction include Lillo and Farmer [2004], Bouchaud et al. [2009], Yamamoto and Lebaron [2010], and, Tóth et al. [2015]. This phenomenon is commonly referred to as market endogeneity (Bouchaud et al. [2018]), reflecting the idea that market activity is influenced by its own past behavior. Our focus lies in

analyzing the conditional probability distribution of these events, given the history of prior events, in order to capture the feedback mechanisms inherent in the dynamics of the LOB. Inspired by Lu and Abergel [2018a], in this section, we study how the occurrence of certain events in the LOB affects the likelihood of other events happening shortly afterward.

Let us denote an event e happening at x ticks distance from the respective top of the LOB by e_x . We use positive (resp. negative) values for x to refer to ask (resp. bid) side events. In the following we will consider four types of events e^4 :

- Passive Limit Orders ($e = LO$), i.e., limit orders which do not change the mid-price
- Active or ‘In-Spread’ Limit Orders ($e = IS$) i.e. limit orders which are posted at an improved price than the respective top price,
- Cancel Orders ($e = CO$),
- and Market Orders ($e = MO$)

Let us note that market orders can only occur at best levels, thus MO_x is only valid for $x = 0$.

Given two consecutive events $e_x^{(1)}$ and $e_y^{(2)}$, we compute the ratio of the conditional probability (i.e., the chance of $e_y^{(2)}$ given $e_x^{(1)}$) to the unconditional probability of $e_y^{(2)}$ occurring i.e. $\frac{P(e_y^{(2)}|e_x^{(1)})}{P(e_y^{(2)})}$. This ratio is known as Conditional Probability Leverage or simply ‘leverage’. We illustrate the results in Figure 3.9 for two assets : a large-tick asset (BAC) on the left and small-tick asset (GOOG) on the right. For some more plots for a medium-tick asset (IBM) and another small-tick asset (AMZN), we refer the reader to Figure C.7 in the Appendix C.5. To facilitate a qualitative analysis, we plot the leverage for all these events on a 2D grid with the x-axis representing an event $e_x^{(1)}$ and the y-axis the successive event $e_y^{(2)}$. As indicated by the colormap, a red (resp. blue) shade corresponds to a leverage higher than one which corresponds to an *excitation* (resp. *inhibition*) of $e_y^{(2)}$ by $e_x^{(1)}$. For the large-tick asset we use a linear scale for x and y with $x, y \in [0, 10]$. For the small-tick asset we use a linear scale for x and y up to $x, y < 10$ and a log scale for $x, y \geq 10$ and smaller than the 95% quantile of the shape of the LOB (this is done to enforce high resolution near the top of the LOB).

⁴Of course we are aware there exist several more order types offered by the exchanges (for e.g., fill-or-kill, stop loss etc). However these four types of events are the most basic ones. And the more exotic ones can generally be thought, mathematically, as a combination of these four basic order types.

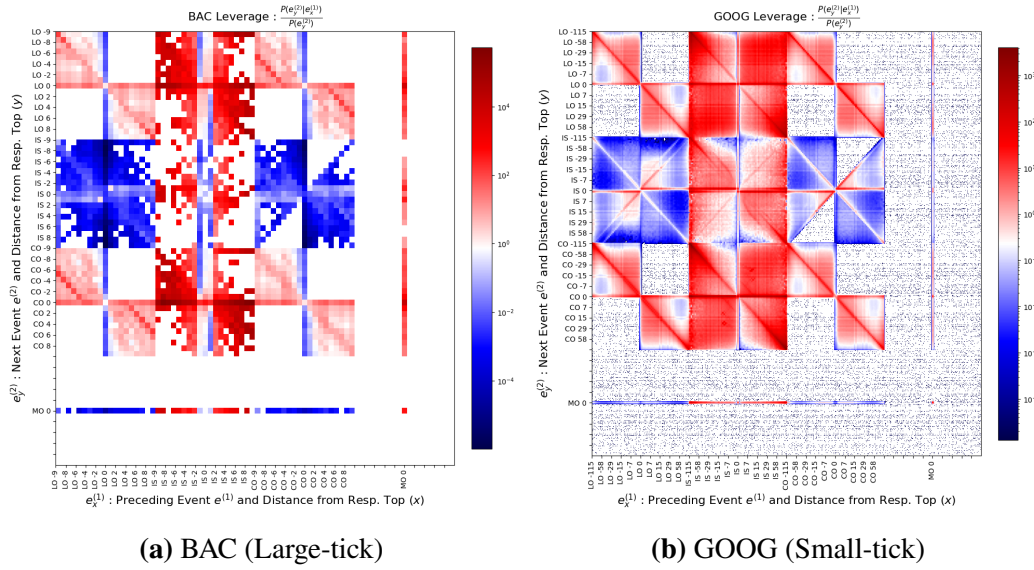


Figure 3.9: Leverage (from top of the LOB): Red shade implies Excitation, White implies Independence and Blue implies Inhibition - The x-axis represents the preceding event, and the y-axis represents the future event. This figure highlights both common and distinct features across different assets. Notably, similar patterns are observed in LO-LO, CO-CO, and MO-MO excitations, while differences are evident in CO-LO and LO-CO blocks between large-tick (Fig. 9a) and small-tick (Fig. 9b) assets. For a detailed analysis, see the main text.

Figure 3.9 shows that most of the main patterns of the leverage do not seem to depend strongly on the relative tick sizes (albeit with different scaling). Thus, the way events influence each other is consistent, up to a linear-log scaling, regardless of how large or small the tick-size is. These findings provide insights about the dynamics of event interactions within the LOB. Let us point out some of the main patterns that do not depend on relative tick-size and some of the main differences that do depend on relative tick-size :

1. Whatever the relative tick-size is, we observe that LOs excite COs and vice-versa. This is not surprising as Cancels are causally related to previously posted LOs, and, Cancels usually imply posting a new LO at a possibly different depth from the mid-price.
2. Whatever the relative tick-size is, we observe that the LOs at the top of the LOB behave remarkably differently from the ones at deeper levels. For instance, in Figure 3.9a, $e_x^{(1)} = LO_0$ mostly inhibits other orders (i.e. the vertical line is blue colored) however $x = LO_2$ excites $e_y^{(2)} = LO_{0/1/2}$. A similar pattern is observed in Figure 3.9b.
3. For small-tick assets, COs and LOs behave very similar to each other except for

the off-diagonal terms which indicates that deeper LOs lead to cancellations nearer to the top. Additionally deeper COs lead to LOs nearer to the top as well. However the pattern is really difference for large-tick assets (see Figure 3.9b). Figure C.7a in the Appendix C.5 shows that the transition is smooth going from small to medium to large-tick assets.

4. ISs are both self excited and cross excited by opposite end LOs and COs in a triangular fashion. This is not visible in Figure 3.9a due to the fact that the probability of a higher than one tick distance from mid-price IS event is very low, but in Figure 3.9b, one can clearly see the self excitation and the cross excitation by LOs and COs in the red anti-diagonal. This is an interesting un-intuitive finding as deeper LOs and COs seem to be affecting opposite side in-spread LOs.
5. ISs excite other events very similarly between small and large-tick assets. They seem to excite themselves both on the same and the opposite side of the LOB, and in general looks to increase all kinds of events' occurrence probability. Notable exceptions are when IS happens very close to the current top of the LOB.
6. As already mentionned, MOs naturally only occur at the top of the LOB, however we can see very similar patterns across all types of assets for market orders. They excite all other kinds of orders and are themselves inhibited by previous LOs and COs.

3.8 Conclusion

In this chapter we outline how, statistically, does the LOB change when the granularity of price levels is kept constant with a constant tick-size while the price of the security increases from a few dollars to a few thousands of dollars. We quantify this change by the change in relative tick-size ϵ which is defined as the ratio between the tick-size and the price of the security. One of our key contributions is to highlight several stylized facts, which are then used to differentiate between large-, medium- and small-tick assets, and define clear metrics to measure them. We also provide a set of cross-asset visualizations of the stylized facts and show how different (or similar) these attributes are with varying relative tick-size. We found different power law relations in these stylized facts. Firstly, the average spread of a security and the average absolute returns of the security are both approximately proportional to the inverse of the relative tick-size. Secondly, the depth at which we find the maximum

liquidity in the LOB follows a power law over the relative tick-size with the exponent roughly equal to $\frac{3}{2}$. In addition to that we see definitive trends in the shape of the spreads distribution as well as the sparsity in the LOB as well decrease the relative tick-size. We make use of a stylized fact known as *leverage*, which seems to be universally similar in structure across all assets we studied, to model the LOB as a Hawkes Process with the top of the LOB as a separate dimension and the rest of the LOB (roughly upto the median of the volume shape of the LOB) as a separate dimension.

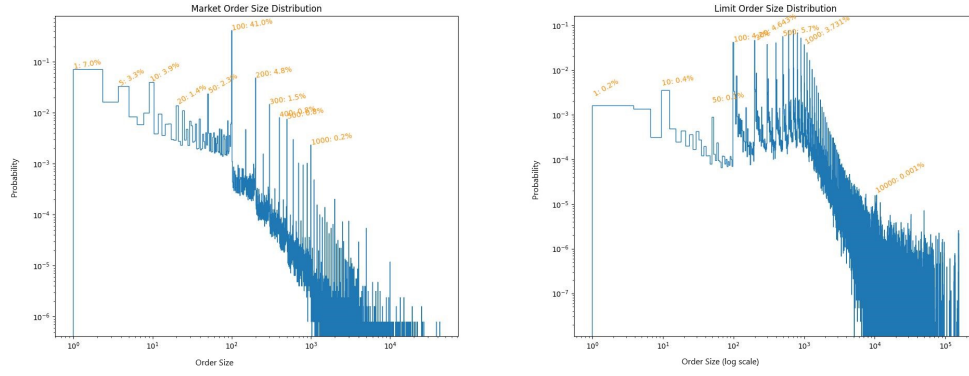
Chapter 4.

Limit Order Book Simulation: The Compound Hawkes Process Methodology

This chapter focuses on the creation of a realistic LOB simulator by taking inspiration from the current state of the art and the literature covered in Chapter 2. Particularly, the simulator is built to be usable by the RL agent and is realistic in particular with respect to Market Impact. This will ensure practical applicability of the RL agent we introduce in Chapter 6. Finally a rigorous goodness of fit study is undertaken in this chapter to make sure the stylized facts mentioned in Chapter 3 are matching between the simulator and the real world.

4.1 Introduction

The Hawkes process, known for its high adaptability, offers a more comprehensive point process methodology for modeling order book arrivals than the Poisson process and its variants, without the need to explicitly model individual traders' behaviors in the market. Its capability to replicate microstructural details such as volatility clustering and correlated order flow makes it a suitable candidate for Limit Order Book (LOB) models. It is important to highlight that these point process models are mathematically descriptive, providing full transparency in their nature and thus are suitable for applications where black-box solutions are not preferred. In their comprehensive review and tutorial, Bacry et al. [2015b] outlay the major ideas of the Hawkes Process, its mathematical theory, some of its crucial properties and finally applications including a detailed review over the Order Book models. Recently, state dependent Hawkes Process have been quite popular (Morariu-Patrichi and Pakkanen [2022], Kirchner and Vetter [2022], Mucciante and Sancetta [2023], Wu et al. [2019]). However, as noted in Rambaldi et al. [2017] and Lu and Abergel [2018b], individual order's size is an important aspect of the LOB which the Hawkes model alone is



(a) Market Orders' Size Distribution (log-log scale) (b) Limit Orders' Size Distribution (log-log scale)

Figure 4.1: Empirical Distribution of Order Sizes

unable to capture.

4.1.1 Empirical Distributions of Order Sizes:

We show here in Figure 1 two trading weeks' of data (10 dates) for Apple and plot the empirical histograms on a log-log scale. We only focus on top of the book cancels and limit orders. As Figures 4.1a and 4.1b show, the market orders and limit orders sizes have several spikes at round numbers indicating the preference of the traders. For example, we see more than 40% of the market orders and 60% of limit orders are of 100 size. Naturally, Cancel Orders' sizes are capped at the size of an individual order. In fact we observe that outright cancels (i.e. full quantity of an order is cancelled) constitute 99.3% of all cancels.

4.1.2 Properties of the Order Book Dynamics:

There have been several variations to the Hawkes model to accommodate for certain properties of the order book in exchanges.

Prop 1: Bid-Ask Spread is always non-negative: The bid-ask spread can never be less than zero being an important one was tackled by using order arrival intensities dependent on spread-in-ticks by Lee and Seo [2022]. Previously, Zheng et al. [2014] have used constrained Hawkes Process to control negative spreads.

Prop 2: Order intensities are dependent on time-of-day: It is well known that trading volumes follow an intraday seasonality which is observed to be stationary across multiple days. Naturally, we observe that the order intensities too exhibit this intraday seasonality (more in Section 2). Hawkes models have been adapted to account for the same in Mucciante and Sancetta [2023], Prenzel et al. [2022], and Kirchner and Vetter [2022].

Prop 3: Cross-excitations can be inhibitory: As noted in Lu and Abergel [2018a], the cross-excitation of events need not necessarily always be catalyzing. For example, we observe that limit orders inside the spread and cancels at top at opposing sides of the book have inhibitory effects on each other. Generally modelling for this can introduce negative intensities in the point process. To avoid such complications, Lu and Abergel [2018a] propose to floor the total intensity of any element of the Hawkes process to zero.

In this work we show some lack of support for the hypothesis that the order arrival intensities are impacted by the past order sizes. Thus we conclude that the Compound Hawkes Process is a suitable candidate for the model. We then create a stationary distribution of the order sizes for each type of order which closely mimics the empirical distribution. This distribution is used to sample the order sizes in the compound Hawkes Process. We create a novel formulation of the Hawkes intensities which satisfies Properties 1, 2 and 3. In Section 3, we show the calibration results for the Apple Inc. stock by using Level 2 data in the NASDAQ exchange. We also provide an analysis for the quality of fit.

4.2 Methodology

4.2.1 Mathematical Formulation:

We consider the task of modelling LOB's current best bid and ask (i.e. Level 1 LOB). We note that since placing order in the spread is a common practice in equities, the reason being that the spread usually has many empty price levels to attract market participants to place a more aggressive order, the Hawkes formulation should contain modelling the intensity of the nearby empty levels as well. Hence we formulate the order book as a queueing system with six different queues which are $\{\text{ask}_{+1}, \text{ask}_0, \text{ask}_{-1}, \text{bid}_{+1}, \text{bid}_0, \text{bid}_{-1}\}$.

Here ask_0 and bid_0 are the respective best ask and bid prices while the subscripts denote the price level distance in ticks from the best ask/bid. for e.g.. ask_{+1} is the price level which is 1 tick more than ask_0 . The 2×6 vector of $((q_k^{(s)}, s_k)_{s \in \{\text{ask}, \text{bid}\}, k \in \{-1, 0, +1\}})$, where $q_k^{(s)}$ denotes the queue size, is what we define as the state of the order book. The motivation to model just these six levels and not the entire order book is two-fold. Firstly, the question of parsimony becomes important when we model more levels than just these six. Secondly, we observe that the LOB state changes are generally events which move the prices by 1 tick. Indeed we observe, in our dataset, 97.4% of all in-spread orders occur 1-tick away from the best respective quote with mean being 1.035 ± 0.285 ticks for 2 million samples, and 97.5% of all price changes

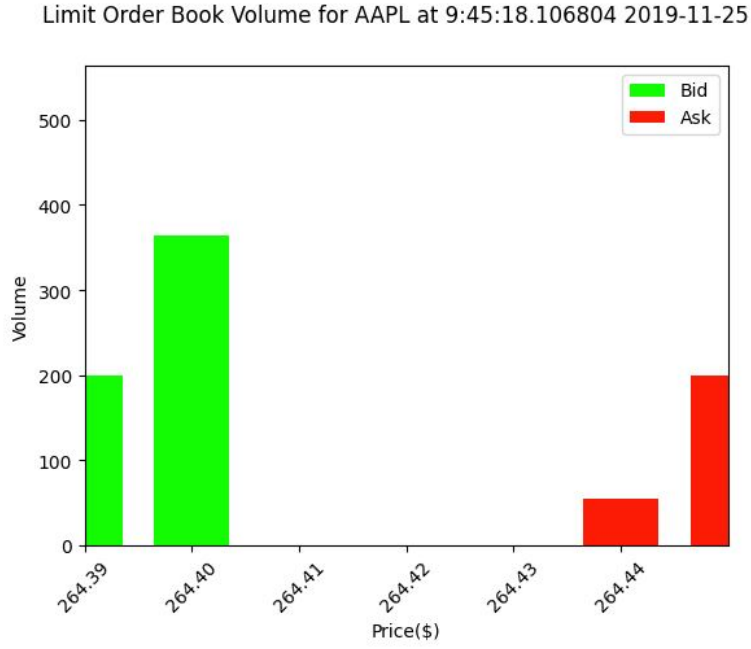


Figure 4.2: Order Book Snapshot

have the next non-empty price level at 1-tick distance as well with the mean being 1.033 ± 0.283 ticks.

Mechanical Constraints: Naturally, since ask_0 and bid_0 are the best ask and bid prices, the queue size at $\text{ask}_{-1}/\text{bid}_{+1}$ is zero. Therefore any incoming limit order at $\text{ask}_{-1}/\text{bid}_{+1}$ (i.e. an in-spread (IS) order) creates a new best ask/bid and so the LOB state transforms from

$$S := \{(q_{+1}^{(\text{ask})}, \text{ask}_{+1}), (q_0^{(\text{ask})}, \text{ask}_0), (0, \text{ask}_{-1}), (0, \text{bid}_{+1}), (q_0^{(\text{bid})}, \text{bid}_0), (q_{-1}^{(\text{bid})}, \text{bid}_{-1})\}$$

to

$$S_{IS} = \{(q_0^{(\text{ask})}, \text{ask}_0), (q_{-1}^{(\text{ask})}, \text{ask}_{-1}), (0, \text{ask}_{-2}), (0, \text{bid}_{+1}), (q_0^{(\text{bid})}, \text{bid}_0), (q_{-1}^{(\text{bid})}, \text{bid}_{-1})\}$$

for an in-spread ask side limit order.

Note that it is possible that the spread is only 1-tick wide which would mean ask_{-1} and bid_{+1} coincides. There is also the possibility of zero spread in which case ask_{-1} and bid_{+1} do not exist. We will put some constraints on the Hawkes process intensities for these levels further in this section to account for these possibilities. On

the other hand, if a queue-depletion (QD) event at best bid/ask happens (for e.g., a large enough market order), the spread widens and therefore the state moves from S to

$$S_{QD} = \{(q_{+2}^{(\text{ask})}, \text{ask}_{+2}), (q_{+1}^{(\text{ask})}, \text{ask}_{+1}), (0, \text{ask}_0), \\ (0, \text{bid}_{+1}), (q_0^{(\text{bid})}, \text{bid}_0), (q_{-1}^{(\text{bid})}, \text{bid}_{-1})\}$$

for a queue-depleting ask side market order/cancel order. Here we observe an unknown quantity $q_{+2}^{(\text{ask})}$. We sample this unknown quantity from a stationary distribution calibrated from empirical data. Finally, we note that Market Orders can only occur at $\text{ask}_0/\text{bid}_0$ and $\text{ask}_{-1}/\text{bid}_{+1}$ cannot have Cancel Orders since by definition the quantity there is zero.

Compound Hawkes Process: A compound point process (CPP) Z_t is defined as

$$Z_t = \sum_{i=1}^{N(t)} Y_i \quad (4.1)$$

where $\{N(t) : t \geq 0\}$ is the counting process associated with a point process $\{Y(t) : t \geq 0\}$ with intensity $\lambda(t)$, and $Y_i \sim P_\Theta(Y_i)$ is a random variable denoting the jump size with the corresponding $P_\Theta(Y_i)$ being the jump size distribution. Compound Hawkes process (CHP) is a special case of the CPP where $N(t)$ is a counting process associated with a Hawkes Process. For a d -dimensional Hawkes process the intensity of the process $\lambda_t^{(i)}$ and the associated counting process $N_t^{(i)}$ for $i = 1, \dots, d$ is defined as:

$$\lambda_t^{(i)} = \mu_t^{(i)} + \sum_{j=1}^d \sum_{t_j \in T_j} \phi^{(j \rightarrow i)}(t - t_j) \quad (4.2)$$

where $T_j := \{t_j : t_j \leq t\}$ denotes the set of past event times in the j dimension of the Hawkes Process. Here, $\mu_t^{(i)}$ is the exogenous intensity of the i -th dimension and $\phi^{(j \rightarrow i)}(t - t_j)$ is the excitation term from j -th dimension to i -th dimension. The excitation terms are a function of the time since the event (generally a decaying function in time like exponential decay or power law decay). An alternate but similar

formulation is the following:

$$\lambda_t^{(i)} = \mu_t^{(i)} + \sum_{j=1}^d \int_0^t \phi^{(j \rightarrow i)}(t-s) dN_s^{(j)} \quad (4.3)$$

Limit Order Book at Compound Hawkes Process: We define a 12D Compound Hawkes Process (in accordance with the mechanical constraints) for the 6 queues in the order book state where each dimension corresponds to the following event types:

$$\{LO_{ask_{+1}}, CO_{ask_{+1}}, LO_{ask_0}, CO_{ask_0}, MO_{ask_0}, LO_{ask_{-1}}, \\ LO_{bid_{+1}}, LO_{bid_0}, CO_{bid_0}, MO_{bid_0}, LO_{bid_{-1}}, CO_{bid_{-1}}\}$$

where LO is Limit Order, CO is Cancel Order and MO is Market Order. Here each of the 12 event types' order size ($\kappa^{(e)}$ for event e) is sample from their own specific calibrated distribution ($\Pi^{(e)}(\Theta)$). We postulate that the order intensities themselves are not impacted by the past order sizes but only the past order event-times as is the general assumption in Hawkes models applied to LOB data. We provide some weak evidence for this claim in the Appendix B.1. We note that our work on the distribution of order size closely follows the work by Lu and Abergel [2018b] however they make use of a Poisson model. Let us also mention that Abergel et al. [2016] too use parametric distributions for order sizes in the Hawkes Process model. For Cancel Order quantities, since we observe almost all cancels in empirical data are outright cancels, we draw randomly, with equal probability, from the available limit orders' quantities in the queue. We thus create a CHP for the queue-sizes at the six queues of the LOB:

$$q_k^{(s)}(t) = \sum_{e \in E_k^{(s)}} \sum_i^{N^{(e)}(t)} \kappa_i^{(e)} \quad (4.4)$$

$$\kappa_i^{(e)} \sim \Pi^{(e)}(\Theta) \quad (4.5)$$

where $i = 1, \dots, N^{(e)}(t)$ and $E_k^{(s)}$ is the set of eligible events for s and k

To control the spread of the process (Prop 1), we formulate the intensities of $LO_{ask_{-1}}$ and $LO_{bid_{+1}}$ in the following manner as a function of the current spread-in-

ticks (s). If $\psi_t^{(\cdot)}$ represents the intensity (either exogenous or excitation), then,

$$\psi_t^{(\cdot)}(s) = (\psi_t^{(\cdot)})s^\beta; \beta > 0, s \in \mathbb{N} \quad (4.6)$$

In this formulation we enforce the mechanical constraint if $s = 0$, $\lambda_t^{LO_{ask-1}} = \lambda_t^{LO_{bid+1}} = 0$ with an extra parameter β . We motivate the choice of a power law dependence of order intensity over spread in Appendix B.2.

We observe a strong dependence between the intensities $\lambda_t^{(\cdot)}$ on the time of day of the trading day (Prop 2). There is a “U”-shape of order intensities with respect to the time of day i.e. right after the open auction and before the close auction, the activity in the market is much higher than the middle of the day. This effect is observed for trading volumes in equities with a number of probable causes but the most commonly accepted one is that because auctions cause a halt in trading, this halt leads to increased activity. To maintain model parsimony, we bin the 6.5 hour trading day of NASDAQ into thirteen 30-minute bins (indexed by $Q_t \in \{1, 2, \dots, 13\}$) for t being a time of day). We model the intensity as a separable function of time-of-day and the Hawkes Intensities:

$$\lambda^{(i)}(Q_t, t, s) = f(Q_t) \times \lambda^{(i)}(t, s) \quad (4.7)$$

Finally, to make sure the inhibitory kernels do not create negative intensities, we floor the total excitation $\lambda^{(i)}(Q_t, t, s)$ to zero.

Price Dynamics: The mid price ($P(t)$) dynamics after an event at t at side ζ (and $\xi = +1$ for $\zeta = \text{ask}$ and -1 for $\zeta = \text{bid}$) is a jump process with jumps defined by:

$$\begin{aligned} P(t^+) - P(t) = & -\frac{\eta}{2}\xi dN^{(IS)}(t) \\ & + \frac{\eta}{2}\xi dN^{(CO)}(t) \mathbb{1}_{\{\#q_0^{(\zeta)}(t)=1\}}(t) \\ & + \frac{\eta}{2}\xi dN^{(MO)}(t) \mathbb{1}_{\{q_0^{(\zeta)}(t) \leq \kappa^{(MO, \zeta_0)}(t)\}}(t) \end{aligned} \quad (4.8)$$

where η is the tick-size, $N^{(IS)}(t)$ is the counting process for IS orders, and $\#q_0^{(\zeta)}(t)$ denotes the number of individual limit orders in the queue.

4.2.2 Calibration:

We follow the non-parametric method of calibration in Kirchner [2017] and take inspiration from Bacry et al. [2016] to create a time grid at both linear and log scales to account for the slowness of the decay kernels. This method has the advantage of

not having any priors on the shape of the kernels themselves (popular choices in the literature include exponential decay and power-law decay) since it is non-parametric.

For an event e (not a spread-dependent event), with the counting process $N^{(e)}(t)$ having an intensity as in (2), we define the binned observed counts in the dataset by $X_{\Delta}^{(e,n)} := N^{(e)}(t_n) - N^{(e)}(t_{n-1})$, $n \in \mathbb{Z}$ where $\Delta := t_n - t_{n-1}$ is the bin size (taken constant). Taking inspiration from Kirchner [2017] we have, for $\mathcal{H}_{n-1}^{(e)}$ denoting the set of previous $n-1$ bins' observed values:

$$\mathbb{E} \left[X_{\Delta}^{(e,n)} \middle| \mathcal{H}_{n-1}^{(e)} \right] = \int_{(n-1)\Delta}^{n\Delta} \mathbb{E} \left[f(Q_t) \lambda^{(e)}(t) \middle| \mathcal{H}_{n-1}^{(e,N)} \right] dt \quad (4.9)$$

$$\begin{aligned} \frac{\mathbb{E} \left[X_{\Delta}^{(e,n)} \middle| \mathcal{H}_{n-1}^{(e)} \right]}{f(Q_t)} &= \int_{(n-1)\Delta}^{n\Delta} \mu_t^{(e)} dt \\ &\quad + \sum_{j=1}^d \int_{(n-1)\Delta}^{n\Delta} \int_0^t \phi^{(j \rightarrow i)}(t-s) N^{(j)}(ds) dt \end{aligned} \quad (4.10)$$

Here we use the fact that $f(Q_t)$ is piecewise constant and construct the bins such that there is perfecting overlapping between the Δ grid and the Q_t grid. Now we define another grid $\delta_k := t_{max}^{(k)} - t_{min}^{(k)}$ with $t_{max}^{(1)} = t$; $t_{min}^{(k)} = t - T$ for $k = 1, \dots, K$ and $T < \infty$ being the cut-off point for the integral to make an approximation:

$$\int_0^t \phi^{(j \rightarrow i)}(t-s) N^{(j)}(ds) \approx \sum_{k=1}^K \frac{X_{\delta_k}^{(e)}}{\delta_k} \int_{t_{min}^{(k)}}^{t_{max}^{(k)}} \phi^{(j \rightarrow i)}(s) ds \quad (4.11)$$

Here $X_{\delta_k}^{(e)}$ is the binned observed events of type e in the bin δ_k . Now we make another approximation, as in Kirchner [2017], that the grid Δ is small enough so that the intensity is constant over $(t_{n-1}, t_n]$. Now with this and substituting (11) in (10), we have:

$$\frac{\mathbb{E} \left[X_{\Delta}^{(e,n)} \middle| \mathcal{H}_{n-1}^{(e)} \right]}{f(Q_t)} \approx \Delta \mu^{(i)} + \Delta \sum_{j=1}^d \sum_{k=1}^K \frac{X_{\delta_k}^{(e)}}{\delta_k} \int_{t_{min}^{(k)}}^{t_{max}^{(k)}} \phi^{(j \rightarrow i)}(s) ds \quad (4.12)$$

As noted in Bacry et al. [2016], a linearly spaced grid to estimate cross-excitation is insufficient for slow decaying kernels like the power-law. Therefore we propose to use the same linear-log spaced grid proposed in Bacry et al. [2016] as the δ_k grid. We set $\delta_1 = \Delta$ and add $\frac{K}{2}$ linearly spaced δ_k . Lastly we add $\frac{K}{2}$ log spaced δ_k to get the final grid

$G = [0, \Delta, \dots, \frac{K}{2}\Delta, \frac{K}{2}\Delta m, \frac{K}{2}\Delta m^2, \dots, T]$ with $m = (\frac{2T}{k\Delta})^{\frac{2}{k}}$. Note that for in-spread events,

the intensity also depends on the current spread. However we note that much like $f(Q_t)$, s_t^β is another piecewise constant function multiplied to the intensity. Therefore the treatment of in-spread events is exactly the same with $f(Q_t)$ being replaced by $f(Q_t) \times \mathbb{E}[s_t^\beta]$.

This formulation is then solved by the least squares method. We get as a result (upon rescaling by $\frac{\delta_k}{\Delta}$) the point estimates for $\int_{t_{min}^{(k)}}^{t_{max}^{(k)}} \phi^{(j \rightarrow i)}(s) ds$ for all $t_{(\cdot)}^{(k)} \in G; i, j \in 1, \dots, d$. We form cumulative sums of these estimates to get point estimates for the function $\Phi(t_{max}^{(k)}) = \int_0^{t_{max}^{(k)}} \phi^{(j \rightarrow i)}(s) ds$. We fit parametric functions on each of these kernels on these point estimates and here we make use of power-law functions and exponential functions as the candidates. Unlike Kirchner [2017], however, we do not calibrate the parameters in 30-minute windows and average over a day's 13 windows, but rather we use the entire day's data to calibrate the parameters and secondly we do not make the approximation $\int_{t_{min}^{(k)}}^{t_{max}^{(k)}} \phi^{(j \rightarrow i)}(s) ds \approx (t_{max}^{(k)} - t_{min}^{(k)}) \phi^{(j \rightarrow i)}(t_{min}^{(k)})$ since the grid size is not uniformly small.

We add two constraints to the least squares problem (from the auto-regressive formulation of Kirchner [2017]). Firstly, we enforce the unidirectional shape of the Hawkes Kernels by constraining the point estimates to be uniform in sign (i.e., either all greater than 0 or all less than 0). For this, we make use of the Hawkes Graph Achab et al. [2018] which directly estimates the kernel norms matrix : $\mathbf{M}_{d \times d}$ (elements of which are defined as $M_{ij} = \int_0^\infty \phi^{j \rightarrow i}(s) ds$ which is the kernel norm of $\phi^{j \rightarrow i}(t)$ for $i, j \in \{1, \dots, d\}$) from the conditional probabilities of the 12D point process. Note that we purely use this method to identify whether a kernel is catalysing or inhibiting in nature (i.e. is the kernel norm positive or negative). Secondly, we constrain the integral of a Hawkes Kernel to be less than 1 to ensure the stability of the Hawkes Process. As noted in Bacry et al. [2015b], a Hawkes Process is stable if the maximum eigenvalue of the Kernel Norm matrix is < 1 . We postulate that the above formulation is an adequate linear approximation to the non-linear constraint of bounded eigenvalues. In practice, we do see the calibrated Hawkes Kernel to occasionally violate this non-linear constraint. We apply a heuristic of multiplying all the kernels by a regularizing term to cap the eigenvalue at 1. We reserve the question of how this heuristic impacts the fit quality for future work.

We calibrate the data for several days individually and test the stationarity (by day) of the calibrated parameters. Since we do see stationarity in the kernel parameters over multiple days (Figure 4.5b), we conclude by using the parameters over multiple days for our final calibrated parameters' estimation.

We note that as mentioned in Kirchner and Bercher [2018], the exogenous

intensity is massively overestimated because of the cut-off error and the distributional errors. Therefore we make use of the result from Bacry et al. [2016] to estimate the exogenous intensities from the average intensity of the events observed (Λ) and the fitted norm matrix M :

$$\mu = (\mathbb{I} - M)\Lambda$$

We use the maximum likelihood method to calibrate the order size distribution. A key observation is the preference of traders of round numbers in their order quantity. This is an important stylized fact of the order book dynamics so we make use of Dirac delta functions to add spikes in the distribution function to account for this. Finally, we use the thinning algorithm to simulate the order book from the calibrated parameters and provide some visualizations of the calibrated parameters in the following section along with the quality of fit results.

4.3 Results

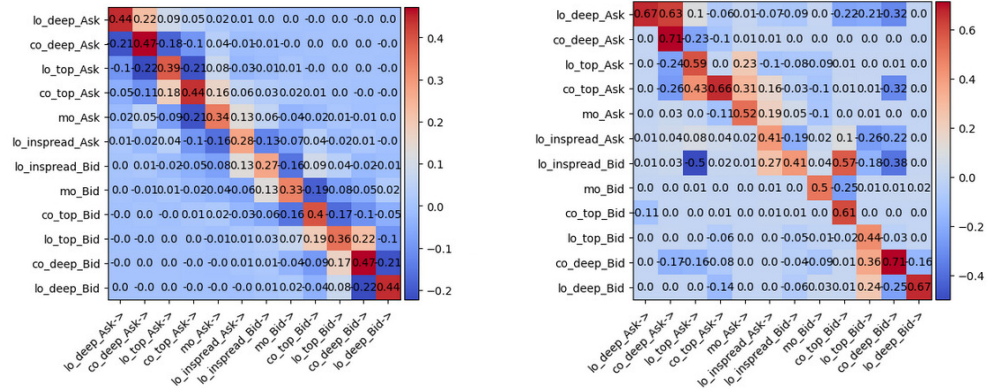
4.3.1 Calibration for Simulated Data:

As evidence of correctness of the fitting method, we construct a 12D Compound Hawkes Process and simulated it for 100 trading days. The parameters of this CHP are chosen randomly from a list of candidate parameters and the norm matrix is chosen to be diagonal with near-diagonal terms having non-zero elements. The real norm matrix is shown in Figure 4.3a. We then transform the simulated order book data into the binned data structure needed for the calibration methodology above and perform the non-parameteric estimation of the kernel point estimates and the exogenous intensities, and finally fit a power law kernel to the points observed over the 100 trading days. We show the fitted norm matrix in Figure 4.3b.

We note that the fitted matrix exhibits similar shape near the diagonal to the real matrix however we do see some off diagonal elements having large inaccurate values. The norm weighted average relative error in estimated parameters across the 144 kernels is 6.0%, 0.9% and 4.4% respectively for α, β, γ parameters of the power law kernel $\phi(t) = \alpha(1 + \gamma t)^{-\beta}$. Hence we conclude that the fitting methodology works reasonably well with a tolerable amount of estimation error.

4.3.2 Calibration for Equities LOB:

We now show the observed intensities conditioned by time of day for a sample dimension (Market Orders at Ask) in Figure 4.4. We observe the common U-shape across all 12 dimensions. In Figure C.1f, we show the the norm of kernels M ,



(a) Norm Matrix: Real, Simulated Data (b) Norm Matrix: Fitted, Simulated Data

Figure 4.3: Simulated Data Calibration

removing the time-of-day multiplier. The x axis shows the excitor and y axis is the excitee. We also show some sample fitted kernel shapes in Figure 4.5b. As we can see here in the translucent blue lines, the kernel shapes fitted over multiple days are stable across days.

Since the number of excitation kernels in a 12D Hawkes Process are 12^2 , in order to maintain model parsimony, we zero-out the effect of small cross-excitations such as $\text{Cancel Bid}_0 \rightarrow \text{Limit Ask}_0$. We set the threshold to select kernels on the basis of their norms to 0.01 as it eliminates several cross-excitation terms to limit the number of kernels to 65.

Note that these estimated kernels are estimated in a non-parametric manner and hence need to be further fitted on a parametric function. We choose between the power-law kernel : $\phi(t) = \alpha(1 + \gamma t)^{-\beta}$ and the exponential kernel : $\phi(t) = \alpha e^{-\beta t}$ to do the parametric fit by comparing the fit's Akaike Information Criteria (AIC). We see that the power-law kernel is selected 100% of the time for this dataset. We show the fitted line in 4.5b in red. As we can see, the power law line fits the point estimates quite well. Indeed we see an average mean square fit error to be $\sim 10^{-3}$. Another noteworthy aspect of the fitting results is the presence of inhibitory kernels.

4.3.3 Calibration of Order Size Distributions

Following Lu and Abergel [2018a], we use Dirac delta at round numbers to account for the stylized facts we observe in Figure 4.1a and 4.1b. We choose the set 1, 10, 50, 100, 200, 500 as the set of round numbers we wish to put spikes in the PDF at. The remainder of the PDF is modelled by a Geometric distribution. We fit this distribution using the maximum likelihood method. We show the final calibrated PDF (smoothened for illustration purposes) in Figure 4.6.



Figure 4.4: Intensities conditioned by Time of Day : we report the average number of events per 30 min bin in a 6 month period

4.3.4 Quality of fit metrics:

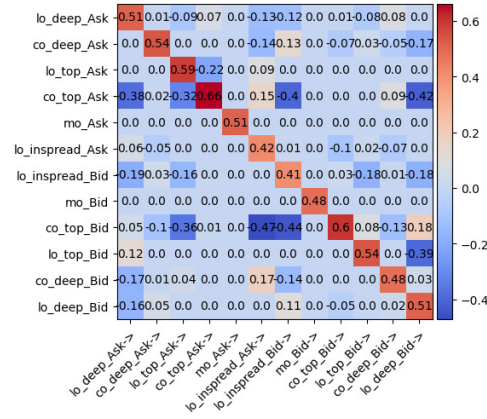
We present the quality of fit metrics in Figure 4.9. We perform a realism quality of fit by comparing some stylized facts of the simulated data to the empirical data. We make use of inter-event durations' distributions, price change time distributions, signature plots, distribution of spread and returns, autocorrelation of returns and sample price paths as our set of stylized facts. As we can see from Figure 4.9, the Hawkes Process is able to replicate the shape of the Signature Plot, the presence of long auto-correlations of absolute returns, the long tail of returns' distribution as well as the two peaks in inter-order arrival times quite well. In addition to this, our particular formulation of the Hawkes Process, with its spread control formulation, is able to match the distribution of the empirical spreads quite well. Figure 4.10 depicts the efficacy of replicating the "U"-shape dynamics of the order book events in our formulation.

4.3.5 Comparison to a similar Poisson Process formulation:

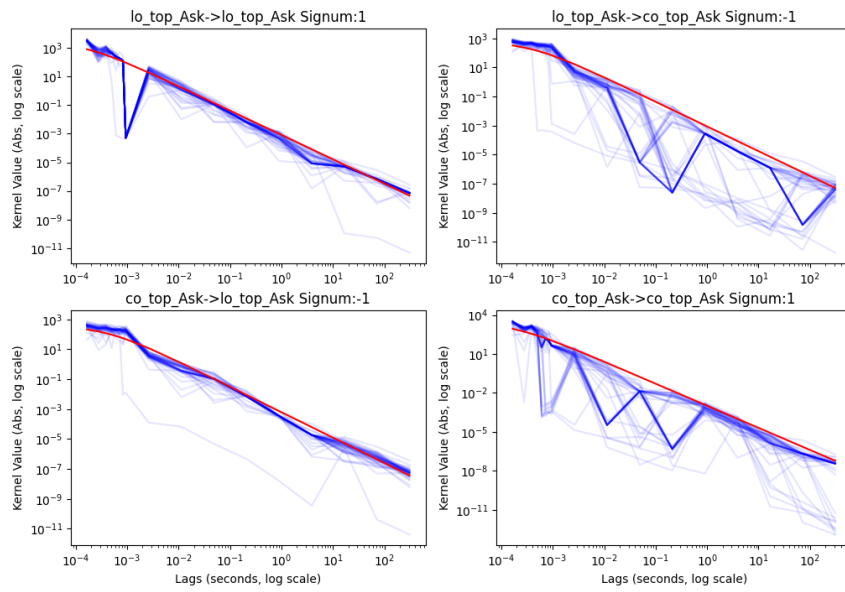
We formulate a 12D Compound Poisson Process with the order intensities of the individual counting processes being Poisson Intensities instead of Hawkes Intensities. The order sizes are sampled from the same distribution as our CHP model. Prop 1 and Prop 2 are enforced by flooring the in-spread order arrival intensity to zero when the spread is 1-tick wide and we multiply the order intensity by the same $f(Q_t)$ time-of-day multiplier. We calibrate the Poisson intensities by the maximum likelihood method and report the quality of fit metrics in Figure 4.11

4.3.6 The effect of mis-specifying the order size distribution:

We now show the necessity to model the order sizes' distribution by considering the calibrated kernels with a constant order size (equal to the median order size). We



(a) Norms of Kernel



(b) Kernel Shape: non-parametric point estimates (blue) and parametric power-law fit (red) - (log-log scale)

Figure 4.5: Excitation Kernels

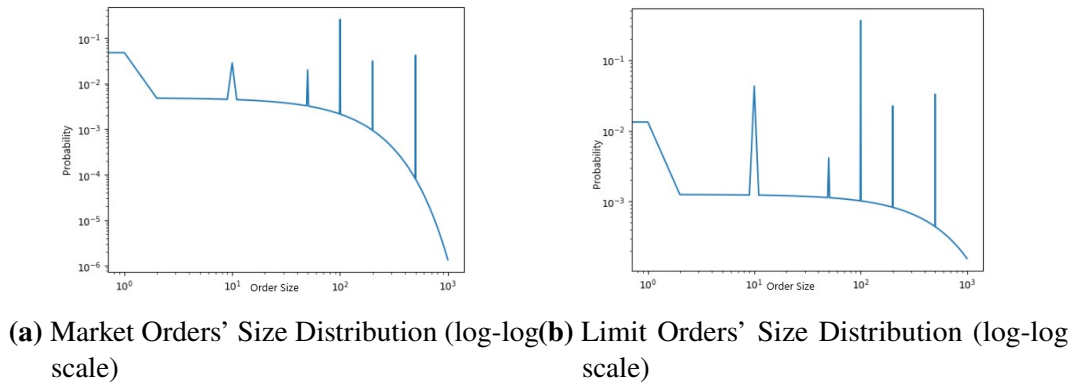


Figure 4.6: Fitted Distribution of Order Sizes

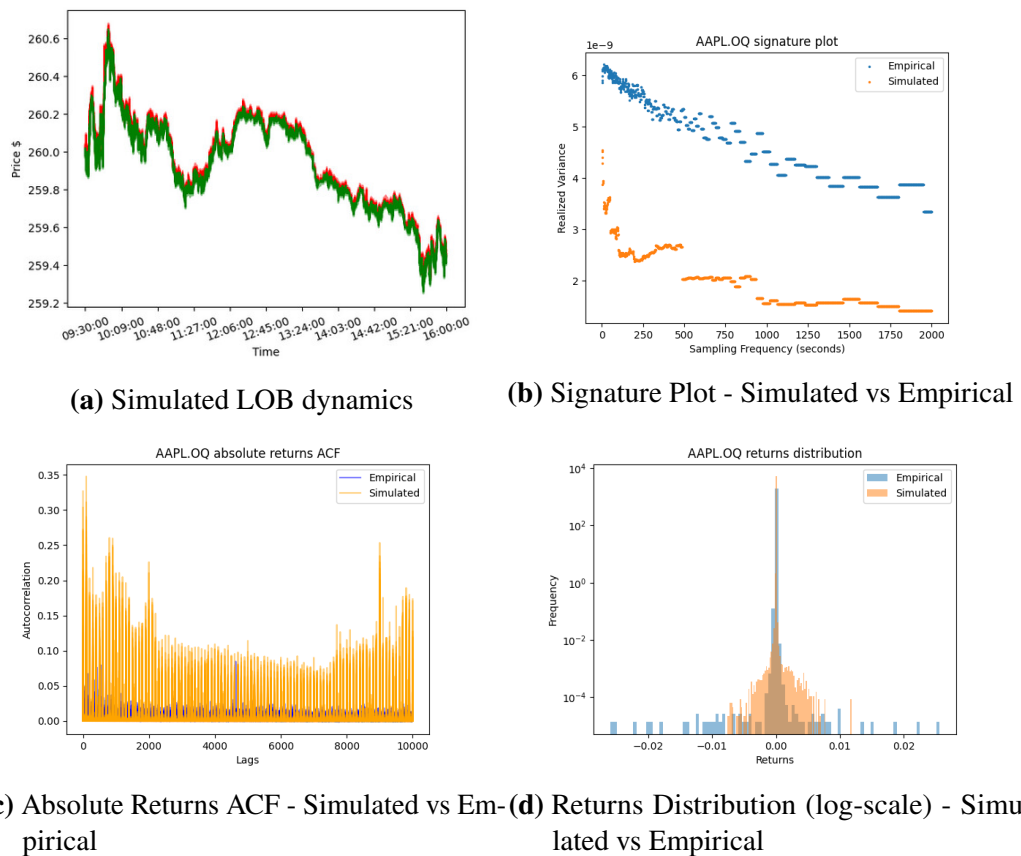
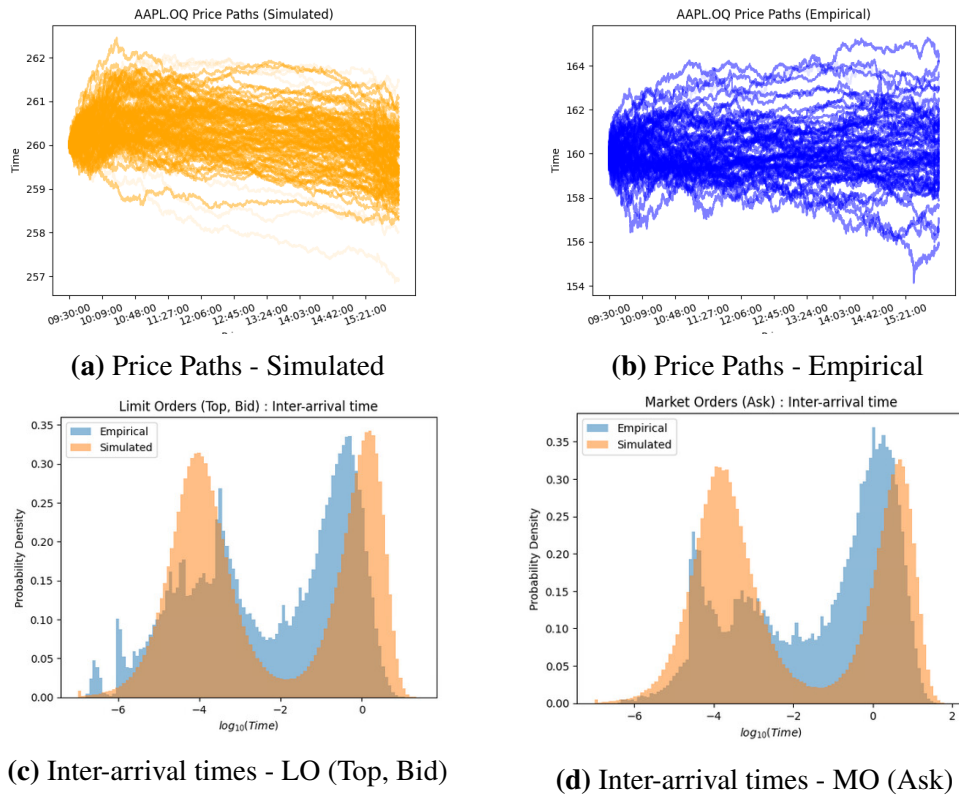
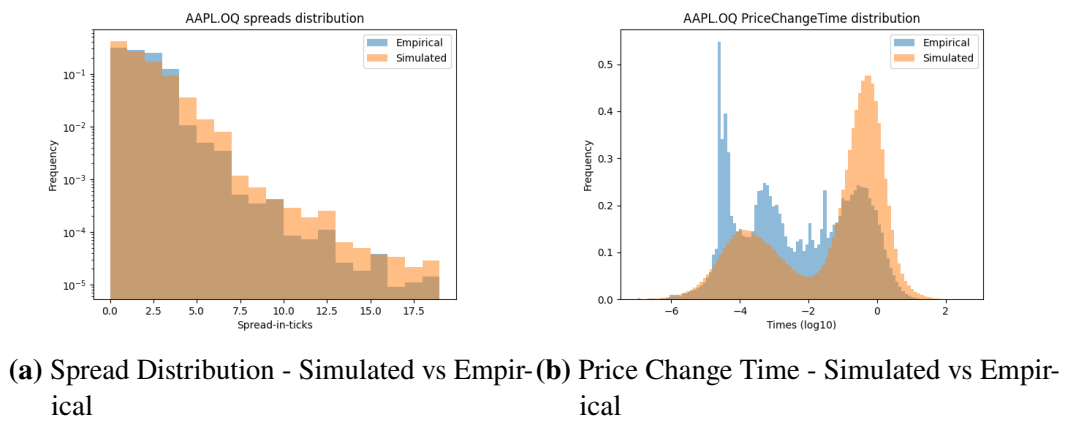


Figure 4.7: Results (full model) - Part 1

**Figure 4.8:** Results (full model) - Part 2**Figure 4.9:** Results (full model) - Part 3

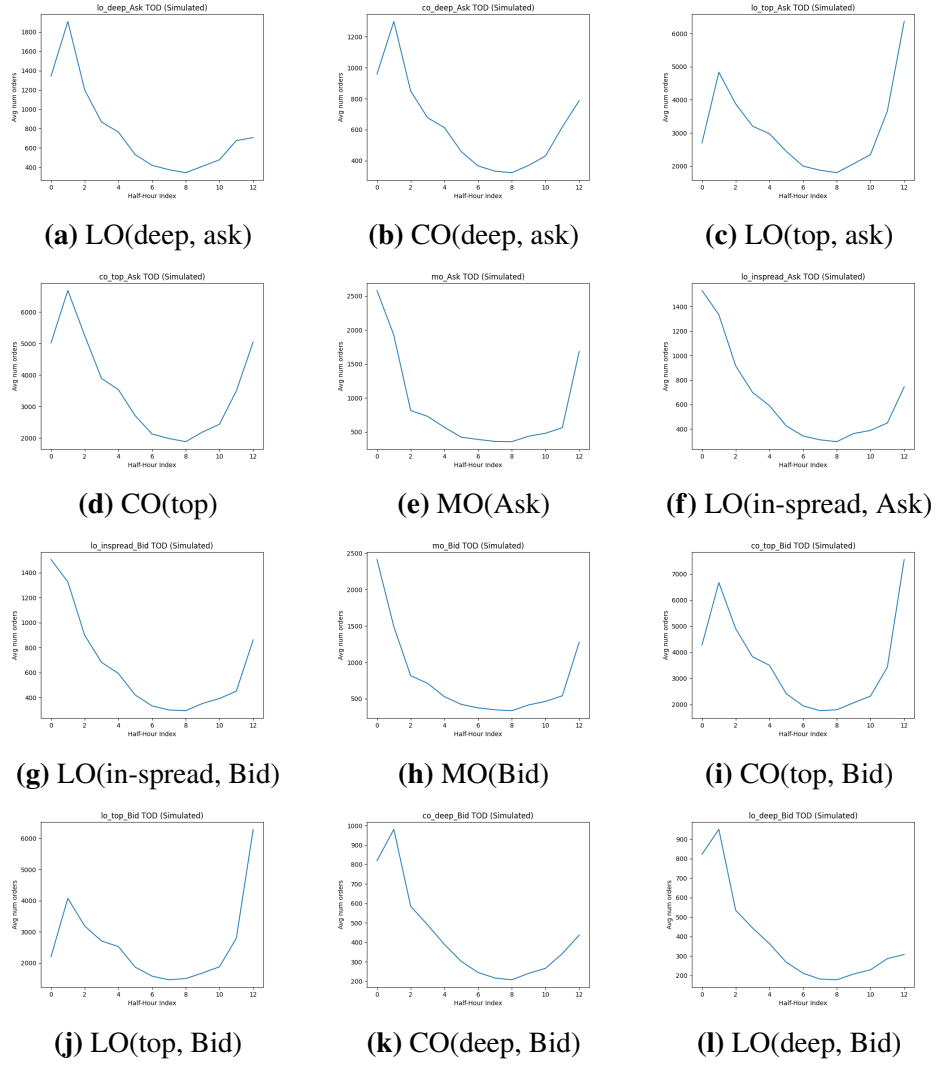


Figure 4.10: Time-of-day shape (simulated)

compare the empirical data's price change event times, the volatility signature plot and the spread distribution with the simulated one for the two types of models. As we can clearly see in Figure 4.12. As we can see here, our model with the calibrated order size distribution fits much better to the empirical data than the constant order size model - the signature plot and the returns distribution show that the volatility is heavily overestimated (by a magnitude of ~ 100 as can be seen in the signature plot), the price changes happen much faster as can be seen in the left tail of the price change time plot, and finally the spreads are much wider in general. Our postulate regarding this observation is that price changes depend on not just order arrival times but also on how fast the queue depletes. Therefore, modelling the order sizes will enable for a more realistic price change event times, returns, spreads and volatility in the simulator.

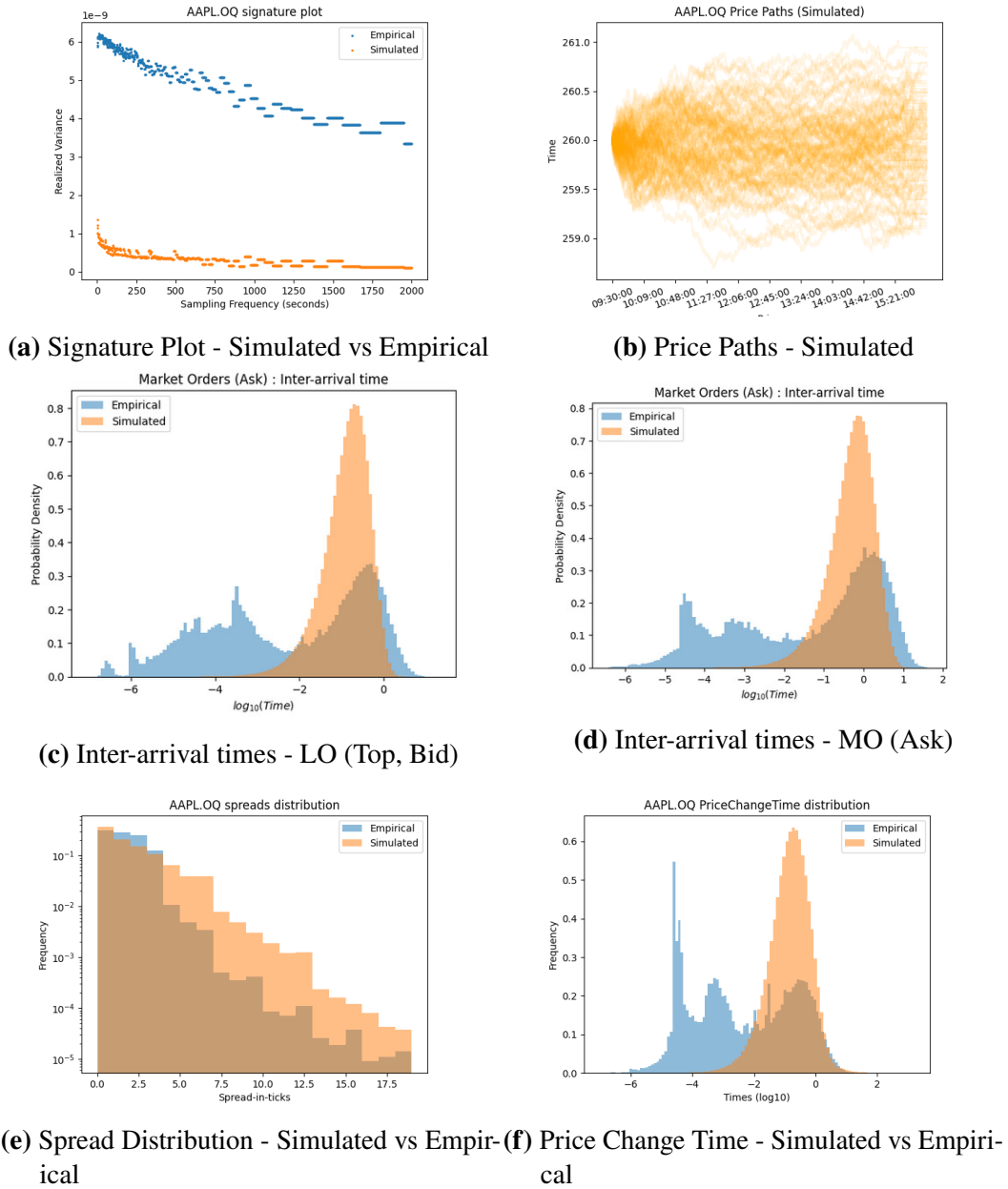


Figure 4.11: Results (Poisson model)

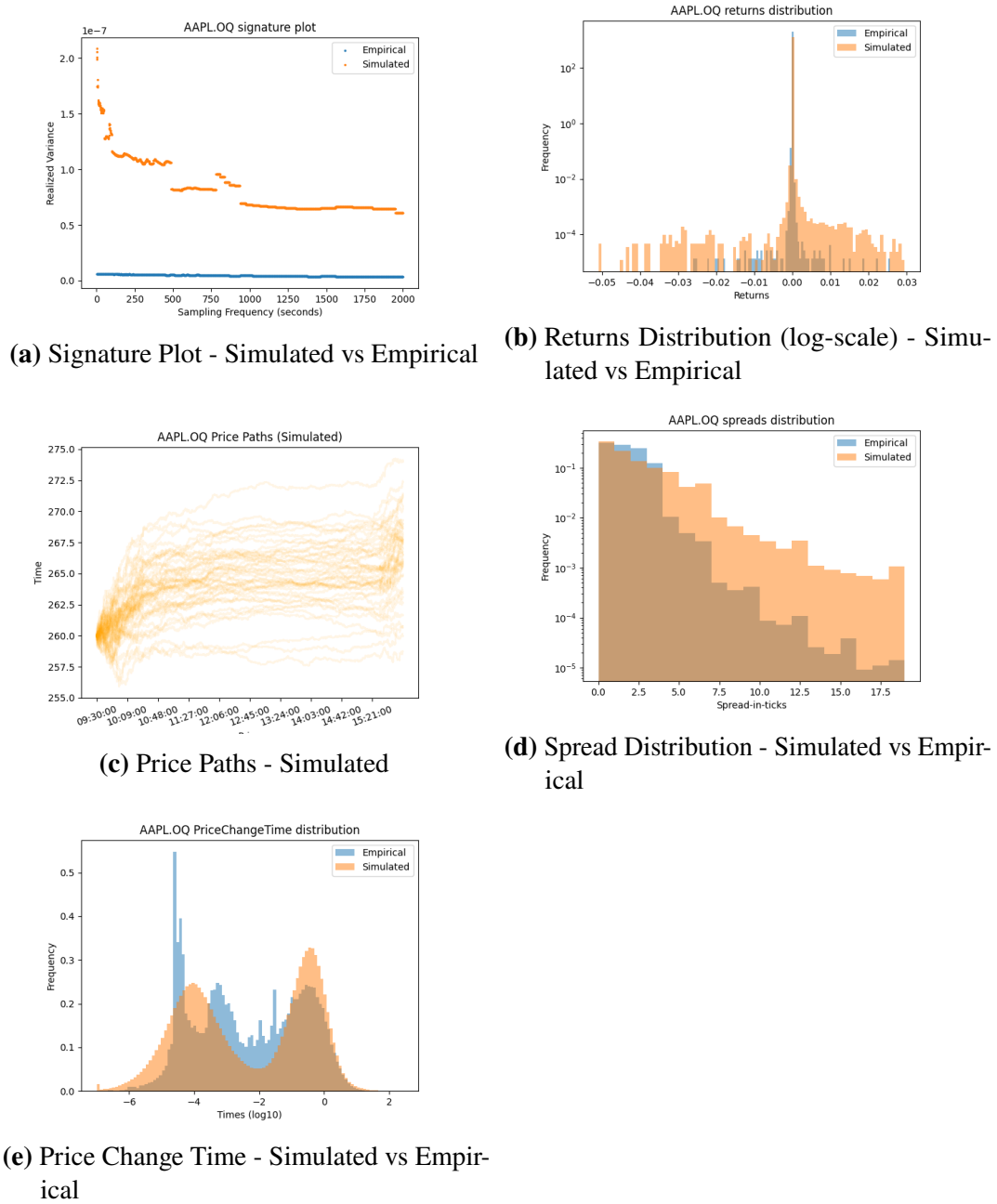


Figure 4.12: Results (Constant Order Size model)

4.3.7 Comparison with Other Hawkes Process Models:

In order to compare our model's performance with the state of the art model, we chose Bacry et al. [2016]'s 8D Hawkes Process model which has the following types of events: {Price Changes, Market Orders (no price change), Limit Orders (top, no price change), Cancel Orders (top, no price change)} $\times$ {bid, ask}. This model takes an alternative approach of modelling the top of the order book than us. We make use of the code made available by their research group Bacry et al. [2017] and implement an Expectation-Maximisation fit over the events on the same dataset that we fit our model. We find that the power law kernels again fit the data better than exponential kernels. We infer the power law parameters from the point estimates fitted by the EM Algorithm using the same MLE method we used in our methodology. We then simulate the Hawkes Process using the code provided by the authors in the tick library Bacry et al. [2017]. Since the simulator is only able to produce event times and not the queue sizes, spreads or the magnitude of the price change, we only use the inter-event arrival times as the stylized fact for comparison since price-paths, returns and spread distributions are not simulated. We show the results in Figure 4.13. As we can see from the simulated distributions, their model is able to capture the right tail of the distribution quite well in all the events we simulate however the model is insufficiently accurate in the high frequency domain (i.e. the left tail).

4.3.8 Market Impact Study:

As noted in Chapter 2, Market Impact (MI) is one of the more crucial characteristics of a realistic order book simulator. We introduce an external agent in the CHP simulator which posts child orders with a quantity $q(Q_T, T, f)$ for a meta order with a quantity Q_T , duration of order T and frequency of orders posted f . We take inspiration from the setup of Cont et al. [2023b] in this study and deploy a TWAP strategy (i.e. $q(Q_T, T, f) = \frac{Q_T}{T \times f}$) with varying parameters $Q_T \in \{3000, 6000, 12000, 24000\}$ shares, $T \in \{15, 30, 60, 120\}$ seconds and $f \in \{1, \frac{1}{3}, \frac{1}{12}, \frac{1}{20}\} \text{ sec}^{-1}$ with order start time fixed. The choice of Q_T is motivated by the fact that average queue size for this stock is found to be around 600 shares, and hence a child order with the most usual configuration in the MI study ($Q_T = 12000, T = 60, f = \frac{1}{3}$) would have a order size equal to the average queue size thus initiating queue depletions. We also conduct a study with varying order start time with other three parameters fixed to $Q_T = 12000, T = 60, f = \frac{1}{3}$. In this way we check for the MI function's sensitivity to each of the 4 parameters we test over. We start the simulation 100 seconds prior to the order start time and observe the mid-price till 120 seconds after the meta order has finished executing. We show

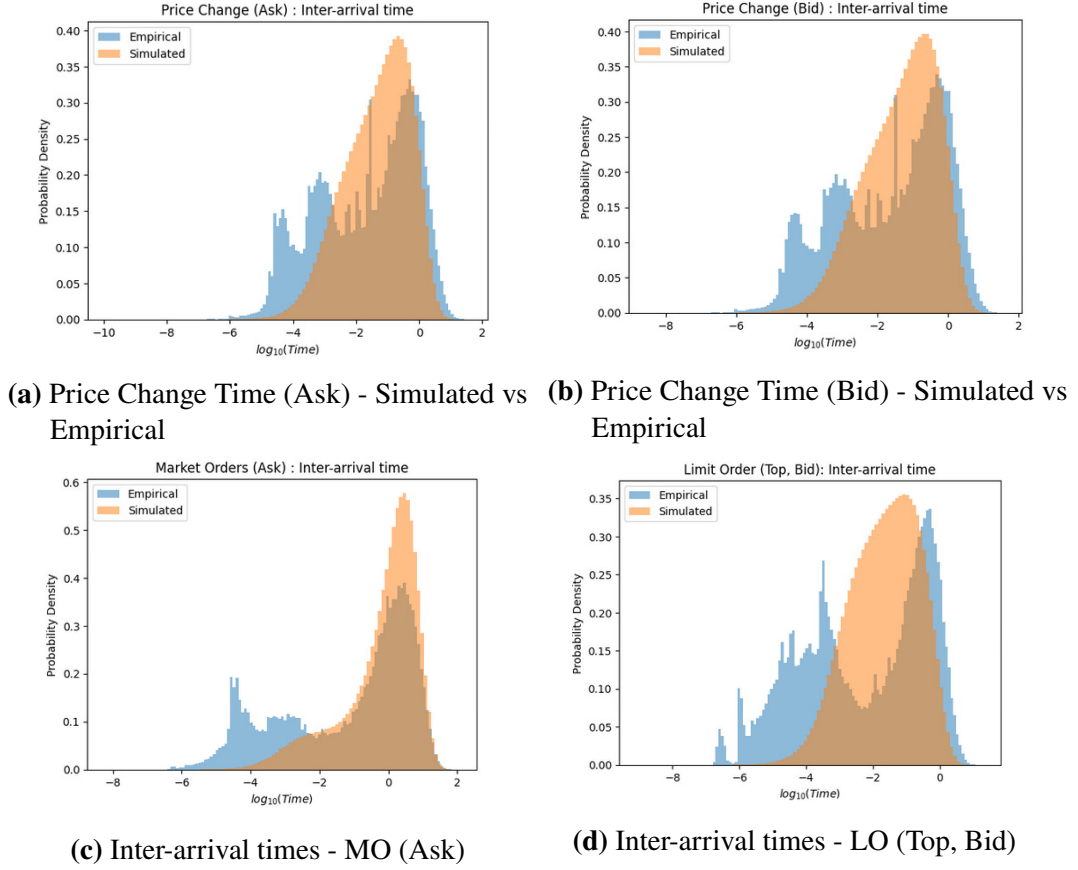


Figure 4.13: Results (Baseline model Bacry et al. [2016])

simulated mid-price paths in blue, green and yellow respectively for before, during and after the execution of the meta order. We show an average price path in orange. We employ three types of order types for the child orders which are market orders, limit orders (top) and limit orders (deep). Finally, we experiment over both the buy and the sell side of the book. We simulate 100 paths per set of parameters and report some results in Figures 4.14, 4.16 and 4.17. We now provide an analysis of the observed MI statistics in the simulator. In Figures 4.14 and 4.16, we focus on MI by Market Orders. We find (Figure 4.14a) that trading during the morning has a higher impact than afternoon. This impact again increases near close. This suggests that the time-of-day conditioning of the model parameters propagates to MI as well. Varying the total quantity in the meta-order we find a concave relation between the MI observed and Q_T (Figure 4.14c). This is inline with the empirically observed concavity in MI with respect to total order quantity. Next testing over the speed of trading, i.e. the percent-of-volume (POV) rate, we vary the time horizon of the meta order T while keeping all other parameters constant. Interestingly we observe a rise in absolute MI ΔP with increasing T (and decreasing POV rate). We note however a

rapid decreasing relation between the rate of change of ΔP which we denote as α in these experiments. This suggests that the behaviour of ΔP could be due to the frequency of trading rather than the POV rate. Indeed in Figure 4.16b, we observe an increase in trading frequency leads to higher ΔP . We also note the price jumps and subsequent relaxation observed in low frequency trading regime (bottom right in Figure 4.15b). This suggests that our simulator is more sensitive to number of trades than volume of trade for price impact. This is ofcourse due to the assumption we made that order sizes are independent of each other and only the order count excites future orders. This suggests that in this simulator a trading engine might be better off by trading big block trades at a lower frequency instead of using smaller batches which is counter-intuitive to the real world phenomenon of adverse selection. This is a potential missing piece in our simulator. However we again note the concavity of the MI function with $\frac{1}{T}, f$ both. Which implies a concavity of the MI function with respect to POV rate and number of order slices. This is again inline with empirically observed facts. In Figure 4.17, we showcase the MI of Limit Orders at various depths in the order book. We note the concavity seen in MI for limit orders at the top of the book in Figure 4.17c. We also observe that limit orders have a much weaker and noisier (but non-zero nonetheless) MI on the LOB than the MOs. The MI weakens to being indistinguishable from noise as we go deeper in the book but at the cost of fill probability. This observation also is inline with observations from empirical data.

Building on these findings, we examine the market impact generated by a TWAP agent within the limit order book simulation. This implementation successfully reproduces key stylised facts associated with meta orders, as documented in Maitrier et al. [2025]. In particular, our subsequent work Jafree et al. [2025] demonstrates that the square root law (SQL) holds during metaorder execution: the price impact, quantified by $\frac{P_t - P_0}{P_0}$, scales as a square root function of the cumulative quantity executed. For our tests, the TWAP agent began trading 100 seconds after simulation start to allow market volume to stabilize, executed over a 1200-second period for a total quantity of 1200, and used a window size of 50 seconds.

Beyond the SQL, the simulation also captures the concave nature of price impact relaxation following metaorder completion. Prior studies Maitrier et al. [2025], Bucci et al. [2018] have shown that post-execution price decay is concave, which can be modeled as $G(t) \approx t^{-\beta}$ Bouchaud et al. [2018]. Empirical research further suggests that, for $z = \frac{t}{T}$ (with $t = 0$ marking the start and $T = 1200$ seconds as the total execution time), the impact follows $\mathcal{I}_{prop}(Q, z) = z^{1-\beta} - (z-1)^{1-\beta}$ Maitrier et al. [2025], Bucci et al. [2018]. This functional form closely matches the post-impact price decay observed for the TWAP agent in our simulation. Notably, our estimated

β of 0.168 aligns closely with the value of 0.19 reported in previous studies Maitrier et al. [2025].

Future work could include a more granular, microstructural analysis of market impact by investigating conditional market impact functionals, as explored in Eisler et al. [2012] and Nagy et al. [2025]. This approach would allow for a deeper understanding of the interplay between market orders, limit orders, and cancellations, and their respective contributions to price dynamics and order book evolution.

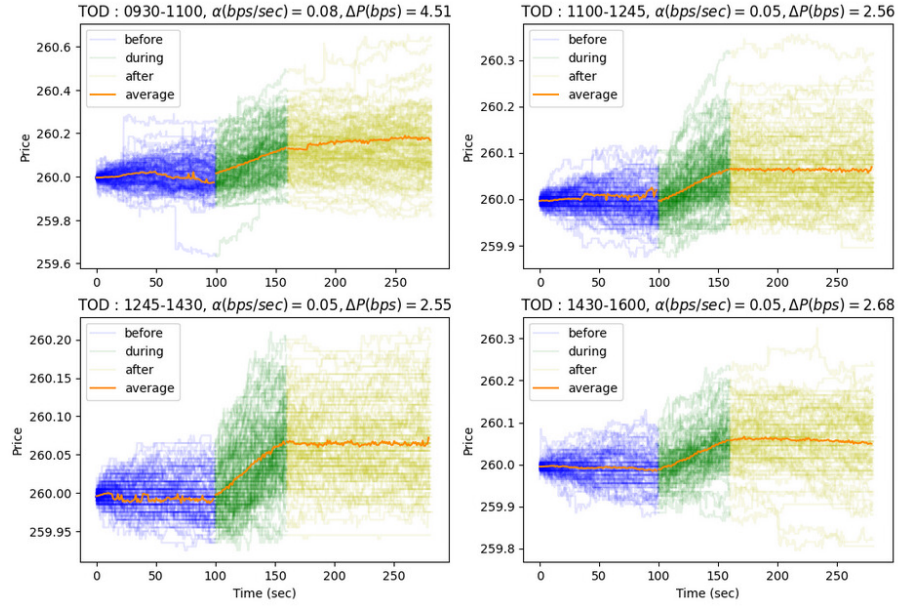
4.4 Conclusion & Future Work

We tackle the problem of simulating a realistic order book by using the Compound Hawkes Process. We particularly focus on building a simulator which is realistic in its order sizes and exhibits the mechanical constraints that a real order book has such as non-negative spreads and positive intensities. We further condition the intensities to be dependent on time-of-day in order for the simulator to not be ignorant of the increase in trading intensities around open and close auctions. We present a number of stylized facts to test the model's efficacy against empirical observations and we compare it our model's empirical facts to several baselines.

In this formulation of the Hawkes Process, we assume nothing about the Price Process of the equity and rather let it rather emerge as a jump process by modest assumptions about the order flow itself. Our contributions can be enlisted as follows:

1. We propose a novel formulation of the Hawkes Process to maintain positive spread.
2. We augment the non-parameteric methods proposed in Kirchner [2017] to work with slow decaying kernels and to be more stable.
3. We make use of calibrated distributions for sampling the order sizes of the events rather than assuming unit order-size.
4. We formulate the in-spread order arrivals as a function of the current spread - utilizing the well known fact that spreads are mean-reverting.
5. We perform a MI study on this simulator along with testing against several baselines and empirical data by making use of a number of popular stylized facts.

Future directions of research includes testing this model on more equities, particularly tackle the problem of small-tick stocks which have a lower amount of information at the top two levels of the LOB which our model assumes. Though our

Market Impact: $Q_T = 12k$, $T = 60s$, $f = 1/3s^{-1}$ 

(a) By TOD (MO, Buy)

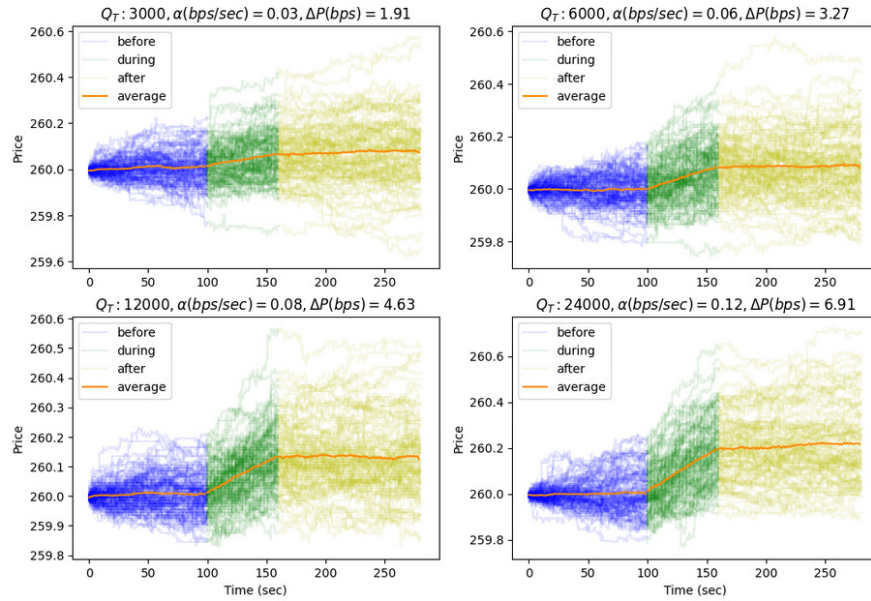
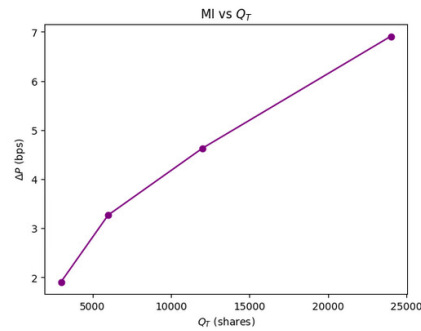
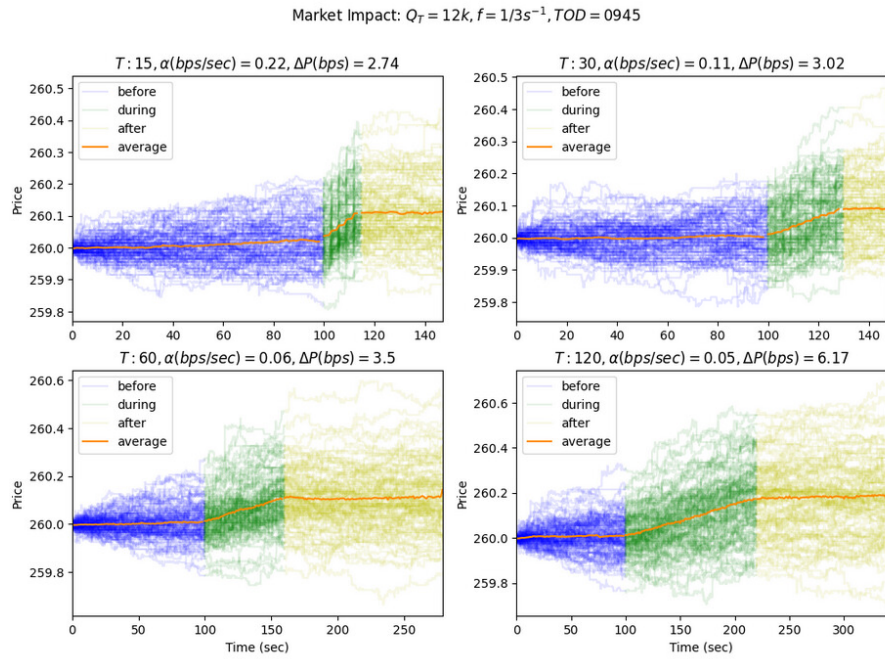
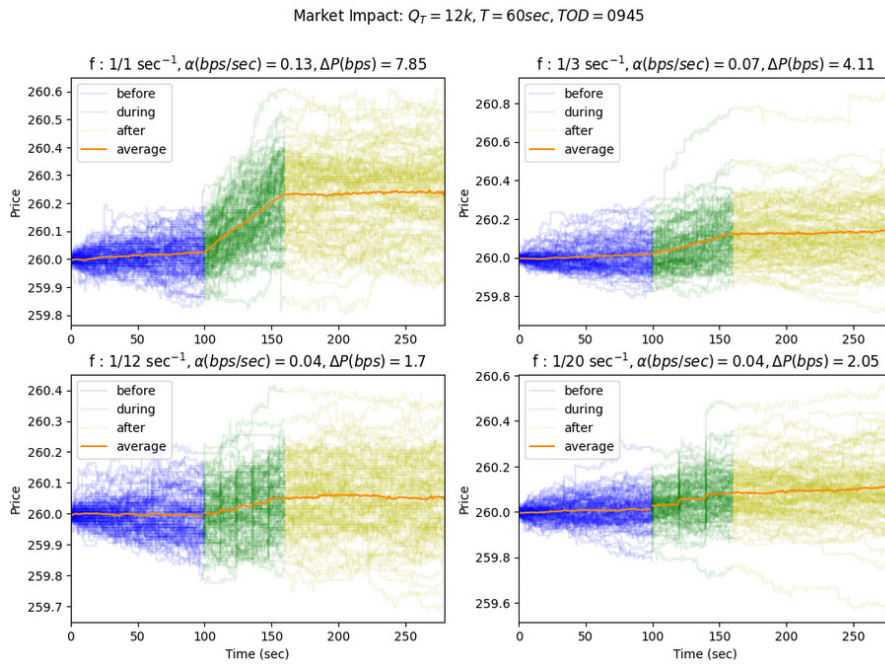
Market Impact: $T = 60s$, $f = 1/3s^{-1}$, $TOD = 0945$ (b) By Q_T (MO, Buy)(c) MI w.r.t. Q_T (MO, Buy)

Figure 4.14: Market Impact Study (Market Orders)

(a) By T (MO, Buy)(b) By f (MO, Buy)**Figure 4.15:** Market Impact Study (Market Orders, continued)

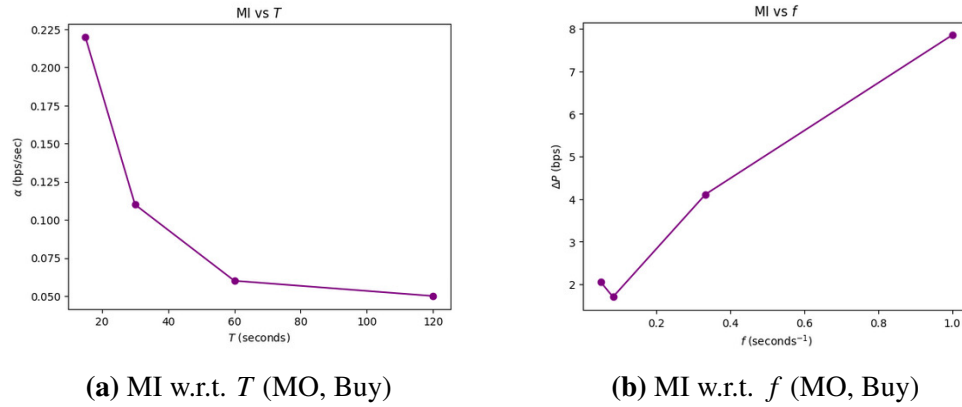
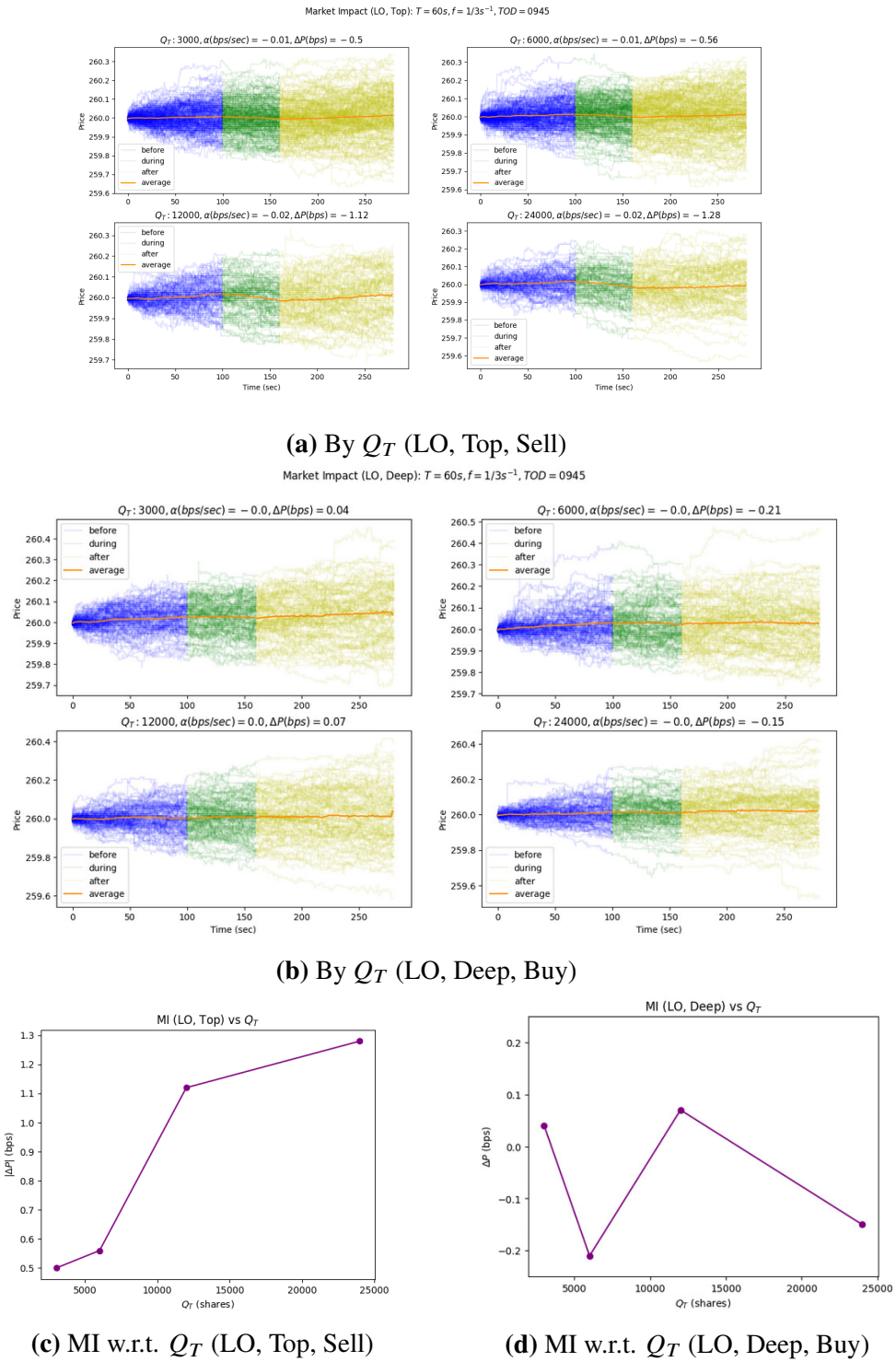


Figure 4.16: Market Impact Study (Market Orders, continued)

formulation theoretically can work with small-tick stocks as well but since it limits its modelling to the top of the book, it will be unsuitable for small-tick stocks. Further, the number of dimensions of this Hawkes Process is quite high, future research could be focused on simulating the order book with lesser number of dimensions.

**Figure 4.17:** Market Impact Study (Limit Orders)

Chapter 5.

Generalizing the Simulator : Small-tick LOBs

We propose a Hawkes Process model that not only fits well for large-tick assets, but also accounts for sparsity, multi-tick level price moves, and the shape of the LOB in small-tick assets. Through simulation studies, we demonstrate the versatility of the model and identify key variables that determine whether a simulated LOB resembles a large-tick or small-tick asset. Our tests show that stylized facts like sparsity, shape, and relative returns distribution can be smoothly transitioned from a large-tick to a small-tick asset using our model. We test this model's assumptions, showcase its challenges and propose questions for further directions in this area of research.

5.1 Introduction

To date, the modeling of the Limit Order Book has focused predominantly on large-tick assets, leaving a significant gap in the literature concerning small-tick assets. We categorize an LOB model as a *full* LOB model if the modelling method is general enough to be applied to deeper dimensions rather than just the top of the LOB. We review the mathematical methodology of some of the state of the art in the following. Abergel and Jedidi [2015], using the exponential kernels of a multivariate Hawkes Process, model a moving frame of the LOB with K price limits on each side of the LOB measured from the opposite best quote. Therefore the frame moves with events at both sides. They develop the generator operator and thereafter long time limits of the price and volume processes from this model. Using a different approach, Cont and Muller [2021] propose an SPDE (Stochastic Partial Differential Equations) model for the centered volume density $u(x, t)$, representing volume at x ticks from the mid-price. Limit orders and market orders are modeled as functions of x , with cancellations handled in three ways: constant multipliers over Limit Order intensities, diffusion for cancel-and-replace near existing levels, and convection towards the mid-price for liquidity seekers. A multiplicative Wiener process is added to capture high-frequency trading activity. Similarly, Horst and Xu [2019] develop

step functions for instantaneous volume density on each side of the LOB, as functions of time and an absolute price coordinate y . Their framework is quite general and they formulate the price and volume dynamics within this framework and study their long time limits. While these SPDE models are useful for calculations of the steady state behaviour of a general LOB, we still stress the gap in literature on simulating a discrete price grid general LOB which is faithful to the stylized facts of the LOB.

In this chapter, we tackle the problem of simulating a general LOB, regardless of its relative tick-size, by modelling the LOB dynamics using a point process. Our contribution can be summarised as:

1. In Section 5.2, we propose an extension to the Compound Hawkes Model in Chapter 4 which is also capable of modeling small-tick assets with a sparse LOB and wide spreads. We introduce the notion “meta-queues” of the LOB in order to maintain model parsimony.
2. In Section 5.3, we showcase the universality of this model (from small to large-tick assets) by depicting various attributes of the LOB by varying the parameters of this model. We further identify certain ‘critical’ parameters which can be controlled to have a large- or small-tick simulated LOB.
3. In Section 5.4, we discuss the calibration techniques and then in the appendix (Appendix D), we challenge the assumptions of our model using empirical evidence. Finally, we propose future research directions to tackle them within our model’s framework.

5.2 A Meta-Queue Hawkes Model for Limit Order Books

In this section, we utilize the information from the study of stylized facts from Chapter 2 to propose a simple extension to the Compound Hawkes Process model proposed in Chapter 4 to account for the nature of both small and large-tick assets.

As seen in Section 3.7 it is well known that that events in an LOB are endogenously excited by themselves (Bacry et al. [2015b], Tóth et al. [2015], Bacry et al. [2016], Lu and Abergel [2018a], Huang et al. [2015]). This stylized fact, along with the general applicability and versatility of the Hawkes Process as discussed in Section 2.1.3.2, is the main motivation of using a self exciting point process as our LOB model. More precisely, we propose a model of LOB dynamics where events are driven by a multidimensional Hawkes process as well as by some state variables

(such as the top of the LOB volume) whose dynamics are itself driven by a point process.

However, as discussed in Chapter 4, a Hawkes model by itself is not really suited to capture dynamics of a small-tick LOB. Indeed, in order to keep track of the activity in each queue of the LOB, a Hawkes model has to have as many components as the number of such active queues. To avoid a model with a dimension of several hundreds (calibration becomes exponentially harder as we increase the dimension of the process Delattre et al. [2014]), one needs to assume that very few queues are really active. Thus, in the literature, Hawkes models for LOB modelling are generally assuming such things like : best bid/ask levels of the LOB containing most of the information of the LOB dynamics, a dense LOB, in-spread orders occurring uniformly one tick away from the previous best quote, etc. As seen in Section 2, all these assumptions basically fit a large-tick asset and are clearly not suitable for a small- or medium-tick asset.

In this section, we propose to create a model of the LOB which not only captures the top of the LOB dynamics but also is able to model the deeper price levels' events without modelling precisely each single queue dynamics (i.e., without increasing the dimension dramatically). This will essentially be made possible by grouping LOB queues into *meta-queues*, regrouping in that way similar dynamics. From our previous sections (and especially from Section 3.7) it appears clearly that there are essentially 5 groups of queues (in each of which orders of a given type have similar dynamics) : the in-spread queues, the best ask (resp. bid) limit queues and the deeper ask (resp. bid) queues.

Therefore we propose to model each of these groups as a meta-queue, i.e., a group of queues of the LOB that are merged together. The exact ranges of queues that are covered by these meta-queues will be associated to dynamic quantities controlled by state variables which evolve with the events in the LOB. In the next section, we fully describe this model.

5.2.1 LOB components and Model State Variables:

Following the previous discussion, our model structures the LOB in 5 different meta-queues (we encourage the reader to look at Figure 5.1 while reading the next paragraph) each of them being characterized by one or several state variables, namely:

- The *spread meta-queue* corresponding to all the queues in the spread. It is characterized by a single state variable :
 - the spread $s(t)$ (i.e., the number of queues in the spread).

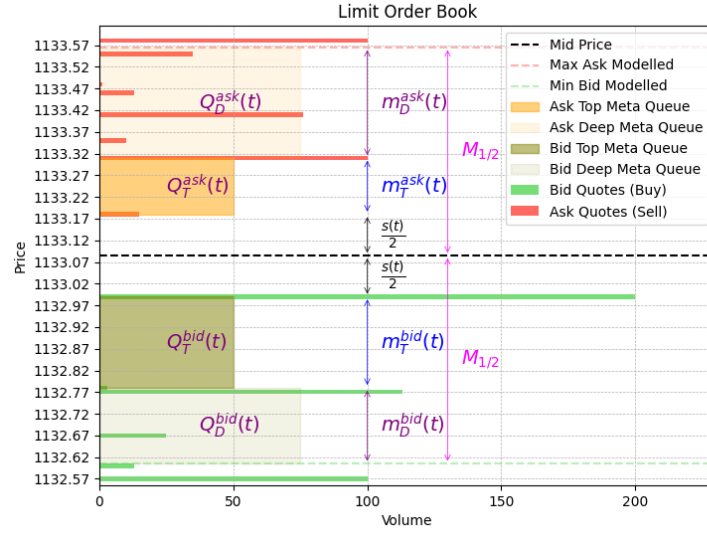


Figure 5.1: Example LOB meta-queues and state variables : Each horizontal line corresponds to the size of a single queue of a snapshot of GOOG's LOB. Let us point the high sparsity of the LOB (i.e., it is full of empty queues). The colored rectangles (not to scale and just for illustrative purposes) indicate the meta-queues that we consider for our model. The corresponding state-variables of each meta-queue are listed on the right side of the figure (in purple).

- The *top ask (resp. bid) meta-queue* that corresponds to a group of queues starting from the best ask (resp. bid) quote and ending just before the second best quote. It is associated to 2 state variables :

- $Q_T^{ask}(t)$ (resp. $Q_T^{bid}(t)$) corresponding to the total volume at the best limit
- $m_T^{ask}(t)$ (resp. $m_T^{bid}(t)$) corresponding to the number of queues in the group (not including the second best quote).

In Figure 5.1, for example, the best ask quote is at 1133.17 \$ and the second best ask is at 1133.31 \$, thus $m_T^{(ask)}(t) = 14$ ticks .

- The *depth ask (resp. bid) meta-queue* corresponding to a group of queues characterized by

- two state variables :
 - * $Q_D^{ask}(t)$ (resp. $Q_D^{bid}(t)$) corresponding to the total volume in the group
 - * $m_D^{ask}(t)$ (resp. $m_D^{bid}(t)$) corresponding to the number of queues in the group (including the second best quote)

– one hyper-parameter

- * $M_{1/2}$: that defines the maximum depth of the meta-queue, assuming orders beyond this point have negligible impact on top-of-book dynamics due to low fill probabilities. This leads to the 4 constraints

$$M_{1/2} \geq \frac{1}{2}s(t) + m_T^{(\zeta)}(t) + m_D^{(\zeta)}(t), \text{ and } m_D^{(\zeta)}(t) \geq 1, \quad \forall \zeta \in \{bid, ask\}. \quad (5.1)$$

In the following we will fix this parameter to be the median of the LOB's average shape (i.e., 50% quartile in Table 3.2) minus 1,

In Figure 5.1, since the the second best ask is at 1133.31 \$ and $M_{1/2}$ has been chosen to be 49 ticks, one gets $m_T^{(ask)}(t) = 26$ ticks .

Let us note that, we do not need to directly model the dynamics of the mid-price. Its changes are a natural mechanical consequence of our events dynamics modelization.

5.2.2 Hawkes Process Model for LOB Events and State Variables Dynamics

Given the structure we just define, there are a total 12 events that can happen in the overall meta-queues, namely : bid or ask LOs and COs in the 4 meta-queues with pre-existing volume (so that sums up to 8 different type of orders), bid or ask in-spread LOs in the spread meta-queue (therefore 2 different types of orders), and bid or ask MOs in the top meta-queues (which correspond to 2 different types of orders). Thus the set of events is:

$$\mathcal{E} := \{LO_{ask_D}, CO_{ask_D}, LO_{ask_T}, CO_{ask_T}, MO_{ask_T}, LO_{ask_{IS}}, LO_{bid_{IS}}, LO_{bid_T}, CO_{bid_T}, MO_{bid_T}, LO_{bid_D}, CO_{bid_D}\}$$

Before describing the Hawkes model that will be in charge of modelling, at any time t , the arrival intensity of each type of orders in each of the 5 meta-queues, we need to first specify the impacts on the state variables of any order arrival.

5.2.2.1 Detailed order arrival impacts on the state variables

For each order we will have to precisely specify:

- how to determine the exact queue (i.e., price) the order arrives at within the meta-queue.

- how to determined the size of the order
- what are the necessary induced updates on the state-variables.

This is the purpose of the next list of items (in the following ζ is used to stand indifferently for *ask* or *bid*) :

1. In-Spread Limit Orders - $LO_{\zeta IS}$:

- **Queue specification.** Let $\eta_{IS} \in \mathbb{N}$ be defined as the number of ticks from the best side ζ where this new in-spread order arrives at. We choose to sample η_{IS} from a calibrated stationary distribution $\Pi_{\eta}(\hat{\eta}_{IS})$ ¹ where $\hat{\eta}_{IS}$ is a calibrated parameter, i.e.,

$$\eta_{IS} \sim \Pi_{\eta}(\hat{\eta}_{IS}) \quad (5.2)$$

η_{IS} is capped at the current spread in ticks and floored by one tick

- **Order size specification.** In the same way, the order size $\kappa_{IS} \in \mathbb{N}$ of this new order is sampled from a calibrated stationary distribution :

$$\kappa_{IS} \sim \Pi_{\kappa}(\hat{\kappa}_{IS}) \quad (5.3)$$

- **State variables update.** When a, IS limit order arrives, one needs to update the new top of the LOB and consequently the deeper meta-queues. Thus, the new width of the top meta-queue is equal to η_{IS} , and the new meta-queue size of the top meta-queue is equal to κ_{IS} of this IS order. The spread moves by η_{IS} . Mathematically, the following changes happen

¹A note on notation: We use $\Pi_a(\hat{a}_e)$ to refer to a given calibrated stationary, parametric probability distribution function with parameter $\hat{a}_e$ for the random variable a_e affecting the event e 's dynamics. We index the distribution by a since it is shared by all a -type random variable (in this case for all the random variables related to a number of ticks η , e.g., η_{IS} , η_T , ...). The choice of each distribution and their calibration will be specified in the next section. These choices will be directly driven by the our analysis of the stylized facts in the previous section.

to the five state variables mentioned above.

$$m_T^{(\zeta)}(t^+) = \eta_{IS} \quad (5.4)$$

$$m_D^{(\zeta)}(t^+) = m_D^{(\zeta)}(t) + m_T^{(\zeta)}(t) \quad (5.5)$$

$$Q_T^{(\zeta)}(t^+) = \kappa_{IS} \quad (5.6)$$

$$Q_D^{(\zeta)}(t^+) = Q_T^{(\zeta)}(t) + Q_D^{(\zeta)}(t) \quad (5.7)$$

$$s(t^+) = s(t) - \eta_{IS} \quad (5.8)$$

An example of such an update is given in Fig. 5.2b.

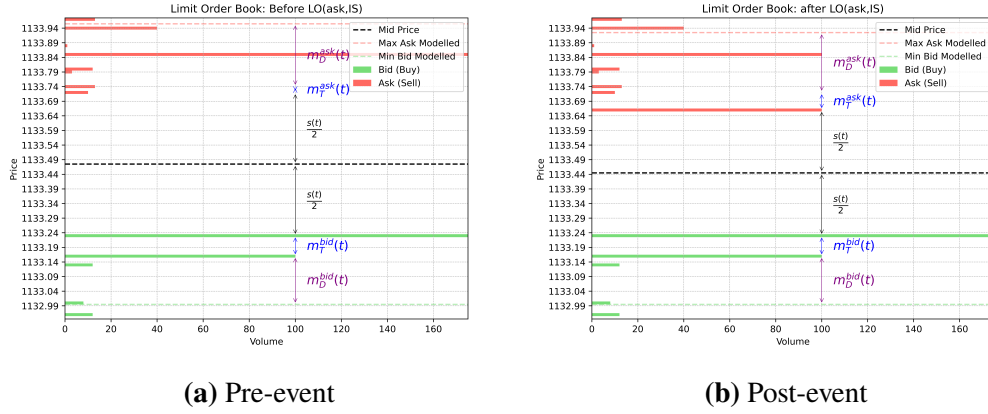


Figure 5.2: $LO_{\zeta IS}$: Ask side In-Spread LO occurs at 1133.66 \$. We remove the overlaid meta-queue rectangles for clarity. $m_T^{(\zeta)}(t^+)$ increases to 6 ticks from the previous 2 ticks following the logic given in Eqs 5.4 and 5.5.

- **Constraint check.** However, since this event is an event which increases $m_D^{(\zeta)}(t)$, there is a chance the first constraint mentioned in Eq. 5.1 is no longer satisfied. If this is the case, we need to “purge” the deep meta-queue to make it less deep. More precisely, we first update the width of the deep meta-queue in order to saturate the constraint, i.e., (we use the t^{++} to refer to an update of the previous update)

$$m_D^{(\zeta)}(t^{++}) = M_{1/2} + 1 - \frac{1}{2}s(t^+) - m_T^{(\zeta)}(t^+) \quad (5.9)$$

then we have to update its corresponding volume $Q_D^{(\zeta)}(t^+)$. Since the model is not tracking each LOB entry individually we have to decide of a somewhat arbitrary rule to make this update.

$$m_D^{(\zeta)}(t^+) = M_{1/2} + 1 - \frac{1}{2}s(t^+) - m_T^{(\zeta)}(t^+) \quad (5.10)$$

$$Q_D^{(\zeta)}(t^+) = Q_D^{(\zeta)}(t) - \Delta Q_D^{(\zeta)}(t) \quad (5.11)$$

$$\Delta Q_D^{(\zeta)}(t) = \text{Partition}(Q_D^{(\zeta)}(t), m_D^{(\zeta)}(t^+) - m_T^{(\zeta)}(t), m_D^{(\zeta)}(t)) \quad (5.12)$$

In order to purge the deep meta-queue, we naturally need to remove certain volume from the meta-queue. The removed volume $\Delta Q_D^{(\zeta)}(t)$ is defined as a partition of the existing $Q_D^{(\zeta)}(t)$. This partition directly depends on the number of levels in the purged partition ($m_D^{(\zeta)}(t^+) - m_T^{(\zeta)}(t)$) and the number of levels in the parent ($m_D^{(\zeta)}(t)$). In the following sections with simulation studies, we make use of a uniform partition function. This partition function can be calibrated according to the shape of the LOB, as outlined in Section 3.5 however for simplicity we choose not to do so.

2. Arrival of Limit Orders at Top - LO_{ζ_T} :

- **Queue specification.** Let $\eta_T(\in \mathbb{N}) \sim \Pi_\eta(\hat{\eta}_T)$ be defined as the number of ticks from the top at side ζ where this new order arrives at. η_T is capped at $m_T^{(\zeta)}(t)$ and floored at zero.
- **Order size specification.** Similar to the previous event, $\kappa_T(\in \mathbb{N}) \sim \Pi_\kappa(\hat{\kappa}_T)$ is the order size of this LO.
- **State variables update.** If η_T is 0, the LO is simply added to the existing meta-queue at the top and no changes occur in the state variables except $Q_T^{(\zeta)}(t^+) = Q_T^{(\zeta)}(t) + \kappa_T$. Otherwise, the LO hits some queue within the the best and the second best quotes. Therefore, the new LO becomes the new second best quote and is absorbed into the deeper meta-queue. We change the width of the top meta-queue according to the distance of the new LO from the top of the LOB and the following changes happen to

the random processes mentioned above:

$$\eta_T \sim \Pi_\eta(\hat{\eta}_T) \quad (5.13)$$

if η_T is 0,

$$Q_T^{(\zeta)}(t^+) = Q_T^{(\zeta)}(t) + \kappa_T \quad (5.14)$$

else,

$$m_T^{(\zeta)}(t^+) = \eta_T \quad (5.15)$$

$$m_D^{(\zeta)}(t^+) = m_D^{(\zeta)}(t) + m_T^{(\zeta)}(t) - \eta_T \quad (5.16)$$

$$Q_D^{(\zeta)}(t^+) = Q_D^{(\zeta)}(t) + \kappa_T \quad (5.17)$$

3. Arrival of Cancel Orders at Top - CO_{ζ_T} :

- **Queue specification.** Since there's only one queue with volume in the top meta-queue, we deplete that queue with this cancellation.
- **Order size specification.** Since we cannot track individual LOs in the existing queues owing to the loss of fine-grained simulation of the LOB in our model, we sample $\kappa_{CO}(\in \mathbb{N}) \sim \Pi_\kappa(\hat{\kappa}_T)$ as the cancellation order size.
- **State variables update.** The top meta-queue size changes as follows $Q_T^{(\zeta)}(t^+) = Q_T^{(\zeta)}(t) - \kappa_{CO}$. If this CO leads to a queue depletion (QD), we need to calculate the new top's width as well as deplete the deeper meta-queue by this new top's total volume. Assuming a sparse structure within the depth meta-queue (informed by section 3.6), we sample a random new top width denoted by η_{T+1} . Here, $\eta_{T+1}(\in \mathbb{N}) \sim \Pi_\eta(\hat{\eta}_{T+1})$ is defined as the number of levels the new top has after the queue depletion event. To calculate the total volume at this new level, we make use of the partition function mentioned earlier to deplete the depth meta-queue. Mathematically, we perform the following to redefine a new top of the

LOB.

$$\text{if QD:} \quad (5.18)$$

$$\eta_{T+1} \sim \Pi_\eta(\hat{\eta}_{T+1}) \quad (5.19)$$

$$m_T^{(\zeta)}(t^+) = \eta_{T+1} \quad (5.20)$$

$$m_D^{(\zeta)}(t^+) = m_D^{(\zeta)}(t) - m_T^{(\zeta)}(t^+) \quad (5.21)$$

$$Q_T^{(\zeta)}(t^+) = \text{Partition}(Q_D^{(\zeta)}(t), m_T^{(\zeta)}(t^+), m_D^{(\zeta)}(t)) \quad (5.22)$$

$$Q_D^{(\zeta)}(t^+) = Q_D^{(\zeta)}(t) - Q_T^{(\zeta)}(t^+) \quad (5.23)$$

$$s(t^+) = s(t) + m_T^{(\zeta)}(t) \quad (5.24)$$

4. Arrival of Market Orders - MO_ζ :

- **Queue specification.** MOs naturally always hit the top meta-queue.
- **Order size specification.** The size of the market order is given by $\kappa_{MO}(\in \mathbb{N}) \sim \Pi_\kappa(\hat{\kappa}_{MO})$
- **State variables update.** MOs, mathematically, behave exactly the same as Cancel Orders at top and we report a trade. If the MO leads to a queue depletion, the same logical waterfall as the one above is followed. If a market order depletes more than one price level, we repeat the process until we completely execute the MO quantity. Thus a MO can deplete multiple price levels using this methodology. This clearly means that the stylized facts related to the price moves induced by Section 3.4 is modelled implicitly.

5. Arrival of Deep Limit Orders - LO_{ζ_D} :

- **Queue specification.** Since we do not keep track of the fine-grained orders or queues in the deeper dimension, we do not specify the exact queue the LO hits at.
- **Order size specification.** The new LO size is $\kappa_D(\in \mathbb{N}) \sim \Pi_\kappa(\hat{\kappa}_D)$.
- **State variables update.** We simply increment the total volume present in the depth meta-queue like so: $Q_D^{(\zeta)}(t^+) = Q_D^{(\zeta)}(t) + \kappa_D$.

6. Arrival of Deep Cancel Orders - CO_{ζ_D} :

- **Queue specification.** Again, since we do not keep track of the fine-grained queues in the deeper dimension, we do not specify the exact queue the cancellation is at.
- **Order size specification.** A similar logic described in the Top CO section applies here. The CO size is $\kappa_{CO_D}(\in \mathbb{N}) \sim \Pi_\kappa(\hat{\kappa}_D)$.
- **State variables update.** Again, if there is no queue depletion event, we simply remove the CO volume from the existing $Q_D^{(\zeta)}(t)$. In case of a QD event, we need to sample the queue size and width from the unmodelled deeper part of the LOB. An important thing to note here is that sampling queue sizes depends on the meta-queue width variable since a wider meta-queue will be expected to have a higher size than a thinner meta-queue. Therefore we introduce this notation for the distribution of unseen meta-queue sizes : $\Pi_Q(\hat{Q}_D, m_D^{(\zeta)}(t^+))$. This has two parameters: first being $\hat{Q}_D$ which is calibrated according to the empirical data and the second being the width of the new meta-queue $m_D^{(\zeta)}(t^+)$.
- **Constraint check**

In the case the width of the deep meta-queue $m_D^{(\zeta)}(t)$ is not violating the first constraint of Eq. 5.1, we must absorb the now empty levels in $m_T^{(\zeta)}(t)$ and create a new depth meta-queue with the following width and volume:

$$m_T^{(\zeta)}(t^+) = m_T^{(\zeta)}(t) + m_D^{(\zeta)}(t) \quad (5.25)$$

$$m_D^{(\zeta)}(t^+) = M_{1/2} - m_T^{(\zeta)}(t) - \frac{1}{2}s(t) \quad (5.26)$$

$$Q_D^{(\zeta)}(t^+) \sim \Pi_Q(\hat{Q}_D, m_D^{(\zeta)}(t^+)) \quad (5.27)$$

In the case the constraint is violated, we set $m_D^{(\zeta)}(t^+)$ to 1, sample $Q_D^{(\zeta)}(t^+)$ from $\Pi_Q(\hat{Q}_D, 1)$ and let the deeper meta-queue have just one non empty level since we cannot go deeper than $M_{1/2}$ in total depth of the LOB.

It is now time to describe the Hawkes process model that is in charge of modeling the arrival intensities of each type of orders.

5.2.2.2 The Hawkes process model for order arrivals

We model the dynamics of these 12 events as a $d = 12$ dimensional counting process $\{N^{(i)}(t)\}_{i \in \mathcal{E}}$ with corresponding intensity $\{\lambda^{(i)}(t)\}_{i \in \mathcal{E}}$. We choose this process to be a Hawkes process in the sense that each intensity function $\lambda^{(i)}$ can be basically seen as the sum of an exogenous intensity $\mu^{(i)}$ and a mutually-exciting term governed linearly by a family of some kernel causal functions $\{\phi^{(j \rightarrow i)}(t)\}_{j \in \mathcal{E}}$ that, for a given j , accounts for the impact of the past events of type j (i.e., the past jumps of $N_j(t)$) on the current intensity $\lambda_i(t)$. Since the kernel functions could take negative values, we use the classical non-linear version of the Hawkes Process (enforcing the intensity to be greater than 0). Moreover in order to take into account that in-spread orders are more likely to happen when the spread is large, for this specific events, we follow the model introduce in Jain et al. [2024b].

More precisely we consider two cases :

- For the intensity of events that are not In-Spread orders :

$$\lambda^{(i)}(t) = \max\left(0, \mu^{(i)} + \sum_{j=1}^d \int_0^t \phi^{(j \rightarrow i)}(t-u) dN^{(j)}(u)\right), \quad \forall i \in \mathcal{E}, i \neq \text{IS} \quad (5.28)$$

- For the intensity of In-Spread orders :

$$\lambda^{(IS)}(t) = \max\left(0, \left(\frac{\delta(s(t)-1)}{\alpha}\right)^\beta \times \left(\mu^{(IS)} + \sum_{j=1}^d \int_0^t \phi^{(j \rightarrow IS)}(t-u) dN^{(j)}(u)\right)\right), \quad (5.29)$$

where $s(t)$ is the spread (in ticks), δ is the tick-size, α (in \$) is a scale parameter and β a shape parameter.

Thus the model we propose is a model that extends to the median of the shape of the LOB, which can be several dozens of ticks from mid-price while being parsimonious : we have a total of 12 types of events spanning 5 meta-queues and 9 state variables. When these events seem to show patterns of self- and cross-excitation, we model the events as a 12D mutually exciting Hawkes Process. In this manner we are able to account for the stylized fact shown in Section 3.7. We now will try to show through differential equations and simulations studies that this general class of meta-queue Hawkes models are in fact able to replicate several stylized facts we studied in Section 2.

5.2.3 Some Salient Features of the Model

In this section we describe the dynamics of the some of the statistical features of a simulated LOB as defined above in Section 3 for empirical LOBs from the above model formulation.

5.2.3.1 Differential Equations for Bid-Ask Spread and Mid-Price Changes of the LOB model

It is easy to see that the spread process is decreased by η_{IS} whenever an in-spread LO occurs and increased by $m_T^{(\zeta)}(t)$ whenever a queue depletion event (QD) occurs at the top queue on side ζ . Therefore we can write the differential equation of spread as:

$$ds(t) = \sum_{\zeta} \left(-\eta_{IS} dN_{LO_{\zeta IS}}(t) + m_T^{(\zeta)}(t) 1_{(QD^{(\zeta)})}(t) (dN_{CO_{\zeta T}}(t) + dN_{MO_{\zeta}}(t)) \right) \quad (5.30)$$

Here $1_{(QD^{(\zeta)})}(t)$ is the indicator function for a QD event at side ζ . As can be seen from the above, the spread process follows a jumpy point process methodology. Similarly the mid-price changes $dp_{mid}(t)$ follow another jumpy point process. The only change is that while the spread is symmetrically changed by events in both the sides of the LOB, price changes happen anti-symmetrically. That is, an in-spread order at ask side would decrease the mid-price but at the bid side, it would increase the mid-price. We define $z \in \{-1, 1\}$ as a binary variable associated with ζ such that when $\zeta = \text{ask}$, $z = 1$ and when $\zeta = \text{bid}$, $z = -1$.

$$dp_{mid}(t) = \sum_{\zeta} \frac{z}{2} \left(-\eta_{IS} dN_{LO_{\zeta IS}}(t) + m_T^{(\zeta)}(t) 1_{(QD^{(\zeta)})}(t) (dN_{CO_{\zeta T}}(t) + dN_{MO_{\zeta}}(t)) \right) \quad (5.31)$$

It is clear that for large and medium-tick assets where price improvements of IS orders happen uniformly at one tick and the width of the top is one tick as well, we can see that the bid ask spread will change by one tick at an event. For small-tick assets we can expect multiple tick moves in the spread and mid-price process governed both by the distribution of η_{IS} as well as $m_T^{(\zeta)}(t)$.

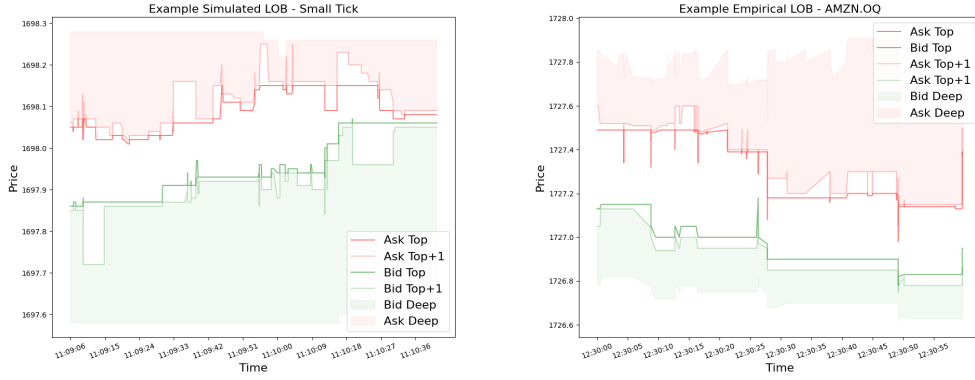
5.2.3.2 Shape and Sparsity of the LOB model:

Let the volume available at a distance x ticks away from the mid price be denoted by $Q^{(\zeta)}(x, t)$ at an instant t .

$$Q^{(\zeta)}(x, t) = Q_T^{(\zeta)}(t) 1_{x=\frac{1}{2}s(t)}(t) + \frac{Q_D^{(\zeta)}(t)}{m_D^{(\zeta)}(t)} 1_{x \geq \frac{1}{2}s(t) + m_T^{(\zeta)}(t)}(t) \quad (5.32)$$

As can be seen from this, the shape of the LOB's maxima will depend on the dynamics of $s(t)$ and $m_T^{(\zeta)}(t)$. Another noteworthy aspect from the above is that the instantaneous value of $Q^{(\zeta)}(x, t)$ will be zero for $x \leq \frac{s(t)}{2}$ which for small-tick assets means that the shape of the LOB will be increasing as we go deeper from the mid-price. In terms of sparsity, again the same fact that the LOB is empty from the top of the LOB till the deeper level starts implies that sparsity should be replicated in the model as well. In the following section we would perform a simulation study to reinforce the fact that the model proposed is general enough to capture several stylized facts in a meaningful way.

5.3 Results from a Simulation Study



(a) Example Simulated LOB Dynamics

(b) Example Empirical LOB Dynamics

Figure 5.3: Simulation Study (Example LOB Dynamic): Figure (a) showcases an example simulated coarse grained LOB. Figure (b) shows an empirical LOB (AMZN.OQ) with similar ‘Meta-Queues’. Notice the similarity in the two figures. Several of the stylized facts we discussed in Section 2 are faithfully being replicated: Bid-Ask Spread is persistently much higher than one tick, the best asks and bids change in multiple tick moves (and therefore so does the mid-price), and the LOB is sparse (notice the white gap between Top and Top+1 i.e. the second best quote).

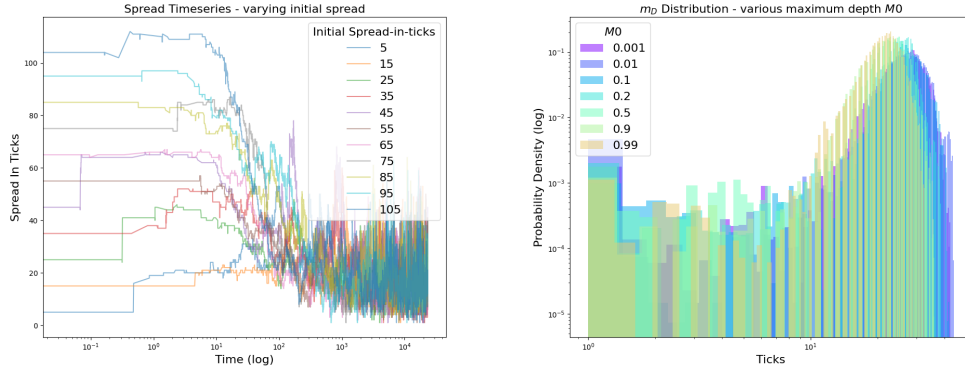
Using the model specified above, we simulate a LOB with random Hawkes and

LOB Parameters. In Figure 5.3a, we show a sample of the simulated LOB for a small-tick configuration of the parameters which is specified in the following. We select the Hawkes Parameters by taking a power law kernel of the form $\phi^{(\cdot)}(t) = a(1 + ct)^{-b}$ with values for parameters a, b, c for the kernels and the exogenous intensities chosen from the calibrated results on AMZN.OQ from Jain et al. [2024b]. The exact parameter specification is given in Appendix C.6. Note that this choice is purely driven by having a baseline LOB which makes logical sense (for e.g., for a stable spread process, we require the rates of MOs and COs to approximately nullify the rates of LOs). Therefore the following discussion can be replicated for any logically sensible set of Hawkes parameters. We reflect the parameters of the Ask side to the Bid side to maintain model's symmetry. For In-spread events, we choose $\alpha = 0.95, \beta = 0.6$ to be corresponding to a small-tick asset for this study. Following Jain et al. [2024b], we choose geometric distributions for order sizes $\kappa_{(\cdot)}$ with spikes at round numbers like 1, 10, and 100. Similarly for unseen queue sizes, we use the geometric distribution with spikes at 1, 10, 100, 500 and 1000. Next, for $\eta_{(\cdot)}$, we choose the geometric distribution again (with no spikes) since empirical observations (Figure 3.7) show that the distance between top and top + 1 (i.e. the second best quote) levels seem to follow a geometric distribution. The similarity in the two figures in Figure 5.3 should be noticed. Several of the stylized facts discussed in Section 2 are being faithfully replicated: the Bid-Ask Spread is persistently much higher than one tick, multiple tick changes are observed in the best asks and bids (and consequently in the mid-price), and the LOB appears sparse (as indicated by the white gap between Top and Top+1, i.e., the second-best quote). In the following sections we first perform an ergodicity study and then move on to identification of critical parameters in our model which decide whether the simulated LOB is a large-, medium- or small-tick asset. Even though we start our analysis with a small-tick configuration, we vary each of these parameters on a range of values to showcase the fact that this model is general enough to be able to simulate characteristics of all kinds of assets regardless of their relative tick-size. Note that this analysis is done by keeping all other parameters to be the same as presented in Appendix B and thus, does not represent a true calibration for any asset. Even so, in this section we will show that controlling certain critical parameters can simulate a LOB very similar to a large-tick asset even with rest of the parameters in the small-tick setting.

5.3.1 Ergodicity

Since proving ergodicity analytically for Equations 5.30 and 5.31 under a Hawkes Counting Process for 12 types of events with various cross-interacting state variable

dynamics is mathematically very difficult and beyond the scope of this paper, we resort to a simulation study to showcase the ergodic properties of the proposed model. In future work, we intend to perform an analysis for these equations in a simpler setting such as a Poisson Process with uniformly unit order sizes. We vary



(a) Spread Timeseries: Varying $s(0)$ in the simulations (b) Width of deep levels: Varying initial sparsity in the simulations

Figure 5.4: Simulation Study (Ergodicity): In (a) we do an ergodicity study for the spread time series by perturbing the initial spread of the simulation. Similarly in (b) we show the invariance of the distribution of $m_D(t)$ with various initial levels of sparsity (denoted by M_0). A similar trend is observed for other state variables' distribution with varying initialisations.

the initialization of the LOB and check for stationarity in the evolution of the LOB dynamics. In Figure 5.4a, we vary the initial spread from 5 ticks to 105 ticks for a small-tick parameter configuration and observe that the spread quickly converges to the mean spread of the LOB process which for this experiment is around 15 ticks.

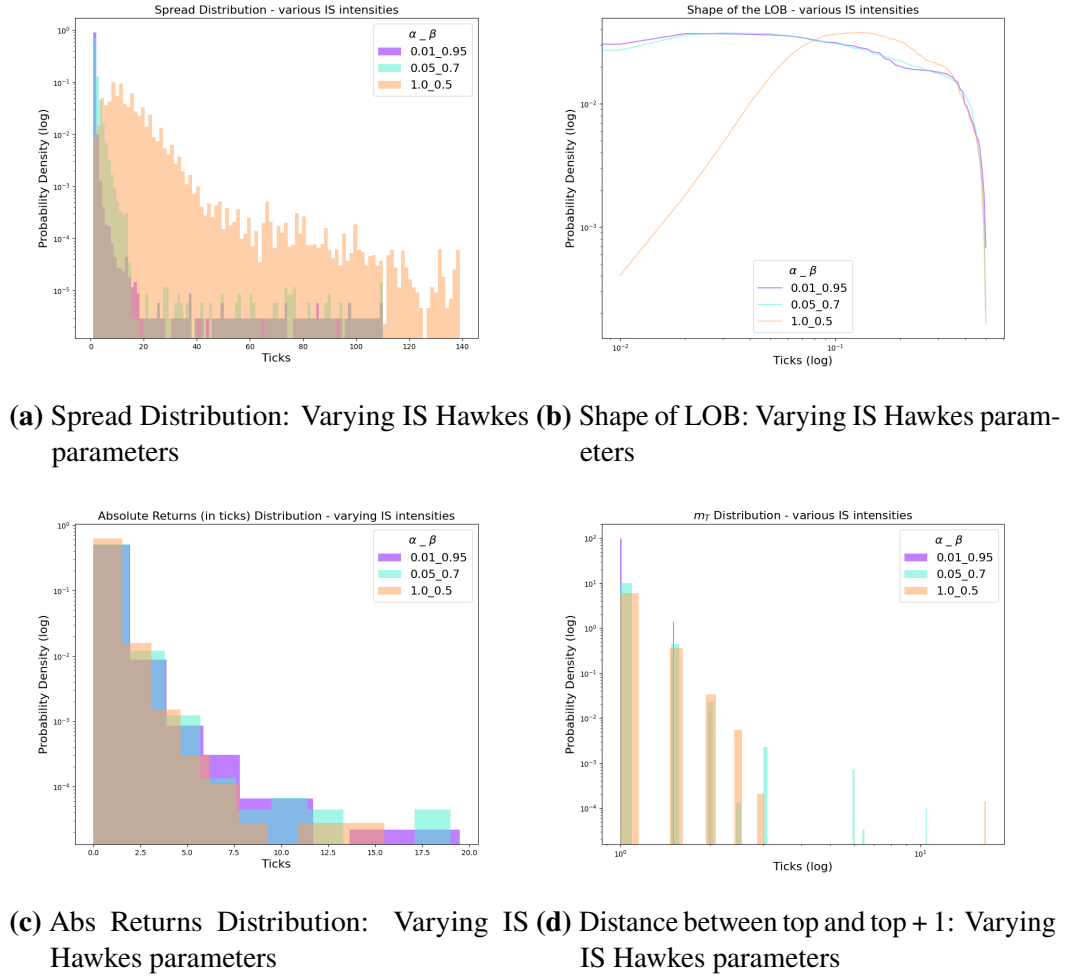


Figure 5.5: Simulation Study (IS Params): In (a) we vary the IS Hawkes Parameters α, β (also known as AvgSpread, SpreadBeta resp.) and we can see the distribution of spread changes from one that looks similar to the exponential distribution to one that looks more like a log-normal distribution (recall from Section 3.3 that exponentially distributed and log-normally distributed spread are characteristics of large- and small-tick assets respectively). The average shape of the LOB (in a coarse grained sense) is plotted in (b) for varying AvgSpread and SpreadBeta. We can clearly see the shift of the maximum liquidity from the top of the LOB (i.e. half tick away from mid price) to much deeper in the LOB as we decrease SpreadBeta and increase AvgSpread (notice the similarities with empirical shapes seen in Section 3.5). In (c) and (d) we show that controlling these two parameters alone does not account for the returns distribution's shape as well as the distribution of the state variable $m_T(t)$ to a very good extent.

In Figure 5.4b, we vary the initial sparsity of the LOB model by varying the parameter $M0 := (M0_T, M0_D)$. We initialise the LOB's $m_T(0), m_D(0)$ using the geometric distribution with parameters $M0_T, M0_D$ respectively. In this experiment we set $M0_T = M0_D$ and we vary this quantity from near zero (which implies a very

dispersed distribution similar to the uniform distribution for the initial sparsity) to near one (which implies a very narrow distribution for the initial sparsity and therefore implies no sparsity at all). We plot the the distribution of $m_D(t)$ and see that the distribution remains invariant of the choice of M_0 . Similarly, for other quantities of interest, we see an ergodic simulation with varying initializations. These two figures reinforce our hypothesis that the process formed from the dynamics illustrated above is ergodic.

5.3.2 Controlling Relative tick-size : Controlling the Spread Distribution and Shape of the LOB

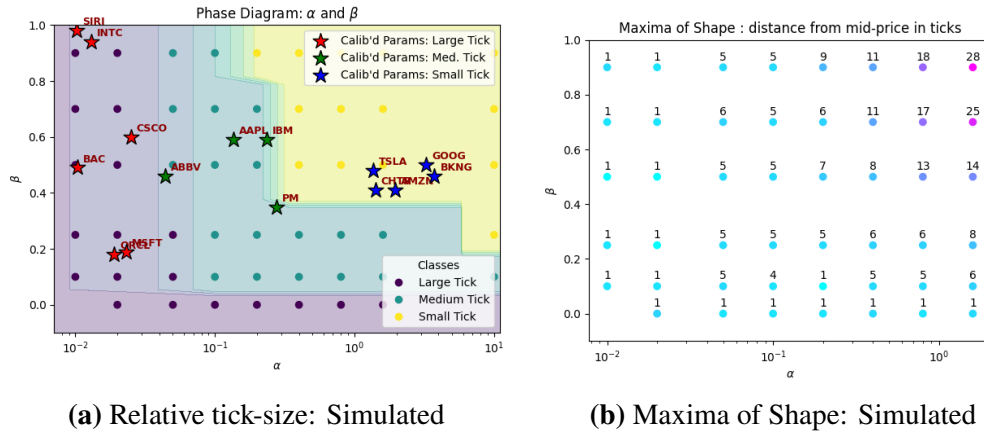


Figure 5.6: Phase Diagrams of varying In-Spread Hawkes Parameters α, β : Previously we noted that several stylized facts depend heavily on the parameter choice of α and β . In these charts we show a grid of varying α and β with a classification between Large and small-tick characteristics in the simulated LOB based on the mean spread as well as based on the depth of the maximum liquidity in the LOB. Clearly, both the parameters are crucial in defining the tick-ness of the asset.

In this section we want to test the model's ability to simulate all kinds of relative tick-sizes of the LOB. To that extent, we are interested in identifying critical parameters of the model which control the stylized facts pertaining to the relative tick-sizes classes as we have discussed in Section 2. As can be seen in Equation 5.29, the in-spread events have two extra parameters α and β . Since these two parameters control the events within the spread and therefore are a part of the process which control the width of the spread, we postulate that controlling this parameter is critical for simulating LOBs of various relative tick-sizes. In Figures 5.5a and 5.5b, we can clearly see that varying these parameters from $\alpha = 0.01, \beta = 0.95$ (i.e. a large-tick setting) to $\alpha = 1.0, \beta = 0.5$ (i.e. a small-tick setting) gives us the characteristics of large, medium and small-tick assets for the Bid-Ask Spread distribution and Shape

of the LOB. One can clearly see the spread distribution transforms from one that looks like an exponential distribution in purple (similar to a large-tick asset), to a log-normal distribution in orange (similar to a small-tick asset). Similarly, we see the shape of the LOB turns from one where most of the liquidity is at the top of the LOB in purple (as is the case for large-tick assets) to one where the maxima actually lies much deeper in the LOB (as is the case for small-tick assets).

However we note that absolute returns and sparsity is largely unchanged in these simulations (Figures 5.5c and 5.5d). This suggests that while α, β are critical parameters for spread distribution and LOB shape, they alone are insufficient to account for the stylized facts we are interested in. Regardless, to classify the simulations into small-, medium- and large-tick based on the average spread, we make use of the classification rule from Section 3.3 by calculating $\bar{s}$ for various simulated LOBs over a grid of α and β . We show this classification using a phase diagram in Figure 5.6a. Additionally, we report the maxima of the LOB shape (in ticks) in line with Section 3.5 for this grid over α, β in Figure 5.6b. We can see the three phases of the simulated LOB in Figure 5.6a: when α, β are high, the LOB simulated resembles a small-tick asset and when not, it is similar to a large-tick one. The borderline cases correspond to a medium-tick asset. Additionally we can see the maxima of the LOB shape shift smoothly from one tick to 20+ ticks in Figure 5.6b for the row with $\beta = 0.7$, for example. Hence, using our model one can smoothly transition from a large-tick asset to a small-tick asset by varying the two IS parameters. Additionally, we can clearly infer from the diagram that if we set β to zero, i.e. make the in-spread orders' intensity independent of the current spread, it is impossible to simulate a small-tick LOB. This finding underscores the importance of a spread-dependent intensity for spread-changing events like the IS limit orders.

5.3.3 Controlling Sparsity and Mid-Price Change Sizes

Continuing our discussion on how to create a more realistic LOB model for small-tick assets where the LOB is sparse and also exhibits multi-tick price move, we analyse the next crucial parameter for changing the relative tick-size of the simulated LOB - $\hat{\eta}_{(.)}$ i.e. the parameter of the distribution of $\eta_{(.)}$. This parameter controls the number of empty levels between two queues in the LOB and therefore is directly related to sparsity. We vary it from 0 to 1, with the Geometric distribution as the candidate for $\Pi_{\eta}(\cdot)$, where near one $\hat{\eta}$ would mean that, for example, the price improvements due to in-spread limit orders is always one tick, i.e. no sparsity at the top, and near zero $\hat{\eta}$ would correspond to price improvements happening randomly from one tick to spread-in-ticks, i.e. very high sparsity at the top. Unlike previous discussions, in

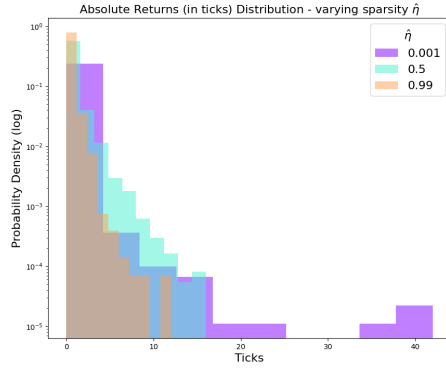
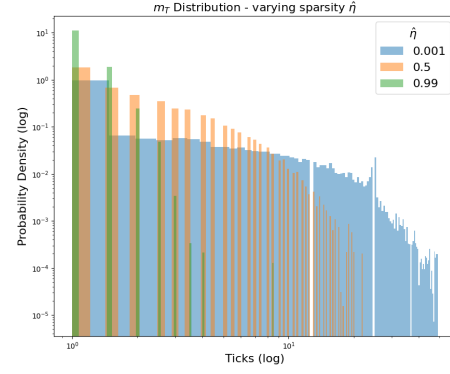
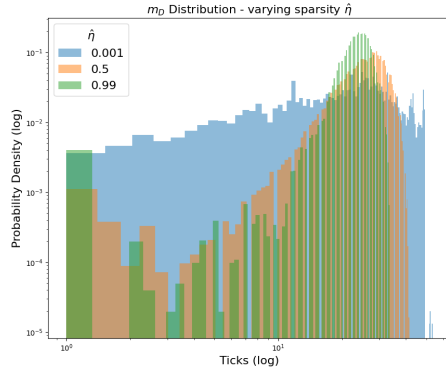
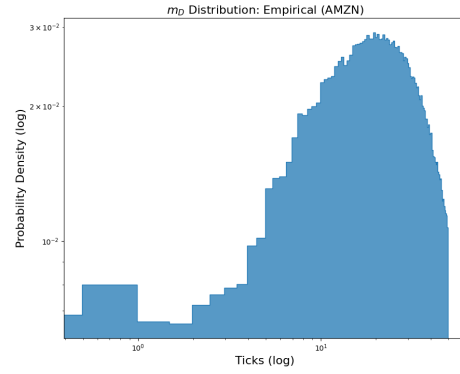
(a) Absolute Returns: Varying $\hat{\eta}$ (b) Distance between top and top + 1: Varying $\hat{\eta}$ (c) m_D : Varying $\hat{\eta}$ (d) m_D : AMZN Empirical

Figure 5.7: Simulation Study (Sparsity Params): In this set of figures we vary the dynamic sparsity parameters $\hat{\eta}_{(\cdot)}$, while setting the α, β parameters to a small-tick asset's setting, and observe the simulated LOB dynamics. We can see that varying the dynamic sparsity can also enable us to vary the returns distributions' tail as well as the state variables' ($m_T(t), m_D(t)$) distribution. For empirical shapes of $m_T(t)$, we refer to the figure 3.7 and we show an example small-tick empirical distribution for $m_D(t)$ in (f).

Figures 5.7a and 5.7b, we show that this parameter can also affect the distribution of returns - note the higher variance with lower $\hat{\eta}_{(\cdot)}$ akin to the small-tick assets shown in Figure 3.4. Thus we can conclude that as we go to lower $\hat{\eta}_{(\cdot)}$, we observe the returns distribution becoming more and more fat tailed. Also notice the sparsity of the LOB indicated by the distribution of the top width variable $m_T(t)$ flattening as we decrease $\hat{\eta}_{(\cdot)}$. Therefore we can conclude that this parameter can also control the sparsity at the top of the LOB. In terms of modelling the deeper part of the LOB, we show in Figures 5.7c and 5.7d, that varying the sparsity levels can vary the variance of m_D and the shape quite closely resembles that of the empirical data. We can see

that varying this parameter leads to varying relative tick-size of the simulated LOBs even though the in-spread order intensity parameters (α, β) are corresponding to that of a small-tick asset i.e. $(0.95, 0.6)$. We again stress that given the choice of $\beta = 0$, it is impossible to simulate a small-tick spread distribution even with high dynamic sparsity. Therefore we conclude that it is important to calibrate all of the critical parameters: α, β and $\hat{\eta}_{(\cdot)}$ accurately.

5.3.4 Comparing MQH LOB Stylized Facts to Empirical Evidence Across Relative Tick-sizes

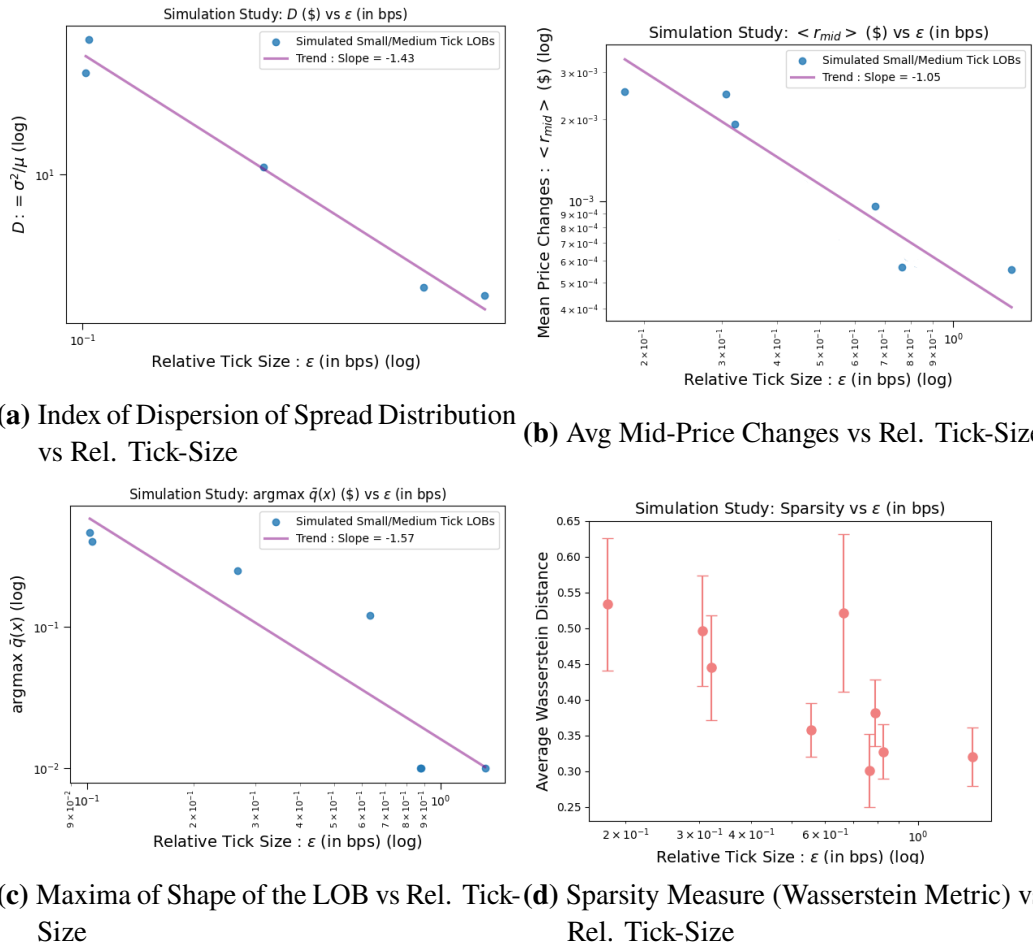


Figure 5.8: Stylized Facts (Simulated): The simulated ϵ is compared to various stylized facts measures from Chapter 3 for small and medium-tick LOBs. On log-log scale, we clearly see the smooth regressions of D (index of dispersion for spread), $\langle r_{mid} \rangle$ (mean mid-price changes), and $\arg\max \bar{q}(x)$ (maxima of shape of the LOB) as we saw in Figs 3.2, 3.3, 3.6, respectively. Further, the slopes of the fitted trends are approximately the same as the slopes we calibrated from empirical data. Finally, we also see the sparsity measure, introduced in Section 3.6, also increases with decreasing ϵ as noticed in Fig. 3.8.

This section assesses whether the MQH model is able to reproduce the key stylized facts of the limit order book documented empirically in Chapter 3.

Before studying how stylized facts vary with the relative tick size ϵ in the simulated MQH model, we must first define a way to estimate ϵ in simulations. An important feature of the MQH model is that the absolute scale of prices plays no role in the MQH model's dynamics. Consequently, it is not possible to compute ϵ directly from Eq. 3.3, which requires a price scale for its definition. Instead, we estimate ϵ in simulations using the empirical power-law relation between the average spread $\bar{s}$ and ϵ established in Eq. 3.4 and illustrated in Fig. 3.1. This relation provides a way to infer relative tick size from observed spread statistics. However, this method is only valid for small- and medium-tick LOBs, as it breaks down in the large-tick regime. Therefore, in what follows, we restrict our comparison to simulated LOBs that fall into the small- and medium-tick classes based on this proxy.

We begin with the spread distribution. As shown in Fig. 5.8a, the index of dispersion D of the spread decreases systematically with increasing ϵ , following a clear power-law trend. This is consistent with the empirical observation in Fig. 3.2, where a similar monotonic dependence was reported. The simulated scaling exponent $\alpha_D^{\text{sim}} = -1.43$ is in close agreement with the empirical value $\alpha_D^{\text{emp}} = -1.36$.

We now turn to mid-price changes. Fig. 5.8b shows that the mean absolute mid-price return $\langle r_{\text{mid}} \rangle$ scales with ϵ in a manner that closely follows the empirical trend shown in Fig. 3.3. The estimated slope from simulations is $\alpha_r^{\text{sim}} = -1.05$, which compares well to the empirical slope of -0.95 .

Next, we study the shape of the order book. Fig. 5.8c displays the variation in the location of the peak of the average LOB shape $\arg \max \bar{q}(x)$ as a function of ϵ . As in the empirical case (Fig. 3.6), the peak shifts deeper into the book as ϵ decreases. The simulated slope $\alpha_q^{\text{sim}} = -1.57$ again aligns well with its empirical counterpart $\alpha_q^{\text{emp}} = -1.50$.

Finally, we examine the sparsity of the LOB. As shown in Fig. 5.8d, the Wasserstein-based sparsity metric introduced in Section 3.6 increases with decreasing ϵ , confirming the empirical trend illustrated in Fig. 3.8. This demonstrates that the MQH model is able to replicate not only return- and volume-based features, but also structural characteristics of queue configurations.

In conclusion, across all metrics considered—spread, mid-price return, LOB shape, and sparsity—the MQH model demonstrates strong consistency with the empirical scaling behaviors documented in Chapter 3. This validates the model's capacity to realistically replicate LOB microstructure for small- and medium-tick assets when appropriately calibrated. We emphasize that this consistency is not

Metric	Symbolic Form	Simulated Slope	Empirical Slope (from Ch 3)
Index of Dispersion	$D \sim \epsilon^{\alpha_D}$	$\alpha_D^{\text{sim}} = -1.43$	$\alpha_D^{\text{emp}} = -1.36$
Mean Mid-Price Change	$\langle r_{\text{mid}} \rangle \sim \epsilon^{\alpha_r}$	$\alpha_r^{\text{sim}} = -1.05$	$\alpha_r^{\text{emp}} = -0.95$
Maxima of Avg. LOB Shape	$\arg \max \bar{q}(x) \sim \epsilon^{\alpha_q}$	$\alpha_q^{\text{sim}} = -1.57$	$\alpha_q^{\text{emp}} = -1.50$

Table 5.1: Comparison of scaling exponents obtained from simulations and empirical analysis (Chapter 3) for key LOB metrics: Table 5.1 summarizes the comparison between scaling exponents measured in real order-book data and those obtained from our MQH simulations. In Chapter 3 we established three power-law relationships between relative tick size and (i) the index of dispersion of the spread, (ii) the average mid-price change, and (iii) the location of the peak in the average LOB shape. Repeating the same regressions on simulated LOBs—using our proxy estimate of relative tick size—yields slopes that closely match the empirical values. This agreement demonstrates that the MQH model faithfully reproduces the key microstructural scaling laws observed in real markets.

solely a consequence of imposed state-space constraints, but reflects data-consistent parameter estimates that align with the theoretically predicted regimes of the model.

5.3.5 Market Impact Study

Following Chapter 4, we validate the meta-queue Hawkes simulator using a buy-side TWAP meta-order composed exclusively of market orders, executed across large-, medium-, and small-tick limit order books. The only test conducted in this work concerns the scaling of market impact with the total traded quantity Q_T . As shown in Fig. 5.9, the simulator reproduces a concave dependence of market impact on Q_T across all tick-size regimes, consistent with empirical evidence and prior simulation results.

While the qualitative concavity is consistent, its magnitude varies with tick-size. Other empirical validations (e.g. limit-order impact, intraday effects, and execution-speed dependence) are not repeated here and are thoroughly documented in Chapter 4.

5.3.6 Comparisons to a Poisson Process Model:

In order to showcase the marginal gains in quality of fit we receive by using a Hawkes Process model for the events' intensities rather than a simpler Poisson Process model, we create a baseline Poisson Process model with the same state variables and their dynamics. We study a few stylized facts and compare them to the Hawkes Model in the following. Keeping all else the same as in Section 5.2, we change the point process of the 12 events from a Hawkes Process to a Poisson Process with in-spread

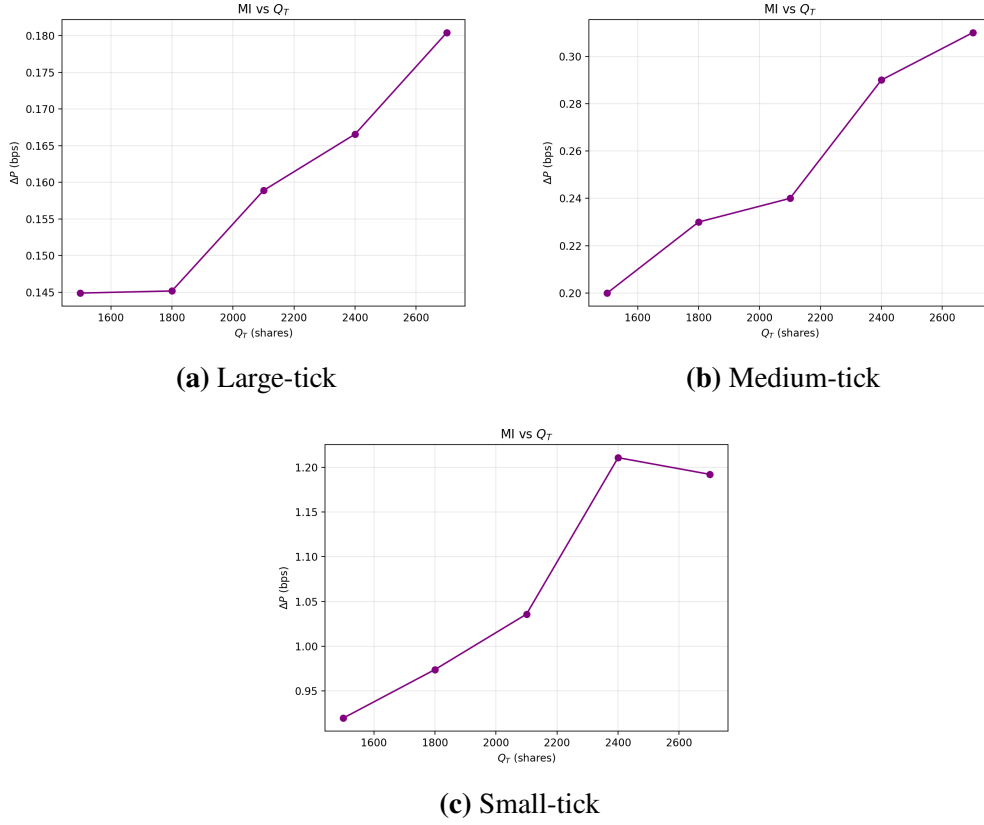


Figure 5.9: Market Impact Study: Market impact of a buy-side TWAP meta-order across large-, medium-, and small-tick LOBs. Average price change measured in bps with varying quantities Q_T executed by Market Orders only during a 5 minute window over 50 simulations per datapoint in the figure. Impact is concave in total traded quantity Q_T consistent with empirical patterns.

intensities again depend on the current spread as a power law (a similar comparison was performed in Chapter 4).

As we can see in Figure 5.10, with the methodology proposed above, one can replicate the spread distribution of a small-tick asset even with a Poisson model of the events' dynamics. However we note that the distribution shape and tail of returns in the Hawkes Process model is significantly different from that of the Poisson model. We also note that while the distribution of Price Change Time (defined as the time between two successive mid price changes) is bimodal in the Hawkes Process, it is unimodal in the Poisson Process. This quantity is of interest here since we do not explicitly model this event in the point process. As noted in Jain et al. [2024b], the shape of returns and price change time distributions in the Hawkes Process method is closer to the empirical distribution shape than the Poisson Process method's. The reason for the bimodal nature of the inter-event times distributions in the Hawkes

model is due to the interaction terms since removing them gives us the low-frequency mode alone. This multimodality in the distributions of the Hawkes events' times is an interesting observation. The exact reason for only one other mode in the distribution and the calculation of this mode's value is reserved for future research.

5.4 Model Calibration:

Large-tick				Medium-tick				Small-tick			
Asset	α	β	$\hat{\eta}$	Asset	α	β	$\hat{\eta}$	Asset	α	β	$\hat{\eta}$
SIRI	0.0102	0.98	0.99	ABBV	0.0440	0.46	0.79	TSLA	1.3610	0.48	0.19
BAC	0.0103	0.49	0.98	PM	0.2750	0.35	0.77	CHTR	1.4100	0.41	0.15
INTC	0.0130	0.94	0.98	AAPL	0.1350	0.59	0.92	AMZN	1.9630	0.41	0.09
CSCO	0.0250	0.60	0.99	IBM	0.2350	0.59	0.71	GOOG	3.2670	0.50	0.09
ORCL	0.0190	0.18	0.97					BKNG	3.7410	0.46	0.03
MSFT	0.0230	0.19	0.98								

Table 5.2: Calibrated critical parameters α , β , and $\hat{\eta}$ for the 15 assets from Table 3.1.

Since the Hawkes Dynamics is assumed to be independent of the LOB state variables, and since the distributions of the order sizes, and order distances from top are unconditional on any state variable and stationary, we can decouple the calibration to a parametric estimation of the distributions and a non-parametric estimation of the Hawkes kernels. We suggest the methodology described in Jain et al. [2024b], combining Kirchner [2017] and Bacry et al. [2016], for the latter. A brief summary of the method is as follows. They approximate the expectation of the intensity's integral with the observed bin counts of each of the 12 events. With certain constraints to control the stationarity of the Hawkes Process and the stability of the fitted kernels, they formulate an AR problem and solve it using a Quadratic Programming solver.

Following Jain et al. [2024b], to calibrate the critical parameters α, β , we analyze empirical high-frequency order book data. Specifically, we focus on the relationship between the order intensity and the spread measured in ticks. We approximate the instantaneous order intensity by counting the number of in-spread limit orders arriving within very short time intervals of 0.01 seconds. Since it is impossible to have in-spread limit orders with one tick spread, we remove this datapoint from the visualization and the regression. In Fig. 5.11, we visualize this relationship using scatter plots, where each point corresponds to the observed order intensity at a given spread value. The distribution of these observations is also represented by translucent red bars, which help illustrate the density and variability of the data points at different spread levels. To uncover the functional form governing this relationship, we perform

a regression analysis on the logarithm of both the spread and the estimated intensity, i.e., a log-log regression (red line).

For the parametric distributions, we suggest making use of the Maximum Likelihood method with a prior that all these distributions are Geometric distributions. In Fig. 5.12, we showcase the empirical η_{IS} distributions (note the shape looks exactly like the Geometric distribution) and their calibrated parameter: $\hat{\eta}_{(\cdot)}$ across three assets: INTC (Large-tick), AAPL (Medium-tick) and GOOG (Small-tick). Optionally, following Jain et al. [2024b], Lu and Abergel [2018a], one can add spikes in the probability distribution function to account for the traders' preference of round numbers in the order sizes and queue sizes. We present the calibrated critical parameters in Table 5.2 for the 15 assets we studied throughout this paper. As we can clearly see, the trend of decreasing ϵ with increasing α and decreasing β and $\hat{\eta}$ remains consistent in empirical calibration as well. Additionally, as mentioned previously, we overlay the calibrated α, β over the phase diagram plot in the previous section (Fig. 5.6a) to show that the calibrated parameters indeed fall consistently into their respective categories. Finally, for the partition functions, Ξ we made use of a simple linear partition function - i.e. we assume that the shape of the LOB in the deeper LOB is flat for purging queues. This is clearly a strong assumption (Section 3.5), and a better choice could be using the average shape of the LOB.

5.5 Discussion and Conclusion

In this work we outline how, statistically, does the LOB change when the granularity of price levels is kept constant with a constant tick-size while the price of the security increases from a few dollars to a few thousands of dollars. We quantify this change by the change in relative tick-size ϵ which is defined as the ratio between the tick-size and the price of the security. One of our key contributions is to highlight several stylized facts, which are then used to differentiate between large-, medium- and small-tick assets, and define clear metrics to measure them. We also provide a set of cross-asset visualizations of the stylized facts and show how different (or similar) these attributes are with varying relative tick-size. We found different power law relations in these stylized facts. Firstly, the average spread of a security and the average absolute returns of the security are both approximately proportional to the inverse of the relative tick-size. Secondly, the depth at which we find the maximum liquidity in the LOB follows a power law over the relative tick-size with the exponent roughly equal to $\frac{3}{2}$. In addition to that we see definitive trends in the shape of the spreads distribution as well as the sparsity in the LOB as well decrease the relative tick-size. We make use of a stylized fact known as *leverage*, which seems to be

universally similar in structure across all assets we studied, to model the LOB as a Hawkes Process with the top of the LOB as a separate dimension and the rest of the LOB (roughly upto the median of the volume shape of the LOB) as a separate dimension.

We propose a meta-queues multivariate Hawkes model extending the one proposed in Jain et al. [2024b] to account for sparsity, multi-tick level price moves and the humped shape of the LOB for small-tick assets. We showcase the versatility of the model by performing several simulation studies. Thereafter, we identify the in-spread Hawkes Parameters and the sparsity variables as the critical variables which define if the simulated LOB will be similar to a large-tick or a small-tick asset. We perform several tests to showcase that a number of stylized facts like sparsity, shape, and relative returns' distribution can be varied smoothly from that of a large-tick to small-tick asset using our model. Further we show that the stylized facts' metrics variation with relative tick-sizes is preserved in the model and is inline with empirical observations.

We note that to maintain analytical tractability and facilitate calibration, several simplifying assumptions were made during model development. Testing the underlying assumptions of any quantitative finance model is essential to ensure its practical relevance and robustness. These assumptions have been explicitly identified and their empirical validity rigorously examined in the Appendix D, providing a critical assessment of the model's foundations.

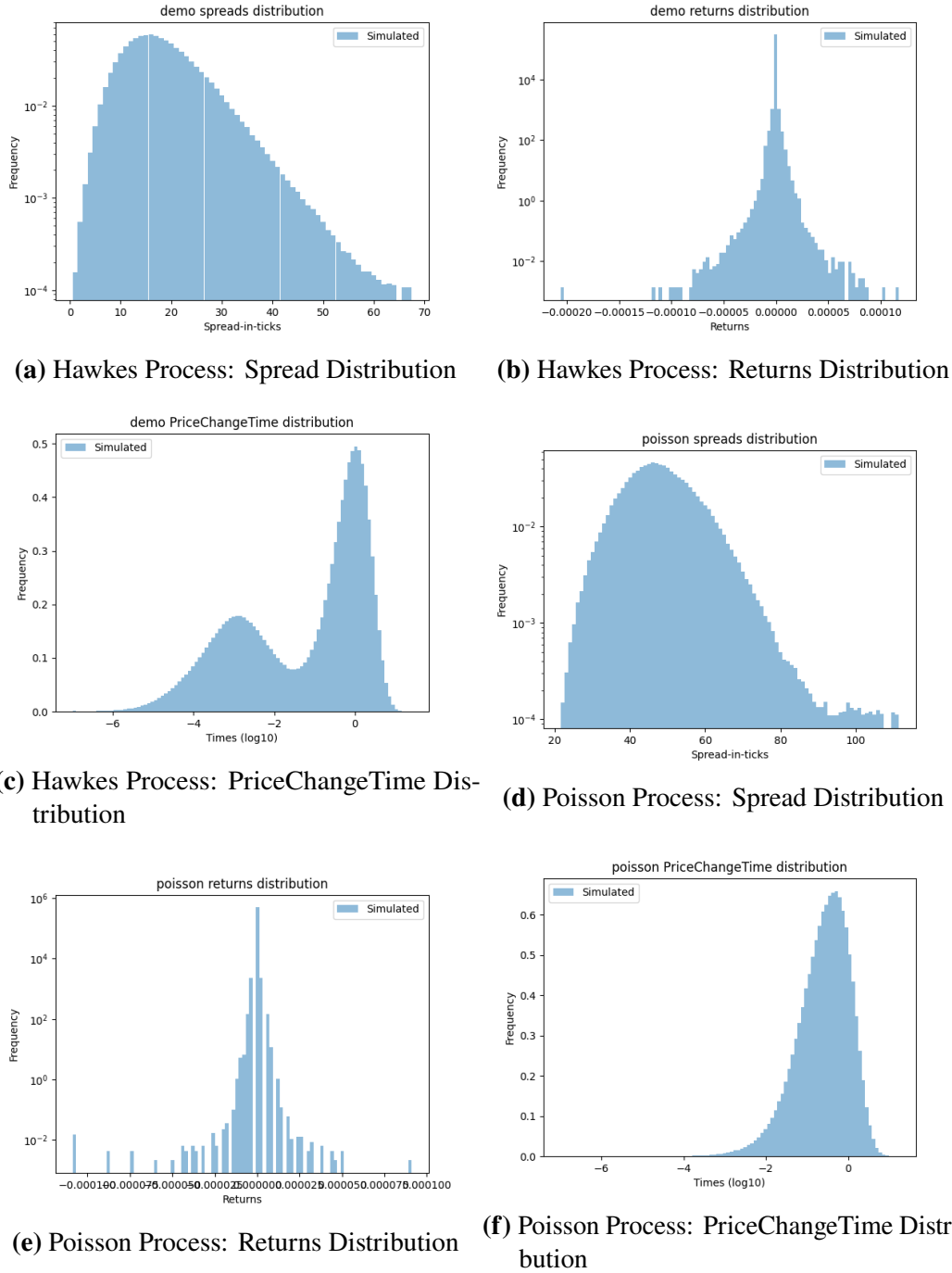


Figure 5.10: Hawkes Process vs Poisson Process: To compare with a simpler baseline, we make use of the Poisson Process (with similar handling of IS intensities as the Hawkes Process). While the spread distribution between the two look similar in shape, the returns distribution does not. Clearly the Hawkes Process simulations have a fatter tail than the Poisson Process. Secondly, we make use of an event ‘PriceChangeTime’ which is not explicitly modelled. Here we can see clearly that the Hawkes Process is able to create a bimodal distribution of inter-event times while the Poisson distribution remains unimodal. As noted in Jain et al. [2024b], the former is more inline with empirical data as it captures the high frequency domain of the inter-event durations.

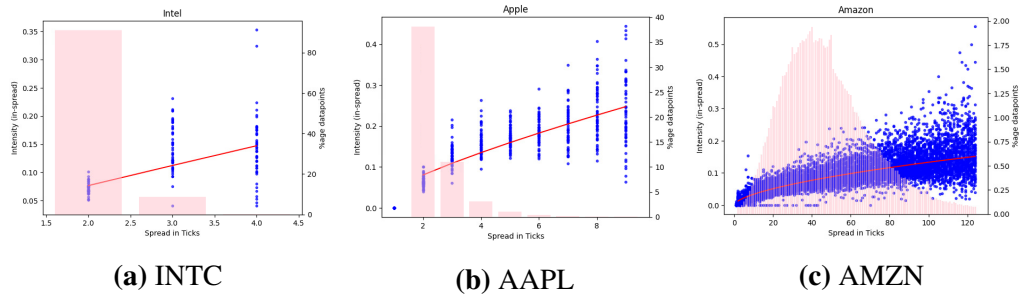


Figure 5.11: α, β Calibration: The order intensity is approximated by counting in-spread order arrivals within 0.01-second intervals and plotted against the current spread in ticks using scatter plots. The translucent red bars indicate the distribution of these observations. A linear regression, depicted by the red line, on the log-log transformed data (excluding spreads of 1 tick) yields a power-law. The fitted coefficients are summarized in Table 5.2. We can clearly see the power-law behaviour persists for all three categories of stocks depicted here.

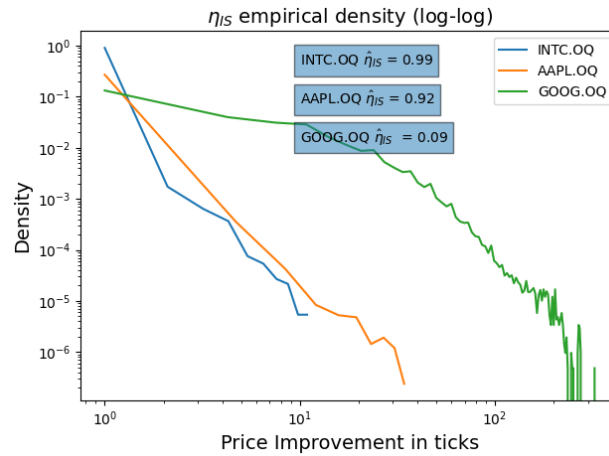


Figure 5.12: $\hat{\eta}_{IS}$ Calibration: $\hat{\eta}_{IS}$ is the parameter which controls the sparsity of the LOB. Here we show three examples, one from each category of assets, of empirical histograms of the sparsity levels. On a log-log scale it can be clearly seen that the distribution is similar to the geometric distribution. Inset we also show the calibrated parameters with this prior for all three assets - we observe the calibrated parameters move from near one to near zero as we decrease the relative tick-size. This indicates an increase in sparsity with a decrease in relative tick-size.

Chapter 6.

Impulse Control and Reinforcement Learning solutions to Market Making

We study optimal market making in a limit order book modeled by a mutually exciting Hawkes process developed in the previous chapters, capturing key microstructural features beyond Brownian price dynamics. The problem is formulated as an impulse control problem, leading to a high-dimensional HJB quasi-variational inequality that is computationally challenging to solve. We address this using deep reinforcement learning approximation, and a deep learning PDE solver of the HJB, demonstrating strong performance in realistic simulated markets.

6.1 Introduction

The previous chapters developed a high-fidelity simulation framework for limit order book dynamics based on mutually exciting Hawkes processes, designed to reproduce key empirical features of modern electronic markets. By explicitly modeling order arrivals, cancellations, and trades as self- and cross-exciting point processes, the simulator captures essential microstructural phenomena such as clustered event arrivals, queue dynamics, and endogenous price formation. This provides a realistic yet tractable environment in which the impact of trading strategies can be studied beyond the assumptions of diffusive price models.

Building on this framework, we now turn to the problem of optimal market making within such a microstructural setting. Unlike classical formulations in which prices evolve continuously and controls can be updated at all times, a market maker interacting with a limit order book faces discrete, event-driven decision opportunities and must account for inventory risk, execution uncertainty, and the feedback effects induced by their own actions. These features naturally lead to a control problem with jumps, path dependence, and non-local intervention effects.

To reflect these practical constraints, we formulate the market making problem as an impulse control problem in a Hawkes-driven LOB. Trading decisions are restricted to discrete interventions—such as posting, cancelling, or executing orders—which results in a high-dimensional Hamilton–Jacobi–Bellman quasi-variational inequality.

While this formulation provides a principled theoretical description of optimal behavior, its direct numerical solution is rendered challenging by the dimensionality of the state space and the non-local nature of the control.

The remainder of this chapter explores approximation-based approaches to this problem, combining insights from stochastic control with modern deep reinforcement learning methods. Using the previously developed simulator as a training and evaluation environment, we assess the effectiveness of these methods in producing robust market making strategies under realistic microstructural dynamics.

Overview This chapter is organized as follows. Section 6.2 situates the present chapter in its relevant theoretical and empirical context. Section 6.3 details the methodology, including the Limit Order Book model, Optimal Market Making formulation, the State–Intervention Operator, and the resulting HJB-QVI, along with its generator, solution approaches, and challenges. Section 6.4 applies the Deep Galerkin Method to solve the HJB-QVI and presents numerical results. Section 6.5 develops a Reinforcement Learning approximation of the impulse control problem, describing the state and action spaces, reward structure, training procedure using PPO and Self-Imitation Learning, and reports simulation results and sensitivity analyses. Section 6.6 discusses the findings and concludes.

6.2 Background

Controlled jump systems: The discrete nature of price movements in modern electronic markets, coupled with the complex interaction dynamics within limit order books, necessitates a more sophisticated modeling approach. While continuous-time diffusion-based models remain well-suited to quote-driven markets such as corporate bond trading, where dealers face pricing problems analogous to those originally modeled by Ho and Stoll [1981], equity markets require explicit modeling of the discrete limit order book structure. The work of Guilbaud and Pham [2013] represents a significant advancement in this direction, providing a discrete-time framework that more accurately captures the microstructural realities of modern electronic trading venues.

The transition from continuous to discrete modeling frameworks raises fundamental questions about the nature of control in high-frequency market making environments, see for example Law and Viens [2019]. In particular, diffusion-based control models implicitly rely on a continuously evolving filtration, which is difficult to justify in the context of limit order books, where no new conditioning information is revealed between successive market events. From the market maker’s perspective, the state of the system remains unchanged between two order book updates.

One related formulation addressing this issue is the jump decision process (or Piecewise Deterministic Markov Decision Process, see for example Bandini and Keller [2024], Bäuerle and Rieder [2011], Abergel et al. [2020]) framework, in which each jump is treated as a decision epoch and controls may be adjusted in response to every market event. While this approach acknowledges the event-driven nature of order book dynamics, it effectively allows for quote adjustments at every jump time, even though no new information arrives between such events.

In contrast, impulse control frameworks restrict interventions to discrete decision times and does not assume that quotes remain unchanged between interventions. This structure aligns more closely with the informational constraints of limit order book markets, where the filtration is constant between consecutive events and the world may be viewed as standing still until the next order book update. From this perspective, impulse control provides a natural modeling paradigm in which actions might only be taken when new information becomes available. Another complementary direction is the control of event intensities, see for instance Bayraktar and Ludkovski [2014].

Optimal Market Making: Market Making in Limit Order Books (LOBs) is a high frequency trading task, where liquidity is provided with the goal of capturing the bid-ask spread. While many classical formulations rely on continuous-time dynamics and control, the true microstructure of the LOB is inherently discrete and driven by a pure jump process.

The distinction between liquidity provision and liquidity consumption represents a fundamental dichotomy in modern market microstructure theory. As noted in Guéant [2016], while optimal execution strategies primarily focus on liquidity-taking behavior—the efficient liquidation of large positions or implementation of hedging strategies—market making fundamentally concerns itself with the provision of liquidity to the marketplace. This shift in perspective necessitates a fundamentally different analytical framework, one that accounts for the unique risks and opportunities inherent in standing ready to transact at posted prices.

Market makers, defined as traders who continuously provide liquidity by quoting bid and ask prices, serve as the backbone of price discovery mechanisms across diverse market structures Guéant [2016]. The heterogeneity of market making arrangements reflects the varied institutional frameworks within which these participants operate. On order-driven markets, official market makers such as Designated Market Makers (DMMs) on the NYSE operate under explicit contractual obligations to maintain fair and orderly markets, including mandatory participation in opening and closing auctions and adherence to National Best Bid and Offer (NBBO) quoting requirements.

Conversely, proprietary market makers, including high-frequency trading firms, provide liquidity without formal obligations, seeking to profit from the bid-ask spread while maintaining flexibility in their market participation.

The fundamental economic proposition underlying market making involves the temporal arbitrage of buying at the bid price and selling at the ask price, with the bid-ask spread representing the market maker's expected compensation for providing immediacy services. However, this seemingly straightforward profit mechanism is complicated by the inevitable inventory risk that market makers must bear. The asynchronous nature of buy and sell transactions means that market makers typically hold non-zero inventory positions, exposing them to adverse price movements during the holding period Guéant [2016].

Optimal Market Making (MM) in Limit Order Book (LOB) settings has been approached primarily through two methodological lenses: stochastic control and reinforcement learning. Early stochastic control models often assume Markovian mid-price dynamics with exogenously specified order flow. The seminal work of Avellaneda and Stoikov [2008] established the theoretical foundation for modern market making models by formulating the problem as a continuous-time stochastic control problem. Their framework, along with the closed-form approximations developed by Fernandez-Tapia et al. [2016], provides elegant analytical solutions for optimal bid and ask quote placement. Some interesting properties of the optimal quoting policy has been observed in several variants of the setup for instance see Barucci et al. [2025], Boyce et al. [2025], Chilenje et al. [2025]. However, these continuous-time models face significant limitations when applied to actual order-driven markets, particularly in their treatment of price discreteness arising from tick size constraints and their inability to capture the granular dynamics of limit order book evolution. For instance, Guilbaud and Pham [2013] model the mid-price as a continuous-time Markov chain and the bid-ask spread as a discrete Markov process, allowing the MM to place limit orders, market orders, and even aggressive internalizing strategies. These controls are optimized through a hybrid control framework involving a Hamilton-Jacobi-Bellman (HJB) equation. More recent efforts have focused on incorporating more realistic LOB dynamics. Ricci [2014] consider a Hawkes-like LOB, where market orders impact a latent alpha signal and fill probabilities, making them informative. The MM, restricted to placing only limit orders, solves a stochastic control problem in this non-Markovian environment. Similarly, Jusselin [2021] study a quote-driven market in which market takers' actions are modeled via Hawkes processes. They derive HJB solutions in the exponential kernel case and extend to more realistic power-law kernels via approximations. A

more discrete approach is taken by Abergel and Jedidi [2011], who model LOB dynamics using a state-dependent Poisson process and then extend the model to an exponential Hawkes process, solving the corresponding Markov Decision Process (MDP). While most of these works rely on analytically tractable models, they tend to assume known dynamics, stationarity, and full observability, which limit their applicability to real-world markets.

Reinforcement Learning: Reinforcement Learning (RL) has emerged as a compelling alternative, capable of handling partial observability, path-dependence, and model uncertainty. A recent survey by Gašperov et al. [2021] challenges classical assumptions made in stochastic control—such as complete knowledge of the environment and fixed time horizons—and organizes RL approaches by control logic (inventory-based, signal-driven, robust) and state observability (tabular vs. deep RL). A second taxonomy classifies methods by algorithmic design, highlighting the shift from analytical and tabular methods to deep actor-critic variants. Most RL frameworks define the agent’s state using features like inventory, PnL, spread, imbalance, and short-term technical signals such as RSI. To tackle sample inefficiency, several works employ simulators or historical replay mechanisms. For instance, Gašperov and Kostanjčar [2022] use an 8-dimensional Hawkes-based simulator in which the MM interacts via limit and market orders in a uni-episodic loop; their Soft Actor-Critic agent outperforms a stochastic control baseline, while DQN and TD3 fail to converge. Other approaches, like those of Spooner et al. [2018], Guo et al. [2023], rely on historical replay with simpler discrete actions and no explicit modeling of market impact. Particularly for jump processes, one example of recent work on reinforcement learning is Dai and Gluzman [2022]. Further, Croissant et al. [2024] develops an approximation of the discrete system at very high intensities to be near-continuous and they develop RL methods in that setting. Recent innovations include adversarial training and auxiliary signal units to improve policy robustness and generalization. Despite RL’s flexibility, fundamental challenges remain—Guéant [2016] highlights issues such as non-stationarity, the absence of feedback loops in historical data, and the inherent difficulty in designing risk-sensitive reward structures. These limitations continue to motivate hybrid models that blend the tractability of control theory with the adaptability of learning-based methods.

Impulse control: Impulse control has been a well studied area of stochastic control with several applications, for instance see Chen et al. [2025], Aïd et al. [2020], Lv and Xiong [2022], Davis et al. [2010], Chen and Guo [2013], Bayraktar et al. [2013], Basei et al. [2022]. Within the impulse control paradigm, market makers are viewed

as making discrete intervention decisions at carefully chosen times, rather than continuously adjusting their positions. This perspective naturally accommodates the technological constraints of modern trading systems while preserving the essential economic intuition underlying market making strategies. The framework recognizes that optimal market making involves not just the selection of appropriate bid and ask prices, but also the strategic timing of order placement and cancellation decisions.

The mathematical formulation of market making under impulse control requires careful consideration of the underlying market dynamics and the constraints faced by market participants. The pure jump nature of limit order book evolution, driven by the arrival of discrete market and limit orders, naturally aligns with the impulse control framework's emphasis on discrete intervention strategies. This alignment suggests that impulse control methods may provide both more realistic and more computationally tractable solutions to the market making problem. Impulse control in jump diffusions has been studied extensively in Davis et al. [2010], Guo and Wu [2009], Chen and Guo [2013].

Our approach builds upon these foundational insights while addressing the practical limitations of existing methodologies. By explicitly modeling the discrete nature of both market dynamics and control decisions, we develop a framework that more accurately captures the reality of modern electronic market making while maintaining analytical tractability. The resulting formulation provides a natural bridge between the theoretical elegance of continuous-time models and the practical requirements of high-frequency trading systems.

Our Contributions: This chapter makes several contributions to the literature on optimal market making in high-frequency electronic markets:

1. We formulate the market making problem in a high-fidelity Hawkes-driven limit order book as an impulse control problem, explicitly accounting for the discrete, event-driven nature of order flow and the realistic constraint that market makers cannot adjust strategies at every market event. This leads to a high-dimensional Hamilton-Jacobi-Bellman Quasi-Variational Inequality whose generator incorporates the full microstructure of queue dynamics, self-exciting order arrivals, and endogenous price impact.
2. We implement and analyze the Deep Galerkin Method Sirignano and Spiliopoulos [2018] for solving this jump HJB-QVI, documenting both its successes in simplified settings and its fundamental limitations in high-dimensional mutually-exciting environments, including convergence to economically unrealistic equilibria.

3. We develop a novel reinforcement learning approximation inspired by the auxiliary control formulation of impulse problems, decomposing the control into timing and action subproblems via a two-network PPO architecture augmented with self-imitation learning. This model-free approach achieves superior empirical performance with limited training data, yielding annualized Sharpe ratios above 30 while learning interpretable, symmetric liquidity provision strategies.
4. We provide comprehensive sensitivity analyses demonstrating the robustness of the RL framework to Hawkes kernel specifications, transaction costs, and state representations, while identifying critical design elements necessary for stable convergence. We further compare the performance to a rule-based baseline agent.

Our findings bridge impulse control theory, neural PDE methods, and modern deep reinforcement learning, providing both theoretical insights and practical algorithms for optimal execution in jump-driven microstructural markets.

6.3 Methodology

We discuss the methodology of the LOB model, the market making problem and its impulse control formulation in this section. We provide the mathematical details of the Hamilton-Jacobi-Bellman Quasi-Variational Inequality of the value function of this control problem. Finally, we state the impulse control problem's solution approaches and their respective challenges.

6.3.1 Limit Order Book Model

We model the LOB using a d -dimensional mutually-exciting Hawkes process as developed in Chapters 4 and 5. This process reproduces stylized facts of LOB dynamics such as realistic spreads, long-memory in returns, and clustered arrival times. Unlike Brownian motion-based models, the mid-price emerges endogenously from queue dynamics and event causality. We refer to Chapter 4 for more details on the LOB setup. The events that form the Hawkes process are as follows.

$$\mathcal{E} := \{LO_{ask_D}, LO_{ask_T}, CO_{ask_T}, MO_{ask}, LO_{ask_{IS}}, LO_{bid_{IS}}, LO_{bid_T}, CO_{bid_T}, MO_{bid}, LO_{bid_D}\}$$

We allow for general kernel types in the Hawkes process, but restrict to exponential kernels for mathematical tractability. The model implicitly includes

a concave price impact due to self-excitation. While there is no adverse selection in this framework due to the absence of information asymmetry, we address this important question in our subsequent paper Jafree et al. [2025].

6.3.2 Optimal Market Making

In this Hawkes LOB, we develop a market making agent which can interact with the LOB by sending impulses at any given time. These impulses are restricted to be one of the 12 events in $\mathcal{E}$ for the purpose of this work. More complex impulses may include a combination of several events in one impulse however to foster discussion we restrict ourselves to the case of one order per impulse. The market making agent observes the LOB's volumes at all the price levels fully i.e. it has level 2 access to the LOB data. The market maker's objective is to maximise her terminal cash and the value of her terminal inventory. While doing so, a market maker prefers to keep as low an inventory at any point of time as possible. One way to achieve this is to penalise the current inventory of the market maker. For discussions around various methods of penalisation, we refer the reader to Cartea et al. [2015]. We make use of the quadratic running cost penalisation method and therefore state the objective of the market maker as below. If the market maker trades from time t to T , while observing the state $\mathbf{S}_t$ and employing the optimal policy $u^*(t)$, we have the objective:

$$J^{(u^*)}(t, \mathbf{S}_t) = \sup_{u \in \mathcal{U}} \mathbb{E} \left[\int_t^T -\eta Y_t^2 dt + X_T + Y_T P_T^{(mid)} - \kappa Y_T^2 + \sum_{t \leq \tau_i \leq T} K(\mathbf{S}(\tau_i), \psi_i) \right] \quad (6.1)$$

where

$$K(\mathbf{S}, \psi) = \begin{cases} 0 & \text{for limit/cancel orders} \\ z p_t^{(\zeta)} & \text{for market orders} \end{cases}$$

The system state $\mathbf{S}_t$ and policy $u(t) \in \mathcal{U}$, where $\mathcal{U}$ denotes the set of admissible policies, at time t is defined as

$$u(t) := \{(\tau_i, \psi_i)\}_{i=1, \dots, N} \text{ where } \tau_N < t \quad (6.2)$$

$$\mathbf{S}_t := \{X_t, Y_t, p_t^{(\zeta)}, q_t^{(\zeta)}, q_t^{(\zeta, D)}, n_t^{(\zeta)}, P_t^{(mid)}, (\lambda_t^{(i)})_{(i=1, \dots, d)}\}_{\zeta \in \{a, b\}} \quad (6.3)$$

Here, X_t is the cash of the market maker, Y_t is the inventory, $n_t^{(\zeta)}$ and $q_t^{(\zeta)}$ denote the queue-priority and size, $\lambda_t^{(i)}$ are the Hawkes intensities. These state variables follow jump equations due to the Hawkes process jumps. The dynamics of the LOB

state-variable are given in the Appendix E. Accordingly, the policy is a set of impulse times τ_i and impulse types $\psi_i \in \mathcal{E}$. We assume that the order sizes of the Hawkes process and that of the agent are uniformly constant. Therefore we are only interested in the optimal impulse time and type. The question of optimal order size is out of scope for this work. Finally, η and κ are the respective running and terminal inventory penalty parameters while $K(\mathbf{S}, \psi)$ is the instantaneous profit made by the market maker by sending an impulse ψ at state $\mathbf{S}$.

6.3.3 The State-Intervention Operator

The market maker's control problem is of the impulse type: at discrete intervention times τ_i , the agent selects impulses ψ_i , which correspond to submitting market orders, limit orders, or cancellations. Unlike continuous control, where adjustments are infinitesimal and ongoing, impulses induce discrete jumps in the state variables, reflecting the event-driven structure of order books.

Formally, the intervention rule is represented by the state-intervention operator:

$$\mathbf{S}(\tau_i) = \Gamma(\mathbf{S}(\tau_i^-), \psi_i),$$

where $\mathbf{S}(\tau_i^-)$ is the pre-impulse state, ψ_i is the chosen control, and Γ encodes how that control affects the state vector. For example, inserting a limit order at the top of the book ($LO_T^{(\zeta)}$) increases both the queue size and possibly modifies the maker's queue position. The intervention can be specified as follows:

$$\begin{aligned} n^{(\zeta)}(\tau_i) &= n^{(\zeta)}(\tau_i^-) \mathbb{1}(n^{(\zeta)}(\tau_i^-) \leq q^{(\zeta)}(\tau_i^-)) + q^{(\zeta)}(\tau_i^-) \mathbb{1}(n^{(\zeta)}(\tau_i^-) > q^{(\zeta)}(\tau_i^-)) \\ q^{(\zeta)}(\tau_i) &= q^{(\zeta)}(\tau_i^-) + 1 \end{aligned}$$

Similarly, cancelling a top order ($CO_T^{(\zeta)}$) removes volume, possibly shifting prices if the best quote disappears. Market orders ($MO^{(\zeta)}$) consume liquidity, changing inventory, cash, and potentially mid-price. Each of these is specified by the difference equations in the Appendix F.

6.3.4 Hamilton–Jacobi–Bellman Quasi-Variational Inequality

One can see from the objective's formulation in Eq. 6.1 that the running cost is $f(S_t) = -\eta Y_t^2$ and the terminal cost is $g(S_T) = X_T + Y_T P_T^{(mid)} - \kappa Y_T^2$. Let $V(t, \mathbf{S}_t)$ denote the value function at time t in state $\mathbf{S}_t$. The problem admits the following quasi-variational inequality (QVI) see Øksendal and Sulem [2005], Fleming and

Soner [2006] for instance:

$$\min \left\{ -\partial_t V - \mathcal{L}V, V(t, \mathbf{S}) - \sup_{\psi \in \mathcal{A}} V(t, \Gamma(\mathbf{S}, \psi)) \right\} = 0, \quad (6.4)$$

with terminal condition

$$V(T, \mathbf{S}_T) = X_T + Y_T P_T^{(mid)} - \kappa Y_T^2 \quad (6.5)$$

We see that the conditions and assumptions for this HJB-QVI's wellposedness and the solutions' existence have been met (Ch. 2 Øksendal and Sulem [2005]). Here:

- The operator $\mathcal{L}$ is the infinitesimal generator of the controlled Hawkes-driven LOB dynamics in the absence of impulses. It accounts for stochastic jumps due to market orders, cancellations, and limit order arrivals.
- The intervention operator $\sup_{\psi \in \mathcal{A}} V(t, \Gamma(\mathbf{S}, \psi))$ encodes the value of optimally choosing an impulse at state $\mathbf{S}$.
- The terminal payoff consists of the cash position plus the liquidation value of inventory (using the mid-price) with κ being the terminal liquidation penalty.

This QVI formulation unifies the continuous-time Markov jump dynamics of the order book with the discrete impulse controls of the market maker. Solving it yields the optimal market-making policy: when to post or cancel limit orders, when to submit market orders, and how to manage queue positions in response to the stochastic evolution of order flow.

Let $\Phi(t, \mathbf{S}_t) = \sup_u J^{(u)}(t, \mathbf{S}_t)$ be the candidate value function. The value-intervention operator is defined as:

$$\mathcal{M}\Phi(t, \mathbf{S}_t) = \sup_{\psi} \{ \Phi(t, \Gamma(\mathbf{S}_t, \psi)) + K(\mathbf{S}_t, \psi) \} \quad (6.6)$$

Introducing a binary control $d_t \in \{0, 1\}$ (impulse or not), we can rewrite the HJB-QVI as the following, see Azimzadeh [2018] for instance:

$$\sup_{d \in \{0, 1\}} \{ (1-d)(\mathcal{L}\Phi + f) + d(\mathcal{M}\Phi - \Phi) \} = 0 \quad (6.7)$$

We note that an alternate approach exists which uses the penalty scheme developed in Reisinger and Zhang [2018, 2020] to represent the QVI.

6.3.5 The Generator

Now for pure jump processes, $\mathcal{L}\Phi$, the generator of Φ , is given by for Poisson Process driven processes:

$$\mathcal{L}\Phi(t, s) = \Phi_t(t, s) + \sum_i \lambda^{(i)}(\Phi(t, T_i(s)) - \Phi(t, s)) \quad (6.8)$$

Where $T_i(\cdot)$ is transition function of state s when an event i happens in the point process. Due to the similarity with the state-intervention operator, we omit its mathematical formulation. If a Hawkes Process drives the point processes, we have the following SDE representation of the Hawkes intensity for exponential kernels:

$$d\lambda^{(i)}(t) = \gamma_i(\mu_i - \lambda^{(i)}(t))dt + \alpha_i dN_t^{(i)}; \lambda^{(i)}(0) = \mu_i \quad (6.9)$$

$$\lambda^{(i)}(t) = \mu_i + \alpha_i \int_0^t e^{-\gamma_i(t-s)} dN_s^{(i)} \quad (6.10)$$

For multidimensional, mutually-exciting Hawkes we have M^2 such equations which when added gives us the final intensity:

$$\lambda_t^{(i)} = \mu_i + \sum_{j=1}^M \alpha_{ij} \int_0^t e^{-\gamma_{ij}(t-u)} dN_u^{(j)} \quad (6.11)$$

We note that there is no SDE representation for general kernels of the Hawkes process. As shown in several studies including Jain et al. [2024b], the more realistic choice of kernels is the power-law function however we lose the Markovian representation of the intensity process. One can approximate these power-law functions by an infinite sum of exponential kernels as shown in Jusselin [2021] however this leads to an infinite dimensional Markov process. In this work, we will restrict ourselves to single exponential kernels unless specified.

Abusing some notation to denote the candidate function by $\Phi(t, \lambda, s)$ instead of $\Phi(t, S_t)$, the generator becomes, see for instance Bensoussan and Chevalier-Roignant [2024]:

$$\begin{aligned} \mathcal{L}\Phi(t, \lambda, s) = & \Phi_t(t, \lambda, s) + \sum_i \left(\lambda^{(i)}(t) (\Phi(t, \lambda + \alpha_{.i}, T_i(s)) - \Phi(t, s)) \right. \\ & \left. + \Phi_{\lambda_i}(t, \lambda, s) \frac{d\lambda_i}{dt}(t) \right) \end{aligned} \quad (6.12)$$

6.3.6 Solution Approaches and Challenges

Solving the optimal market making problem under the proposed Hawkes-driven LOB dynamics presents considerable analytical and numerical challenges. First, obtaining a closed-form solution to the resulting quasi-variational inequality (QVI) (see for instance Ricci [2014]) is generally infeasible due to the high dimensionality and nonlinearity of the system. The state space, which includes variables such as the MM's inventory, cash, active order queue positions, best bid/ask prices, spread, imbalance, time since last event, and a multi-dimensional Hawkes kernel history, is of several dimensions. This renders traditional grid-based numerical techniques (see for instance Azimzadeh [2018], Cleynen and Saporta [2023]) impractical due to the curse of dimensionality. To overcome these obstacles, deep learning-based PDE solvers have been explored for HJBs in continuous stochastic control literature, see Hu and Lauriere [2023], Capponi and Lehalle [2023] for a recent review. Another interesting approach taken in Chevalier et al. [2024] was to approximate the impulse control problem as a series of optimal stopping problem and solving it with a deep learning method. Deep learning solutions for optimal stopping problems has been widely studied in the literature, for example see Reppen et al. [2023], Dai et al. [2024]. For a more general deep learning framework of solving empirical risk minimization problems, see Reppen et al. [2022], Reppen and Soner [2023] for instance.

In particular, we attempted to apply the Deep Galerkin Method (DGM) introduced by Sirignano and Spiliopoulos [2018], which approximates solutions to high-dimensional PDEs using neural networks trained by minimizing the PDE residual over a sampled domain. Despite its success in solving parabolic PDEs arising in finance and physics, DGM struggled to converge in our setting. The primary bottleneck lies in the non-local and discontinuous structure of the impulse control operator in the QVI, which poses significant challenges for sampling-based learning methods. Extensions such as those proposed by Al-Arabi et al. [2022] to better handle discontinuities and boundary conditions were also tested, but their effectiveness was limited by the irregular jump structure induced by the impulse control. These challenges highlight the need for more specialized neural-PDE solvers

that can handle the specific structure of control problems with non-local operators and event-driven dynamics. Hybrid methods that combine analytical insights from stochastic control with data-driven approximators remain a promising yet largely unexplored avenue in this setting. Nevertheless in the below we showcase the DGM methodology and the training results we achieved.

6.4 Deep Galerkin Method to solve the HJB-QVI

In order to circumvent the problems with traditional approaches to solve the HJB-QVI, we employ deep learning approximations to the value function and the control function(s). Our implementation uses the Deep Galerkin Method (DGM) of Sirignano and Spiliopoulos [2018] to solve a jump Hamilton-Jacobi-Bellman (HJB) equation arising in optimal market making with limit order books. The DGM method is a mesh-free method of fitting a neural network to the value function of the optimal control problem. The value function, the decision policy, and the control policy respectively are represented by neural networks with parameters θ, χ, ξ respectively:

$$\phi(t, \mathbf{S}_t) = \phi_\theta(t, \mathbf{S}_t)$$

$$d(t, \mathbf{S}_t) = d_\chi(t, \mathbf{S}_t)$$

$$u(t, \mathbf{S}_t) = u_\xi(t, \mathbf{S}_t)$$

It fits this neural network by minimizing a loss function constructed from a candidate value function using the HJB-QVI. As the HJB-QVI involves several partial derivatives of the value function, the well known method of automatic differentiation is employed to compute them in a mesh-free way again. That is, time and lambda derivatives are computed using automatic differentiation:

$$\frac{\partial \phi_\theta}{\partial t}(t, s) = \nabla_t \phi_\theta(t, s)$$

The DGM method first involves sampling time and space points from given distributions both in the interior domain as well as the boundary of the state space. The sampling strategy is to generate states from stationary distributions for example:

$$Y \sim \mathcal{N}(0, 4) \text{ (rounded to integers)}$$

$$P^{mid} \sim \mathcal{N}(200, 100)$$

$$\text{Spreads} \sim \text{Geometric}(0.8) \times 0.01$$

The state space consists of 23 dimensions representing market microstructure

variables. As discussed earlier, the value function $\phi(t, \mathbf{S}_t)$ satisfies the jump HJB-QVI equation:

$$\sup_{d \in \{0,1\}} \left\{ (1-d) \mathcal{L}\phi(t, \mathbf{S}_t) + d \sup_{\psi \in \mathcal{A}} \mathcal{M}^\psi \phi(t, \mathbf{S}_t) \right\} = 0 \quad (6.13)$$

subject to terminal condition:

$$\phi(T, \mathbf{S}_T) = X_T + Y_T P_T^{(mid)} - \kappa Y_T^2 \quad (6.14)$$

Next the value function and the HJB-QVI is evaluated at these points and the loss function is calculated. The DGM loss function combines interior and boundary losses:

$$\mathcal{L}_{DGM} = \mathcal{L}_{interior} + \mathcal{L}_{boundary} \quad (6.15)$$

Interior Loss:

$$\mathcal{L}_{interior} = \mathbb{E} \left[\left| \left\{ (1-d_\chi(t, \mathbf{S}_t)) \mathcal{L}\phi_\theta(t, \mathbf{S}_t) + d_\chi(t, \mathbf{S}_t) \mathcal{M}^{u_\xi} \phi_\theta(t, \mathbf{S}_t) \right\} \right|^2 \right] \quad (6.16)$$

$$\mathcal{M}^{u_\xi} \Phi(t, \mathbf{S}_t) = \Phi(t, \Gamma(\mathbf{S}_t, u_\xi(t, \mathbf{S}_t))) + K(\mathbf{S}_t, u_\xi(t, \mathbf{S}_t)) \quad (6.17)$$

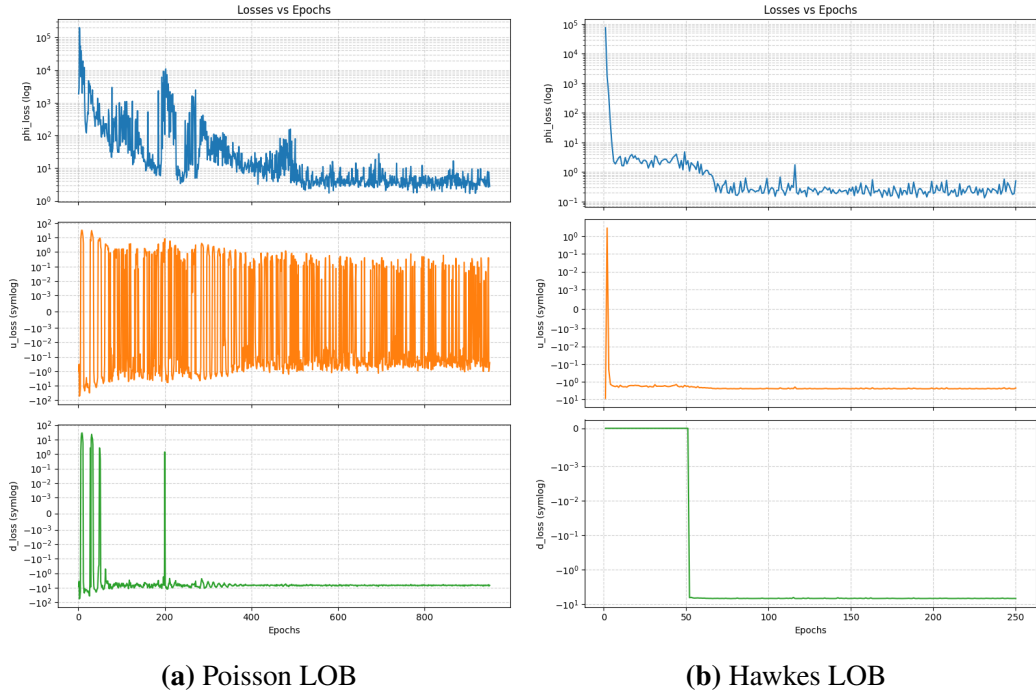
Boundary Loss:

$$\mathcal{L}_{boundary} = \mathbb{E} \left[\left| \phi_\theta(T, \mathbf{S}_T) - (X_T + Y_T P_T^{(mid)} - \kappa Y_T^2) \right|^2 \right] \quad (6.18)$$

The objective for the value function is to minimise these costs however as noted in Al-Aradi et al. [2022], the objective for the control functions is to maximise these costs. Therefore it becomes a minimax optimization problem similar to an actor-critic setup. We use the ADAM backpropagation algorithm for learning the weights of the three neural networks. We used the architecture mentioned in Sirignano and Spiliopoulos [2018] i.e. an LSTM for all three networks (ϕ, d, u). We used the relu activation function and a fixed learning rate of 10^{-3} .

6.4.1 Results:

Training logs are shown in Figure 6.1 for two settings of the LOB setup - the Poisson process setting and the Hawkes process setting. In particular, we restrict ourselves to the Hawkes process setup where only Market Orders are Hawkes processes

**Figure 6.1:** DGM Training Logs

and the remaining events are Poisson processes. This is done to show a proof of concept of the Deep Galerkin Method in the Hawkes setting. We observe the convergence is oscillatory due to the opposing updates of value and policy function. The networks used were 3 layers deep and 20 neurons wide for the Poisson setting and 10 layers deep and 50 neurons wide for the Hawkes setting. These network architecture hyperparameters were established by a large scale grid search-like method of comparing losses achieved after 200 epochs of training. In the Hawkes setting we saw the d network was quickly converging to a local optima where it always decided not to act in the first few training epochs. Therefore we chose to freeze the weights of the d network for the first 50 epochs to enable learning in the ϕ and u networks. We note that in the full Hawkes LOB setting (i.e. all 12 events form a mutually exciting Hawkes Process), the model did not converge. This is primarily due to the extremely high dimensional nature of the problem since 12 mutually exciting Hawkes process require a 144 dimensional Markovian representation for the stochastic control framework.

On a practical note, we tried several methodological and hyperparametric variants of the DGM for the full 12D Hawkes setting however without any success. We tried, for example, neural net depth upto 50 layers and 300 layers wide, MLP, Transformers as the core unit instead of the LSTM suggested in the DGM literature,

model gradient clamping and clipping, and training for more than a thousand episodes with various initial LOB and various settings of the Hawkes parameters.

Table 6.1: Out-of-sample Testing Results

LOB Model	Sharpe Ratio	Mean Abs. Inventory
Poisson	4.54	0.891
Hawkes - Market Orders Only	0.78	21.56
Full 12D Hawkes	Did Not Converge	Did Not Converge

We test the policy networks on OOS data by running them through our simulator for 5 minutes over several episodes. We calculate the annualized sharpe ratio from these simulations and the mean absolute inventory over the trajectories. The results are summarised in Table 6.1. We observe positive annualized sharpe ratios in both the Poisson and Hawkes setting however we note that the Hawkes setting has a very high mean absolute inventory. We further investigate this in Figure 6.2. We observe that over multiple tests (light lines) the strategy of the agent remains exactly the same. Since this seems to rely on pumping the market by sending aggressive orders initially and liquidating the inventory on a slow scale, we name this strategy ‘pump and dump’. It seems to be suboptimal as we see the profit to be near zero with a quite high variance. It is interesting that this obviously illegal strategy has been learnt by solving the HJB-QVI. Indeed it is mathematically allowed to have this strategy but it is neither market making nor legal by regulations. We note that in the setting of Hawkes with Market Orders being the only self-exciting events, with exponential kernels there is a dynamic arbitrage as reported by Alfonsi and Blanc [2015]. This dynamic arbitrage theoretically allows for these ‘pump and dump’ strategies to be profitable on average. This could be the reason why the HJB-QVI solver converged to this strategy in this case.

It has been shown in Jusselin and Rosenbaum [2020] that in this Market Orders only Hawkes setting, no-arbitrage conditions imply power law kernels of the Hawkes process. This choice of power law kernels has also been widely supported in the literature, for instance see Jain et al. [2024b] for a discussion on the same. We however note that power law kernels as it is do not allow for a Markovian representation of the intensity process. We therefore were not able to formulate the intensity process’s dynamics in a SDE representation. This poses a significant challenge. Jusselin [2021] show that the power law kernel can be approximated as a sum of infinite number of exponential kernels. Unfortunately this does not solve our numerical problem however it does give us an approximation technique. We can indeed approximate the power law kernel to a reasonable extent by finite number of exponential kernels

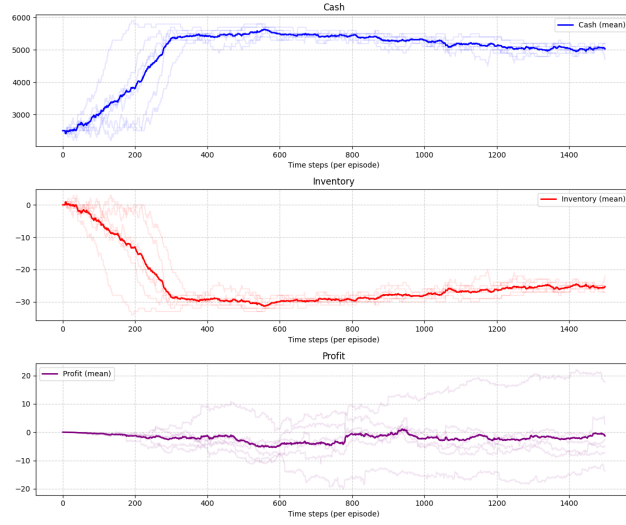


Figure 6.2: DGM Testing: Pump and Dump strategy of the Market Orders only Hawkes setting

however this has the same curse of dimensionality as the full 12D Hawkes setting mentioned before.

These challenges motivate the use of a model free methodology instead of the model based stochastic control methodology we employed before. In the next section, we develop a reinforcement learning framework to learn the optimal strategy from the principles of policy gradient instead of trying to solve the value function's HJB-QVI.

6.5 Reinforcement Learning Approximation

The analytical intractability of the HJB-QVI derived in Section 3 necessitates alternative solution methodologies. As established in Section 3.5, the generator $\mathcal{L}$ encompasses a high-dimensional state space including queue sizes, queue priorities, Hawkes intensities, and the complete bid-ask structure across multiple price levels. The presence of impulse controls introduces non-local operators into the partial differential equation, rendering standard finite-difference schemes or backward-induction methods computationally prohibitive even under aggressive discretization. The curse of dimensionality, particularly acute in the 23-dimensional state space of our full Hawkes-driven LOB model, fundamentally limits the applicability of grid-based numerical techniques as discussed in Azimzadeh [2018].

To circumvent these computational barriers, we adopt a model-free reinforcement learning approach that learns optimal policies through simulated interaction with the Hawkes-driven limit order book environment, thereby avoiding the need to solve the QVI directly. This methodological pivot aligns with recent developments in the

algorithmic trading literature reviewed in Section 1, where reinforcement learning has emerged as a compelling alternative.

6.5.1 Decomposition of the Impulse Control Problem

Drawing inspiration from the auxiliary control formulations explored in Mguni et al. [2023], we approximate the impulse control problem by decomposing it into two interacting reinforcement learning agents. This decomposition directly mirrors the structure of the QVI presented in Section 3.4, which inherently distinguishes between the continuation region (no impulse) and the intervention region (apply an impulse).

Specifically, we introduce:

- **Decision network d_χ (timing):** A policy that determines *when* to intervene in the limit order book. This network approximates the binary control $d_t \in \{0, 1\}$ introduced in the reformulated HJB-QVI of Section 3.4, thereby capturing the stopping aspect of the impulse control problem. The decision network effectively learns to identify states where the value of intervention exceeds the value of continuation.
- **Action network u_ξ (impulse selection):** A policy that determines *what* order action to execute, conditional on the decision to intervene. This network approximates the impulse selection operator $\sup_{\psi \in \mathcal{A}} V(t, \Gamma(\mathcal{S}, \psi))$ from the QVI formulation, where ψ_i corresponds to submitting market orders, limit orders, or cancellations as enumerated in the event set $\mathcal{E}$ defined in Section 3.1.

This architectural separation preserves the natural structure of the quasi variational inequality while enabling gradient-based learning through policy optimization. Unlike the Deep Galerkin Method discussed in Section 3.6, which attempts to directly parameterize the value function $V(t, \mathcal{S}_t)$ and solve the PDE residual, the reinforcement learning formulation learns the optimal policy implicitly through interaction with the environment, thereby bypassing the non-local operator evaluation challenges that hindered DGM convergence in high-dimensional settings.

6.5.2 State Space Construction

The state representation for the reinforcement learning agent must capture both the instantaneous market conditions relevant for execution risk and the historical information necessary for predicting future order flow dynamics. Recognizing that the Hawkes process intensities $\lambda_t^{(i)}$ provide a sufficient statistic for the infinitesimal arrival rates conditional on the filtration generated by past events (as established in Section 3.6), we construct the augmented state space as:

$$\mathbf{S}_t := \left\{ X_t, Y_t, s_t, \frac{n_t^{(\zeta)}}{q_t^{(\zeta)}}_{\zeta \in \{a,b\}}, \lambda_t^{(i)}, \mathcal{H}_{t-\tau_H:t} \right\} \quad (6.19)$$

The components of this state vector warrant detailed justification:

- $X_t \in \mathbb{R}$: The market maker's cash position, which evolves according to the jump equations specified in Appendix E.
- $Y_t \in \mathbb{Z}$: The inventory position, which appears in both the running cost $f(S_t) = -\eta Y_t^2$ and the terminal liquidation value.
- $s_t := p_t^{(a)} - p_t^{(b)} \in \mathbb{R}_+$: The bid-ask spread, which determines the maximum profit capturable per round-trip transaction and influences both fill probabilities and the mid-price evolution $P_t^{(mid)}$ that enters the terminal condition.
- $\frac{n_t^{(\zeta)}}{q_t^{(\zeta)}} \in [0, 1]$ for $\zeta \in \{a, b\}$: The relative queue position of the market maker's limit orders on each side of the book. As detailed in Section 3.1, the queue dynamics $(n_t^{(\zeta)}, q_t^{(\zeta)}, q_t^{(\zeta,D)})$ govern the fill probabilities and therefore critically impact the expected profit from posted limit orders. The normalization by total queue size $q_t^{(\zeta)}$ ensures scale invariance across different liquidity regimes.
- $\lambda_t^{(i)} \in \mathbb{R}_+$ for $i = 1, \dots, d$: The Hawkes process intensities corresponding to each event type in $\mathcal{E}$.
- $\mathcal{H}_{t-\tau_H:t}$: A truncated window of the recent event history spanning the interval $[t - \tau_H, t]$. While the Hawkes intensities provide a sufficient statistic for the infinitesimal conditional intensity, finite memory windows can capture additional predictive information about market microstructure regimes not fully summarized by the exponential kernel parameterization. This augmentation addresses the limitation noted in Section 3.6 regarding the restriction to exponential kernels, as more realistic power-law kernels do not admit finite-dimensional Markovian representations.

6.5.3 Objective Function and Reward Structure

To enable gradient-based policy learning, we discretize the continuous-time objective function presented in Equation 6.1 over a uniform grid $\{t_0, t_1, \dots, t_N\}$ with $t_0 = t$, $t_N = T$, and timestep $\Delta t = t_{i+1} - t_i$. The discrete-time approximation of the value function becomes:

$$J^{(u)}(t, \mathbf{S}_t) = \mathbb{E} \left[\sum_{i=0}^{N-1} \left(\int_{t_i}^{t_{i+1}} -\eta Y_s^2 ds + \Delta X_{t_i} + \Delta(Y_{t_i} P_{t_i}^{(mid)}) - \kappa \Delta Y_{t_i}^2 + \mathbb{1}_{\{\tau_j \in (t_i, t_{i+1}]\}} K(\mathbf{S}(\tau_j), \psi_j) \right) \right] \quad (6.20)$$

where $\Delta X_{t_i} := X_{t_{i+1}} - X_{t_i}$ represents the change in cash due to order fills during the interval $[t_i, t_{i+1}]$, and $\Delta(Y_{t_i} P_{t_i}^{(mid)})$ captures the marked-to-market change in inventory value. The indicator function $\mathbb{1}_{\{\tau_j \in (t_i, t_{i+1}]\}}$ equals one if the agent chose to intervene at some impulse time τ_j within the interval, with the instantaneous cost/profit $K(\mathbf{S}(\tau_j), \psi_j)$ defined in Section 3.2.

The reward signal at each timestep thus aggregates:

1. The instantaneous inventory penalty $-\eta \int_{t_i}^{t_{i+1}} Y_s^2 ds \approx -\eta Y_{t_i}^2 \Delta t$, which discourages the accumulation of directional positions and aligns with the risk-aversion objective discussed by Cartea et al. [2015].
2. The incremental profit/loss from inventory changes and order executions, reflecting both the bid-ask spread capture mechanism emphasized in Section 1 and the adverse selection risks inherent in providing liquidity.
3. The intervention cost $K(\mathbf{S}(\tau_j), \psi_j)$, which for limit and cancel orders equals zero, while for market orders equals the immediate liquidity consumption cost. This asymmetry naturally discourages excessive aggressive trading while permitting strategic use of market orders for inventory management.

6.5.4 Training Methodology

6.5.4.1 Proximal Policy Optimization

We implement the training procedure using Proximal Policy Optimization (PPO) Schulman et al. [2017], selected for its empirical robustness to noisy gradient estimates and its ability to handle the continuous-time approximation inherent in our episodic simulation framework. PPO optimizes policies through a clipped surrogate objective that constrains policy updates to remain within a trust region, thereby mitigating the variance and instability that plague vanilla policy gradient methods in high-dimensional action spaces.

The PPO objective for the decision network d_χ and action network u_ξ takes the form:

$$\mathcal{L}^{PPO}(\chi, \xi) = \mathbb{E}_{\tau \sim \pi_{\chi, \xi}} \left[\min \left(r_t(\chi, \xi) \hat{A}_t, \text{clip}(r_t(\chi, \xi), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (6.21)$$

where $r_t(\chi, \xi) := \frac{\pi_{\chi, \xi}(a_t | s_t)}{\pi_{\chi_{old}, \xi_{old}}(a_t | s_t)}$ denotes the probability ratio between the current and previous policies, $\hat{A}_t$ is an estimate of the advantage function, and ϵ (typically set to 0.2) controls the size of the trust region. The clipping operation ensures that the policy does not change too drastically in a single update, which is particularly important given the instability observed in Section 3.6 when training the DGM-based value and policy networks with opposing gradient updates.

6.5.4.2 Simulation Environment

The training environment is instantiated using the high-fidelity Hawkes-driven limit order book simulator developed in Jain et al. [2024b], which reproduces the stylized facts of LOB dynamics discussed in Section 3.1, including realistic spreads, long-memory in returns, and clustered arrival times. Each training episode spans a trading horizon of $T = 300$ seconds for computational feasibility of training over several episodes.

The agent's decision frequency is discretized at intervals of $\Delta\tau = 0.1$ seconds, at which points both the decision network d_χ evaluates whether to intervene and, conditional on intervention, the action network u_ξ selects the specific order type to execute. This discretization acknowledges the realistic constraint, emphasized in Section 1, that market makers cannot update strategies at every LOB event due to latency constraints and the discrete nature of order book updates. The chosen frequency represents a conservative estimate of achievable reaction times in modern electronic trading systems, falling well within the impulse control framework's applicability regime while remaining computationally tractable for training.

6.5.4.3 Action Space Restriction

To reduce the complexity of the learning problem and focus on the fundamental market making mechanisms, we restrict the action space $\mathcal{A}$ to a subset of the full event set $\mathcal{E}$ defined in Section 3.1:

$$\mathcal{A}_{restricted} := \{LO_T^{(a)}, LO_T^{(b)}, CO_T^{(a)}, CO_T^{(b)}\} \quad (6.22)$$

This restriction includes:

- $LO_T^{(\zeta)}$ for $\zeta \in \{a, b\}$: Placement of limit orders at the top of the book on either the ask or bid side.
- $CO_T^{(\zeta)}$ for $\zeta \in \{a, b\}$: Cancellation of the agent’s existing limit orders at the top of the book.

Market orders ($MO^{(\zeta)}$) are explicitly excluded from $\mathcal{A}_{restricted}$ at this stage to focus policy learning on queue management and symmetric liquidity provision rather than aggressive liquidity taking. This design choice contrasts with the behavior observed in Section 3.6, where the DGM-trained agent converged to a “pump and dump” strategy heavily reliant on market order submission.

The restriction to top-of-book orders reflects the empirical reality that, for large-tick assets, the vast majority of trading volume occurs at the best bid and offer. Orders placed deeper in the book face significantly lower fill probabilities while providing limited additional option value, particularly given the inventory constraints typical of high-frequency market makers.

6.5.4.4 Transaction Costs

To ensure that learned policies remain profitable under realistic trading conditions, we impose a transaction fee structure on all inventory liquidations. Specifically, when computing the reward, we apply a transaction cost of 1 basis point (0.01%) on the absolute value of terminal inventory:

$$\hat{r}(S_T) = r(S_T) - 0.0001 \cdot |Y_T| \cdot P_T^{(mid)} \quad (6.23)$$

This fee structure serves multiple purposes. First, it prevents the emergence of unrealistic strategies that rely on costless round-trip transactions to generate artificial profits, addressing a common criticism of simulation-based reinforcement learning in finance noted by Guéant [2016]. Second, it incentivizes the agent to minimize unnecessary inventory turnover, thereby encouraging stable quoting behavior rather than rapid order cycling. Third, it approximates the exchange fees and market impact costs that real market makers face, as discussed in the context of Designated Market Makers and proprietary trading firms in Section 1.

The magnitude of 1 basis point reflects a conservative estimate of combined exchange fees, clearing costs, and micro-scale market impact for a medium-liquidity equity. While actual fee structures vary significantly across venues and participant types—with some designated market makers receiving rebates for providing liquidity—

our choice ensures that any profitable strategy identified by the RL agent would remain viable under realistic cost assumptions.

6.5.5 Self-Imitation Learning

Standard policy gradient methods, including PPO, suffer from high variance and poor sample efficiency when applied to financial trading problems. This inefficiency stems from the sparsity of high-reward trajectories in the policy distribution, particularly during early training phases when the agent has not yet discovered profitable trading patterns. The problem is exacerbated in our setting by the complexity of the Hawkes-driven dynamics and the high dimensionality of the state space, which create a vast exploration space with sparse reward signals.

To address these challenges, we augment the PPO training procedure with self-imitation learning (SIL) following Oh et al. [2018]. The key insight underlying SIL is that the agent should explicitly imitate its own past trajectories that achieved returns exceeding its current value function estimate, thereby accelerating the propagation of successful behaviors throughout the policy.

6.5.5.1 Mechanism

The SIL augmentation modifies the PPO loss function by adding a cross-entropy term that encourages the current policy to replicate actions from a replay buffer of high-performing trajectories:

$$\mathcal{L}^{SIL}(\chi, \xi) = -\mathbb{E}_{\tau \sim \pi_{\chi, \xi}} \left[\mathbb{1}_{\{R_t > V_\theta(s_t)\}} \log \pi_{\chi, \xi}(a_t | s_t) \right] \quad (6.24)$$

where $\mathbb{1}_{\{R_t > V_\theta(s_t)\}}$ denotes the indicator function for the replay buffer containing past ‘good’ experiences, R_t is the empirical return-to-go from state s_t , and $V_\theta(s_t)$ is the current value function estimate. The indicator function $\mathbb{1}_{\{R_t > V_\theta(s_t)\}}$ ensures that only trajectories exceeding current expectations contribute to the imitation loss.

The combined training objective becomes:

$$\mathcal{L}^{total}(\chi, \xi, \theta) = \mathcal{L}^{PPO}(\chi, \xi) + \beta_{SIL} \mathcal{L}^{SIL}(\chi, \xi) - \beta_{entropy} \mathcal{H}(\pi_{\chi, \xi}) \quad (6.25)$$

where β_{SIL} controls the strength of the self-imitation signal, $\mathcal{H}(\pi_{\chi, \xi})$ is the policy entropy that encourages exploration, and $\beta_{entropy}$ balances exploration with exploitation. The hyperparameters $(\beta_{SIL}, \beta_{entropy})$ are tuned to ensure that self-imitation does not prematurely collapse the policy distribution before sufficient exploration has occurred.

6.5.5.2 Benefits in the Market Making Context

The integration of self-imitation learning provides several critical advantages in our impulse control setting:

1. **Avoidance of catastrophic forgetting:** By explicitly maintaining and imitating past successes, SIL ensures that profitable patterns, once discovered, remain accessible to the policy.
2. **Accelerated convergence:** The sparsity of highly profitable trajectories means that vanilla policy gradients provide weak learning signals, particularly during early training when the agent’s exploration is largely undirected. SIL effectively amplifies the learning signal from rare successful experiences by repeatedly reinforcing the actions that led to those outcomes, thereby accelerating the convergence to profitable strategies.
3. **Variance reduction:** By focusing policy updates on trajectories with high returns rather than the full distribution of explored behaviors, SIL reduces the variance of gradient estimates.

The combination of PPO’s trust-region optimization with SIL’s experience replay creates a training framework that is simultaneously stable, sample-efficient, and capable of discovering complex temporal strategies in high-dimensional state spaces—properties essential for learning optimal impulse control policies in the realistic, jump-driven market microstructure environment established in Section 3.1.

6.5.6 Results

With an initial cash balance of \$2000, inventory penalty $\eta = 10$, and trading horizon $T = 300$ seconds, the trained agent achieves an annualized Sharpe ratio of 31.54 within only 60 training episodes. Despite the high dimensionality of the state space and the non-locality of the jump-PDE, the RL formulation converges rapidly and consistently to a policy that resembles genuine market making: the agent symmetrically provides liquidity at the top of the book and dynamically balances inventory. To benchmark the RL approximation, we compare it against the Deep Galerkin Method (DGM) implementation of the QVI from Section 3. As discussed earlier, the DGM training logs (Figure 6.1) display oscillatory convergence, reflecting the opposing updates of the value and policy networks. In contrast to the pump and dump strategy learned by the DGM method, RL-based approximation avoids such degenerate equilibria:

- The PPO agent with self-imitation learning converges within ~ 60 episodes,



Figure 6.3: Policy Statistics for $\eta = 10$

while the DGM-based approach requires orders of magnitude more iterations without escaping local minima.

- The RL agent learns symmetric liquidity provision strategies rather than exploiting transient price impact.
- Sharpe ratios under the RL approach (31.54 annualized) are both significantly higher and far more stable than the noisy near-zero returns observed under the DGM-trained agent.

Overall, the comparison suggests that while the QVI formulation admits mathematically feasible but economically unrealistic solutions, the reinforcement learning approximation with an explicit inventory penalty guides the agent towards economically meaningful and stable market making strategies.

We did not apply this model to historical replay data, even though the absence of market impact and fees would likely result in higher Sharpe ratios, because our framework assumes a full information setting where all state variables, including intensities, are observable, which would require the development of a filtration technique for historical replay. Moreover, this was not the primary objective of this work, as we are aware of the overfitting biases associated with such approaches (as discussed in Chapter 1), and we believe that our ablation and sensitivity analyses, along with comparison to a probabilistic baseline, provide sufficient evidence for the robustness of our results.

6.5.7 Sensitivity and Ablation Study

We perform a hyperparameter sensitivity and state ablation study to understand some reasons behind the good performance of the RL agent. Indeed such results of a high sharpe strategy require rigorous assumption and realism checks to be useful to drive production insights.

Table 6.2: Sensitivity Analysis of Model Performance Across Parameters and Kernel Types

Parameter	Value	Kernel Type	OOS Sharpe
Standard		Exponential	31.54(*)
		Power-Law	28.81
Self-Imitation	Off On	Exponential	Pump & Dump *
Probabilistic Agent	Section 6.5.7.1	Exponential Power-Law	7.73 20.12
Inventory Penalty η	0.1	Exponential	Pump & Dump
	1.0		Pump & Dump
	10		*
	100		21.32
Transaction Costs	1bps	Exponential	*
	2bps		3.02
	4bps		-5.67
	8bps		-19.48
	1bps	Power-Law	28.81
	2bps		2.40
	4bps		-25.22
	8bps		-107.08

To this end, we systematically vary key model components and environment parameters to assess the robustness and interpretability of the learned policy. The sensitivity analysis focuses on hyperparameters influencing the temporal structure of the Hawkes process, the regularization of inventory risk, and the impact of transaction costs, under both Exponential and Power-Law kernel specifications. Complementarily, the ablation study evaluates the informational contribution of each state variable by selectively removing components of the agent’s observation space. Together, these experiments aim to identify which design elements—both architectural and environmental—are essential for stable learning, realistic execution behaviour, and sustained profitability out-of-sample.

6.5.7.1 Probabilistic Agent Baseline

The Probabilistic Agent extends a standard limit order book (LOB) trading agent by incorporating probabilistic reasoning about the timing and direction of the next market order (MO) event, as inferred from the estimated Hawkes intensities.

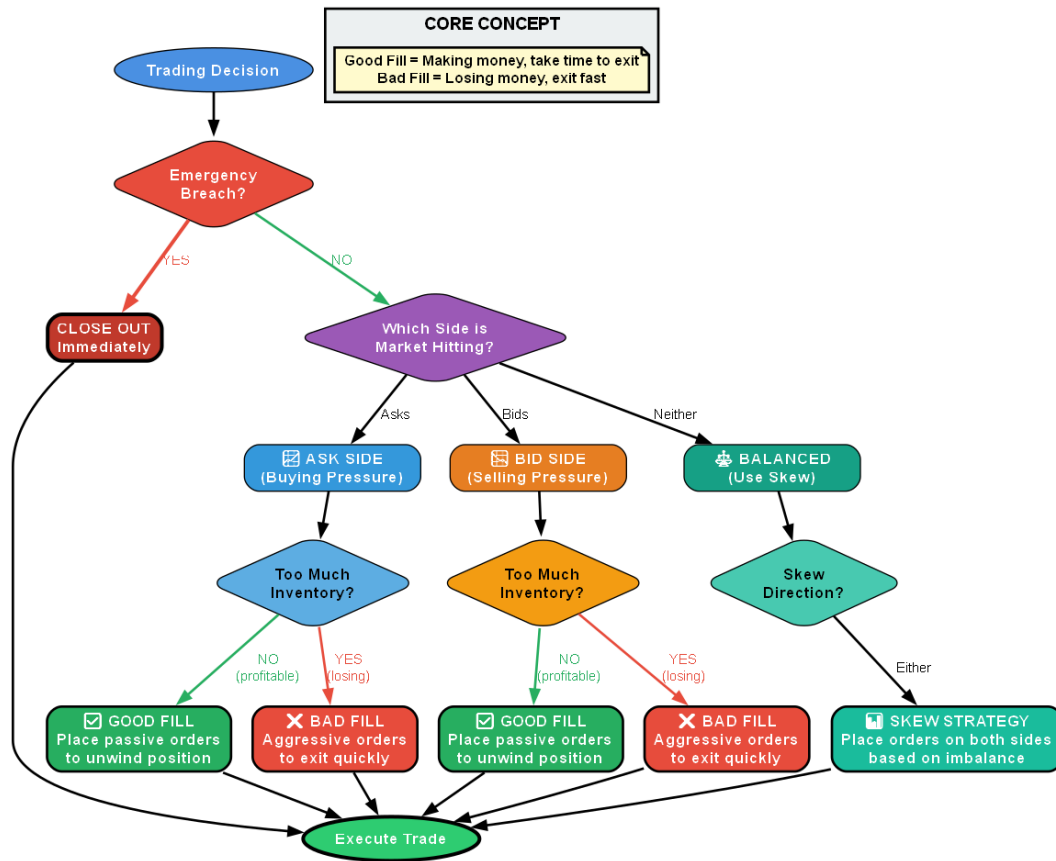


Figure 6.4: Baseline - Probabilistic Agent

At each decision time, the agent normalizes the Hawkes intensities to form a probability distribution over event types, thus estimating which side of the book is most likely to be hit next. The decision logic, as summarised in Fig. 6.4, then adjusts its quoting or market-taking behaviour accordingly. For instance, if the next most probable event is a market buy (MOBid), the agent anticipates upward price pressure and may reduce short inventory, place liquidity on the bid side, or cancel vulnerable asks. Conversely, if a market sell (MOAsk) is likely, the agent increases bid-side exposure or unwinds long positions.

Inventory management is incorporated through threshold-based controls on inventory imbalance. When the absolute inventory exceeds a fixed threshold, the agent enforces corrective market orders to re-centre its position. The action selection process also accounts for posted quotes' skew—the relative imbalance between bid

and ask quotes of the market maker—which influences whether the agent improves quotes within the spread or cancels existing orders at deeper levels.

6.5.7.2 Discussion

Table 6.2 indicate that the RL agent’s performance is broadly robust but exhibits sensitivity to kernel structure, transaction costs, and inventory penalization. The Exponential kernel consistently outperforms the Power-Law, suggesting that short-memory excitation provides a more accurate representation of transient order flow dependencies, while the Power-Law kernel offers marginally greater stability under partial state information. Stable convergence further requires the inclusion of Self-Imitation, whose removal induces pathological “Pump & Dump” dynamics. Comparing to the Probabilistic Agent, the RL agent consistently outperforms this baseline.

Moderate inventory penalization ($\eta = 10$) achieves the best balance between stability and responsiveness, whereas excessive penalization suppresses profitability. The strategy remains viable up to transaction costs of 2bps but collapses beyond that threshold, underscoring its dependence on low-friction environments.

The ablation study in Table 6.3 highlights the essential role of intensity and relative position features, whose exclusion results in unprofitable or unstable policies. Overall, the most effective configuration combines Exponential Hawkes dynamics, moderate regularization, and comprehensive state representations that capture both queue positioning and order flow intensity.

Table 6.3: State Ablation Study and Kernel Types

Parameter Removed from State	Kernel Type	OOS Sharpe
None (Standard)	Exponential	31.54
	Power-Law	28.81
History $\mathcal{H}_{t-\tau_H:t}$	Exponential	Pump & Dump
	Power-Law	30.85
Intensity $\lambda_t^{(\cdot)}$	Exponential	-20.23
	Power-Law	-51.21
Spread s_t	Exponential	Pump & Dump
	Power-Law	17.66
Relative Position $\frac{n_t^{(\zeta)}}{q_t^{(\zeta)}} \zeta \in \{a,b\}$	Exponential	Pump & Dump
	Power-Law	Pump & Dump (unprofitable)

6.6 Conclusion

This work formulates optimal market making in a high-fidelity limit order book environment as an impulse control problem, departing from traditional continuous-time Brownian frameworks to explicitly capture microstructural realities including queue dynamics, clustered arrivals, and endogenous price impact through a mutually-exciting Hawkes process. The resulting Hamilton-Jacobi-Bellman Quasi-Variational Inequality poses substantial computational challenges due to high dimensionality and non-local operators, rendering classical finite-difference methods impractical. We explored two solution approaches: a Deep Galerkin Method inspired by Sirignano and Spiliopoulos [2018] and a model-free reinforcement learning approximation based on Proximal Policy Optimization with self-imitation learning. While the DGM framework successfully converged in simplified settings—achieving positive Sharpe ratios of 4.54 in the Poisson case and 0.78 when only market orders exhibited Hawkes dynamics—it encountered fundamental limitations. In the market-orders-only Hawkes setting, the learned strategy converged to an economically unrealistic “pump and dump” behavior exploiting the dynamic arbitrage conditions identified by Alfonsi and Blanc [2015] under exponential kernels. Moreover, the DGM approach failed to converge entirely in the full 12-dimensional mutually-exciting Hawkes environment, highlighting the curse of dimensionality inherent to neural PDE solvers in non-local impulse control settings. In contrast, the reinforcement learning approximation, which decomposes the impulse control problem into timing and action subproblems via a two-network PPO architecture, demonstrated substantially superior performance. Within only 60 training episodes, the RL agent achieved an annualized Sharpe ratio of 31.54 while learning symmetric liquidity provision strategies consistent with genuine market making behavior. The integration of self-imitation learning proved critical in accelerating convergence and reducing variance by focusing policy updates on rare but profitable trajectories. The comparison between methods reveals a fundamental trade-off: while the HJB-QVI formulation admits mathematically feasible solutions, it lacks intrinsic mechanisms to exclude economically unrealistic equilibria such as manipulative strategies. The RL framework, augmented with explicit inventory penalties and transaction costs, naturally guides learning toward stable and interpretable policies. These findings underscore the promise of combining impulse control theory with modern deep reinforcement learning to address optimal execution problems in jump-driven microstructural markets, while also highlighting the need for specialized neural-PDE solvers capable of handling non-local operators and event-driven dynamics in high-dimensional settings. The sensitivity study shows

that the agent is robust to moderate inventory penalties but performance deteriorates sharply under high transaction costs or extreme penalization. The ablation study highlights that intensity and relative position features are critical for maintaining profitability and stability. Notably, the Exponential kernel often leads to profitable “Pump & Dump” patterns, whereas this is rarely observed under the Power-Law kernel, providing some evidence that the DGM’s convergence to such behaviour may be driven by the Exponential kernel specification. Finally, we use the Probabilistic Agent as an interpretable baseline to benchmark and contextualize the performance of the black-box RL strategy.

Chapter 7.

Conclusions & Future Work

7.1 Summary

This thesis set out to bridge the gap between high-frequency market microstructure modelling and reinforcement learning (RL) for optimal decision-making in electronic trading. By combining point process-based stochastic models of Limit Order Book (LOB) dynamics with modern RL techniques inspired by stochastic optimal control theory, the work contributes to both the theoretical understanding and practical modelling of how adaptive agents operate in highly stochastic, low signal-to-noise ratio (SNR) financial environments.

7.1.1 Summary of Contributions:

The first part of the thesis (Chapter 3) established a detailed empirical characterization of LOB microstructure across assets with varying relative tick sizes. By analysing tick-by-tick order book data from the LOBSTER dataset, we identified a set of robust stylized facts that systematically vary with the relative tick size ϵ , defined as the ratio between the tick size and the asset price. In particular, we showed that both the average bid–ask spread and the average absolute mid-price returns scale approximately as ϵ^{-1} across assets, revealing a clear power-law relationship between price granularity and short-term price fluctuations. Furthermore, we demonstrated that the depth at which liquidity is maximized in the LOB follows a power law in ϵ with an exponent close to $3/2$, indicating a non-trivial reorganization of liquidity as markets transition from large- to small-tick regimes. In addition to these scaling laws, we documented systematic changes in the shape of spread distributions, the degree of sparsity in the book, and the spatial distribution of volume away from the best quotes. These findings provide quantitative criteria for distinguishing between large-, medium-, and small-tick assets and form a unifying empirical framework for cross-asset microstructure analysis.

Building on these empirical insights, the second stage of the thesis (Chapter 4) focused on the construction of a realistic yet analytically tractable simulator of the limit order book based on a Compound Hawkes Process (CHP). Rather than prescribing an exogenous price process, prices in the CHP framework emerge endogenously

from modest structural assumptions on order flow. Several methodological advances were introduced to ensure mechanical consistency and empirical fidelity: the model enforces non-negative spreads and positive intensities by construction; order sizes are sampled from calibrated empirical distributions rather than assumed to be unit-sized; in-spread order arrivals are modelled as an explicit function of the prevailing spread, reflecting its empirically observed mean-reverting behaviour; and intraday seasonality is incorporated through time-of-day–dependent intensities. To facilitate calibration, existing non-parametric estimation techniques were extended to handle slowly decaying kernels and improve numerical stability. Extensive validation against empirical data and multiple baselines demonstrated that the CHP simulator reproduces a wide range of stylized facts, including clustered inter-arrival times, heavy-tailed volume distributions, realistic spread dynamics, and return distributions, thereby establishing it as a high-fidelity experimental environment for market microstructure research.

Extending this framework to small-tick assets (Chapter 5) posed additional challenges due to the increased dimensionality required to model multi-tick price moves, sparsity, and the characteristic humped shape of the LOB. To address this, we introduced the Meta-Queue Hawkes (MQH) model, which aggregates microstructural interactions across price levels into a lower-dimensional representation while preserving the essential excitation structure of the order flow. This approximation represents a significant methodological contribution, as it enables the calibration and simulation of realistic small-tick order books without resorting to prohibitively high-dimensional Hawkes models. Through systematic simulation studies, we identified the in-spread Hawkes parameters and sparsity mechanisms as the key drivers governing transitions between large- and small-tick regimes. Crucially, the MQH model preserves the empirical scaling of stylized fact metrics with relative tick size, allowing smooth interpolation between different market regimes and providing a unified modelling framework across assets.

The CHP and MQH simulators provide a realistic and versatile experimental environment for studying trading strategies and market dynamics. The simulator captures endogenous market impact through mutually exciting order flow, responds dynamically to the agent’s actions via state-dependent intensities, and naturally reproduces feedback effects such as spread widening, queue depletion, and liquidity regeneration. Because prices are not imposed but arise from order book interactions, the environment is inherently reactive: aggressive trading decisions directly influence future order arrivals, liquidity conditions, and price movements. At the same time, the framework remains sufficiently tractable to be calibrated across a wide range

of assets and market regimes, making it suitable not only for market making, but also for optimal execution, order placement, stress testing, and policy evaluation in high-frequency markets.

The final part of the thesis (Chapter 6) addressed the problem of optimal market making within these high-fidelity simulated environments. The market maker's decision process was formulated as an impulse control problem, explicitly capturing the discrete and event-driven nature of trading actions, including limit order placements, cancellations, and market orders. This formulation led to a Hamilton–Jacobi–Bellman Quasi-Variational Inequality (HJB–QVI) that provides a principled theoretical characterization of optimal behaviour, but also highlights the computational challenges associated with high dimensionality and non-local operators. Two numerical solution approaches were investigated. A mesh-free Deep Galerkin Method (DGM) solver was shown to converge in simplified settings, achieving positive risk-adjusted returns, but struggled in fully interacting Hawkes environments, sometimes converging to economically unrealistic equilibria. In contrast, a reinforcement learning approximation based on a two-network Proximal Policy Optimization architecture augmented with self-imitation learning demonstrated rapid and robust convergence. Within a limited number of training episodes, the RL agent learned symmetric liquidity provision strategies consistent with genuine market making behaviour, achieved substantially higher risk-adjusted performance, and maintained effective inventory control. Sensitivity and ablation studies further showed that access to intensity-based features and relative queue position information is critical for stability and profitability, and that the choice of excitation kernel plays an important role in suppressing manipulative dynamics.

Taken together, the contributions of this thesis form a coherent methodological pipeline for studying optimal participation in stochastic, event-driven financial markets. Starting from empirical measurement and stylized fact extraction, the work progresses through the construction of structurally grounded stochastic simulators and culminates in the learning of adaptive trading strategies within realistic microstructural environments. Beyond the specific applications considered, the results illustrate how point process models and impulse control–inspired reinforcement learning can be combined to produce interpretable, robust, and practically relevant decision-making frameworks for high-frequency trading.

7.1.2 Limitations and Critical Overview:

Nonetheless, several limitations merit discussion.

1. **Absence of exogenous shocks:** Despite its realism, presence of concave market

impact and liquidity's endogenous feedback effects, the simulator remains an abstraction of market dynamics and cannot fully replicate unobservable, exogenous market conditions such as news, volatility shocks, liquidity shocks, cross-asset effects, cross-venue interactions, or the impact of hidden liquidity and dark pools.

2. **Calibration Challenges:** The calibration of Hawkes process parameters—especially in high-dimensional settings—remains computationally intensive, and parameter instability across time or assets poses challenges for generalization.
3. **HJB-QVI Remained Unsolved in Closed Form:** Although the impulse control formulation yields a principled Hamilton-Jacobi-Bellman Quasi-Variational Inequality, the high dimensionality and non-local intervention terms render analytical solutions intractable. The numerical approaches explored therefore remain approximations rather than exact solutions.
4. **No Latency Assumption:** The market maker is assumed to act without latency, implicitly observing and reacting to the LOB state instantaneously. This abstraction neglects delays arising from order transmission, processing, and queue positioning, which can materially affect execution outcomes in real markets.
5. **Explainability Challenges of the Learned Policy:** While the reinforcement learning agent achieves strong empirical performance, the resulting policies are represented by neural networks, limiting interpretability. This makes it difficult to extract explicit economic or structural insights from the learned decision rules.
6. **Existence of Locally Optimal but Undesirable Policies:** The reinforcement learning optimization landscape admits locally optimal strategies that may exhibit aggressive or unstable behavior. Although profitable in simulation, such policies may be undesirable or prohibited under realistic risk management or regulatory constraints.
7. **Full Information Assumption in the RL Setting:** The learning framework assumes access to the full simulated LOB state, including latent variables driving the Hawkes dynamics. In practice, market participants operate under partial observability, which may limit the direct transferability of the learned policies.

Addressing these challenges would require further advances in efficient inference for point processes, improved domain adaptation techniques, and the integration of hybrid model-based/model-free RL paradigms.

In summary, this thesis contributes to the growing literature at the interface of stochastic microstructure modelling and reinforcement learning for financial decision making. By combining theoretical rigor with empirical validation and computational innovation, it takes a step toward closing the gap between realistic market simulation and adaptive control, thereby laying the groundwork for future developments in autonomous trading systems and quantitative financial research.

7.2 Future Work

Building on the contributions and limitations of this thesis, several promising directions emerge for future research. One natural extension involves developing a multi-asset LOB simulator to investigate cross-asset interactions, cross-impact effects, and portfolio-level optimal control strategies. Another avenue is to incorporate latent variable or state-space models capable of capturing unobserved drivers of order flow, thereby enhancing an agent's ability to reason about hidden liquidity and complex market dependencies.

The analysis in this thesis also highlights that several modelling assumptions are violated in empirical data, which in turn motivates a number of theoretical and methodological questions for future investigation. In particular, it remains an open problem to develop mathematically tractable frameworks that jointly capture the interaction between queue sizes, the relative width of the top and deeper regions of the LOB, and the resulting dynamics of order arrival rates. Closely related is the challenge of modelling order sizes in a manner that preserves the rich correlation structures observed empirically, both across price levels and through time. Finally, the calibration of such models in the presence of realistic levels of parameter uncertainty remains largely unexplored and represents an important direction for further research.

Building on these foundations, future work will aim to enhance the adaptability and accuracy of LOB models, with the ultimate goal of improving the robustness and practical applicability of algorithmic trading strategies in complex, event-driven markets.

From an algorithmic perspective, future work could explore the development of risk-sensitive and distributional reinforcement learning methods, as well as offline RL approaches leveraging large-scale historical data to improve robustness and generalization of trading agents. Furthermore, the convergence of generative modelling techniques, such as diffusion models, with stochastic control offers a

compelling pathway for constructing adaptive, self-consistent market simulators that co-evolve with the agents operating within them.

While this thesis demonstrates the effectiveness of deep reinforcement learning for high-dimensional Hawkes-driven market making, several avenues remain to advance both theoretical understanding and computational efficiency.

From a methodological standpoint, alternative analytical and numerical approaches could further improve solution quality and scalability. For instance, stochastic maximum principle (SMP) methods Chahim et al. [2012] may provide semi-analytical characterizations of optimal impulse controls in jump-driven LOBs. Forward-backward stochastic differential equation (FBSDE) techniques could decouple forward state dynamics from backward value propagation, enabling scalable approximations of the HJB-QVI Hu and Lariere [2023], Carmona [2016], Anderson et al. [2024]. Similarly, stochastic partial differential equation (SPDE) solvers Lord et al. [2014] could generalize neural-PDE approaches to fully accommodate the non-local operators arising from mutually-exciting Hawkes dynamics. Delay differential equation frameworks Balachandran et al. [2009], Cartea and Sánchez-Betancourt [2023] offer a tractable means of capturing intrinsic lagged dependencies in the LOB and order flow, while emerging methods in non-Markovian control Posamaï and Tangpi [2024] and Volterra control Agram and Øksendal [2015] provide promising directions to address long-memory effects induced by power-law kernels, enabling the design of computationally tractable control strategies in non-Markovian environments.

Appendix A.

Comparative Study of LOB Modelling Techniques

The following table compares several works on LOB Modelling from Chapter 2.1

Glossary:

1. PDF : Probability Distribution Function
2. Exp(.) : Exponential Distribution
3. DL : Deep Learning
4. ABM : Agent Based Model
5. MI : Market Impact
6. SPDE: Stochastic Partial Differential Equations
7. ACF : Auto-Correlation Function

Model	Category	Stylized Facts Tested	Quality of Fit Comments
Bouchaud et al. [2002]	Poisson	Order flow statistics and average order book shape	Qualitative tests performed
Farmer et al. [2005] (tests Smith et al. [2003])	Zero-intelligence	Average Spread, Price Diffusion rate	All stylized facts compared between empirical and predicted.
Luckock [2003]	Poisson	PDFs of trade and best ask/bid prices; Density of unexecuted orders w.r.t. price	Qualitative tests performed
Cont et al. [2010]	Poisson	Arrival rates w.r.t. distance from opposite quotes; Average LOB Shape; Probability of price increase w.r.t. Queue Size; Probability of execution (one side and both sides) before mid-price movement	Tests done to compare empirical simulated results for probability of price increase w.r.t Queue Size and across various levels. Comparisons are made with the empirical data as well as theoretical results by the Laplace transform method.
Abergel and Jedidi [2011]	Poisson	Average Depth Profile, PDF of Spread-in-ticks; ACF of price increments, Price Paths, Signature Plots, Average Depth, Spread, long term volatility	Qualitative Tests Performed on all stylized facts; Q-Q tests are performed on mid-price increments.
Cont and de Larrard [2013]	Poisson	Joint distribution of best bid and ask size, Probability of price increase conditional on current bid/ask sizes	Diffusive coefficient of Price from simulation and from empirical data is compared
Huang et al. [2015]	Poisson	Order intensities distribution by queue size; PDF of queue size for 3 levels of bid/ask; Joint distribution of first two levels queue sizes; Joint distribution of best bid and ask queue sizes	Qualitative tests performed
Lu and Abergel [2018b]	Poisson	Order intensities and Order size distribution by queue size; Conditional distributions and other statistics of various types of orders and their arrival times; PDF of best bid/ask sizes	Qualitative tests performed using Monte Carlo simulations
Toke [2010]	Hawkes	PDF of spread, inter-arrival times, variance of mid-price	Qualitative tests performed
Zheng et al. [2014]	Hawkes	Signature Plots (Bid1, Ask1, Mid)	Qualitative tests performed
Da Fonseca and Zaatour [2014]	Hawkes	PDF and histograms of inter-arrival time; Autocorrelation function of number of trades in a time window; Signature plots	Q-Q plot comparison made of inter-arrival time to the exponential distribution. Qualitative tests upon the stylized facts is performed. Comparisons of the fitted parameters with an MLE baseline is done along with testing of robustness of the params by investigating the standard deviations.
Bacry et al. [2016]	Hawkes	Event cross and self-excitation versus time	N/A
Rambaldi et al. [2017]	Hawkes	PDF of inter-arrival times, order volumes	They show a number of plots and calibration results for their estimated Hawkes Process
Lu and Abergel [2018a]	Hawkes	Conditional probabilities of events; Signature Plots (mid-price); PDF of inter-event times	Residuals' Q-Q Plot tested to follow Exp(1) distribution; Qualitative tests performed
Mounjid et al. [2019]	Hawkes	No Tests Performed	N/A
Wu et al. [2019]	Hawkes	PDF of queue size	Q-Q plot of inter-arrival times are compared with empirical data as well as a Queue-reactive Poisson model baseline. Calibrated results are showcased for the model. Qualitative tests performed over stylized facts.
Kumar [2021]	Hawkes; DL	PDF and Auto-correlation function of returns	Qualitative tests performed
Kirchner and Vetter [2022]	Hawkes	Average order intensity by time of day, unconditional transition probabilities between all orders, Market Orders against Imbalance	Properties of calibrated excitation kernels is plotted against time of day
Morariu-Patrichi and Pakkanen [2022]	Hawkes	State transition probabilities	Residuals' Q-Q Plot tested to follow Exp(1) distribution. Comparison made between generic Hawkes and state-dependent Hawkes models.
Lee and Seo [2022]	Hawkes	Bid/Ask Price plots	Stationarity of estimated parameters is checked, comparisons between normal Hawkes and spread conditioned Hawkes (proposed model) is also done. Tests are also performed in further model selection between the 5 proposed models. Residuals' Q-Q Plot tested to follow Exp(1) distribution.
Nyström and Zhang [2022]	Hawkes	Price change in ticks frequency; Price Paths; PDFs of number of jumps	Price Paths are compared between exponential and power law kernels. Statistical significance tests as well Qualitative tests on estimated parameters are also done. Further the computational time for both types of Kernels' fitting process is reported. Distributions of number of jumps are compared qualitatively between empirical data and simulated paths.

Model	Category	Stylized Facts Tested	Quality of Fit Comments
Byrd et al. [2019]	ABM	Price Paths; Trade Paths;	Price Paths are compared qualitatively. Errors and accuracy Out of Sample are also reported.
Belcak et al. [2020]	ABM	N/A	Performance comparisons are made with the ABIDES Byrd et al. [2019] platform
Shi and Cartlidge [2023]	ABM; Hawkes; DL	Hurst exponents for absolute returns; Autocorrelation functions; Order flow imbalance impact; Price Impact function; Spread w.r.t. time	A sensitivity analysis of the parameters with respect to the various stylized facts mentioned is performed to check the robustness of the parameters.
Li et al. [2020]	DL	PDFs of mid-price;	Quantitative tests performed like KS test, Jarque-Bera test, Student t-test, t-statistic and p-values calculated for volatility measures (realized volatility, realized volatility per trade and intraday volatility)
Shi and Cartlidge [2021], Shi et al. [2021]	DL	N/A	Several baseline models like Ridge Regression, SVR, Random Forests, 1-layer feedforward Neural Networks etc. are used to test the performance of the model against other Machine Learning techniques. Further tests are conducted on the effectiveness of using an ODE-RNN by comparing the performance against LSTMs and GRUs. An ablation study is performed as well.
Lim and Gorse [2021]	DL	PDFs of mid-price returns	Quantitative tests performed like KS test, Jarque-Bera test, Student t-test, t-statistic and p-values calculated for volatility measures (realized volatility, realized volatility per trade and intraday volatility)
Coletta et al. [2022]	DL	PDF of log returns. PDFs of order type, time to fill, top of the book volumes, spread by time of day. Price Paths, Autocorrelation of returns and ACF of square returns.	The simulation's unconditional and conditional PDFs of the stylized facts is compared with empirical PDFs and two baseline models.
Prenzel et al. [2022]	DL	PDFs (unconditional, conditioned by time of day, conditioned by market volatility) of order intensities by each order type, Price Paths	Qualitative tests performed for all stylized facts between real and simulated data.
Cont et al. [2023b]	DL	PDF of Queue Size at best Ask/Bid; PDF of 3 levels of bid/ask queues; Average LOB Shape; Correlation of 3 levels of bid/ask queues; Price Paths; Price change probabilities	Qualitative tests performed to test real vs simulated data for all stylized facts. Comparisons are also made between the GAN model the authors propose, a Poisson Process model and a Hawkes Process model.
Nagy et al. [2023a]	DL	PDFs of returns, order type, arrival times	Perplexity scores are used to test the LLM. The simulation's unconditional and conditional PDFs of the stylized facts is compared with empirical PDFs.
Cont and de Larrard [2011]	SPDE	Q-Q plot of inter-arrival times compared with exponential distribution, number of shares per event (to showcase clustering), Spread's (1 tick vs $\frac{1}{2}$ tick) lifetimes, Joint PDF of bid and ask volumes, ACF of absolute order sizes, inverse of inter-arrival times, PDF of inter arrival times.	Stylized facts are used to create priors on the model
Korolev et al. [2015]	SPDE	PDF of order arrival rate; ratio of intensities at bid/ask	Qualitative tests performed
Chávez-Casillas and Figueroa-López [2017]	SPDE	Aymptotics of mid-price process; PDF of price; PDF of time for price change; Probability of price change conditional on current state	N/A
Gao and Deng [2018]	SPDE	Average shape of the order book,	Shapes of order book at various timestamps are compared qualitatively.
Rojas et al. [2020]	SPDE	Joint distribution of best bid and ask size; PDF of spread; PDF of lifetime of spread ;Price Paths; Autocorrelation Function	Price Paths are compared qualitatively.
Hambly et al. [2020]	SPDE	Price Paths; Average LOB Shape	Qualitative tests performed and volatility between empirical and simulated results is compared. Further this volatility is decomposed into two sources: exogenous movements and local imbalance.
Cont and Muller [2021]	SPDE	Average LOB Shape	Intraday Price Volatility are compared with empirical observations qualitatively

Table A.1: Comparative Study

Appendix B.

Compound Hawkes Process Calibration:

B.1 Previous Order Sizes do not impact Future Order Arrival Rates

We calculate the next order intensity in our dataset for a past order size by counting the number of future events in a window of $t = 0.01\text{sec}$. The joint scatter plot for Market Orders at Bid is shown in Figure B.1. Qualitatively the distribution looks to be quite uniform. We make use of the Hoeffding Independence Test to calculate the distance between the observed joint distribution of these two random variables and the distribution if they were independent. The test statistic is observed to be 0.00068 which is sufficiently low for us to conclude, albeit with weak evidence, that the two variables are independent. We observe similar scatter plots and Hoeffding statistics for all other events.



Figure B.1: Previous Order Size vs Future Arrival Rates

B.2 In-Spread order intensity depends on current spread

We plot the order intensity against the current spread-in-ticks in empirical data (3 months data) in Figure B.2. Here we show a scatter plot since the intensities are approximated by the number of in-spread orders in the next 0.01 seconds and therefore are random. We show the distribution of data points in the translucent red bars. We fit a linear regressor on log-log transformation of these data points (excluding 0 and 1 spread) and find the best exponent to be 0.7479. The red line in the plot shows the fitted line. We observe an R^2 of 0.85 for this regression.

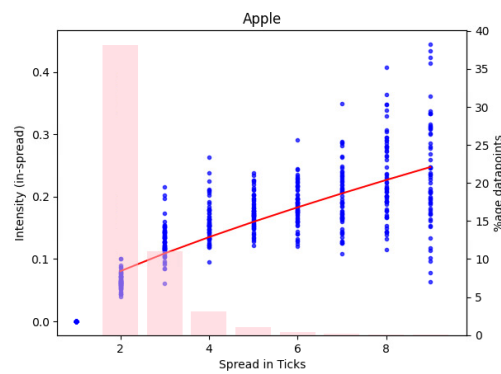


Figure B.2: In-Spread Arrival Rates vs Current Spread

Appendix C.

Stylized Facts additional illustrations

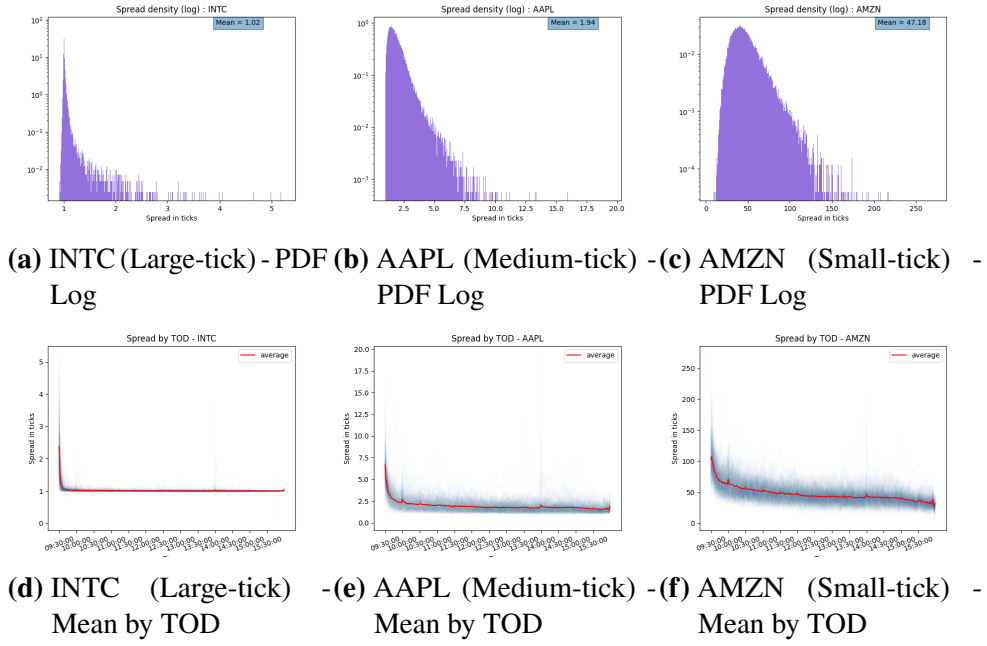


Figure C.1: Spread log empirical probability distribution function (PDF, figures on the first column) and average spread as a function of the time of the day (TOD, figures on the second column) for 3 different assets (resp. large, medium and small-tick) Bid-Ask Spread

In this appendix we include some additional illustrations of the different stylized facts discussed in the paper

C.1 Additional illustrations on the spread

In Fig. C.1, we display statistics on the spread $s(t)$ of three assets INTC, AAPL, and AMZN, resp. a large, medium and small-tick asset. Each row in the figure corresponds to one of the three assets. In each row, the left figure displays the empirical distribution of the spread $s(t)$ whereas the right figure displays the average spread $\bar{s}$ as a function of the time of the day as defined by (3.2).

Unsurprisingly, we clearly see that spread is higher at open time and for large-tick assets, except for open time, the spread is basically equal to its constrained minimum

(1 tick). As far as the empirical distribution is concerned, the shape appears to be very different for small, medium and large-tick assets. In order to measure the difference in shapes we make use of the index of dispersion (i.e., variance divided by mean) in Section 3.3.

C.2 Additional illustrations on the mid-price change

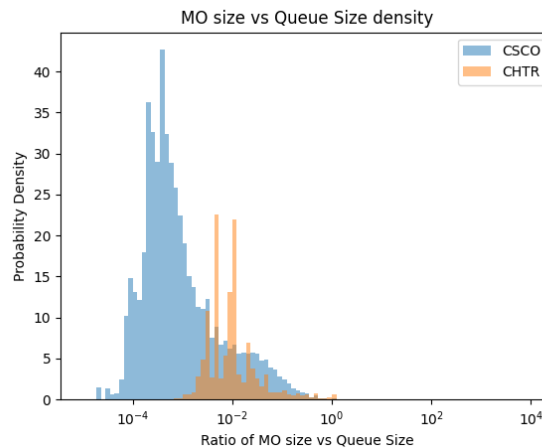


Figure C.2: Market Order Size vs Instantaneous Volume at Best: Probability density of the ratio of the average market order size with the average corresponding best limit for different tick-size assets. This ratio is clear higher for small-tick assets, implying thin quoting at the top of the LOB leading to multiple tick level mid-price moves.

In Section 3.4 we saw that the average mid-price change induced by a market order is much larger for small-tick assets than for large-tick assets. An indirect way of illustrating one of the probable causes of the same stylized fact consists of studying the ratio of the average market order size with the average corresponding best limit for different tick-size assets. This quantity clearly drives the queue-depletions which lead to the so-called trade-induced mid-price changes. Fig. C.2 shows the distribution of this ratio for two assets (one small-tick : CHTR and one large-tick : CSCO). It clearly displays that for small small-tick assets market orders deplete a larger portion of the top of the LOB quoted volume than for large-tick assets. This implies that the quoted volume at the top for small-tick assets is relatively thinner than that of large-tick names. Large-tick assets typically have several queued up orders at the best, as we will illustrate in Section 3.5, which leads to lower depletions of the top level of the LOB by incoming market orders.

C.3 Additional illustrations on the shape of the LOB

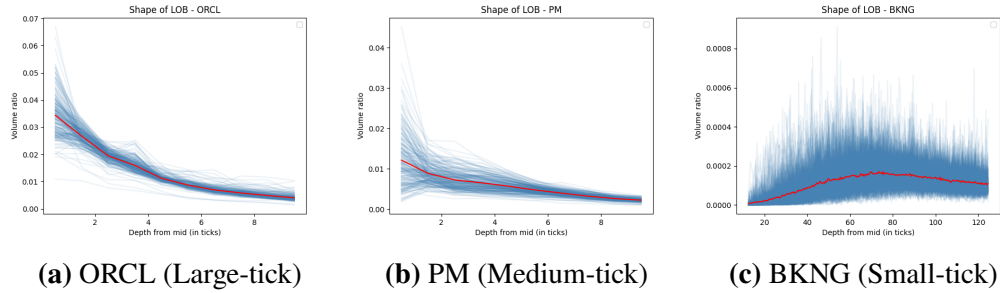


Figure C.3: Per-asset Shape of the LOB: light blue lines are per day shapes and red is the average shape across one calendar year. As we can clearly see in this set of figures, the shape of the LOB both per-day and over the year are quite similar leading to the notion of stationarity of the shape across calendar days. Secondly, we note that the trend we observed in Section 3.5 of the shape of the LOB getting flatter as we decrease the relative tick-size can be again seen here.

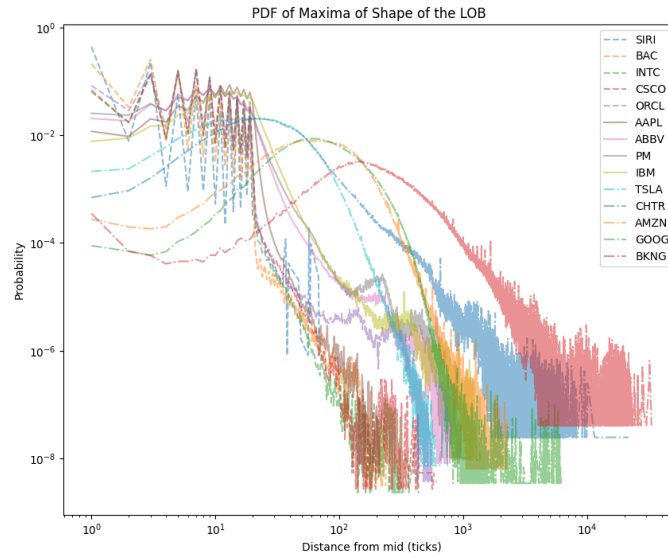


Figure C.4: Distribution of Shape Maxima (log-log): this figure shows the empirical density of liquidity maxima, with greater variance as relative tick-size decreases. Despite frequent maxima near the mid-price in small-tick assets, Fig. C.1 shows a low probability of a one-tick wide spread.

In Fig. C.3, we show the per-day and per-year average shape of three assets : ORCL, PM and BKNG resp. examples of large-, medium- and small-tick assets. First observation is that the yearly aggregated shape of the LOB seems very close to

the per-day shape and therefore we can assume that the average shape of the LOB as shown in Section 3.5, is stationary. Secondly, the shape of the LOB decays from its maximum at half a tick from the mid-price for large and medium-tick assets, as illustrated in Figs C.3a and C.3b. Notably, the shape profile decays more rapidly for large-tick assets compared to medium-tick ones. On the other hand, for small-tick assets such as the one shown in Fig. C.3c, the liquidity distribution is wider, with the maximum liquidity lying several ticks below the mid-price.

The empirical density of the maxima of the liquidity as a function of distance from the mid-price, shown in Fig. C.4, indicates that the variance in the location of maximum liquidity increases as the relative tick-size decreases. The variability in the location of this maximum provides insights into how deep the liquidity is distributed in the LOB. Even in small-tick assets, there are significant instances where the maximum liquidity lies just half a tick from the mid-price. However, as depicted in Fig. C.1, the time-weighted likelihood of a one-tick wide spread is quite low.

C.4 Additional illustrations on the sparsity of the LOB

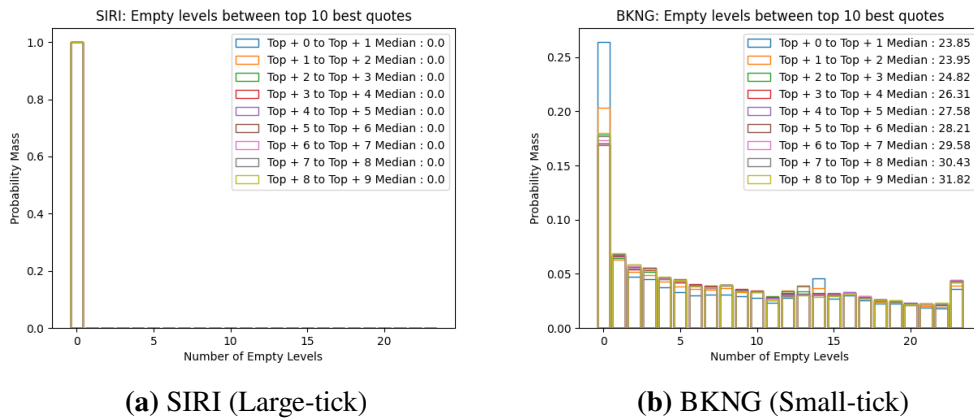


Figure C.5: Sparsity of the LOB of a given asset : time-weighted distribution of the values $n_e(i, t)$ for the top 10 quotes ($i \in [0, 10[[]]$) for (a) a large-tick asset (SIRC) and (b) a small-tick asset (BKNG). We see the distributions hardly depend on the level of the quotes and that the smaller the relative tick-size, the sparser the LOB : the distributions are somewhat flat for small-tick assets and very concentrated around 0 for large-tick assets.

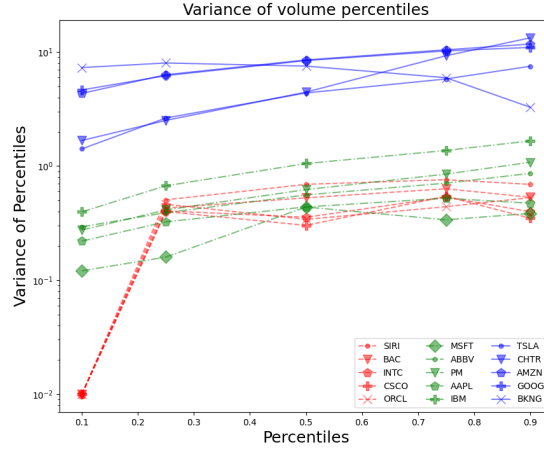


Figure C.6: Variance of shape of 4 LOB quantiles (10%, 25%, 75%, and 90%) (log scale) for the 15 assets. The smaller the tick, the higher the variance, the sparser the LOB.

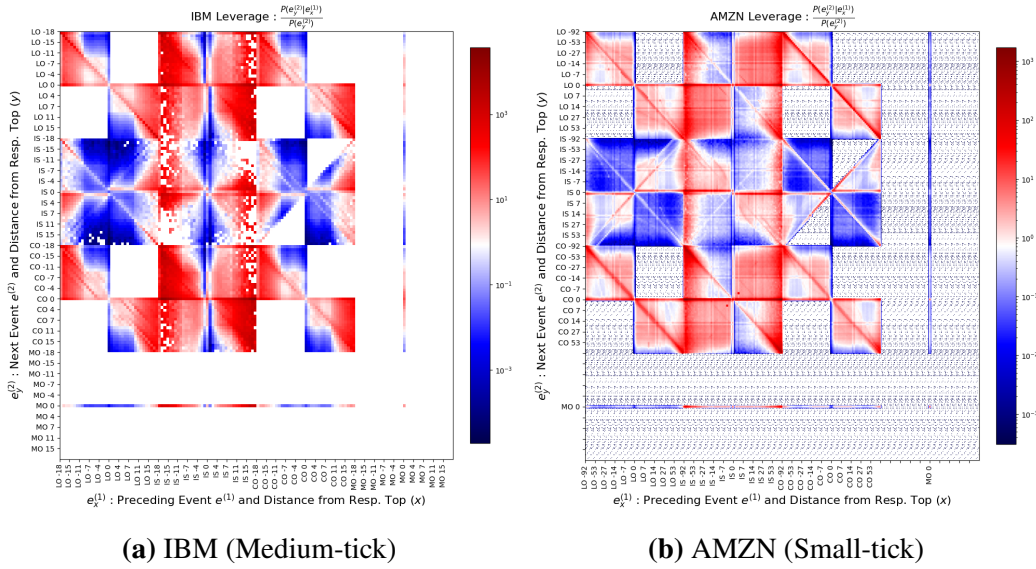


Figure C.7: Leverage (from top): Darker the hue, more the leverage. In continuation with Fig. 3.9, we compare the leverage plots of a medium-tick asset, IBM, to a small-tick asset, AMZN. Note the remarkable similarity in the structure of these two plots as well as its similarity to Fig. 3.9.

In this Section we show that the sparsity pattern that was illustrated in Fig. 3.7 studying $n_e(1, t)$ (i.e., the number of empty queues between the best quote and the next non empty quote) is confirmed for deeper levels in the LOB.

For doing so, we compute $n_e(i, t_n)$ for the top ten quotes ($i \in [1, 10[)$) at each time t_n this quantity is changing. Fig. C.5 shows the empirical distributions of this quantity for each level i for the SIRI stock (large-tick) and for the BKNG stock

(small-tick asset). We see that for a given asset the distribution of empty levels is similar across all depths of the LOB, moreover, we confirm that the smaller the relative tick-size, the sparser the LOB : the distributions are somewhat flat for small-tick assets and very concentrated around 1 for large-tick assets.

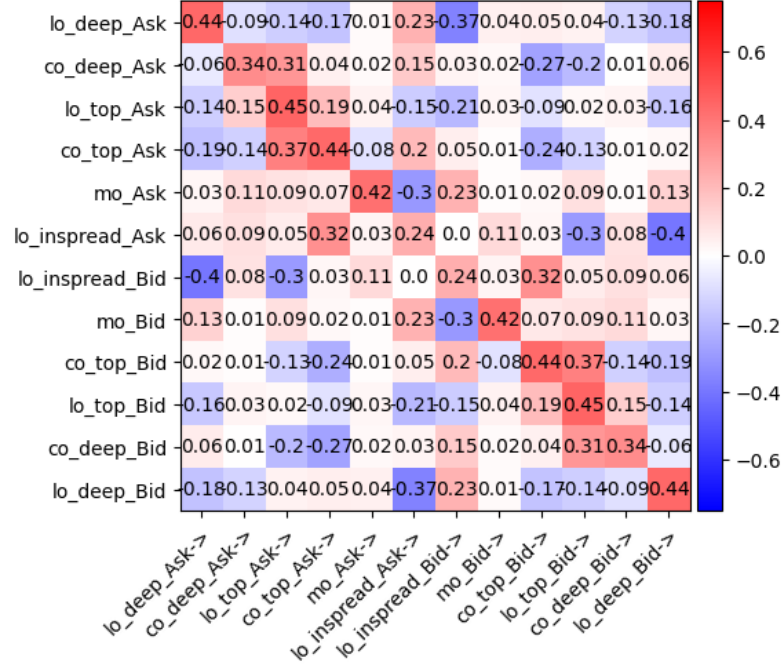


Figure C.8: Norms of Kernel: AMZN.OQ

A sparse LOB induces a lot of variability in its shape. Thus another natural proxy for the sparsity of the LOB is to measure this variability. We choose to empirically estimate the variance of four quantiles (10%, 25%, 75%, and 90%) of the shape of the LOB. Fig. C.6 shows clearly that the larger the relative tick-size the smaller the variances.

C.5 Additional illustrations on the endogeneity of the LOB

In Fig. C.7, we continue the depiction from Section 3.7 of uniformity of leverage across all assets albeit with different scaling. We can see here the observation that medium-tick $LO \rightarrow CO$ leverage plot has both diagonal and off-diagonal elements. This implies that the transition of the structure of $LO \rightarrow CO$ from large-tick to medium- and thereafter small-tick is smooth.

Event	$\mu^{(\cdot)} (s^{-1})$	Event	$\mu^{(\cdot)} (s^{-1})$	Event	$\mu^{(\cdot)} (s^{-1})$
LO_{Ask_D}	0.86	$LO_{Ask_{IS}}$	0.47	CO_{Bid_T}	0.48
CO_{Ask_D}	0.32	$LO_{Bid_{IS}}$	0.47	LO_{Bid_T}	0.33
LO_{Ask_T}	0.33	MO_{Bid}	0.02	CO_{Bid_D}	0.32
CO_{Ask_T}	0.48	MO_{Ask}	0.02	LO_{Bid_D}	0.86

Table C.1: Exogenous Intensity

C.6 Simulation Study Parameters (continued):

In Table C.1, we provide the exogenous intensity part of the Hawkes Process parameters ($\mu^{(\cdot)}$) and in Fig. C.8, we depict the Kernel Norm matrix according to standard research practice (Bacry et al. [2016], Jain et al. [2024b]). Since the number of parameters is quite large (3×72 for the 72 power law kernels), we do not report them in detail.

Appendix D.

Assumptions and Their Empirical Evidence or Counter-Evidence:

In Section 5.3 we underlined the universality of the model by illustrating, with numerical experiments, the fact that the model we propose is general enough so that controlling three parameters α, β and $\hat{\eta}_{(\cdot)}$ enables us to simulate an LOB with Large or small-tick characteristics. However we made a number of simplifying assumptions in the model construction to foster a straightforward analysis, an easier calibration, and an understandable simulation study. In the following we check if some of these assumptions are realistic as respect to empirical observations.

Assumption D.1. *Hawkes Parameters ($\alpha, \beta, \mu^{(\cdot)}, \phi^{(\cdot)}(t)$) and the LOB variables (i.e. LOB state variables ($m_{(\cdot)}^{(\cdot)}(t), Q_{(\cdot)}^{(\cdot)}(t)$), order sizes ($\kappa_{(\cdot)}$), and order arrival distances from top ($\eta_{(\cdot)}$) are independent of each other.*

Queue reactiveness in the LOB, i.e. the notion that event intensities depend directly on the current sizes of the queues in the LOB ($Q_{(\cdot)}^{(\cdot)}(t)$), has been relatively well established in literature (Morariu-Patrichi and Pakkanen [2022], Wu et al. [2019], Huang et al. [2015]). It is evident from these papers that the intensities of LOB events depend on the current queue sizes of the LOB. However, it is unclear whether the number of past events alone (i.e., without considering individual quantities) is a reasonable proxy for the cause of this dependence. We would like to note that the Hawkes Process methodology implicitly makes the intensity depend on the count of past events.

We perform tests on this assumption in Fig. D.1 where we check the dependence of the LOB state variables to the events' likelihood by plotting the ratio of the joint probability of an event happening with the LOB state variable v i.e. $P(e, v)$ with the product of the individual unconditional probabilities i.e. $P(e)P(v)$, against various categories of the width variable. This ratio informs us whether the two random variables are independent or not. While this test is not exhaustive or strict, it does inform us of the interdependence of two random variables in a simplified manner. If the ratio is near one, we have some albeit insufficient evidence to consider the two

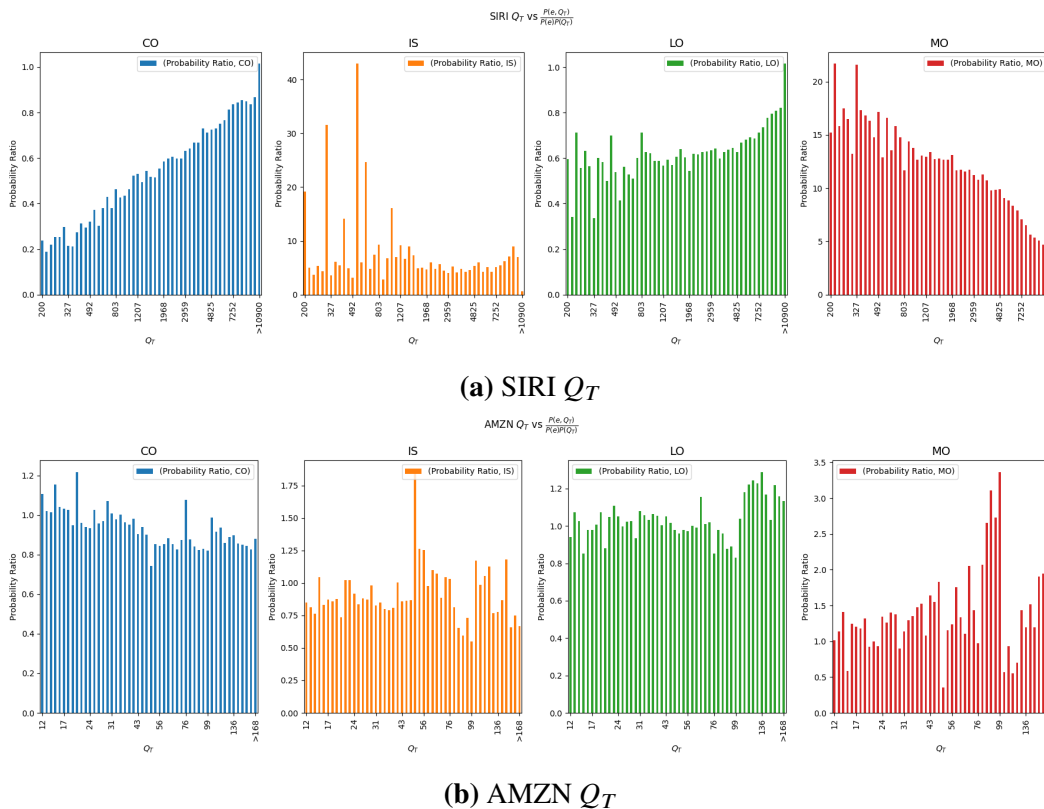


Figure D.1: Testing Assumption 1: Queue Reactiveness: This independence test shows that for large-tick assets like SIRI, the intensity of events like COs, LOs and MOs, and the queue sizes at the top of the LOB are not independent variables. Unsurprisingly, the same is not true for the deeper volumes in a large-tick asset. For a small-tick asset like AMZN, the picture becomes less clear for Q_T however for MOs, one can observe a clear dependence among the variables.

variables independent. However if the ratio is significantly different from one, we can conclude that the two variables are not independent. From this figure, we see that the queue reactivity is very evident for top of the LOB queue size in large-tick assets however it is very less significant in the small-tick situation.

Further we show the same analysis for Q_D instead of Q_T in the Fig. D.3. Unsurprisingly, large-tick assets dynamics seem to be unreactive to the deeper quoted volumes but there is a definite structure, particularly in MOs, for small-tick assets. Additionally, we see that events of the type IS, CO and MO have a definitive trend with increasing m_T (Fig. D.4a). There is a U-shape trend observed in IS and MO order types with increasing m_D (see Fig. D.4b). Finally we observe a strong dependence of MO's probability with increasing η_{IS} (Fig. D.4c). However we must note that the trend becomes significant only in the right tail of the quantities which are low probability states. We do not see any clear trend when we do the same analysis for

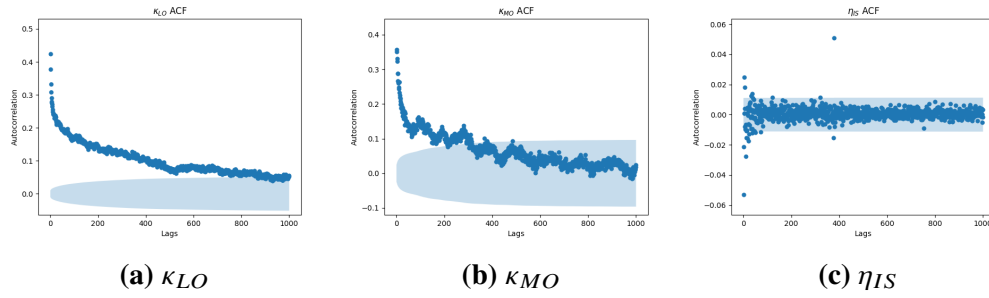


Figure D.2: CHTR: Testing Assumption 2 (i), (ii) and (iii) - Clearly the ACF of order sizes ($\kappa_{(.)}$) is non negligible for 10s of lags which means that the order sizes are probably not i.i.d. However we do see the ACF of price improvements by IS orders (η_{IS}) quickly dying down to 0, giving some support to the assumption 3.

κ_{LO}, κ_{MO} (not illustrated). We observe similar characteristics in medium-tick assets and all other small-tick assets as well.

Therefore we conclude that though for LOs and COs, we can assume the LOB state variables except Q_T to be independent of their intensity functions, there exists a clear dependence of intensities of in-spread orders on the current m_T, m_D . MOs seem to also be very dependent on the state variables. Thus we identify Assumption 1 to be too strong for empirical data and reserve the direction of relaxing this assumption for future work.

Assumption D.2. (i) Order sizes, (ii) Order arrival distances and (iii) Unseen Queue sizes on queue depletion, are i.i.d. random variables.

To test the i.i.d. assumption, we measure the autocorrelations at various lags of the timeseries of the quantity. It is clear from Fig. D.2 that Assumption 2 (i) is clearly violated by the empirical data. Assumption 2 (ii), however, seems to have some support from the data since the autocorrelations are absent. With regards to Assumption 2 (iii), we note that this is a modelling choice.

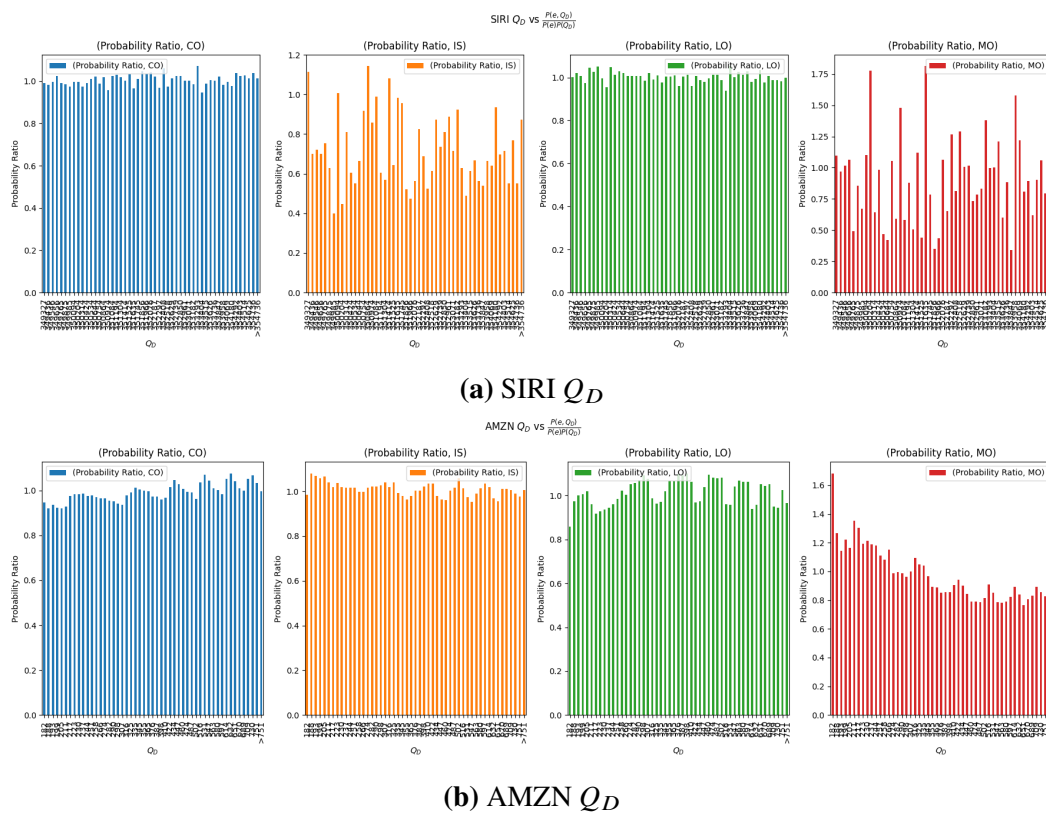


Figure D.3: Testing Assumption 1: Queue Reactiveness: For Q_D too we note that the MOs intensity is quite clearly not independent in the case of small-tick assets.



Figure D.4: AMZN: Testing Assumption 1: We do the same analysis for a small-tick asset AMZN to check the dependence between the state variables and the intensity of various events. It is evident from figures (a) and (b), that all the order intensities are related to the current width of the top queue ($m_T(t)$) as well as the deeper queue ($m_D(t)$). However for the dynamic sparsity variable, we note that there seems to be a lot of evidence (except for when η_{IS} is very large which is a low probability event) for independence between the intensity and the previous η_{IS} which is the most recent price improvement (in ticks) made by an in-spread LO.

Appendix E.

Dynamics of State Variables:

$$\text{Policy} \equiv u(t) := \{(\tau_i, \psi_i)\}_{i=1, \dots, N} \text{ where } \tau_N < t \quad (\text{E.1})$$

$$\text{State} \equiv \mathbf{S}_t := \{X_t, Y_t, p_t^{(\zeta)}, q_t^{(\zeta)}, q_t^{(\zeta, D)}, n_t^{(\zeta)}, P_t^{(mid)}\}_{\zeta \in \{a, b\}} \quad (\text{E.2})$$

$$\text{Cash} \equiv X_t \text{ s.t. } dX_t = \sum_{\zeta} -z \mathbb{1}_{(\zeta)}(t) (P_t^{(mid)} + z p_t^{(\zeta)}) dN_t^{(MO_{\zeta})} \quad (\text{E.3})$$

$$\text{Inventory} \equiv Y_t \text{ s.t. } dY_t = \sum_{\zeta} -z \mathbb{1}_{(\zeta)}(t) dN_t^{(MO_{\zeta})} \quad (\text{E.4})$$

$$\begin{aligned} \text{Best price at } \zeta \equiv p_t^{(\zeta)} \text{ s.t. } dp_t^{(\zeta)} = & z(-dN_t^{IS_{\zeta}} \\ & + \mathbb{1}(q_t^{(\zeta)} = 1)(dN_t^{(MO_{\zeta})} + dN_t^{(CO_{\zeta})})) \end{aligned} \quad (\text{E.5})$$

$$\begin{aligned} \text{Best quote at } \zeta \equiv q_t^{(\zeta)} \text{ s.t. } dq_t^{(\zeta)} = & (1 - q_t^{(\zeta)}) dN_t^{IS_{\zeta}} + dN_t^{(LO_{\zeta})} \\ & - (\mathbb{1}(q_t^{(\zeta)} > 1) + \mathbb{1}(q_t^{(\zeta)} = 1)(q_t^{(\zeta, D)} - q_t^{(\zeta)})) \\ & \times (dN_t^{(CO_{\zeta})} + dN_t^{(MO_{\zeta})}) \end{aligned} \quad (\text{E.6})$$

$$\begin{aligned} \text{2nd quote at } \zeta \equiv q_t^{(\zeta, D)} \text{ s.t. } dq_t^{(\zeta, D)} = & (q_t^{(\zeta, D)} - q_t^{(\zeta)}) dN_t^{IS_{\zeta}} \\ & + dN_t^{(LO_{\zeta}^D)} + dN_t^{(CO_{\zeta}^D)} \end{aligned} \quad (\text{E.7})$$

$$\text{Mid-Price} \equiv P_t^{(mid)} \text{ s.t. } dP_t^{(mid)} = \frac{dp_t^{(a)} + dp_t^{(b)}}{2} \quad (\text{E.8})$$

$$\begin{aligned} \text{Queue Priority at } \zeta \equiv n_t^{(\zeta)} \text{ s.t. } dn_t^{(\zeta)} = & dN_t^{IS_{\zeta}} - dN_t^{(MO_{\zeta})} - \frac{n_t^{(\zeta)}}{q_t^{(\zeta)}} dN_t^{(CO_{\zeta})} \\ & - \mathbb{1}(n_t^{(\zeta)} > q_t^{(\zeta)}) \frac{n_t^{(\zeta)} - q_t^{(\zeta)}}{q_t^{(\zeta, D)}} dN_t^{(CO_{\zeta}^D)} \\ & + \mathbb{1}(n_t^{(\zeta)} = 0) \tilde{n}_t^{(\zeta)} dN_t^{(MO_{\zeta})} \end{aligned} \quad (\text{E.9})$$

Appendix F.

State-Intervention Operator

1. $LO_T^{(\zeta)}$

$$\begin{aligned} n^{(\zeta)}(\tau_i) &= n^{(\zeta)}(\tau_i^-) \mathbb{1}(n^{(\zeta)}(\tau_i^-) \leq q^{(\zeta)}(\tau_i^-)) \\ &\quad + q^{(\zeta)}(\tau_i^-) \mathbb{1}(n^{(\zeta)}(\tau_i^-) > q^{(\zeta)}(\tau_i^-)) \end{aligned} \quad (\text{F.1})$$

$$q^{(\zeta)}(\tau_i) = q^{(\zeta)}(\tau_i^-) + 1 \quad (\text{F.2})$$

2. $LO_D^{(\zeta)}$

$$\begin{aligned} n^{(\zeta)}(\tau_i) &= n^{(\zeta)}(\tau_i^-) \mathbb{1}(n^{(\zeta)}(\tau_i^-) \leq q^{(\zeta)}(\tau_i^-) + q^{(\zeta,D)}(\tau_i^-)) \\ &\quad + (q^{(\zeta)}(\tau_i^-) + q^{(\zeta,D)}(\tau_i^-)) \\ &\quad \times \mathbb{1}(n^{(\zeta)}(\tau_i^-) > q^{(\zeta)}(\tau_i^-) + q^{(\zeta,D)}(\tau_i^-)) \end{aligned} \quad (\text{F.3})$$

$$q^{(\zeta,D)}(\tau_i) = q^{(\zeta,D)}(\tau_i^-) + 1 \quad (\text{F.4})$$

3. $LO_{IS}^{(\zeta)}$

$$n^{(\zeta)}(\tau_i) = 0 \quad (\text{F.5})$$

$$q^{(\zeta)}(\tau_i) = 1 \quad (\text{F.6})$$

$$q^{(\zeta,D)}(\tau_i) = q^{(\zeta)}(\tau_i^-) \quad (\text{F.7})$$

$$p^{(\zeta)}(\tau_i) = p^{(\zeta)}(\tau_i^-) - z^{(\zeta)} \quad (\text{F.8})$$

$$P^{mid}(\tau_i) = P^{mid}(\tau_i^-) - \frac{z^{(\zeta)}}{2} \quad (\text{F.9})$$

4. $CO_T^{(\zeta)}$: only possible when there are some orders in the LOB that are the

agent's.

$$n^{(\zeta)}(\tau_i) = \tilde{n}^{(\zeta)}(\tau_i^-) \quad (\text{F.10})$$

$$\begin{aligned} q^{(\zeta)}(\tau_i) &= (q^{(\zeta)}(\tau_i^-) - 1)\mathbb{1}(q^{(\zeta)}(\tau_i^-) > 1) \\ &\quad + q^{(\zeta,D)}(\tau_i^-)\mathbb{1}(q^{(\zeta)}(\tau_i^-) = 1) \end{aligned} \quad (\text{F.11})$$

$$p^{(\zeta)}(\tau_i) = p^{(\zeta)}(\tau_i^-) + z^{(\zeta)}\mathbb{1}(q^{(\zeta)}(\tau_i^-) = 1) \quad (\text{F.12})$$

$$P^{mid}(\tau_i) = P^{mid}(\tau_i^-) + \frac{z^{(\zeta)}}{2}\mathbb{1}(q^{(\zeta)}(\tau_i^-) = 1) \quad (\text{F.13})$$

5. $MO^{(\zeta)}$: disabled if the top order on the resp. side is the agent's.

$$n^{(\zeta)}(\tau_i) = n^{(\zeta)}(\tau_i^-) - 1 \quad (\text{F.14})$$

$$\begin{aligned} q^{(\zeta)}(\tau_i) &= (q^{(\zeta)}(\tau_i^-) - 1)\mathbb{1}(q^{(\zeta)}(\tau_i^-) > 1) \\ &\quad + q^{(\zeta,D)}(\tau_i^-)\mathbb{1}(q^{(\zeta)}(\tau_i^-) = 1) \end{aligned} \quad (\text{F.15})$$

$$p^{(\zeta)}(\tau_i) = p^{(\zeta)}(\tau_i^-) + z^{(\zeta)}\mathbb{1}(q^{(\zeta)}(\tau_i^-) = 1) \quad (\text{F.16})$$

$$P^{mid}(\tau_i) = P^{mid}(\tau_i^-) + \frac{z^{(\zeta)}}{2}\mathbb{1}(q^{(\zeta)}(\tau_i^-) = 1) \quad (\text{F.17})$$

$$X(\tau_i) = X(\tau_i^-) + z^{(\zeta)}p^{(\zeta)}(\tau_i^-) \quad (\text{F.18})$$

$$Y(\tau_i) = Y(\tau_i^-) - z^{(\zeta)} \quad (\text{F.19})$$

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